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UNIVERSIDADE Ð COIMBRA

Vasco Tomás Ramos Rodrigues

A Clinician-Validated LLM Pipeline for Structuring Oncology Data and Clinical Trial Eligibility Matching

Dissertation in the context of the Master in Informatics Engineering, Artificial Intelligence, advised by Professor Pedro Furtado and presented to the Department of Informatics Engineering of the Faculty of Sciences and Technology of the University of Coimbra.

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Uma Pipeline de LLM Validada por Clínicos para Estruturar Dados Oncológicos e Avaliar a Elegibilidade de Doentes para Ensaios Clínicos

Dissertação no âmbito do Mestrado em Engenharia Informática, especialização em Inteligência Artificial, orientada pelo Professor Pedro Furtado e apresentada ao Departamento de Engenharia Informática da Faculdade de Ciências e Tecnologia da Universidade de Coimbra.

julho 2026

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I would also like to thank the Department of Digital Transformation and Innovation of ULS Coimbra for providing me with this opportunity and for continuously supporting the project, showing confidence in its potential. In particular, I would like to express my sincere appreciation to Dr. Margarida Marques and Eng. Adelino Pais, who participated in almost every project meeting, helping me rethink decisions, providing valuable insights, and ensuring that the developed solution remained aligned with the real needs and constraints of the hospital environment.

A special acknowledgement goes to Dr. Judy Paulo, who was undoubtedly an essential part of this project's success. Her constant availability, clinical expertise, and genuine enthusiasm towards the project were invaluable throughout its development. Her involvement not only contributed significantly to improving the final solution but also motivated me to continue striving for a better outcome.

I would also like to thank my family and friends, whose support was fundamental throughout this journey. Their encouragement, trust, and constant belief in my abilities helped me overcome challenges and reminded me that this project represented a valuable opportunity to apply all the effort, knowledge, and dedication I have built throughout my academic path.

My sincere gratitude also goes to everyone at ULS Coimbra for this internship opportunity, which contributed significantly to my personal and professional growth, providing me with valuable knowledge and a deeper understanding of the realities and challenges of the professional world.

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Without the support and contributions of everyone mentioned above, this project could have been completed, but certainly not with the same level of quality and success that I believe it achieved. For that, I am truly grateful. Thank you all.

Statement on the use of artificial intelligence tools

This document was reviewed and refined with the assistance of Large Language Models, such as ChatGPT, or similar tools, which helped check grammar, correct typos, and enhance clarity. Any Artificial Intelligence-generated text or content is clearly indicated within the document and is used only to a limited extent. The overall content and ideas remain solely the responsibility of the author.

Abstract

Oncology departments generate large volumes of unstructured clinical data, complicating patient–trial matching and contributing to the under-enrolment of eligible patients in clinical trials. While Large Language Models (LLMs) offer promising solutions, existing approaches often rely exclusively on probabilistic reasoning, limiting transparency and reproducibility. This work proposes an integrated, hybrid system that leverages LLMs to transform unstructured oncology notes into structured, clinician-validated profiles and support transparent eligibility assessment.

The system operates in two core phases: (i) an LLM-based pipeline that extracts and normalizes clinical parameters from free-text notes for expert validation, and (ii) an automated module that converts trial criteria into machine-readable logical rules. Eligibility is then evaluated through a hybrid mechanism, combining deterministic rule-based execution for structured fields with LLM-assisted reasoning for complex, unstructured clinical context. This architecture balances the auditability of rule-based logic with the flexibility of generative models.

The system was evaluated using three open-source 8B-parameter LLMs across five pipeline tasks. Structured extraction yielded strong results, with the best model achieving 92.3% accuracy in criteria extraction and 78.3% in patient profile extraction. Transitioning clinical language into formal logic proved the most challenging task, with strict accuracies between 52.2% and 64.1%. Nevertheless, the final patient–trial matching stage demonstrated clinically reliable performance; the top models achieved up to 84.0% accuracy and expert-rated justification quality scores of 4.93 out of 5, ensuring decisions were grounded in clinical evidence. These findings demonstrate that modular pipelines combining structured representations, deterministic logic, and lightweight LLMs can support transparent, efficient, and clinically aligned patient recruitment in real-world oncology settings.

Keywords

Large Language Models; Information Extraction; Clinical Trial Eligibility; Unstructured Clinical Data; Clinician-in-the-Loop; Deterministic Eligibility Evaluation; Survival Analysis; Structured Patient Data

Resumo

Os serviços de oncologia produzem grandes volumes de dados clínicos não estruturados, o que dificulta o processo de correspondência entre doentes e ensaios clínicos e contribui para a sub-inscrição de doentes elegíveis. Embora os Large Language Models (LLMs) ofereçam soluções promissoras, muitas abordagens existentes dependem exclusivamente de raciocínio probabilístico, limitando a transparência e a reprodutibilidade. Este trabalho propõe um sistema híbrido e integrado que utiliza LLMs para transformar notas clínicas não estruturadas em perfis estruturados e validados por clínicos, apoiando uma avaliação de elegibilidade transparente.

O sistema funciona em duas fases principais: (i) uma pipeline baseada em LLMs que extrai e normaliza parâmetros clínicos a partir de texto livre para validação por especialistas, e (ii) um módulo automatizado que converte critérios de ensaios clínicos em regras lógicas legíveis por máquina. A elegibilidade é depois avaliada através de um mecanismo híbrido que combina execução determinística baseada em regras para campos estruturados com raciocínio assistido por LLMs para contexto clínico complexo e não estruturado. Esta arquitetura equilibra a auditabilidade da lógica determinística com a flexibilidade dos modelos generativos.

O sistema foi avaliado utilizando três LLMs open-source de 8 mil milhões de parâmetros, aplicados a cinco tarefas da pipeline. A extração estruturada apresentou resultados sólidos, com o melhor modelo a atingir 92,3% de accuracy na extração de critérios e 78,3% na extração de perfis de doentes. A conversão de linguagem clínica em lógica formal revelou-se a tarefa mais desafiante, com strict accuracies entre 52,2% e 64,1%. Ainda assim, a fase final de correspondência entre doente e ensaio demonstrou desempenho clinicamente fiável; os melhores modelos alcançaram até 84,0% de accuracy e pontuações de qualidade de justificação avaliadas por especialistas de 4,93 em 5, garantindo que as decisões estavam fundamentadas em evidência clínica. Estes resultados demonstram que pipelines modulares que combinam representações estruturadas, lógica determinística e LLMs leves podem apoiar processos de recrutamento de doentes transparentes, eficientes e alinhados com a prática clínica em contextos reais de oncologia.

Palavras-Chave

Modelos de Linguagem de Grande Escala; Extração de Informação; Elegibilidade para Ensaios Clínicos; Dados Clínicos Não Estruturados; Clínico-no-Ciclo; Avaliação Determinística de Elegibilidade; Análise de Sobrevivência; Dados Estruturados de Pacientes

Contents

Acronyms

AI Artificial Intelligence.

DB DataBase.

DR Data Requirement.

ECOG-PS Eastern Cooperative Oncology Group Performance Status.

FFN Feedforward Neural Network.

FR Functional Requirement.

IE Information Extraction.

IR Interface Requirement.

JSON JavaScript Object Notation.

KM Kaplan–Meier.

LLM Large Language Models.

ML Machine Learning.

NER Named Entity Recognition.

OS Overall Survival.

PCA Principal Component Analysis.

PDF Portable Document Format.

PFS Progression-Free Survival.

RNN Recurrent Neural Network.

SDoH Social Determinants of Health.

SPSS Statistical Package for the Social Sciences.

UI User Interface.

ULS Unidade de Saúde Local.

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Chapter 1

Introduction

This dissertation reports the development of a clinical decision-support system carried out in collaboration with the Unidade Local de Saúde (ULS) de Coimbra, specifically within the oncology department. The project was conducted in a real-world hospital context, addressing practical challenges faced by oncologists in their daily clinical and research activities. Its primary objective is to support the structuring of unstructured patient clinical data and to enable its systematic comparison with clinical trial eligibility criteria, with the aim of improving efficiency in patient screening and trial enrolment processes.

The work was developed in close collaboration with an oncology specialist, Dra. Judy, who acted as the main clinical stakeholder. Her expertise and continuous feedback were fundamental in shaping the system requirements, validating the clinical relevance of the extracted data, and ensuring alignment with real clinical workflows. The target users of the proposed system are oncology clinicians involved in patient follow-up and clinical research, particularly those responsible for identifying and enrolling eligible patients into clinical trials.

By being grounded in an operational oncology department and informed by direct clinical input, this project seeks to bridge the gap between advanced Artificial Intelligence techniques and their practical applicability in hospital environments, ensuring that the proposed solution remains clinically meaningful, usable, and ethically aligned with existing medical decision-making processes.

1.1 Motivation

Oncology departments handle a vast number of patients on a daily basis, generating large volumes of complex and unstructured clinical data. This reality, common across many healthcare systems, is characterized by the extensive use of free-text clinical notes, resulting in a highly unstandardized set of medical records. This poses significant challenges for information management, as critical data required for patient monitoring and research support remains embedded in formats that are difficult to process and analyze.

Oncologists, who already operate under high cognitive and emotional demands, face a substantial administrative burden due to the manual extraction and interpretation of this information. This contributes to operational inefficiency and clinician burnout. One of the most critical consequences of this scenario is the persistent under-enrollment of eligible patients in clinical trials. When patient information recorded in clinical diaries cannot be effectively matched with structured trial eligibility criteria, institutions risk financial penalties and miss opportunities to advance clinical research.

This project aims to address these challenges through the use of Large Language Models (LLMs), which offer a promising approach to processing unstructured textual data. Unlike traditional solutions that rely exclusively on structured inputs, LLMs are capable of interpreting free-text clinical narratives, despite variations in writing style and the presence of inconsistencies. By leveraging LLMs to support the extraction, normalization, and analysis of unstructured oncology data, this project seeks to assist clinicians in the identification of patients eligible for clinical trials. The ultimate goals are to reduce the burden of manual patient screening and to contribute to a more efficient and data-driven oncology research workflow.

1.2 As-Is Flow

Currently, the process of identifying eligible patients for clinical trials is highly repetitive and predominantly manual. Clinicians are required to compare each patient against multiple clinical trials, each containing extensive inclusion and exclusion criteria. This task is often performed alongside routine clinical activities and is frequently interrupted by consultations and other duties. Such interruptions force clinicians to repeatedly revisit trial documentation, further reducing efficiency.

The lack of standardized clinical records significantly exacerbates this problem. As a result, clinicians spend a considerable amount of time performing administrative tasks that could otherwise be allocated to direct patient care or research-related activities.

Given this context, there is a clear need to standardize patient clinical data and to support the automated comparison between patient profiles and clinical trial eligibility criteria.

All that was described can be seen on the Figure 1.1, it is important to mention that the not filled circle represents the initiation of the process, and the filled circle represents the end of the process.

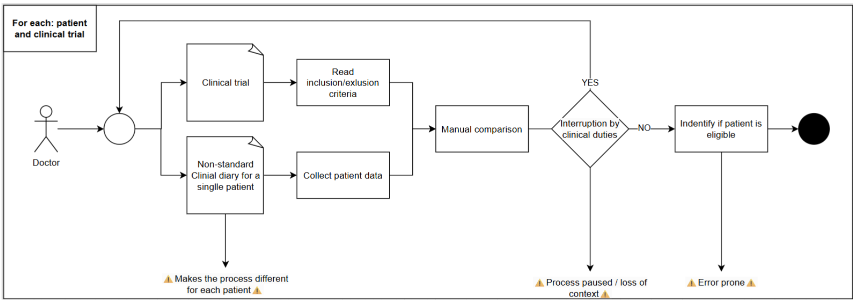


Figure 1.1: Current workflow for patient screening

1.3 To-Be Flow

The proposed solution addresses the limitations of the current workflow by introducing a modular, semi-automated pipeline supported by Large Language Models and clinician validation at key stages. The system decomposes the complex patient–trial matching problem into a sequence of specialized tasks, each with well-defined inputs, outputs, and evaluation criteria.

The pipeline begins with the transformation of unstructured clinical diaries into structured patient profiles through an LLM-based extraction process. These extracted profiles are reviewed and validated by the clinician before being stored in a centralized database, eliminating the need for repeated manual interpretation in subsequent analyses.

In parallel, clinical trial documentation is processed to identify its structure and to extract inclusion and exclusion criteria. These criteria are subsequently formalized into machine-interpretable logical representations, also subject to clinician validation to ensure fidelity to the original trial requirements.

Patient–trial matching is performed through a hybrid mechanism that combines deterministic rule-based evaluation for criteria that can be directly assessed against the structured patient schema, with targeted LLM-assisted reasoning for criteria that require interpretation of broader clinical context. This design allows structured and objectively measurable criteria to be evaluated transparently while preserving the flexibility to handle complex or schema-independent eligibility requirements.

The system outputs a list of potentially eligible patients, accompanied by detailed justifications explaining the reasons for inclusion or exclusion. These results are made available to clinicians through the system interface, supporting transparency, traceability, and informed decision-making.

All that was described can be seen on the Figure 1.2. it is important to mention that the not filled circle represents the initiation of the process, and the filled circle represents the end of the process.

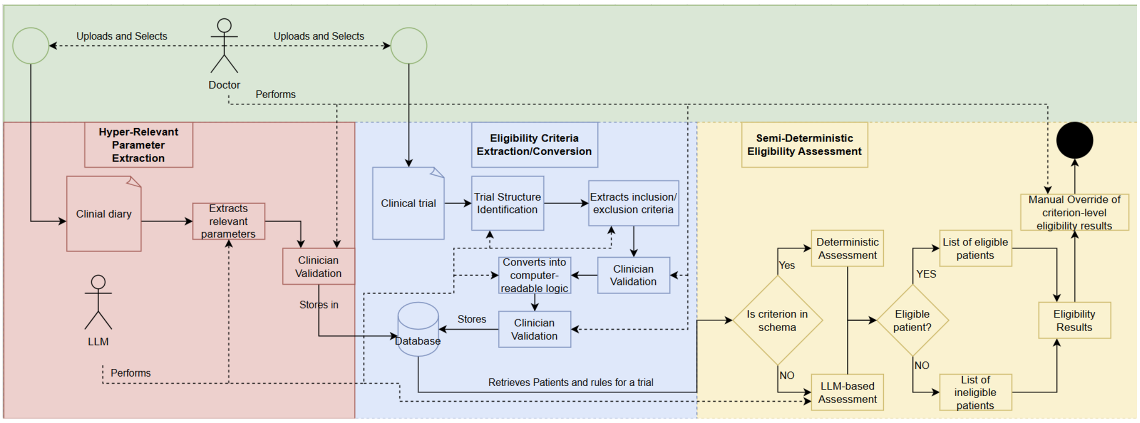


Figure 1.2: Proposed solution workflow

By decomposing the problem into specialized stages, incorporating clinician-in-the-loop validation, and combining deterministic evaluation with targeted LLM assistance, the proposed workflow significantly reduces manual workload, improves transparency and auditability, and streamlines the clinical trial enrollment process, while preserving full clinician control over all critical decisions.

Importantly, the solution is designed strictly as a clinical decision-support system and does not perform autonomous clinical decision-making.

1.4 Contributions

The main contributions of this dissertation can be summarized as follows:

- The design and implementation of a modular LLM-based pipeline for the identification of potentially eligible patients for clinical trials, decomposing the patient–trial matching problem into a sequence of specialized tasks with well-defined inputs, outputs, and evaluation criteria.
- An LLM-based extraction pipeline for transforming unstructured oncology clinical diaries into structured, clinician-validated patient profiles, including the identification and normalization of clinically relevant parameters.
- A semi-automated process for extracting and formalizing clinical trial inclusion and exclusion criteria from free-text documentation into machine-interpretable logical representations, subject to clinician validation.
- A hybrid patient–trial matching mechanism that combines deterministic rule-based evaluation for criteria directly assessable against the structured patient schema with targeted LLM-assisted reasoning for criteria requiring interpretation of broader clinical context, ensuring transparency, reproducibility, and auditability of eligibility decisions.
- The integration of clinician-in-the-loop validation at multiple stages of the pipeline, ensuring clinical correctness and interpretability throughout the workflow.

- An experimental evaluation of lightweight open-source LLMs (approximately 8B parameters) across all pipeline components in a real hospital context, demonstrating feasibility and identifying task-specific performance patterns, with results indicating strong performance in structured extraction tasks, greater difficulty in logical formalization of eligibility criteria, and competitive performance in the final eligibility assessment stage when grounded in explicit clinical context.

- The development of a clinically validated synthetic dataset based on real oncology clinical documentation patterns, enabling realistic evaluation under data privacy constraints while maintaining clinical realism and reflecting the deployment conditions of healthcare environments.

1.5 Structure

The remainder of this document is organized as follows.

Related Work presents an analysis of the state of the art relevant to the problem addressed in this dissertation.

Methodology and Planning details the methodological approach adopted for the development of the system, including the project management framework employed. It also describes the planning and management of the project, the tasks carried out during its development, and their temporal organization, illustrated through a Gantt diagram, as well as the activities planned for the next semester. Furthermore, it identifies the main risks associated with the project and outlines the mitigation strategies defined to ensure its successful completion.

Approach describes the proposed solution in detail and the design decisions that guided its implementation.

Requirements presents the requirements analysis, discussing the needs, constraints, and expectations of the main stakeholder, as well as the formal specification of the system's requirements.

Architecture introduces the C4 model for the software architecture, formalizing it from the context level down to the component level.

Experimental Setup defines and guides the experiments and exploratory analysis planned for the next semester.

Conclusion provides a brief summary of the work completed and the activities to be carried out in the following semester.

Chapter 2

Methodology and Project Planning

In this chapter, the methodology adopted for the development of the project is formally presented. The structure of the sprints, the management tools used to monitor progress, and the overall workflow are described in detail. Additionally, a clear overview of the work completed during the first phase is provided, supported by an illustrative diagram. This chapter also outlines the planning for the upcoming phase of the project and concludes with an analysis of the main project risks and the corresponding mitigation strategies.

2.1 Methodology

2.1.1 Sprint structure and ceremonies

The development process followed an iterative workflow inspired by agile principles, where tasks were organized into incremental development cycles supported by regular progress meetings. These meetings provided a structured environment for presenting the work completed, discussing encountered challenges, and defining the priorities for the following development period. Each meeting represented an opportunity to validate the current direction of the project, receive feedback from the stakeholders, and ensure that the developed solution remained aligned with both technical objectives and clinical requirements.

Throughout the project, the supervisor and stakeholders maintained an active involvement in the development process. Professor Pedro Furtado continuously reviewed and challenged the technical decisions made, helping to refine the proposed solutions and maintain a critical perspective over the implemented approaches. Additionally, Eng. Adelino Pais, Dr. Margarida Marques, and Dr. Judy Paulo provided valuable feedback from both organizational and clinical perspectives, ensuring that the system evolved according to the practical needs of the hospital environment.

The development process was structured around successive cycles of analysis, implementation, validation, and refinement. Initial iterations focused on understanding the problem domain, defining the system requirements, and establishing the overall architecture. As development progressed, the focus shifted towards implementing the main system modules, evaluating different approaches, and incorporating feedback obtained from stakeholder interactions. This iterative process allowed previously defined components to be continuously improved as new requirements and insights emerged.

During the final phase of the project, three major validation meetings were conducted with the participation of Dr. Margarida Marques and Dr. Judy Paulo. These meetings focused on presenting and discussing the developed modules: Hyper-Relevant Parameter Extraction, Clinical Trial Eligibility Criteria Extraction and Conversion, and Semi-Deterministic Eligibility Assessment. These validation sessions were essential for assessing the practical relevance of the implemented functionalities, identifying potential improvements, and guiding the remaining development activities according to stakeholder feedback.

2.1.2 Tools and Tecnologies

The following tools and technologies were used to develop the proposed project:

- Django structured the backend and overall project, following the Model-Template-View (MTV) pattern. It was a supervisor requirement, as the company was already familiar with it; alternatives considered included Flask, Ruby, and Laravel.
- Visual Studio Code served as the main IDE, chosen for its project directory navigation and error-highlighting extensions for HTML, Django, and Python; alternatives included Atom, PyCharm, and Sublime Text.
- Python was the backend language, used by necessity since Django is Python-based.
- HTML structured the platform's page templates, while CSS handled styling, allowing reuse of existing code to speed up development.
- JavaScript was used alongside Bootstrap mainly to support the application's responsiveness.
- Bootstrap accelerated interface design through pre-built styles and components. It was chosen due to prior familiarity from both academic experience and the company team's recommendation; alternatives included Foundation, Materialize, and Bulma.
- Trello was used for task tracking and project management, helping organize work and monitor progress throughout development.
- GanttProject was used to plan and visualize the task schedule for efficient time and resource management.

- Obsidian supported requirements analysis through quick, intuitive note-taking, chosen for the development team's familiarity with it; alternatives included Notion and Windows Notepad.
- Github was used for version control and for tracking every change made to the project's codebase, ensuring collaboration, traceability, and safe iteration.

2.2 Work Conducted in the First Semester

During the first phase of this thesis, a substantial amount of work was carried out, primarily focused on formalizing documentation and specifications to guide the development of the project in the following semester. The initial weeks of the project were mainly dedicated to defining the project scope and studying the relevant related work.

While the project definition was being refined, an initial study on parameter extraction from unstructured clinical diaries was already underway. Once the project scope was clearly defined, the topic of patient–clinical trial matching was explored in more depth, leading to the creation of the related work document. In parallel with this second phase of the related work, a study on Large Language Models was conducted, focusing on their evolution and capabilities. During this period, several meetings were held with the stakeholder to better understand the current clinical workflow and the underlying motivation for the project.

Based on these insights, the LLM-based workflow of the proposed solution was conceived and defined. Following this, three refined modules were established: the unstructured patient parameter extraction workflow, the clinical trial matching workflow, and a study on how patient care monitoring should be supported within the proposed solution. To provide a tangible sense of progress to the stakeholder, interface mockups were developed.

As a result of these activities, the overall software architecture was formally documented. Throughout this entire phase, the specification of requirements was continuously updated, as it evolved alongside the project definition and required constant refinement. Finally, a generalized structural analysis of both clinical diaries and clinical trial documents was performed. The final segment of this first phase was fully dedicated to the formalization of the thesis document.

All the activities described above are summarized in the Gantt diagram presented in Figure 2.1.

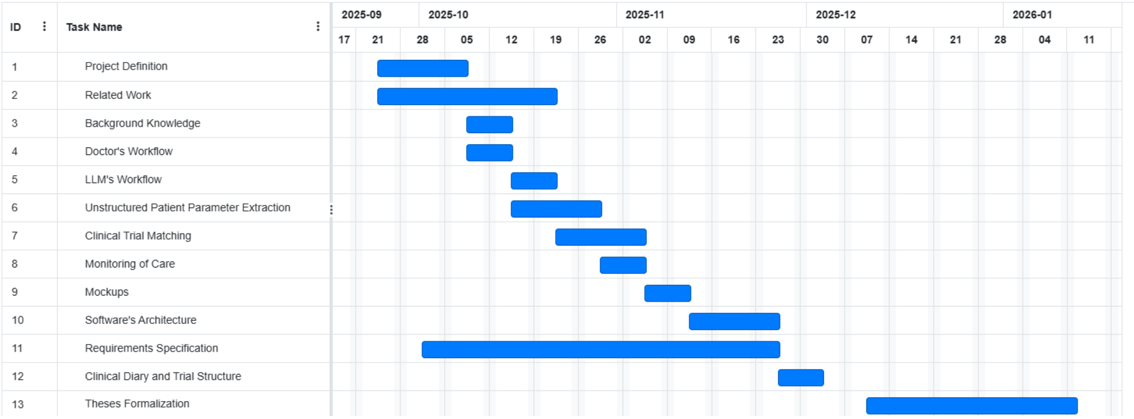


Figure 2.1: Gantt Diagram of first semester

2.3 Work Conducted in the Second Semester

During the second phase of this thesis, the main focus shifted from project definition and specification towards system implementation, experimentation, and validation. As initially planned, the finalization and confirmation of the requirements were completed, with the adoption of a MoSCoW prioritization methodology to guide the development efforts and ensure that the most relevant functionalities were addressed first.

Following the confirmation of the requirements, synthetic data generation was conducted to provide the necessary resources for the development and evaluation phases. This process focused on creating representative clinical diaries that would support the implementation, testing, and experimentation of the proposed solution.

With the requirements finalized and the dataset prepared, software development could begin. The five main modules of the system were progressively implemented, covering the core functionalities defined throughout the project. Alongside development, a comprehensive unit testing strategy was established, allowing the identification and correction of implementation issues and ensuring the reliability of the developed components.

In parallel with the testing activities, the experimental phase was designed and conducted. Three different Large Language Models were evaluated in order to assess their capabilities across the five main tasks supported by the application. These experiments provided relevant insights regarding the applicability of LLM-based approaches within the proposed clinical workflow.

Simultaneously, the scientific article associated with this work was developed (see Appendix **??** for the full manuscript). Although this activity was not considered in the initial planning, it contributed positively to the overall thesis development process. The experimental results obtained during this phase required additional analysis and documentation, but the preparation of the article also facilitated the revision and updating of several thesis sections, ultimately simplifying the finalization process.

Throughout the semester, significant effort was also dedicated to producing supporting technical documentation, including the Entity Relationship diagram specification, unit testing documentation, and web application specification. These artifacts contributed to improving the traceability, maintainability, and overall organization of the developed system.

Finally, after completing the implementation and experimentation phases, the finalization of the thesis document was performed. This included the revision and update of previously written sections to ensure consistency with the final version of the implemented solution.

Although some deviations occurred, they were mainly related to changes introduced during the development process and contributed to improving the final outcome.

The experimentation phase was postponed compared to the initial planning in order to prioritize the delivery of tangible system results to the stakeholder at an earlier stage. This decision ensured continuous stakeholder engagement and allowed the developed application to be validated before conducting the final experimental evaluation.

Additionally, the development of the web application required more time than initially anticipated. This was mainly due to the introduction of additional functionalities identified during requirements refinement and implementation. In particular, the identification of trial structures required a significant refactoring of both the criteria extraction process and the patient matching workflow and the Eligibility Criteria Conversion was much more complex than previously anticipated. Furthermore, the initial assumption of a fully deterministic patient matching approach was reconsidered, leading to the adoption of a semi-deterministic strategy after identifying the existence of complex eligibility criteria that could not be directly represented within the patient schema.

The preparation of the scientific article also represented an additional activity not included in the initial planning. However, this effort proved beneficial, as it encouraged continuous refinement of the experimental analysis and facilitated the final consolidation of the thesis document.

Every document and line of code produced is present on the following Github Repository.

Everything describe above, can be seen on the Figures 2.2 and 2.3.

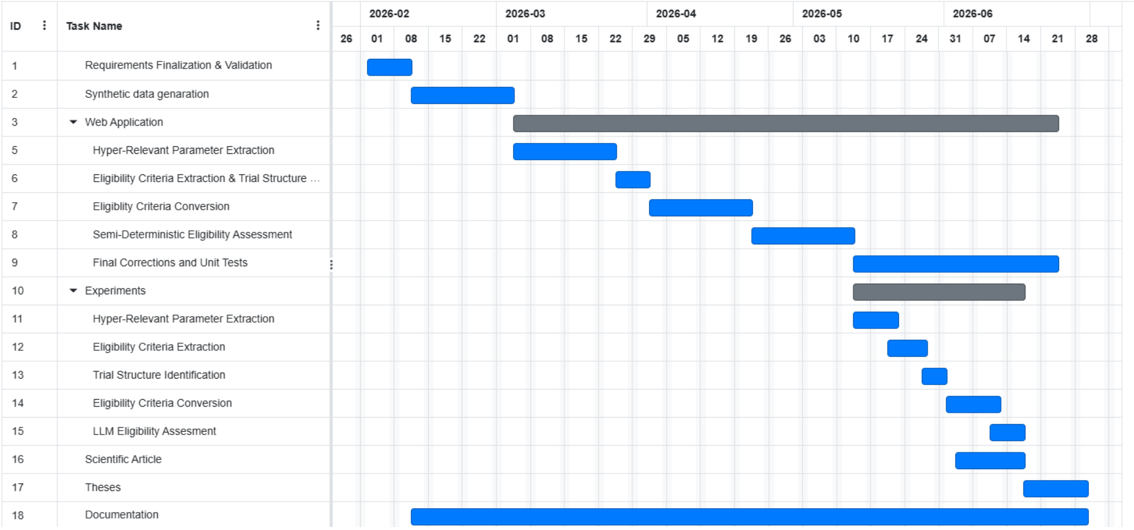


Figure 2.2: Gantt Diagram of second semester - Real

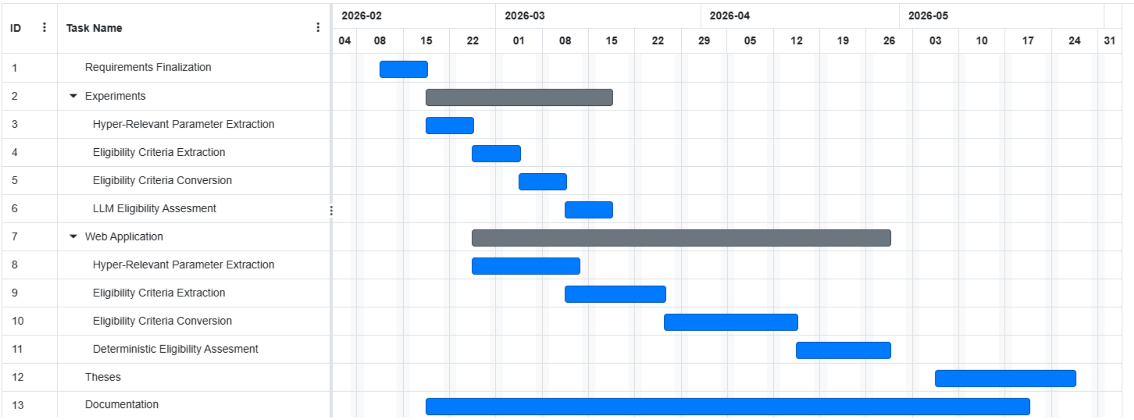


Figure 2.3: Gantt Diagram of second semester - Predicted

2.4 Risk Management

The development of this project was subjected to a set of inherent risks to the exploratory nature and technological context associated. This risk are related with technical factors, limitations of resources or external dependencies, being able to impact the project's success and progress.

The following risk table has the below legend for its columns:

- ID, unique identifier for the Risk
- Risk, the risk itself, formalized on a affirmation → consequence manner
- I, impact that the risk will have on the project if it happens, this metric is evaluated on a 1-3 scale, where 1 corresponds to a negligible risk and 3 to a critical effect

- P, probability of the risk happening, this metric is evaluated on a 1-3 scale, where 1 corresponds to a unlikely event and 3 to a highly likely event
- Sev, severity of the risk calculated as Impact × Probabilty, making severity a 1-9 scale

Bellow there is a comprehensional table of the scales for the 3 last described columns, Table 2.1:

Metric	**Class**	**Criteria**
Impact	1	Negligible, minimal disruption, no effect on deliverables
	2	Moderate, noticeable impact requiring corrective action
	3	Critical, severe impact threatening project success
Probability	1	Very unlikely, low chance of occurring
	2	Moderate, could occur during the project
	3	Very likely, high chance of occurring
Severity	1–2	Low, limited effect on project execution
	3–4	Moderate, requires management attention
	6–9	High, potential to compromise project success

Table 2.1: Risk Assessment Metrics and Classification Criteria

Bellow the Table 2.2 presents the main risks identified, along with their impact, probability of occurrence, severity and mitigation plan defined to bridge the risk.

ID	Risk	I	P	Sev	MP
R1	Limited access to real clinical data for experimentation. Because medical records are subject to strict ethical, legal, and privacy constraints, obtaining large and diverse datasets of unstructured clinical diaries and patient profiles may not be possible.	3	2	6	Synthetic clinical data will be generated to simulate realistic patient profiles and clinical trial scenarios. This synthetic data will be designed to reflect real-world distributions, clinical constraints, and oncology-specific characteristics such as staging, treatment timelines, and survival outcomes.
R2	Dependence on Large Language Models to extract structured information from clinical diaries and to convert clinical trial criteria into logical rules may produce incorrect, incomplete, or ambiguous outputs, especially when dealing with complex medical terminology or poorly structured text.	3	3	9	Clinician validation steps for both extracted patient information and translated eligibility criteria must be mandatory. These checkpoints ensure that no automated decision is made without expert review.
R3	Large Language Models may struggle with highly specialized oncology terminology, ambiguous phrasing, or institution-specific writing styles, affecting the accuracy of patient parameter extraction and the translation of eligibility criteria.	2	2	4	Conduct preliminary tests with representative clinical content, refine prompt templates iteratively, and incorporate domain-specific examples to improve extraction consistency.
R4	Running large models locally may exceed available computational resources, leading to slow inference times or making certain experiments infeasible.	2	2	4	Use optimized execution environments, select model sizes appropriate to the available hardware, and offload heavier experiments to cloud-based environments when necessary.
R5	The project and experimentation require clinician involvement to validate extracted parameters, assess the correctness of eligibility criteria translations, and ensure alignment with real clinical workflows. Limited clinician availability may delay validation cycles.	2	1	2	Schedule validation sessions in advance, define objective validation criteria, and maintain short, focused review iterations to minimize clinician workload.

Table 2.2: Project Risks and Mitigation Strategies

Chapter 3

Related Work

In this chapter, the relevant scientific literature underpinning the project is reviewed. Each selected paper is briefly summarized, highlighting its main contributions, limitations, and results, thereby establishing a clear foundation for identifying the gaps addressed by the proposed solution. The chapter is organized around the two core components of the system: the extraction of unstructured clinical information using large language models (LLMs) and the matching of patients to clinical trials through LLM-based methods. For each of these areas, a concise synthesis is provided, outlining the key insights and requirements that inform the design of the proposed solution.

Before starting the study it is important to have well defined these concepts, since they will be of major importance for the following description:

- Open-Source Models, an open-source model, is a completly transparent model, whose architecture, source code, training data and wieghts are made available for public use, therefore, emphasizing collaboration and redistribution, [xen];
- Open Models, an open model is a publicly accessible machine learning model that supports transparency, reuse, and collaborative improvement. although it doesn't guaranty full rights for modification and redistribution, [lyz, 2025];
- Closed/Proprietary Models, this is a model which is owned by a company, using private training data and resources. They are not publicly available and can only be acces via API's, [pro].

3.1 Extraction of Unstructured Clinical Information through LLMs

The extraction of meaningfull information from unstructured clinical data such as electronic health records (EHRs) is a fundamental challenge in medical informatics. Recent studies have explored the potential of large language models (LLMs) to automate and enhance this process, focusing on diverse tasks including entity recognition, summarization, and classification of clinical text. The following works provide an overview of the methods, results and limitations associated with these approaches

3.1.1 Evaluating LLM Training and Prompting Strategies

The paper [Vithanage et al., 2024] investigated various training and prompting methods for improving LLM accuracy in clinical contexts. Techniques such as zero-shot and few-shot prompting were tested in combination with Parameter-Efficient Fine-Tuning (PEFT) and Retrieval-Augmented Generation (RAG). The results showed that few-shot prompting outperformed zero-shot prompting when no additional techniques were applied, and that PEFT produced the largest performance gains, improving accuracy, precision, recall, and F1 by over 30% in the zero-shot setting. RAG offered smaller but still meaningful improvements, particularly in the few-shot configuration. Overall, PEFT proved to be the most impactful technique, offering strong guidance for future fine-tuning approaches to unstructured clinical text analysis.

3.1.2 Comparative Evaluation of LLMs for Information Extraction

The study [Vithanage et al., 2025] proposed a holistic evaluation framework for information extraction (IE) from free-text clinical notes. It systematically assessed different LLMs on Named Entity Recognition (NER) and summarization tasks using real-world unstructured data. In its optimal configuration, LLaMA 3.1-8B achieved high NER performance (≈88% accuracy and 90% F1) and strong summarization quality (≈88% F1) when combining RAG with few-shot prompting. The combined RAG setup (LangChain + LlamaIndex) consistently outperformed isolated pipelines, improving both NER and summarization metrics. Health domain specialized models showed only marginal advantages over general-purpose ones, with differences not reaching statistical significance. Despite strong scores, the study reported notable robustness challenges, with error rates rising from 38% to 49% under noisy conditions and false negatives representing the majority of errors. These findings highlight the need for domain adaptation and hybrid architectures to ensure reliable performance in sensitive medical settings.

3.1.3 Efficiency of LLMs Compared to Manual Review

In [Bhagat and Mackey], the efficiency of ChatGPT 3.5 was compared against manual data review for extracting clinical information from unstructured records. Despite being an out-of-the-box model, ChatGPT exhibited high specificity and sensitivity across several conditions—for example, it reached 1.00 sensitivity and 0.95 specificity in depression, 1.00 sensitivity and 0.94 specificity in heavy-smoker detection, and 0.85 sensitivity with 1.00 specificity for cancer-related information, while performance dropped for family history of cardiac disease (0.58 sensitivity, 0.97 specificity). These results show that the model could reliably identify most clinically relevant concepts, although nuanced semantic distinctions remained challenging. The study also reported stable outputs across repeated runs and noted that traditional systems such as MedTagger previously achieved 0.98 sensitivity and 0.96 specificity for cardiac-risk extraction. Overall, the findings suggest that LLMs can outperform classical NLP pipelines that require extensive manual engineering, and that targeted fine-tuning and structured extraction rules could further enhance their reliability in clinical settings.

3.1.4 Large-Scale Model Benchmarking for Data Extraction

The work [Ntinopoulos et al., 2025] assessed the performance of eighteen different LLMs using synthetic medical notes validated by specialists. The evaluation covered four entity-extraction tasks and five binary-classification tasks under a zero-shot setting. Output consistency was measured across repeated prompt iterations. Advanced LLMs such as Claude 3.0 Opus, Claude 3.0 Sonnet, Claude 2.0, and GPT-4 achieved overall accuracies above 0.98, with Claude 3.0 Opus reaching 0.995, and nine models achieving perfect accuracy (1.0) in entity extraction. In binary classification, Claude 3.0 Opus failed only 2 out of 250 tasks, while GPT-4 and Claude 2.0 failed 5 out of 250. These top models also showed strong detailed metrics—for example, Claude 3.0 Opus achieved an F1 of 0.992, and GPT-4 an F1 of 0.979—and high consistency, with Claude 2.0 reaching a Krippendorff's alpha of 1.0 across all iterations. Overall, these results demonstrate that modern LLMs substantially outperform baseline systems such as RoBERTa (accuracy 0.742) and can reliably extract structured information from EHRs, reducing clinicians' data-processing workload.

3.1.5 Extraction of Social Determinants of Health (SDoH)

The study [Gu et al., 2025] explored whether open-source LLMs could effectively extract Social Determinants of Health (SDoH) from clinical text. Two pipelines, Default and Refined, were implemented, with the latter incorporating prompt engineering to mitigate hallucinations. Manual annotations by human reviewers served as gold standards, with inter-annotator agreement reaching 93%. Results showed that LLMs outperformed the traditional pattern-matching baseline, which achieved 77% overall accuracy but only 39% accuracy on explicit mentions. In contrast, openchat 3.5 was the only model to surpass the baseline across all nine SDoH categories, improving direct-mention accuracy by over 40% and outperforming the baseline Macro-F1 score of 0.53–0.54. Models such as Llama-2 performed significantly worse than the baseline, while the refined pipeline reduced hallucinations and invalid outputs. Quantized variants remained robust under a 2048-token limit, although none of the models detected SDoH elements missed by human reviewers. These findings highlight the importance of prompt design for improving accuracy in unstructured clinical text extraction without fine-tuning.

3.1.6 Oncology Information Extraction with Expert-Annotated Data

The [Sushil et al., 2024] paper introduced a new annotated dataset (CORAL) for evaluating LLMs in oncological information extraction. The corpus contained 40 progress notes annotated hierarchically by oncology specialists, capturing entities, attributes, and relationships. Three models: GPT-4, GPT-3.5-turbo, and FLAN-UL2, were tested in a zero-shot setting on extracting clinical narrative elements sucha as symptoms, test results, and treatment plans.GPT-4 achieved the best overall progression, with manual evaluation confirming 68% accuracy. Nevertheless, hallucinations and inconsistent causal reasoning persisted, indicating that even state-of-the-art models are not yet suitable for direct clinical application. Moreover, the study did not explore the potential benefits of prompt engineering, leaving room for future enhancement

3.1.7 Summary

Across these studies, LLMs consistently demonstrate strong capabilities in extracting clinical information from unstructured data, frequently outperforming traditional NLP and rule-based systems, offering greater flexibility and reduced development effort. Techniques such as PEFT, RAG, and carefully engineered prompts emerge as key factors for improving accuracy and reliability. Hybrid frameworks combining multiple techniques have also demonstrated improved robustness and extraction accuracy when evaluated against expert-annotated gold standards. However, performance often degrades in noisy or highly contextual real-world clinical environments, and limitations persist in terms of hallucination control, inconsistent outputs, and limited robustness, which remain significant barriers to clinical deployment. The need for domain-specific fine-tuning, evaluation frameworks, and interpretability remains central to ensuring trustworthy clinical application.

3.2 Matching Patients for Clinical Trials with LLMs

The automation of patient-trial matching represents a major frontier in clinical informatics, aiming to accelerate recruitment, reduce operational costs, and ensure equitable access to medical research. Traditional matching pipelines are labor-intensive and constrained by the unstructured nature of electronic health records (EHRs). Recent studies have demonstrated that large language models (LLMs) can address these challenges by interpreting clinical narratives and eligibility criteria, offering explainability and scalibility through zero-shot and few-shot learning. This section presents a synthesis of recent works exploring LLM-based frameworks for clinical trial matching.

3.2.1 Zero-Shot Matching and the TrialGPT Framework

The paper [Jin et al., 2024] introduces TrialGPT a modular framework employing LLMs for zero-shot patient-trial matching. It consists of three interconnected modules:

1. TrialGPT-Retrieval, which filters a large pool of trials by extracting keywords from EHRs;
2. TriaçGPT-Matching, which evaluates patient eligibility criterion-by-criterion, generating natural language explanations, localizing relevant EHR phrases, and producing an eligibility classification; and
3. TrialGPT-Ranking, which aggregates eligibility predictions at the trial level to rank potential matches.

Two ranking strategies were proposed: Linear Aggregation, based on direct percentage computations for inclusion and exclusion criteria, and LLM Aggregation, which leverages LLMs to infer global relevance and eligibility. TrialGPT-Retrieval recalled 90% of relevant trials using only 6% of the total dataset, with the generated keywords surpassing those manually produced by experts. TrialGPT-Matching achieved an 87.3% accuracy, nearly matching human specialists (0.887-0.900), and demonstrated an F1 score of 88.6% in identifying relevant phrases. The outperforming competing systems by 43.8% in trial inclusion and exclusion classification. The study also estimated a 42.6% reduction in screening time, establishing TrialGPT as an efficient foundation for clinical trial automation.

3.2.2 Retrieval Pipelines and LLM-Based Re-Ranking

In [Rybinski et al., 2024], a retrieval-based pipeline was developed to match patients with suitable clinical trials. The process began with query formulation, representing each patient's profile as a query, using either a template-based or free-text representation generated by GPT-3.5-turbo. A BM25 model performed initial retrieval of 1,000 candidate trials, indexing both individual and aggregated fields of trial descriptions. This was followed by a re-ranking stage, where several strategies were applied:

- SCT-BERT, a fine tuned BERT variant specialized for clinical trial text;
- TCRR, inspired by curriculum learning, which first learns topical relevance and then fine-tunes on true eligibility;
- LLM-based embeddings, using semantic similarity between the patient and trials texts;
- LLM-based relevance scoring, where GPT-3.5-turbo directly produces numerical ranking scores;
- LLM-based filtering, applying binary yes/no filtering to the top candidates.

The results showed that fine-tuned LLM-based re-ranking achieved the highest retrieval precision, with the BM25-ES + TCRR + fine-tuned GPT pipeline reaching 0.777 nDCG@10, 0.697 P@10, and 0.783 RR, clearly outperforming the BM25 baseline (0.619 nDCG@10, 0.330 P@10). Zero-shot filtering with GPT achieved 0.720 nDCG@10 and 0.592 P@10, while GPT-4o delivered competitive zero-shot results, including 0.785 nDCG@10 and 0.844 RR, though with substantially higher latency. Latency varied widely across components, from 0.15 s for BM25 to 46.48 s for LLM re-ranking, with GPT-4o CoT reaching 242 s. The study highlighted the effectiveness–efficiency trade-off, noting that fine-tuning required around 10,000 training samples and increased latency roughly 15× for a 12% gain in nDCG@10. Overall, PLM-based re-rankers offered a more cost-efficient balance, while LLMs remained computationally demanding despite their superior matching quality.

3.2.3 Ontology-Enhanced Prompt-Baed Matching

The work [Rahmanian et al.] proposed an approach combining LLMs with structured ontologies such as SNOMED CT to refine eligibility extraction. Using GPT-3.5-turbo in a deterministic configuration (temperature = 0), the system automatically annotated clinical records with MedCAT, linking text to SNOMED CT and UMLS terminologies. Phrases containing relevant concepts were extracted to reduce input length while preserving essential information. Each eligibility criterion was converted into a question-based prompt for binary classification ("yes" or "no"), forming a zero-shot evaluation pipeline.

The model achieved micro and macro F-measures of 0.9061 and 0.8060, respectively placing it among the top-performing systems on the n2c2 dataset. without summarization, the performance dropped considerably (micro: 0.7126; macro: 0.5277), underscoring the value of preprocessing and semantic abstraction. Limitations included high false-negative rates caused by non-standard abbreviations, ontology gaps (e.g., missing synonyms and street drug names), and class imbalance in certain criteria. Despite these challenges, the study demonstrated the viability of ontology-enhanced prompting for improving interpretability and precision in patient eligibility classification.

3.2.4 Distilling Open-Source Models for Patient-Trial Matching

The paper [Nievas et al., 2023] presented the first systematic evaluation of open-source versus proprietary LLMs in this domain. The study compared GPT-3.5, GPT-4, and LLAMA-2 (7B, 13B, 70B) models, with a focus on refining open-source systems to rival proprietary performance. Each model performed multi-step reasoning: generating chain-of-thought explanations, identifying evidence phrases, and categorizing patients as "included", "excluded" or "no relevant information". The trial ranking metric was computed as the difference between inclusion and exclusion fulfillment percentages.

To improved open-source adaptability, a synthetic dataset of 2,000 patient-trial pairs was generated by GPT-4, including explanations and evidence. LLAMA variants were fine-tuned using supervised loss on this dataset, resulting in the Trial-LLAMA 70B model. Evaluation metrics included NDCG@10, AUROC, and Criterion-Level Accuracy (CLA) for explicit and implicit reasoning, along with Evidence Faithfulness (precision, recall, and F1). GPT-4 remained the top-performing out-of-the-box model, while Trial-LLAMA 70B achieved performance parity with GPT-3.5 and even exceeded it in criterion-level accuracy after fine-tuning. The results confirmed that open-source models can achieve near-proprietary performance with task-aligned fine-tuning.

However, implicit reasoning remained challenging across all models, and the reliance on GPT-4-generated training data introduced potential bias. Furthermore, improvements in criterion-level accuracy did not always translate to higher trial-level ranking performance, revealing a non-linear relation between component accuracy and end-task success.

3.2.5 End-to-End Matching with Real-World Validation

The study [Gupta et al., 2024] represents a significant advance, as it offers the first empirical end-to-end validation of a trial matching system using real-world unstructured EHRs. The PRISM framework formulates eligibility matching as a question-answering problem, where inclusion and exclusion criteria are decomposed into simplified question with possible answers ("yes", "no", or "N/A"). Logical structures such as Disjunctive Normal Form (DNF) are employed to manage multi-question criteria.

PRISM introduces OncoLLM, a fine-tuned LLM based on Qwen-1.5 (14B parameters), specialized in oncological data and augmented with Chain-of-Thought reasoning. To handle long EHRs, the pipeline includes a chunking and retrieval module using spaCy for tokenization and chronological arrangement of patient notes. The final decision-making follows a logical tree, and probabilistic evaluation is used for missing data. Trials are ranked using a Weighted Tier method prioritizing clinically significant criteria.

OncoLLM achieved 63% criterion-level accuracy, approaching GPT-4's 68%, while outperforming GPT-3.5-turbo and similarly sized models. It also excelled in both patient-centric and trial-centric rankings, with 95% of top-3 recommendations judged clinically appropriate and exclusive use of unstructured data (omitting structured records such as lab results), dependence on imperfect retrievers, and annotation inconsistencies among human experts. These findings underline the need for hybrid systems combining structured and unstructured data with consistent expert validation.

3.2.6 Scalable Zero-Shot Matching Systems

Finally, [Wornow et al., 2024] demonstrated the feasibility of scalling patient-trial matching through a zero-shot architecture. The system evaluated patient eligibility by comparing unstructured clinical histories to trial criteria using models such as GPT-4, LLAMA-2-70B-32k, and Mixtral-8x7B. Lightweight embedding models (BGE and MiniLM) were used to preliminary retrieval in a two-stage pipeline. Four prompting strategies were explored: All Criteria, All Notes (ACAN); All Criteria, Individual Notes (ACIN); Individual Criteria, All Notes (ICAN); and Individual Criteria, Individual Notes (ICIN); With ACIN identified as the most cost-efficient.

Using GPT-4 under the ACIN strategy, the system achieved state-of-the-art results on the n2c2 benchmark, with a Micro-F1 of 0.93 and Macro-F1 of 0.81, surpassing prior rule-based systems and other LLMs. It also achieved dramatic cost reductions, screening 86 patients in two hours at only $1.55 per patient. Clinican evalutation confirmed 97% coherence in model explanations. However, limitation included dataset simplification (English-only, limited criteria coverage), prompt dependence, and potential hallucinations. Privacy and compliance concerns also persist for proprietary models under healthcare regulations such as HIPAA.

3.2.7 Summary

3.2.8 Summary

Across all studies, LLMs demonstrate exceptional potential in automating patient-to-trial matching through zero-shot reasoning, retrieval-augmented architectures, and prompt-based interpretability. Results consistently show that LLMs can match or surpass expert performance, particularly when enhanced with ontologies, retrieval-augmented architectures, or task-aligned fine-tuning. Open-source models have also demonstrated competitive performance relative to proprietary systems, while offering substantial reductions in screening time and computational cost. However, key challenges persist: high false-negative rates caused by non-standard abbreviations and ontology gaps, reliance on synthetic datasets that may introduce bias, sensitivity to prompt design, and challenges related to privacy and regulatory compliance. Furthermore, real-world deployment must address interpretability and integration with structured clinical data.

More recent studies have begun to explore hybrid pipelines integrating extraction and matching within unified frameworks, typically following an extraction-first paradigm or employing multi-agent setups that combine extraction and reasoning. However, most hybrid approaches rely on end-to-end probabilistic LLM reasoning or assume access to structured electronic health records, blurring the distinction between deterministic rule-based evaluation and LLM-assisted interpretation. As a result, few systems explicitly separate the interpretation of unstructured clinical text from the transparent, reproducible evaluation of eligibility criteria.

3.3 Conclusion

Having considered the evidence across the reviewed studies, the literature highlights several gaps that remain insufficiently addressed. Clinical information extraction and patient–trial matching are often treated as independent tasks, and evaluations frequently rely on curated or synthetic datasets that do not reflect the complexity of real-world oncology workflows. Eligibility assessment is commonly performed through probabilistic LLM reasoning, limiting transparency, auditability, and reproducibility. Only a small number of studies propose integrated workflows that combine clinician-validated extraction, structured patient representation, formalization of eligibility criteria into machine-readable rules, and hybrid eligibility assessment mechanisms capable of combining deterministic evaluation with LLM-assisted reasoning when necessary.

These gaps motivate the approach adopted in this dissertation, which employs LLM-based parameter extraction from unstructured clinical diaries while explicitly addressing the well-documented risk of hallucinations through clinician-in-the-loop validation at multiple pipeline stages. In addition, clinical trial criteria are treated as rule-based constraints, translated into machine-interpretable logical representations and validated by clinical experts before being used in a hybrid matching pipeline that combines deterministic evaluation with targeted LLM-assisted reasoning. Overall, this methodology aims to bridge the flexibility of LLMs with the transparency of rule-based systems while operating directly on real-world unstructured oncology data.

Chapter 4

Approach

This chapter presents the methodological approach adopted to address the challenges identified in the oncology clinical workflow. The proposed solution is based on the combination of Large Language Models (LLMs), clinician validation mechanisms, and structured data representations to transform unstructured clinical information into actionable knowledge.

The chapter introduces the overall strategy followed by the system, including the extraction of relevant clinical information from free-text documents, the transformation of clinical trial requirements into machine-interpretable representations, and the evaluation of patient eligibility through a hybrid deterministic and LLM-assisted approach.

Additionally, the principles guiding the solution design are discussed, including the use of clinician-in-the-loop validation, traceability of intermediate results, and the preference for deterministic evaluation whenever possible. The detailed technical implementation of these concepts is presented in Chapter 7.

4.1 Method Overview

The proposed approach aims to reduce the manual effort associated with oncology data management and clinical trial screening by combining automated information extraction with clinician supervision.

The central idea is to employ Large Language Models as assistants capable of interpreting complex clinical text, while avoiding their direct use as autonomous decision-makers. Due to their probabilistic nature, all clinically relevant outputs generated by the models must be reviewed and validated by clinicians before being used in subsequent processes.

The proposed methodology is composed of three sequential phases:

- **Phase 1 — Hyper-Relevant Parameter Extraction**: extraction and normalization of relevant clinical information from free-text patient diaries, producing structured patient representations;

- **Phase 2 — Clinical Trial Criteria Formalization**: extraction of eligibility requirements from clinical trial documentation and transformation of these requirements into machine-interpretable representations;
- **Phase 3 — Semi-Deterministic Trial–Patient Matching**: evaluation of patient eligibility by combining deterministic rule-based assessment with LLM-assisted reasoning when required.

The following sections describe the rationale behind each phase and the methodological decisions that support the proposed approach.

4.2 Hyper-Relevant Parameter Extraction

Clinical information in oncology is frequently recorded as free-text notes, which makes direct computational analysis difficult. The first component of the proposed approach addresses this challenge by transforming unstructured clinical diaries into structured patient representations.

Instead of attempting to extract all available information from clinical documents, the methodology focuses on identifying hyper-relevant parameters that provide the greatest value for downstream tasks, namely clinical trial matching and quality-of-care analysis.

The extraction process relies on a predefined clinical schema developed from representative examples provided by the stakeholder. This schema acts as a guideline for the LLM, ensuring that the model focuses on clinically meaningful information while reducing unnecessary extraction variability.

Since automated extraction may introduce errors or incomplete information, clinician validation is incorporated as a mandatory step before extracted information can influence subsequent analyses.

The detailed extraction workflow, patient schema, and generated artifacts are described in Section 7.2.

4.3 Extraction and Conversion of Clinical Criteria

Clinical trial eligibility criteria are commonly expressed in natural language, making direct computational evaluation difficult. The second component of the proposed approach addresses this limitation by transforming clinical trial requirements into structured representations suitable for automated assessment.

The methodology separates the interpretation of trial documentation into two complementary tasks. First, relevant eligibility requirements are identified from clinical trial documents. Second, these requirements are converted into formal representations that enable their evaluation against structured patient information.

This separation aims to reduce the complexity of LLM-based reasoning by restricting model usage to tasks requiring language understanding, while relying on deterministic mechanisms whenever eligibility conditions can be formally represented.

When a requirement cannot be represented through available structured patient attributes, LLM-assisted reasoning is maintained as a fallback mechanism. This hybrid strategy allows the system to preserve flexibility without sacrificing transparency.

The detailed criteria extraction workflow, logical representation format, and conversion mechanisms are described in Section 7.3.

4.4 Semi-Deterministic Trial–Patient Matching

The final component of the proposed approach focuses on identifying potentially eligible patients for clinical trials by comparing structured patient representations with formalized eligibility requirements.

A purely LLM-based matching strategy was avoided due to the lack of reproducibility and explainability associated with probabilistic outputs. Instead, the proposed methodology follows a semi-deterministic approach.

Whenever eligibility requirements can be expressed using structured patient attributes, they are evaluated through deterministic rules. However, clinical trial criteria may contain complex conditions or information that cannot be captured within a predefined schema. In these cases, LLM-assisted assessment is employed to support interpretation while maintaining clinician oversight.

This combination enables transparent eligibility decisions while preserving the flexibility required to handle the complexity of real clinical documentation.

The detailed matching algorithm, evaluation strategy, and decision aggregation mechanisms are described in Section 7.4.

4.5 Clinical Documents generalization

Two main types of clinical documents will be manipulated in this solution, clinical diaries and clinical trials. Below it is possible to find a generalized template of how both clinical documents are organized in practice, independent of therapeutic are.

The generalized template refers to a canonical data model developed by identifying recurring information patterns across both clinical diaries and trials. It acts as a standardized schema that provides a predefined slots to normalize this information into a logic-based format. By defining these parameters upfront, the template serves as a structural guide for the Large Language Model (LLM) via prompt engineering, ensuring consistent data extraction and the deterministic conversion of clinical criteria into machine-readable rules.

These abstract representations were made and structured having as base real documents provided by the stakeholder to use as example.

Starting with the Clinical Trial protocol. Although clinical trial protocol are written as narrative documents, they follow a well-defined structural organization as can be seen the example bellow. Among these sections, eligibility criteria represent the only on that directly decides the patient selection. While expressed in natural language, inclusion and exclusion criteria are inherently constraint-based, specifying admissible values, thresholds, or conditions over patient attributes.

This structure allows for a systematic separation between descriptive trial information and eligibility logic. By isolating this eligibity within the trial template, the proposed methodology can safely apply Large Language Models to extract and normalize patient selection criteria without affecting other trial metadata. The extracted criteria can then be deterministically translated into machine-readable rules, allowing automated comparison against structured patient records.

This template can be found below on the Figure 4.1

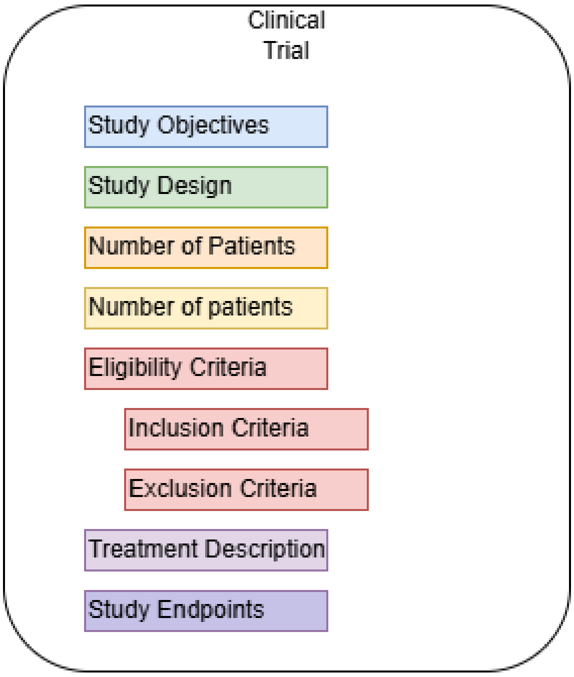


Figure 4.1: Clinical Trial Abstraction

The design of the extraction pipeline is fundamentally driven by a structured template derived from four representative clinical diaries provided by stakeholders. By establishing this foundational template, the system ensures that the diverse nature of free-text clinical notes is mapped to a consistent format. The selection of specific extraction parameters, including age, ECOG-PS, diagnosis details, molecular status, staging, and longitudinal treatment dates, serves as a targeted guidance system. This prevents "information overload" for the model and focuses the computational resources of the Large Language Model (LLM) on high-value clinical data points required for trial matching and survival analysis.

This template can be found below on the Figure 4.2

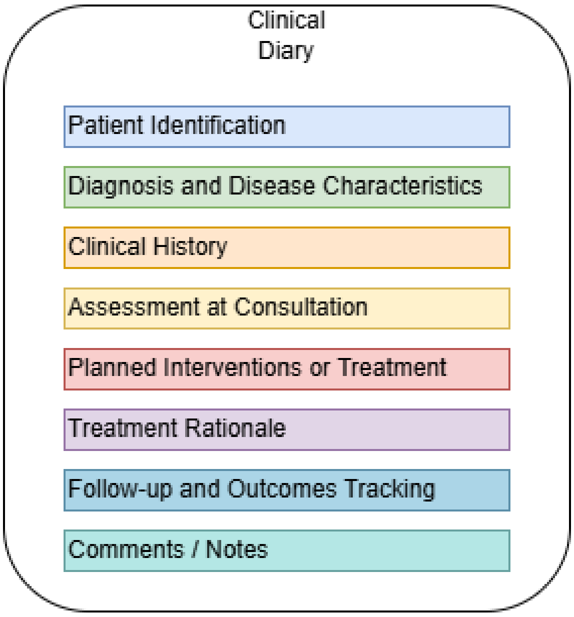


Figure 4.2: Clinical Diary Abstraction

4.6 Design Principles

The selection of methodologies and tools employed in the current project directly arise from the nature of the available clinical data, defined functional requirements and the necessity of ensuring reproducibility, auditability and clinical rigor. The proposed solution combines Natural Language Processing (NLP) techniques, classical static methods and a robust data structure, allowing the transformation of unstructured clinical information in actionable knowledge for decision support.

4.6.1 Usage of Language Models

The choice of LLMs as the core of the extraction and normalization process is justifiable because of the high linguistic variability present in the clinical diaries. These documents present:

- Free writing and not standardized;
- Orthographic errors and abbreviations;
- Lack of formal structure

Traditional NLP tools based on rules or dictionaries can not efficiently deal with this heterogeneity. LLMs on the other hand can:

- Interpret clinical text;
- extract entities and relations without previous annotations;
- significantly reduce manual effort from the clinicians

That's why LLMs were chosen as the most adequate methodology for the compliance of requirements FR3.2 - Patient Retrieval-**??** and FR1.1 - Clinical Trial Text Pre-processing-FR1.2 - Trial Structure Identification.

4.6.2 Semi-Deterministic Evaluation

The matching phase employs a semi-deterministic approach that combines deterministic rule evaluation with LLM-assisted reasoning. This design choice is motivated by two complementary requirements: the need for transparency and reproducibility in clinical decision support, and the practical necessity of handling eligibility criteria that cannot be formally encoded against the patient profile schema.

Whenever a criterion can be mapped to a structured patient field, it is evaluated deterministically. This ensures that the assessment is fully reproducible, auditable, and independent of the non-deterministic behaviour of LLMs. For criteria that reference information outside the schema, such as complex laboratory conditions or narrative clinical judgements, LLM-assisted assessment is employed, with the model providing a justification that the clinician can review and override.

This separation is fundamental: it prevents the introduction of probabilistic behaviour into decisions that can be made deterministically, while preserving the flexibility required to handle the full complexity of real clinical trial criteria.

4.6.3 Clinician-in-the-loop Validation

Given the critical nature of clinical decision support, the proposed approach does not consider LLM outputs as definitive decisions. Instead, clinicians remain responsible for validating extracted information, formalized criteria, and eligibility assessments.

This design choice ensures that automation reduces administrative workload while preserving clinical responsibility, interpretability, and accountability.

4.6.4 JSON structure

The JSON format was chosen as the representation of the LLM's output because it leverages really well structure and readability (FR1.2 - Trial Structure Identification and FR3.2 - Patient Retrieval). Since every extracted profile must later be reviewed and validated by a clinician (FR1.3 - Criteria Extraction and FR3.4 - Assessment Validation), it is essential that the output remains human-readable and easy to interpret. JSON naturally supports this, with its key–value structure making the information explicit, organized, and simple to modify whenever the clinician needs to correct or complete a field. At the same time, JSON is simple to manipulate programmatically, which is important for the downstream steps of normalization, validation, and storage. Its structured nature also simplifies the process of persisting these objects in the database, ensuring that each field can be mapped cleanly into the corresponding schema without ambiguity.

4.6.5 Pipeline

The system architecture follows a modular, sequential pipeline in which each phase produces clinician-validated artifacts that serve as inputs to the next. The overall pipeline can be summarized as:

$$\text{Extraction} \rightarrow \text{Pre-Processing} \rightarrow \text{Validation} \rightarrow \text{Storing} \rightarrow \text{Evaluation}$$

This modular design facilitates the independent evolution of each component, for example, replacing the underlying LLM, without requiring changes to the rest of the pipeline. It also ensures full traceability through intermediate persistence, and guarantees that only validated data is consumed by downstream processes such as quality-of-care analysis. This combination of modularity, traceability, and mandatory validation gates is essential for systems operating in clinical environments.

Chapter 5

Requirement Analysis

In this chapter, a clear formalization of the stakeholders' needs and constraints is presented, establishing the foundation for the system's functional and non-functional requirements. Additionally, the three core workflows of the solution are described in detail, automated structured data extraction, identification of eligible patients for clinical trials, and monitoring of care, each supported by an illustrative diagram. Together, these elements define the thematic scope of the project and outline the essential processes that the proposed system must support.

5.1 Context and Stakeholder Needs

As previously discussed, the current oncology setup is far from standardized. Clinical diaries are written in free-text and vary widely between clinicians, increasing the cognitive load on oncologists. Patient–trial matching is still performed manually, and relevant information is dispersed across multiple unstructured notes, making the process inefficient and error-prone.

This situation increases the likelihood of screening errors, which may lead to financial losses if the hospital fails to meet recruitment targets. These limitations highlight a major workflow problem and motivate the need for a software pipeline capable of connecting unstructured patient data with the manual screening process, reducing errors and preventing unnecessary costs.

The primary stakeholders of this project are the oncologists at ULS Coimbra, particularly Dra. Judy. Through several meetings, a set of key needs were identified:

- Structured patient profiles, since the reliance on free-text documentation creates variability and makes clinical-trial screening slow and error-prone. A tool capable of extracting relevant parameters and organizing them into structured profiles is essential to simplify this workflow.
- Explainable eligibility decisions, as clinicians must verify that each logical rule reflects the trial requirements and understand why a patient was in-

cluded or excluded. Even with deterministic matching after LLM extraction and clinician validation, transparency and auditability remain mandatory.

- Clinician validation, which is critical to ensure correctness of all LLM-generated outputs. Given the stochastic nature of LLMs and their susceptibility to hallucinations, human oversight is required to prevent erroneous information from propagating through the system.

Beyond these functional needs, the system must comply with strict constraints inherent to clinical environments. These constraints shape the requirements analysis and must be considered throughout system design.

First, data privacy is a fundamental restriction. Clinical records contain highly sensitive information and must be handled according to ethical and institutional policies. The system must therefore ensure controlled access, secure storage, and full traceability of all operations.

Second, auditability is essential. Systems used in clinical contexts must produce verifiable and justified results. All processes influencing trial eligibility must be transparent and reproducible, requiring retention of intermediate results, decision outcomes, and clinician interventions for retrospective analysis.

Lastly, the non-deterministic nature of LLMs introduces an additional constraint. Although effective for processing unstructured text, LLMs may generate inconsistent or incorrect outputs. To mitigate this, the system incorporates human in the loop validation, ensuring that all extracted information is clinically reviewed before use. LLMs are therefore employed strictly as decision-support tools, assisting in extraction and interpretation while all clinically relevant decisions remain under clinician control. This preserves authority, ensures accountability, and minimizes unintended clinical impact.

Taking into account the identified needs and constraints, three main categories of system requirements were defined

5.2 Automated Structured Patient Data Extraction

Clinical information required for patient characterization is currently stored in unstructured clinical diaries. Therefore, the system must provide mechanisms to support the transformation of free-text clinical documentation into structured patient information.

The system should allow clinicians to select clinical documents for processing, review automatically extracted information, modify incorrect values, and store validated patient profiles.

The main functional requirements associated with this capability are summarized in Table 5.1, for further specification refer to Appendices Hyper-Relevant Parameter Extraction and Oyster Level Requirements.

Table 5.1: Hyper-Relevant Parameter Extraction – Functional Requirements

ID	Requirement	MoSCoW	Description
FR5.1	Select Diaries for Extraction	M	The system must allow for the selection of clinical diaries to be submitted for extraction.
FR5.2	Clinical Diary Text Pre-processing	S	The system must perform the pre-processing of the clinical diary text for the removal of headers and footers to reduce context and computational overhead.

ID	Requirement	MoSCoW	Description
FR5.3	Information Extraction	M	The LLM is instructed to identify and extract clinically relevant parameters according to a predefined schema: • Age: Patient age or date of birth, depending on the information available in the clinical diary; • Gender: Patient gender; • ECOG-PS: ECOG Performance Status score (0–5), describing the patient's functional capacity; • Diagnosis: Primary diagnosis, typically corresponding to the patient's cancer type or major clinical condition; • Diagnosis Date: Date on which the diagnosis was formally established; • Molecular Status: Relevant molecular and biomarker information; • Stage: Disease staging information; • Pathology Group: Clinical category grouping diagnoses according to the organ or system of origin; • Treatments: List of administered treatments (Name, Start Date, End Date); • Control: Information related to disease control, metastases, comorbidities, and other relevant clinical follow-up observations. If a field does not exist in the clinical diary, it must be defined as `None`.

ID	Requirement	MoSCoW	Description
FR5.3.1	Information Extraction – Normalization of Diagnosis	S	After identifying the diagnosis field, the LLM must perform a normalization step according to the diagnosis map provided by the stakeholder.
FR5.4	Retrieving Analysis	S	The system must be able to retrieve analyses associated with the respective patient, allowing for improved eligibility assessment further in the pipeline.
FR5.5	Validation of Extracted Parameters	M	After the LLM extracts the parameters, the clinician must be able to validate said parameters, being able to edit, create, and remove parameters.
FR5.6	Persistence of Structured Patient Profiles	M	The system must persist all artifacts: parameters, treatments, and analyses associated with the new profile.
FR5.7	Filtering of Clinical Diaries	C	The system must allow the clinician to search clinical diaries by their document name and filter them by file type and extraction status.
FR5.8	Removal of Clinical Diaries	S	The system must allow the clinician to remove any clinical diaries and associated artifacts: patient profiles and versions.
FR5.9	Filtering of Structured Patient Profiles	C	The system must allow the clinician to search patient profiles by their molecular status or diagnosis and filter them by pathology group, molecular status, diagnosis, and stage.
FR5.10	Removal of Structured Patient Profiles	M	The system must allow the clinician to remove patient profiles and associated artifacts (treatments and analyses). Furthermore, the diary that generated the removed profile must be reverted to its original status, allowing for a new extraction.

5.3 Extraction and Conversion of Clinical Criteria

Clinical trial eligibility assessment relies on comparing patient characteristics against requirements defined in clinical trial documentation. However, these requirements are commonly expressed in natural language, making direct computational evaluation difficult. Therefore, the system must support the transformation of unstructured trial documentation into validated and machine-interpretable eligibility representations.

The system must allow clinicians to submit clinical trial documents and obtain a structured representation of their eligibility requirements. This process includes the identification of relevant trial information, extraction of inclusion and exclusion criteria, and conversion of validated criteria into logical representations suitable for automated evaluation.

Since eligibility requirements may contain complex clinical conditions, the system must preserve criteria whose meaning depends on their original composition rather than forcing an artificial decomposition. Additionally, all generated artifacts must be reviewed by clinicians before being used in subsequent eligibility assessments.

The functional requirements associated with this capability are summarized in Tables 5.2 and 5.3. For further specification refer to Appendices Eligiblity Criteria Extraction, Eligibility Criteria Conversion and Oyster Level Requirements

ID	Requirement	MoSCoW	Description
FR1.1	Clinical Trial Text Pre-processing	S	The system must perform the removal of sections that are not relevant to eligibility assessment.
FR1.2	Trial Structure Identification	M	The LLM must analyse the clinical trial document to identify its internal structure, namely the existence of trial cohorts.
FR1.3	Criteria Extraction	M	The LLM must extract inclusion and exclusion criteria from a clinical trial individually.
FR1.4	Extracted Criteria Validation	M	The clinician must be able to validate the LLM-extracted criteria, being able to edit, create, and remove these artifacts.
FR1.5	Persistence of Validated Criteria	M	Once approved, validated criteria must be persisted in the database.
FR1.6	Clinical Trial Filtering	C	The clinician must be able to search clinical trials by study name and filter them based on their metadata: pathology group, trial status, extraction status, date range, and type of file.
FR1.7	Clinical Trial Removal	C	The clinician must be able to remove any clinical trial and its associated artifacts: versions, matches performed, criteria, and logical rules.

Table 5.2: Eligibility Criteria Extraction – Functional Requirements

ID	Requirement	MoSCoW	Description
FR2.1	Retrieval of Extracted Criteria	M	The system must retrieve the respective validated criteria for the trial for further processing.
FR2.2	Eligibility Criteria Conversion	M	The LLM must perform the translation of these criteria into a structured logical representation.
FR2.3	Logical Representation Validation	M	The clinician must be able to validate the generated logical representations, being able to edit, create, and remove these artifacts.
FR2.4	Persistence of Validated Logical Representations	M	Once approved, validated logical representations must be persisted in the database for use during eligibility assessment.

Table 5.3: Eligibility Criteria Conversion – Functional Requirements

5.4 Semi-Deterministic Trial–Patient Matching

The system must support the identification of potentially eligible patients by comparing validated patient profiles against formalized clinical trial requirements.

Due to the complexity and variability of eligibility criteria, a fully deterministic or fully LLM-based approach presents important limitations. Therefore, the system adopts a semi-deterministic evaluation strategy, combining rule-based assessment for criteria that can be represented using structured patient information with LLM-assisted interpretation for requirements that cannot be directly encoded.

The eligibility assessment process must provide transparent results at criterion level, allowing clinicians to understand the factors influencing each decision and validate the generated outcomes. Any automatically generated assessment must remain subject to expert review before being considered final.

The functional requirements associated with this capability are summarized in Table 5.4, for further specification refer to Appendices Semi-Deterministic Trial-Patient Matching and Oyster Level Requirements.

ID	Requirement	MoSCoW	Description
FR3.1	Trial Selection	M	The clinician must be able to select a clinical trial to perform eligibility assessment.
FR3.2	Patient Retrieval	M	The system must retrieve a subset of patient profiles based on the clinical trial's metadata.
FR3.3.1	Eligibility Assessment – Deterministic	M	Each patient retrieved must be evaluated against every validated criterion. If the criterion references a field available in the patient profile schema, the corresponding logical rule must be evaluated directly against the structured patient data.
FR3.3.2	Eligibility Assessment – LLM-Assisted	M	If the criterion references information not represented within the structured schema, the LLM must receive sufficient context to determine eligibility.
FR3.3.3	Eligibility Assessment – LLM Justification	M	When an eligibility assessment is performed by the LLM, the model must provide a justification to support expert review.
FR3.4	Assessment Validation	M	The clinician must be able to review all criterion-level assessments and modify any automatically generated decision.
FR3.5	Persistence of Assessment Results	M	The system must persist all criterion assessments, manual corrections, and final eligibility decisions.

Table 5.4: Semi-Deterministic Trial–Patient Matching – Functional Requirements

5.5 Monitoring the quality of care

This final category, focuses on evaluating clinical outcomes through survival analysis. Using chronologically validated data such as: diagnosis dates, treatment initiation, and progression or death events the system supports the computation of key outcome metrics, including Overall Survival (OS) and Progression-Free Survival (PFS). These analyses enable clinicians to monitor treatment effectiveness and assess the quality of care provided. The results are presented through a dashboard that includes survival curves and cohort level summaries. Additionally, the system supports cohort selection and the exportation of results in formats compatible with external statistical tools or universally readable formats such as PDF and CSV.

This module, due to time constraints won't be included in final system, this decision was discussed with the stakeholder as well as with the supervisor and tutors. Reaching the conclusion that it is more a complementary module than a core module and as such it is dispensable and won't affect the overall working of the pipeline.

Everything described above can be seen as requirements on the below table 5.5, for further specification refer to appendixes Monitoring the Quality of Care and Oyster Level Requirements.

ID	Requirement	MoSCoW	Description
FR4.1	Retrieve Treatment Data	WH	The system must retrieve patient treatment records, including: Diagnosis date; First cycle date; Progression date / End of treatment; Death date; Status (0/1): 0 - censured (patient did not experience the event before the end of the study); 1 - event occurred (progression or death).
FR4.2	Compute OS and PFS	WH	The system must calculate Overall Survival (OS) and Progression-Free Survival (PFS).
FR4.3	Apply Kaplan-Meier Analysis	WH	The system must use Kaplan-Meier survival estimation to analyse outcomes.
FR4.4	Visualize Results	WH	The system must generate survival curves and statistical summaries.

Table 5.5: Monitoring the Quality of Care – Functional Requirements

5.6 Non-Functional Requirements

Beyond functional capabilities, the clinical context introduces additional requirements related to reliability, transparency, and data management.

- **Traceability** - The system must maintain records of generated outputs, clinician modifications, and final decisions to support retrospective analysis.
- **Human Oversight** - The system must allow clinicians to validate and override automatically generated information.
- **Explainability** - Eligibility decisions must provide sufficient information to allow clinicians to understand the factors influencing each result.
- **Data Privacy** - The system must ensure controlled handling of clinical information according to institutional requirements.

Chapter 6

Architecture

In this chapter there will be a clear description of the proposed system architecture, following the C4 Model alongside the system's data layer design. The objective of this description is to provide a structured and progressive view of the system, from its high-level external interactions down to its internal component organization and Entity-Relationship (ER) data structures. Only the first three abstraction levels of the C4 Model were designed (Context, Container, and Component levels), accompanied by a comprehensive ER diagram mapping the relational database.

6.0.1 Database Design

To support the proposed clinical workflow, a relational database model was designed to ensure data consistency, traceability, and efficient management of all artifacts generated throughout the system pipeline. The database design was guided by three main principles: document version control, separation between clinical documents and patient data, and preservation of the complete history of automated and clinician-validated information.

Since the system relies on Large Language Models to generate intermediate artifacts, maintaining traceability between raw outputs, clinician modifications, and final validated information is essential. Therefore, the database stores not only the final entities required for the application workflow, but also the intermediate results generated during extraction, conversion, and matching processes. This approach ensures accountability and allows every decision to be reconstructed whenever necessary, which is particularly important in a clinical decision-support environment.

The final database model is composed of 11 tables, including both core clinical entities and relationship entities required to represent the connections between different artifacts. Together, these tables support the complete system workflow, from document ingestion and information extraction to clinical trial eligibility assessment and patient–trial matching.

The main entities represented in the database are:

- **Document and Version** - responsible for storing the clinical documents introduced into the system and maintaining their version history, ensuring reproducibility and traceability of processed information.
- **Clinical Trial and Trial Cohort** - represent clinical trial information and its internal organization, including cohort-specific eligibility requirements when applicable.
- **Patient Profile** - stores the structured representation of patient information extracted from clinical diaries and validated by clinicians, acting as the central entity for patient-related data.
- **Treatment and Analysis** - store additional patient-related information required for eligibility assessment, namely treatment history and laboratory measurements.
- **Trial Criteria and Logic Criteria** - represent the eligibility requirements extracted from clinical trial documentation and their corresponding machine-interpretable logical representations.
- **Patient Trial Match and Criterion Evaluation** - store the results of the matching process, including patient-level decisions, criterion-level evaluations, automatic results, clinician overrides, and supporting justifications.

The relationships between these entities allow the system to maintain complete traceability between the original clinical documents, generated structured information, clinician validations, and final eligibility decisions. This structure supports both the operational requirements of the application and the reproducibility needs of clinical decision-support systems.

Everything described above, tables, relationships and constraints can be seen on the Figure 6.1.

The complete Entity-Relationship diagram, including all attributes, constraints, and cardinalities, is presented in Appendix G.

6.1 Context

6.2 Context

At this level, the context diagram presents the system as a black box and describes how it interacts with users and external systems, [Vazquez-Ingelmo et al., 2020]. The objective of this diagram is to clearly define the system boundaries and to identify which actors and external dependencies are involved in its operation.

The primary user of the system is defined as the clinician, who directly interacts with the system in order to perform patient screening and clinical trial analysis. The system itself, represented as CTESS, aggregates all functionalities previously

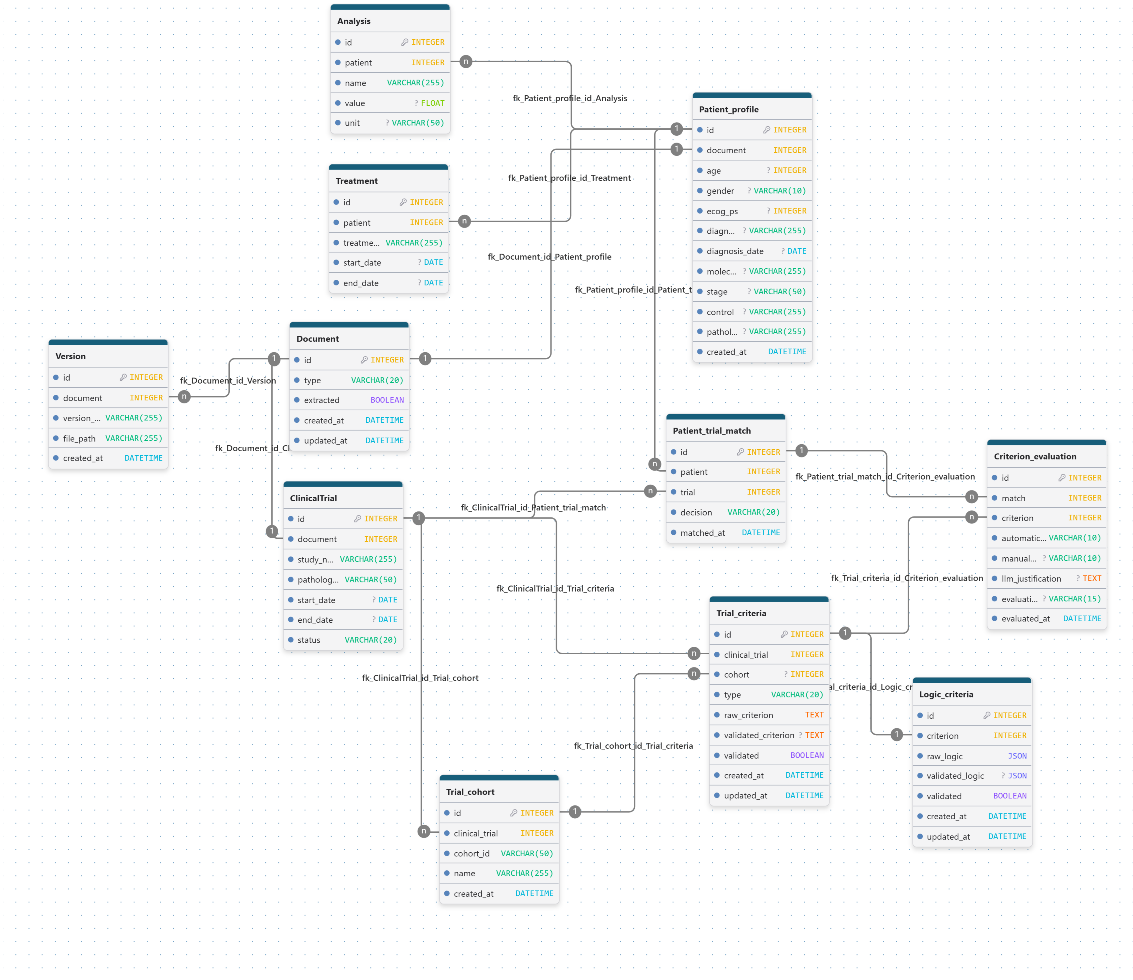


Figure 6.1: Overall ER Diagram

described, including structured data extraction, patient-trial matching, and eligibility assessment.

The initial conception of the solution included a direct integration with SClinico, the platform used by ULS Coimbra for storing free-text clinical diaries. Through this integration, CTESS would retrieve clinical records directly from the existing hospital infrastructure, avoiding manual data transfer and allowing the automated processing of real clinical information.

However, during the implementation phase of this thesis, this integration could not be performed. The required Zero Trust environment, necessary to enable secure communication between external applications and hospital systems, was not available within the project timeframe. Consequently, and to allow the development and evaluation of the proposed pipeline, synthetic clinical diaries were generated and used as the input source for the system.

Therefore, although SClinico represents the intended external data source in the proposed architecture, the implemented prototype operates with synthetic clinical documents that reproduce the structure and characteristics of the real clinical information. This approach allowed the validation of the complete processing pipeline while respecting institutional security constraints.

No modifications are performed on SClinico by the proposed system, ensuring that the solution remains non-intrusive with respect to existing clinical infrastructures.

Everything discussed above is present in Figure 6.2.

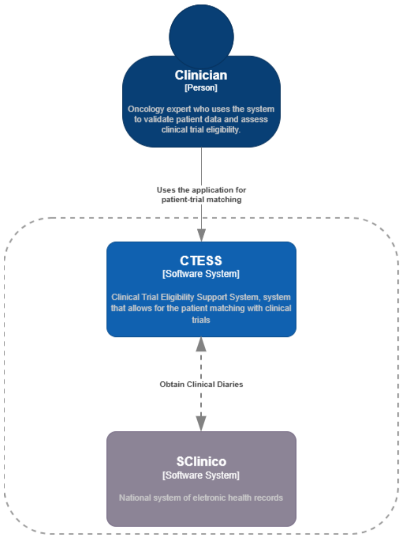


Figure 6.2: Context Diagram of the system

6.3 Container

At this abstraction level, the system is decomposed into its main runtime units, each one having a well-defined responsibility and interaction pattern, [Vazquez-Ingelmo et al., 2020].

- Core Application, which represents the central backend of the system. It exposes the main API consumed by the user interface and orchestrates all system functionalities, including data extraction, patient-trial matching, and monitoring of care.
- Web Application, responsible for delivering the static user interface assets. It presents the extracted parameters, extracted criteria, logical rules, eligibility assessments, and Patient matching to the user in a structured and readable format.
- Single-Page Application, which runs in the user's web browser and provides the interactive functionality of the system. This component acts as a bridge between the Web Application and the Core Application, handling user input and API communication.
- Database, which stores all relevant system data. This includes clinical documents provided either through the planned hospital integration or through synthetic datasets used during the prototype implementation, structured patient profiles, extracted and validated clinical trial criteria, eligibility results, and monitoring outcomes.

Everything discussed above is represented in Figure 6.3, which illustrates the separation of concerns and the communication paths between the different containers.

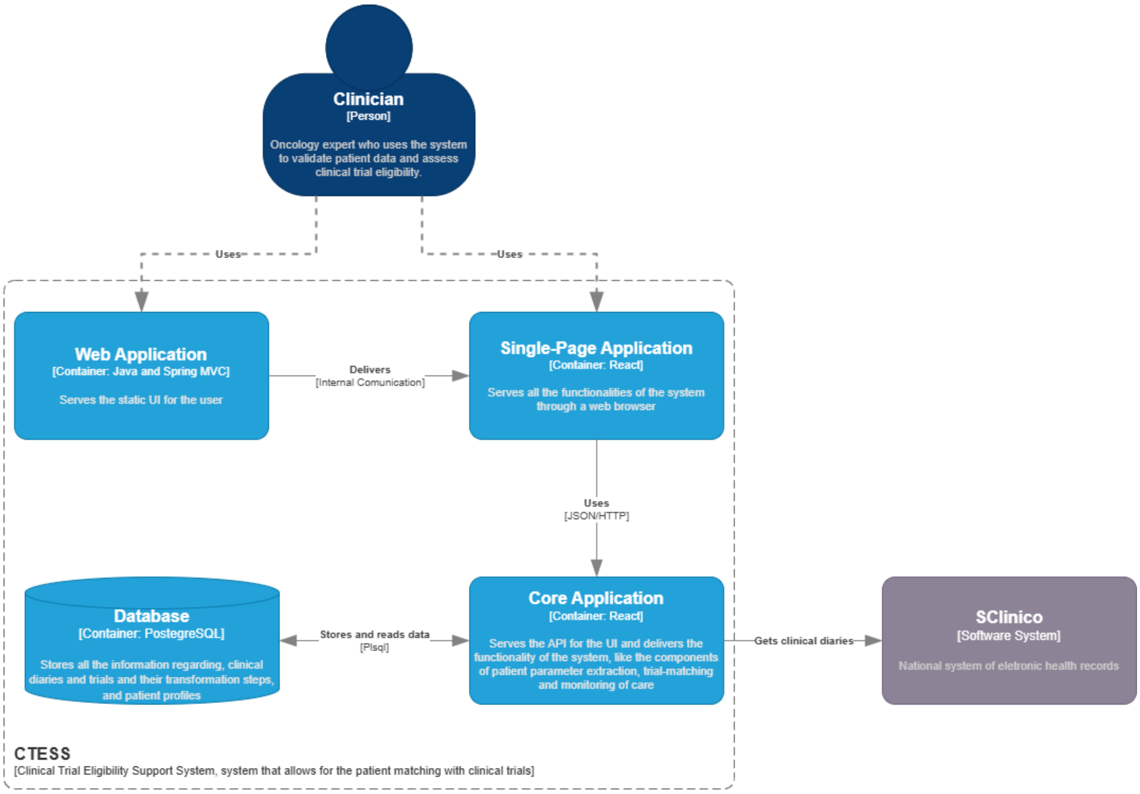


Figure 6.3: Container Diagram of the system

6.4 Component

This level focuses on the internal organization of the Core Application, identifying the main components that implement the system's functional requirements, [Vazquez-Ingelmo et al., 2020]. It is composed of:

- Main component, which is responsible for centralizing and orchestrating all other components. It exposes the API consumed by the user interface and manages the overall execution flow of the system.
- **Hyper-Relevant Parameter Extraction** - which encapsulates all functionalities related to the Hyper-Relevant Parameter Extraction module. This component is responsible for processing unstructured clinical diaries, extracting relevant parameters, and storing the resulting structured data.
- **Eligibility Criteria Extraction and Trial Structure Identification** - which implements part of the Extraction and Conversion of Clinical Criteria module, namely the Eligibility criteria extraction and trial structure identification. It handles the identification of trial cohorts and the extraction and validation of clinical trial criteria

- **Eligibility Criteria Conversion** - which corresponds to the final part of Extraction and Conversion of Clinical Criteria module. This component its responsible for the retrieval of extracted and validated criteria for a given trial and their conversion into logical rules.

- **Semi-Deterministic Eligibility Assessment** - which encapsulates all functionalities related to the Semi-Deterministic Trial–Patient Matching module. It handles the retrieval of patients based on the trial's metadata and the eligibility assessment of patients against that trials criteria, deterministically, when the criterion field belongs to the patient's schema, described on Patient Profile Schema, or LLM-based when the field is not present.

Everything discussed above is represented in Figure 6.4, which provides a detailed view of how responsibilities are distributed within the Core Application.

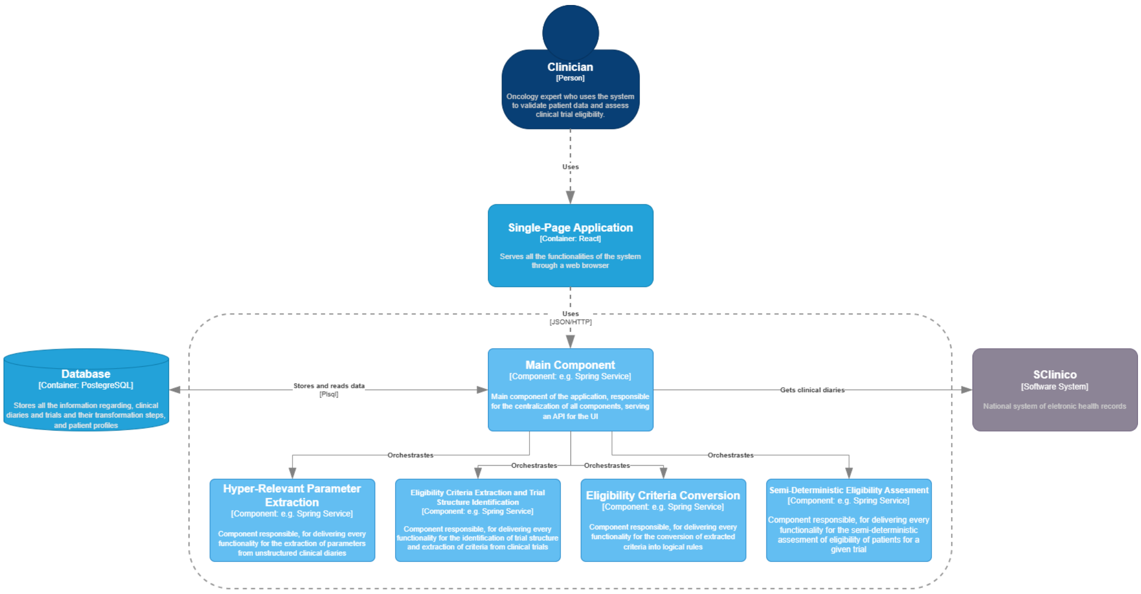


Figure 6.4: Component Diagram of the system

Chapter 7

System Design

This chapter describes the detailed technical design of the proposed system, covering the three main processing phases and the artifacts they produce. While Chapter 4 established the methodological rationale behind each design decision, this chapter focuses on how each phase is concretely realized, detailing the data schemas, processing steps, logical representations, and evaluation strategies that together constitute the system pipeline. The chapter concludes with a description of the overall flow connecting the three phases through the system database.

7.1 Overall Flow

The proposed system follows a sequential workflow in which each phase produces artifacts required by subsequent stages. The Hyper-Relevant Parameter Extraction phase generates clinician-validated structured patient profiles from free-text clinical diaries. In parallel, the Extraction and Conversion of Clinical Criteria phase processes clinical trial documentation to produce validated eligibility criteria and their corresponding machine-readable logical representations.

Communication between phases is performed through the system database, where all intermediate artifacts are persisted after clinician validation. This design ensures traceability and reproducibility, and allows experts to review and modify the outputs of each phase before they are consumed by downstream components.

Once both structured patient profiles and validated eligibility rules are available, the Semi-Deterministic Trial–Patient Matching phase can be executed. This final phase retrieves the artifacts produced by the previous stages and evaluates patient eligibility for a selected clinical trial, producing a list of potentially eligible patients together with criterion-level assessment results.

7.2 Hyper-Relevant Parameter Extraction

The objective of this phase is to transform unstructured free-text clinical information into normalized and structured patient data. The input consists of clinical diaries written in free-text format, which are selected by the clinician through the system interface.

The extraction process comprises the following steps:

1. **Text Pre-processing** - a clinical diary uploaded to the system is selected and pre-processed. During this stage, irrelevant sections such as headers and footers are removed to reduce unnecessary context and computational overhead.

2. **Information Extraction and Normalization** - the pre-processed text is provided to the LLM together with a task-specific prompt. The model is instructed to identify and extract clinically relevant parameters according to a predefined schema. During the same step, diagnosis normalization is performed using a stakeholder-provided mapping to ensure consistency across patient records.

3. **Validation and Structured Output Generation** - the extracted information is presented to the clinician for expert validation and potential correction. Once approved, the validated patient profile is persisted in the database and becomes available for subsequent phases of the workflow.

A formalization of this workflow is presented in Figure 7.1.

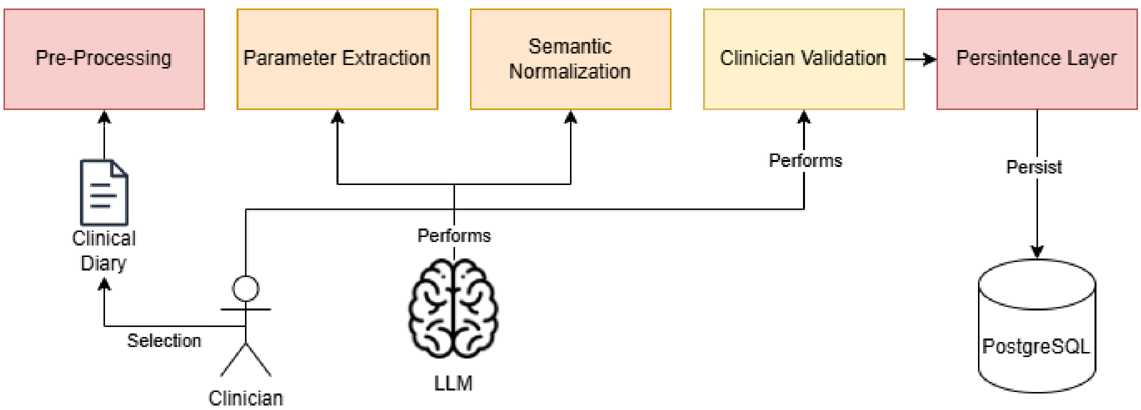


Figure 7.1: Hyper-Relevant Parameter Extraction Workflow

Upon completion of this phase, the system produces a structured repository of validated patient profiles containing the key clinical parameters required for eligibility assessment.

7.2.1 Patient Profile Schema

The extraction schema follows a JSON structure composed of the following fields:

- **Age** - patient age or date of birth, depending on the information available in the clinical diary;
- **Gender** - patient gender;
- **ECOG-PS** - ECOG Performance Status score (0–5), describing the patient's functional capacity;
- **Diagnosis** - primary diagnosis, typically corresponding to the patient's cancer type or major clinical condition;
- **Diagnosis Date** - date on which the diagnosis was formally established;
- **Molecular Status** - relevant molecular and biomarker information;
- **Stage** - disease staging information;
- **Pathology Group** - clinical category grouping diagnoses according to the organ or system of origin;
- **Treatments** - list of administered treatments, each comprising:
  - Name - treatment name;
  - Start Date - date on which the treatment started;
  - End Date - date on which the treatment ended;
- **Control** - information related to disease control, metastases, comorbidities, and other relevant clinical follow-up observations.

All schema fields are provided to the LLM as target extraction parameters. However, as clinical diaries may not contain all required information, the model is allowed to return null values for unavailable fields. Since all extracted information undergoes clinician validation before persistence, missing or uncertain values can be reviewed and corrected when necessary.

7.2.2 Error Handling and Robustness

To improve robustness, several error scenarios are explicitly handled during the extraction step:

- **Invalid JSON Structure** - if the LLM generates explanatory text together with the JSON output, the system attempts to isolate and recover the valid JSON structure. If recovery is unsuccessful, the extraction request is automatically repeated.

- **Missing JSON Output** - if no valid JSON structure is generated, the extraction request is repeated.
- **Truncated Responses** - if the generated output is incomplete due to context or generation limits, the extraction request is repeated.

The objective of these recovery mechanisms is to ensure that a valid and structurally consistent JSON representation is always obtained before the extracted information is presented for clinician validation.

7.2.3 Produced Artifacts

Upon completion of this phase, a single primary artifact is generated and made available through the system interface:

- **Structured Patient Profile** - a clinician-validated JSON representation of the patient containing the extracted clinical parameters required for subsequent eligibility assessment.

The resulting structured profiles provide a normalized representation of patient information that can be consistently queried and evaluated during the matching process, while maintaining full traceability through clinician validation.

7.3 Extraction and Conversion of Clinical Criteria

The objective of this phase is to transform clinical trial eligibility requirements into machine-interpretable representations that can be evaluated during patient eligibility assessment. The input consists of clinical trial documentation uploaded, managed, and selected through the system interface.

The process comprises the following steps:

1. **Text Pre-processing** - the clinician selects a clinical trial document, which undergoes pre-processing to remove sections not relevant to eligibility assessment. Similar to the clinical diary pipeline, this step reduces unnecessary context and computational overhead.
2. **Trial Structure Identification** - before extracting eligibility criteria, the LLM analyses the clinical trial document to identify its internal structure — namely, the existence of trial cohorts. This information is subsequently used during criteria extraction and patient eligibility assessment.
3. **Criteria Extraction** - the LLM receives the processed trial documentation together with the identified trial structure and extracts inclusion and exclusion criteria. Whenever possible, criteria are extracted individually and

associated with their corresponding cohort. However, some eligibility requirements must remain grouped to preserve their original clinical meaning. The system therefore supports both simple and compound criteria.

4. **Extracted Criteria Validation and Persistence** - the extracted criteria are presented to the clinician for expert validation. Once approved, they are persisted in the database, ensuring traceability and providing the input required for the next step of the pipeline.

5. **Criteria Conversion** - following validation, each criterion is translated into a structured logical representation. The objective of this step is to maximize the use of deterministic evaluation and reduce reliance on probabilistic LLM reasoning during eligibility assessment.

6. **Logical Representation Validation and Persistence** - the generated logical representations are presented to the clinician for validation to ensure they accurately reflect the meaning of the original eligibility criteria. Once approved, they are stored in the database and become available for patient eligibility assessment.

A formalization of this workflow is presented in Figure 7.2.

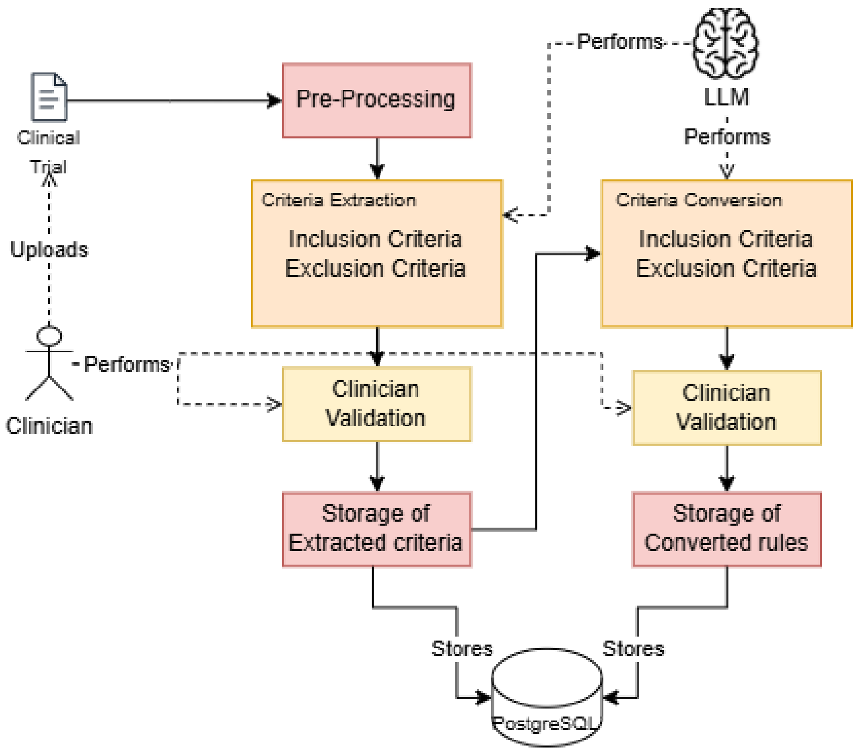


Figure 7.2: Extraction and Conversion of Clinical Criteria Workflow

Initially, criteria extraction and logical conversion were designed as a single task. However, previous studies have shown that decomposing complex tasks into simpler sequential steps often leads to improved performance and reliability []. For this reason, structure identification, extraction, and conversion were implemented as separate steps, each with its own clinician validation gate.

7.3.1 Conversion of Criteria

The LLM is provided with a prompt that guides it to perform the best possible conversion for the system. It attempts, while maintaining semantic equivalence, to map each criterion to a valid field available in the patient profile schema or in the laboratory values schema. Only when such mapping is not possible is the LLM instructed to generate an unmapped representation. These unmapped criteria are handled during the Semi-Deterministic Matching phase, as described in Section 7.4.

7.3.2 Complex Criteria Handling

The identification of simple and compound criteria is performed by the LLM. The model receives the complete textual description of each eligibility criterion and determines whether it can be safely decomposed into independent sub-criteria without altering its clinical meaning. If decomposition preserves the original semantics, the criterion is divided into multiple individual criteria. Otherwise, it is maintained as a compound criterion and processed as a single logical unit.

7.3.3 Logical Representation of Eligibility Criteria

To enable machine-interpretable and partially deterministic eligibility assessment, each validated criterion is converted into a JSON-based logical representation.

Simple criteria are represented as atomic rules composed of a field, an operator, and a value:

```
{
  "field": "age",
  "operator": ">=",
  "value": "18"
}
```

More complex criteria involving multiple conditions are represented through Boolean operators (`AND` / `OR`), where each operator groups a set of sub-conditions:

```
{
  "operator": "AND",
  "conditions": [
    {
      "field": "ecog_ps",
      "operator": "<=",
      "value": "1"
    },
    {
```

```
      "field": "diagnosis",
      "operator": "=",
      "value": "NSCLC"
    }
  ]
}
```

When a criterion cannot be automatically mapped to the predefined logical schema, the following representation is generated, allowing the clinician to manually define the corresponding logical rule or leaving it for LLM-assisted assessment during matching:

```
{
  "unmapped": true,
  "source_text": "unmapped part of the criterion"
}
```

Finally, all converted criteria are grouped according to their category, producing the following top-level structure:

```
{
  "inclusion_criteria": [
    {
      "id": integer,
      "text": string,
      "logic": object
    }
  ],
  "exclusion_criteria": [
    {
      "id": integer,
      "text": string,
      "logic": object
    }
  ]
}
```

7.3.4 Produced Artifacts

Upon completion of this phase, two primary artifacts are generated and made available through the system interface:

- **Clinical Trial Criteria** - a validated list of individually extracted eligibility criteria, organized into inclusion and exclusion categories;

- **Logical Rules** - the corresponding machine-readable logical representations of those criteria, also organized into inclusion and exclusion categories.

This representation allows the system to uniformly encode both simple and compound eligibility requirements. Criteria that reference fields available in the patient profile schema can subsequently be evaluated deterministically, while criteria that cannot be represented through the schema remain available for LLM-assisted assessment during the matching phase. This hybrid approach increases transparency and auditability while preserving the flexibility required to handle complex clinical requirements.

7.4 Semi-Deterministic Trial–Patient Matching

The objective of this phase is to identify potentially eligible patients for a given clinical trial through a semi-deterministic eligibility assessment process. This phase consumes the structured patient profiles generated during the first phase and the validated logical representations of eligibility criteria produced during the second phase.

The matching process comprises the following steps:

1. **Trial Selection and Patient Retrieval** - the clinician selects the clinical trial to be assessed. Based on the metadata defined during trial registration, a subset of patient profiles is retrieved from the database. Patient selection is initially restricted to the pathology group associated with the clinical trial, reducing the number of comparisons and LLM calls required during eligibility assessment.

2. **Eligibility Assessment** - each patient is evaluated against every validated eligibility criterion through one of two complementary assessment paths:

    - **Deterministic Assessment** - if the criterion references a field available in the patient profile schema, the corresponding logical rule is evaluated directly against the structured patient data.
    - **LLM-Assisted Assessment** - if the criterion references information not represented within the structured schema, the LLM receives the patient profile, clinical diary, laboratory results, and the corresponding criterion to determine eligibility. In addition to the eligibility decision, the model provides a natural-language justification to support expert review.

    Although different evaluation mechanisms are employed, both paths produce the same output: a criterion-level eligibility assessment that is subsequently used during final eligibility determination.

3. **Assessment Validation and Persistence** - all criterion-level assessments are presented to the clinician for review. Experts may modify any automatically

generated decision, regardless of whether it originated from deterministic evaluation or LLM-assisted reasoning. Following validation, all criterion assessments, manual corrections, and final eligibility decisions are persisted in the database to ensure traceability and auditability.

A formalization of this workflow is presented in Figure 7.3.

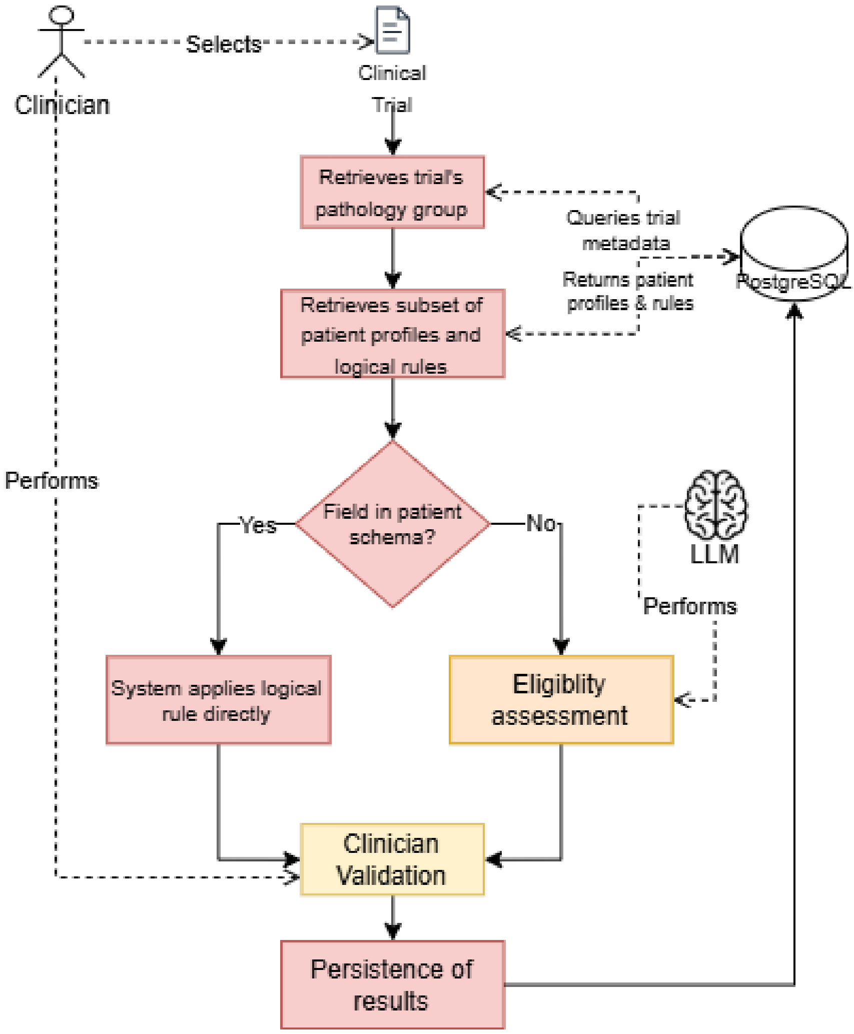


Figure 7.3: Semi-Deterministic Trial–Patient Matching

7.4.1 Matching Strategy

The proposed matching approach is semi-deterministic. Whenever a criterion references information available in the structured patient profile schema, eligibility is assessed through deterministic logical evaluation. Conversely, criteria that cannot be directly represented within the schema are evaluated through LLM-assisted reasoning.

This design maximizes the use of transparent and reproducible rule-based assessment while preserving the flexibility required to evaluate complex clinical requirements that cannot be formally encoded.

7.4.2 Eligibility Aggregation

The final patient eligibility decision is computed through a strict rule-based aggregation of all criterion-level assessments. Inclusion and exclusion criteria are evaluated independently before aggregation.

A patient is considered eligible if and only if:

- all inclusion criteria are satisfied; and
- no exclusion criterion is triggered.

Formally:

```
Eligible = (All Inclusion Criteria Satisfied)
           AND
           (No Exclusion Criteria Triggered)
```

No weighting, ranking, or probabilistic scoring mechanism is employed. The final decision relies exclusively on logical consistency across all eligibility criteria.

For clinical trials containing multiple cohorts, eligibility is evaluated at both the trial and cohort levels. A patient is considered eligible for the trial if the general eligibility requirements are satisfied and at least one cohort-specific eligibility assessment also returns an eligible result. Formally:

- the general trial criteria are satisfied; and
- at least one cohort satisfies its respective inclusion and exclusion requirements.

7.4.3 Inconclusive Assessments

Certain criteria may be impossible to evaluate reliably. This can occur due to missing clinical information, unavailable laboratory values, unit mismatches, parsing failures, ambiguous criterion definitions, or criteria that cannot be resolved through either deterministic evaluation or LLM-assisted assessment. In these situations, the corresponding criterion is classified as inconclusive.

To ensure a conservative clinical decision-support strategy:

- inconclusive inclusion criteria are treated as not satisfied;
- inconclusive exclusion criteria are treated as not triggered.

This approach prioritizes patient safety by avoiding positive eligibility decisions when essential information is unavailable or uncertain.

7.4.4 Produced Artifacts

Upon completion of this phase, two primary artifacts are generated and made available through the system interface:

- **Criterion Assessments** - individual eligibility decisions for each criterion, together with supporting natural-language justifications whenever LLM-assisted assessment is employed;
- **Eligibility Results** - a list of patients classified according to their eligibility status for the selected clinical trial and, when applicable, for individual cohorts.

The resulting assessments provide full traceability from the final eligibility decision down to the individual criterion evaluations, allowing clinicians to review, validate, and audit every step of the matching process.

7.5 User Interface Design

The elaboration of this study would not be complete without integrating the entire pipeline into a system capable of connecting every core component of the proposed solution. For this reason, a web application was developed to support the study and to provide a tangible interface for stakeholders, ensuring their full engagement throughout the process.

The application itself was guided by the requirements previously described, which proved to be an excellent roadmap for the system's development. These requirements helped structure the work, ensuring that the application would be robust and of high quality. In addition, the ER diagram presented in earlier sections played an important role by offering a clear overview of the system even before it was built. It guided not only the design of the models but also the overall reasoning behind the views, ensuring alignment with the database structure defined from the start.

The system was designed as a web application because this format best suits the clinical context, offering portability and integrating naturally with the workflow already in place. The application was developed using Django, a Python-based framework used to build both the frontend and backend. This technology was chosen due to its simplicity, its smooth learning curve, and prior experience with the framework.

All templates were created following the previously designed low-fidelity prototypes, which again proved to be a significant time-saver, since the layout and structure of each page had already been planned. Some adjustments were made during development, as the prototypes were created early in the project. A UI library, Bootstrap, was also used to improve the visual appearance of the system, reducing the effort required to refine the interface.

Every page in the platform follows the same structure to ensure a simple and coherent visual experience, improving usability and making the learning process of the system more intuitive. The web application was designed to accommodate every necessity of the stakeholder; constant contact and feedback ensured that the initial requirements clearly guided its development.

Consequently, the core structure of the application is organized into five main navigation areas, which grant access to all the system's functionalities and pipelines:

- **Home** - Serves as the central entry point to the entire system, providing access to every feature and offering an immediate overview of the platform's current state. This page presents several key metrics that summarize the progress and activity of the system, alongside quick access to the most recently uploaded or produced artifacts. This ensures that clinicians can begin working immediately upon entering the application.
- **Developer Tools** - Dedicated to system administration rather than clinical use, this page provides valuable metrics regarding the overall health of the platform. These metrics include the number of stored documents, versioning health, and the performance of the matching engine. It also features a system overview to guide users who may feel unfamiliar with the application's structure.
- **Clinical Diaries** - Presents a centralized list of all uploaded clinical diaries, allowing users to verify their processing status, filter documents, and manually upload new files. This view also serves as the gateway to detailed artifact inspections and the parameter extraction pipeline.
- **Clinical Trials** - Displays all uploaded clinical trials within the system, tracking whether they are pending or successfully processed. It includes robust filtering and search mechanisms by study name or pathology. From this area, users can access detailed trial views, manage criteria extraction/conversion pipelines, and trigger patient-trial matching.
- **Patients** - Displays the structured patient profiles generated by the extraction pipeline. It allows clinicians to filter patients by stage, pathology group, or molecular status. Selecting a patient opens a comprehensive profile containing clinical overviews, treatments, and laboratory analyses, while also offering the ability to reset the patient state if reprocessing is required.

For comprehensive details regarding the interface layout, step-by-step pipeline views (including criteria extraction, logical conversion, and patient matching screens), and the complete set of application screenshots, please consult Appendix E.

Chapter 8

Materials and Methods

This chapter presents the materials and methods used to develop and evaluate the proposed LLM-assisted framework for clinical trial recruitment. It describes the characteristics and abstractions of the input documents, specifically clinical trial protocols and clinical diaries, followed by a detailed overview of the datasets and manually curated gold standards used for evaluation. Furthermore, this chapter outlines the core pipelines of the proposed semi-deterministic matching method, its guiding research hypotheses, and the comprehensive experimental setup, including the evaluation scenarios, metrics, and open-source large language models analyzed in this study.

8.1 Documents as Inputs

This section describes the two document types processed by the proposed system and the abstractions adopted to model them within the pipeline.

The proposed framework operates on two primary document sources: clinical diaries and clinical trial protocols. Although both are written in free-text format, they differ substantially in their structure, purpose, and degree of standardization. Understanding these characteristics is essential for the design of the extraction and matching pipelines.

8.1.1 Clinical Trials

Clinical trial protocols are predominantly narrative documents describing study objectives, methodology, treatment arms, safety procedures, and participant selection requirements. Despite variations across studies, trial protocols generally follow a well-defined organizational structure.

Among all protocol sections, eligibility criteria constitute the only component directly involved in patient selection. Inclusion and exclusion criteria are expressed in natural language but fundamentally represent constraints over patient attributes, such as demographic characteristics, diagnoses, biomarkers, treatment history, laboratory values, or performance status.

However, eligibility structures vary considerably between trials. Some studies define a single set of eligibility requirements, while others contain multiple cohorts, each with its own inclusion and exclusion criteria. Furthermore, eligibility requirements may range from simple atomic conditions to highly complex compound criteria involving multiple logical dependencies.

To enable automated processing, the proposed system abstracts clinical trial documents into two components: trial structure and eligibility criteria. This separation allows descriptive protocol information to remain independent from the patient selection logic. Consequently, LLM-based extraction can focus exclusively on eligibility requirements, which are subsequently converted into machine-readable logical representations suitable for automated evaluation.

All clinical trials used throughout this study were obtained in English.

8.1.2 Clinical Diaries

Clinical diaries represent the primary source of patient information used by the system. Unlike clinical trial protocols, these documents exhibit substantial variability in writing style, organization, terminology, and level of detail. Such variability arises from differences between institutions, clinical specialties, individual physicians, and documentation practices.

The design of the extraction pipeline was guided by a structured template provided by the clinical stakeholder and derived from four representative oncology clinical diaries routinely used within the Oncology Department of Coimbra University Hospital Centre (HUC). Rather than defining a fixed document structure, these diaries were analysed to identify recurring clinical concepts and information categories required for patient eligibility assessment.

From this analysis, a patient profile schema was defined, containing the hyper-relevant clinical parameters considered essential for trial matching. These parameters serve as the target representation of the extraction pipeline and provide a consistent structured format regardless of how information is expressed in the original diary.

To avoid overfitting the system to a specific documentation template, the abstraction process focused on preserving semantic information rather than document layout or wording. Structural elements, section ordering, formatting conventions, and stylistic patterns were intentionally disregarded whenever they were not clinically relevant. Only information categories consistently required for eligibility assessment, such as diagnosis, staging, treatments, biomarkers, ECOG performance status, and disease evolution, were retained in the final schema.

This abstraction enables the system to process heterogeneous clinical documentation while producing a standardized representation suitable for automated eligibility assessment. Although the proposed schema was derived from documents used within a specific oncology department, it was designed to capture clinically meaningful concepts rather than institution-specific formatting conventions.

All clinical diaries used throughout this study were written in European Portuguese.

8.2 Datasets

This section describes the datasets used to evaluate the components of the proposed system, including synthetic oncology clinical diaries, clinical trial documentation, and manually curated gold standards. A quantitative summary of these normalized datasets is provided in Table 8.1. All data used in this study were fully anonymized; no identifiable patient information was used, and every synthetic diary contains no real clinical content.

8.2.1 Synthetic Oncology Clinical Diaries

The system evaluation was conducted using a dataset of 30 synthetic free-text oncology clinical diaries. These diaries were generated using the open-source `gpt-oss-120b` model, selected for its high generation capacity, and were grounded on four representative clinical diaries provided by the clinical stakeholder to serve as references for writing style, structure, and clinical content. The dataset size balances clinical diversity with the feasibility of manual annotation.

To ensure representativeness, 30 unique patient profiles were first created, each describing a distinct oncological case. The generation pipeline consisted of two sequential stages designed to reproduce the variability observed in real-world clinical documentation:

1. **Idealized Generation** - The LLM generated a clinically coherent and internally consistent Portuguese oncology note from the patient profile, maintaining alignment between diagnosis, staging, treatments, imaging findings, ECOG performance status, comorbidities, and disease evolution.

2. **Noise and Artifact Insertion** - To resemble routine hospital records, the generated diary was transformed by introducing realistic imperfections while preserving complete clinical correctness. For each diary, two generation factors were randomly assigned: a *length mode* (short, medium, or long) and a *style mode* (narrative-dominant, telegraphic hospital style, exam-and-imaging focused, toxicity focused, or psychosocial emphasis). The model introduced variations in sentence length, irregular formatting, heterogeneous abbreviation usage, minor spelling imperfections, structural asymmetries (e.g., missing headers), and administrative workflow notes.

To guide this process, the prompts incorporated examples of authentic clinical documentation exclusively for stylistic learning, with explicit instructions not to reproduce original clinical content. Following generation, an experienced oncologist reviewed and validated all diaries for clinical plausibility and consistency.

After normalization (removing headers, footers, and irrelevant sections), the clinical diary dataset contained a total of 64,953 tokens across the 30 documents, averaging 2,165 tokens per diary.

8.2.2 Clinical Trials Dataset

A dataset of 30 clinical trials was assembled from *ClinicalTrials.gov* to evaluate the criteria extraction, conversion, and patient–trial matching pipelines, seeking to maximize clinical diversity and minimize selection bias.

The trial collection covers a broad range of tumor types (including breast, lung, pancreatic, colorectal, melanoma, among others), disease stages, molecular profiles, and therapeutic settings. The selected trials encompass different designs (with and without cohort structures) and exhibit substantial variability in eligibility complexity, ranging from simple demographic or diagnostic constraints to highly specialized molecular biomarkers and prior treatment histories.

After normalization, the clinical trial dataset contained a total of 120,786 tokens across the 30 protocols, averaging 4,026 tokens per trial, reflecting the complexity of oncology requirements. The eligibility criteria corpus comprises a total of 892 individual criteria (480 inclusion and 412 exclusion criteria), averaging approximately 30 eligibility criteria per trial.

8.2.3 Manually Curated Gold Standards

Gold standards were manually annotated to provide reliable reference data for system evaluation across three core tasks. For the remaining two tasks, no gold standard was defined in advance; instead, model outputs were assessed directly through manual review.

- **Hyper-Relevant Parameter Extraction** - All fields defined in the patient profile schema were manually identified and annotated within the 30 clinical diaries to enable automated, deterministic comparisons with the LLM outputs.
- **Trial Structure Identification** - The organization of eligibility requirements and the presence of cohorts were manually mapped following the system's formal representation.
- **Criteria Extraction** - All 892 inclusion and exclusion criteria were manually extracted as individual requirements to evaluate the completeness of the automated extraction.
- **Criteria Conversion** - No gold standard was produced; instead, the correctness of each generated logical formalization was directly and manually assessed against the original eligibility criterion.

- **Patient-Trial Matching** - No gold standard was produced; instead, each LLM-generated eligibility decision was manually assessed together with its accompanying justification to determine whether the decision was correct.

Table 8.1: Quantitative summary of the normalized datasets used in this study

Dataset Component	Total	Average per Document
Clinical Diaries (tokens)	64,953	2,165
Clinical Trials (tokens)	120,786	4,026
Inclusion Criteria	480	16
Exclusion Criteria	412	14
Total Criteria	892	30

8.2.4 Proposed Method

The proposed method aims to support clinical trial recruitment by combining clinical information extraction and patient–trial matching within a single LLM-assisted framework. Rather than relying exclusively on probabilistic reasoning, the approach seeks to maximize deterministic processing whenever possible by transforming both patient information and trial eligibility criteria into structured representations that can be automatically compared.

The method is composed of three sequential tasks. First, relevant clinical information is extracted from free-text oncology clinical diaries and transformed into structured patient profiles. Second, eligibility criteria are extracted from clinical trial documentation and converted into machine-interpretable logical representations. Finally, patient profiles and trial criteria are combined through a semi-deterministic matching process to identify potentially eligible patients.

The central premise of the proposed approach is that eligibility assessment can be significantly improved by separating complex reasoning into multiple specialized tasks. Instead of asking a model to directly determine patient eligibility from raw documents, the system decomposes the problem into information extraction, criteria formalization, and eligibility assessment stages. This decomposition aims to improve interpretability, traceability, and clinician oversight while reducing reliance on end-to-end LLM reasoning.

The overall system was designed with real clinical environments in mind and was iteratively discussed with domain experts to ensure practical applicability. As a consequence, every LLM-assisted step was built to guarantee interpretability and auditability by allowing direct comparison between outputs and their source documents, while storing and clearly labeling all intermediate results. Throughout the pipeline, the clinician remains the final approver and editor of all outputs, ensuring clinical safety while leveraging LLMs to accelerate the workflow.

A secondary objective of this study is to evaluate whether relatively small, non-proprietary open-source language models can achieve satisfactory performance across the different stages of the pipeline. Demonstrating competitive performance with such models would increase the feasibility of deploying the proposed framework in environments with limited computational resources and stricter privacy requirements.

The proposed method is evaluated through the following tasks, each task is evaluated independently to isolate error sources and quantify the contribution of each module:

- Extraction and normalization of hyper-relevant patient information from free-text clinical diaries;
- Identification of clinical trial eligibility structures, (cohort detection);
- Extraction of individual inclusion and exclusion criteria;
- Conversion of eligibility criteria into machine-interpretable logical representations;
- Semi-deterministic patient–trial matching through the combination of deterministic and LLM-assisted assessments.

The study is guided by the following research hypotheses:

- H1: Large Language Models can accurately extract and normalize hyper-relevant clinical information from heterogeneous oncology clinical diaries while adhering to a predefined structured schema.
- H2: Large Language Models can accurately identify, extract, and formalize clinical trial eligibility criteria while preserving their original clinical meaning and logical relationships.
- H3: A hybrid eligibility assessment strategy combining deterministic rule evaluation and LLM-assisted reasoning can effectively support patient–trial matching while maintaining interpretability and clinician control.
- H4: Small open-source language models can achieve performance levels sufficient to support the proposed workflow across the evaluated tasks.

8.3 Experimental Setup

As previously described, the proposed framework consists of three LLM-assisted pipelines:

- Hyper-Relevant Parameter Extraction;
- Extraction and Conversion of Clinical Criteria;
- Semi-Deterministic Patient–Trial Matching.

The experimental evaluation was designed to assess the performance of the LLM-dependent components of each pipeline independently. Evaluating each stage separately allows the identification of potential error sources and provides a clearer understanding of the contribution of each component to the overall system performance.

The experiments aim to evaluate both the effectiveness and robustness of the proposed approach, as well as its suitability for integration into a semi-automated clinical decision support workflow. Particular emphasis is placed on accuracy, reliability, interpretability, and the ability of the system to operate under clinician supervision.

8.3.1 Evaluation Scenarios

To evaluate the different stages of the proposed framework, two primary experimental scenarios were defined.

Scenario 1 – Clinical Diary Processing

This scenario evaluates the Hyper-Relevant Parameter Extraction pipeline. Each model receives an individual clinical diary and is tasked with extracting and normalizing the information required to populate the predefined patient profile schema.

The objective is to assess the ability of the models to identify clinically relevant information from heterogeneous free-text oncology documentation while generating outputs that comply with the required structured representation.

Scenario 2 – Clinical Trial Processing

This scenario evaluates the Extraction and Conversion of Clinical Criteria pipeline. Each model receives an individual clinical trial protocol and performs the following tasks:

- Identification of the trial eligibility structure, including cohort detection when applicable;
- Extraction of inclusion and exclusion criteria;
- Conversion of extracted criteria into machine-interpretable logical representations.

The objective is to assess whether the models can preserve the clinical meaning of eligibility requirements while transforming them into structured representations suitable for automated evaluation.

Scenario 3 - Patient-Trial Matching This scenario evaluates the LLM-assisted component of the Semi-Deterministic Patient–Trial Matching pipeline. In cases where eligibility criteria cannot be directly evaluated through deterministic logical rules, either due to incomplete formalization or the presence of schema-independent requirements, the model receives the structured patient profile, the original clinical diary, relevant laboratory information, and the eligibility criterion represented as a logical rule. The model is then tasked with determining whether the available patient information satisfies the given criterion and providing a clinically grounded justification for the decision.

The objective is to assess the ability of the models to perform reliable eligibility assessments when operating under constrained and explicit clinical context, as well as the quality and transparency of the generated clinical justifications.

All models are evaluated under identical experimental conditions to ensure fair and reproducible comparisons.

To further improve reproducibility, prompts are version-controlled and preserved throughout the experiments, and all model configuration parameters remain fixed across evaluation runs.

8.3.2 Evaluation Metrics

Different evaluation metrics are employed depending on the task under analysis.

Hyper-Relevant Parameter Extraction

The extraction of structured patient information is evaluated using:

- Field-level Accuracy: proportion of correctly extracted fields relative to the manually annotated gold standard;
- Missing Field Rate: proportion of schema fields returned as null or omitted;
- Schema Compliance Rate: percentage of outputs that fully comply with the predefined JSON schema.

Trial Structure Identification

The identification of clinical trial structure is evaluated through:

- Cohort Detection Accuracy: correctness of identifying whether cohort structures are present;
- Structure Description Correctness: expert assessment of the generated trial structure representation.

Eligibility Criteria Extraction

The extraction of inclusion and exclusion criteria is evaluated through:

- Criteria Extraction Correctness: manual and expert assessment of whether extracted criteria faithfully represent the original trial requirements.

Eligibility Criteria Conversion

The conversion of eligibility criteria into logical rules is evaluated through:

- Logical Representation Correctness: Manual assessment indicating whether the generated logical representation accurately reflects the clinical meaning of the original criterion.

Patient–Trial Matching

The semi-deterministic matching process is evaluated through:

- Eligibility Assessment Accuracy: correctness of the generated eligibility decision relative to the reference documentation of the patient (clinical diary, laboratory values, and patient profile);
- Justification Quality: expert evaluation of the clinical coherence and usefulness of generated explanations using a five-point Likert scale.

8.3.3 Evaluated Models

The experiments compare multiple open-source Large Language Models representing different model families and architectural approaches:

- Qwen3 8B Instruct;
- Llama 3.1 8B Instruct;
- DeepSeek-R1 8B.

These models were selected because they provide a representative sample of state-of-the-art open-source LLMs that can be deployed with relatively modest computational resources.

All models are evaluated using deterministic inference settings to minimize output variability and maximize reproducibility. No model fine-tuning is performed, and all experiments are conducted under a zero-shot setting. This configuration reflects the intended deployment scenario of the proposed framework, where task-specific training data may not be available and rapid adaptation to new clinical contexts is desirable.

Each experiment was executed once under deterministic inference settings, on the datasets of each 30 clinical diaries and 30 clinical trials.

8.4 Unit Testing

To guarantee that the web application functions as intended, an extensive suite of unit tests was implemented. These tests were designed to validate the correctness, stability, and error-handling capabilities of every component within the system, ranging from underlying database structures and utility functions to the core view components governing the execution pipelines. In total, 388 unit tests were executed, ensuring predictable behavior and enabling the early detection and mitigation of edge-case bugs.

The overall testing strategy was divided into four foundational pillars, each targeting a specific layer of the application architecture:

- **Database Models** - Validates the structural backbone of the system's database. These tests ensure accurate data persistence, enforce entity relationship integrity (such as cascade deletes and foreign keys), and verify uniqueness constraints. This layer is crucial since downstream pipeline components rely heavily on the architectural consistency of previously stored artifacts.
- **Helpers & Auxiliary Functions** - Focuses on isolated utility functions that support the main pipelines (e.g., text normalization, JSON extraction, and conditional evaluation algorithms). Testing these components independently ensures system modularity and guarantees that mathematical, parsing, or formatting operations do not introduce side effects into the core pipelines.
- **Application Views & Dashboard Routing** - Assesses the system's request-response lifecycle, template rendering, and HTTP status codes (e.g., handling missing resources or invalid form submissions). This covers everything from the central metrics dashboard to specialized listing, deletion, and file upload views, securing user interaction boundaries.
- **Core Pipeline Modules** - Validates the complete execution flow of the system's intelligent pipelines. Given their complexity and reliance on Large Language Models (LLMs), these modules underwent granular validation across specific subcomponents, as summarized in Table 8.2.

For a comprehensive breakdown of each unit test, including specific input/output test vectors, example execution traces, and detailed evaluations of individual helper functions, please consult Appendix F.

Table 8.2: Summary of Unit Test Scopes across Core Pipeline Modules.

Pipeline Module	Target Testing Scopes
Parameter Extraction	Guard clauses, happy path execution, clinical profile instantiation, retry mechanics on corrupted LLM outputs, and laboratory value/unit parsing.
Criteria Extraction & Structure	Input validation boundaries, single vs. multi-cohort branching logic, textual chunking/splitting mechanisms, and criteria deduplication arrays.
Criteria Conversion	Request interception boundaries, multi-rule batch processing workflows, logical constraint rendering, and rule dependency ordering utilities.
Eligibility Assessment	Deterministic logical evaluations, operator boundary testing ($>=$, IN, CONTAINS), lab parameter evidence extraction, and expert override persistence.

Chapter 9

Results

As mentioned before the selected language models were evaluated in 5 tasks: Hyper-Relevant Parameter Extraction; Trial Structure Identification; Eligibility Criteria Extraction; Eligibility Criteria Conversion; and Patient–Trial Matching.

Hyper-Relevant Parameter Extraction

The first experimental task evaluated the capability of the LLMs to extract clinically relevant information from free-text oncology clinical diaries while adhering to a predefined JSON schema. The evaluation focused on three main aspects: field-level extraction accuracy, schema compliance, and missing field rate. Field-level accuracy was computed by manually comparing the generated patient profiles against the manually annotated gold standards. Two values are reported: strict accuracy, considering only fully correct extractions, and a partial accuracy score, which also accounts for partially correct extractions.

As shown in Table 9.1, all evaluated models were able to consistently follow the predefined output structure, achieving 100% schema compliance. This demonstrates that the evaluated models were capable of generating structurally valid outputs according to the required JSON representation, regardless of their differences in extraction performance.

Regarding extraction accuracy, Qwen3-8B achieved the highest overall performance, obtaining a strict accuracy of 78.3% and a partial accuracy score of 83.3%. DeepSeek-R1-8B followed with 74.3% strict accuracy and 80.5% partial accuracy. Llama3.1-8B presented the lowest performance, achieving 62.3% strict accuracy and 70.3% partial accuracy, with a higher number of incorrect and partially extracted fields. These results indicate that although all models were capable of producing valid structured outputs, their ability to correctly identify and populate clinical information varied considerably. Figure 9.1c illustrates the comparison between strict and partial accuracy across models.

The missing field rate followed a similar trend. Qwen3-8B and DeepSeek-R1-8B presented lower rates of missing information (11.6% and 13.3%, respectively), whereas Llama3.1-8B showed a considerably higher missing field rate of 21.7%.

This suggests that weaker models had more difficulty identifying whether relevant clinical information was present within the diary and subsequently extracting it into the required schema fields.

Table 9.2 provides a more detailed field-level analysis across the patient profile schema. Some fields were consistently extracted with high accuracy across all models. Gender and ECOG-PS achieved 100% accuracy for every evaluated model, indicating that these parameters are generally explicit and unambiguous within oncology clinical documentation. Figure 9.1a further illustrates the performance distribution across schema fields.

Conversely, more complex fields requiring interpretation, normalization, or structured reasoning presented lower performance. Diagnosis extraction represented one of the most challenging fields, with DeepSeek-R1-8B achieving the highest accuracy at 46.7%. However, manual inspection revealed that most errors were not related to failure in identifying the diagnosis itself, but rather to difficulties in applying the required normalization mapping. In several cases, models extracted the correct clinical diagnosis but failed to convert it into the expected normalized representation.

Treatment extraction was another challenging task due to its hierarchical structure, requiring the identification of multiple sub-fields, including treatment name, start date, and end date. Qwen3-8B achieved the best performance in this field, with 70.0% strict accuracy and an 85.0% partial accuracy score, while Llama3.1-8B achieved substantially lower results.

Disease staging also proved challenging, particularly due to the need to preserve the complete staging representation. A recurrent error pattern was the extraction of only the final stage classification (e.g., "IVB") instead of the complete TNM representation (e.g., "cT3N2M1b (IVB)"). Although the clinically relevant information was partially captured, these outputs were considered incomplete according to the defined schema.

Date-related fields, particularly diagnosis dates and treatment periods, also represented a source of errors across models. These errors consisted mainly of incorrect extraction, incomplete temporal information, or returning null values when dates were present in the clinical diary.

Overall, the results demonstrate that lightweight LLMs are capable of performing structured clinical information extraction from heterogeneous oncology documentation. However, performance is strongly influenced by the complexity of the target field, with explicit clinical attributes being extracted reliably, while fields requiring normalization, temporal reasoning, or hierarchical interpretation remain more challenging. Figure 9.1b summarises the schema compliance rate across all evaluated models.

Trial Structure Identification

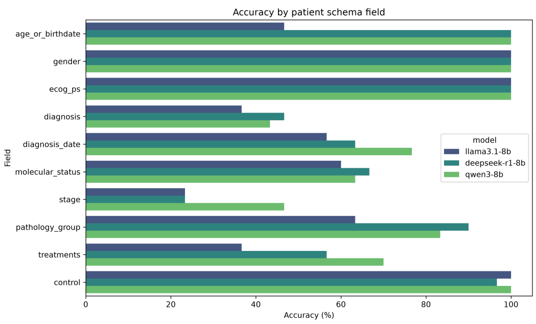
The second experimental task evaluated the capability of the LLMs to identify and represent the internal structure of clinical trials according to a predefined

Table 9.1: Global performance of the evaluated models (Hyper-Relevant Parameter Extraction)

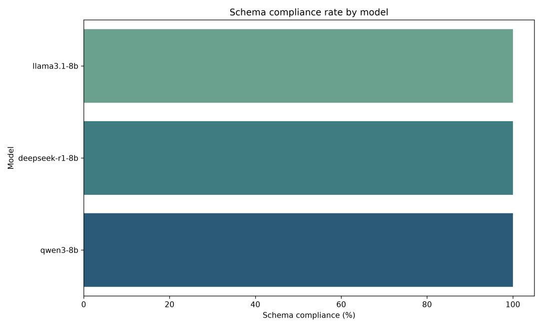
Models	Total	Correct	Partial	Wrong	Acc./Partial (%)	Schema (%)	Missing (%)
Llama3.1-8b	300	187	48	65	62.3/70.3	100.0	21.7
Deepseek-r1:8b	300	223	37	40	74.3/80.5	100.0	13.3
Qwen3-8b	300	235	30	35	78.3/83.3	100.0	11.6

Table 9.2: Field-level performance across schema components

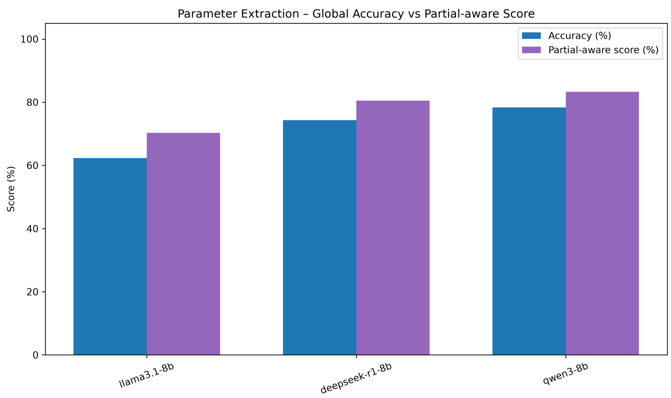
Schema Fields	Llama3.1-8b	Deepseek-r1:8b	Qwen3-8b
age_or_birthdate	46.6/46.6	100.0/100.0	100.0/100.0
gender	100.0/100.0	100.0/100.0	100.0/100.0
ecog_ps	100.0/100.0	100.0/100.0	100.0/100.0
diagnosis	36.7/36.7	46.7/46.7	43.3/43.3
diagnosis_date	56.7/56.7	63.3/63.3	76.7/76.7
molecular_status	60/71.7	66.7/73.3	63.3/71.7
stage	23.3/61.7	23.3/61.7	46.7/73.3
pathology_group	63.3/63.3	90.0/90.0	83.3/83.3
treatments	36.7/66.7	56.7/73.3	70.0/85.0



(a) Accuracy by schema field



(b) Schema compliance rate



(c) Strict vs partial accuracy

Figure 9.1: Hyper-Relevant Parameter Extraction — Accuracy by field, schema compliance, and strict vs partial accuracy.

JSON schema. This step is required for subsequent eligibility assessment, as cohort-specific requirements may define different patient populations within the same clinical trial.

The evaluation considered two aspects: (i) identification of whether formal cohort structures were present, and (ii) correctness of the generated cohort representation. The generated outputs were compared against manually annotated gold standards. Strict accuracy, considering only fully correct outputs, and partial accuracy, accounting for partially correct representations, were reported.

As shown in Table 9.3, substantial differences were observed between models. Deepseek-r1:8b achieved the best performance, correctly identifying 58 out of 60 evaluated fields, corresponding to 96.7% strict accuracy. In contrast, Llama3.1-8b achieved considerably lower performance (25.0%), mainly due to difficulties distinguishing formal trial cohorts from descriptive patient groups. Qwen3-8b obtained intermediate results, achieving 60.0% strict accuracy and 75.0% partial accuracy.

Table 9.3: Global performance of the evaluated models (Trial Structure Identification)

Model	**Total**	**Correct**	**Partial**	**Wrong**	**Acc./Partial (%)**
Llama3.1-8b	60	15	0	45	25.0/25.0
Deepseek-r1:8b	60	58	0	2	96.7/96.7
Qwen3-8b	60	36	18	6	60.0/75.0

Further analysis showed that the main limitation of Llama3.1-8b was the incorrect interpretation of descriptive clinical groups as formal cohorts. Phrases describing patient characteristics, such as disease stage or molecular status, were frequently interpreted as independent cohort structures, leading to false cohort identification.

For Qwen3-8b, most errors were related to schema compliance rather than semantic understanding. In several cases, the model correctly identified the absence of cohorts but returned incomplete JSON structures, omitting required fields such as the empty cohort array. Conversely, when explicit cohort structures were present, all models demonstrated substantially higher performance, indicating that the main challenge was distinguishing cohort absence from descriptive patient categories.

Beyond structural correctness, a qualitative expert assessment was conducted on a subset of four clinical trials (Trials 5, 16, 26, and 30) to evaluate whether generated cohort descriptions preserved clinically relevant information. The assessment focused on the maintenance of diagnosis, molecular alterations, previous treatment exposure, and therapeutic requirements. The summarized expert evaluation of the evaluated models is presented in Table 9.4.

The expert evaluation demonstrated that correct cohort identification does not necessarily guarantee preservation of clinical meaning. Across the analyzed trials, the main errors were related to omission of eligibility-defining information rather than incorrect cohort structures.

Table 9.4: Expert assessment of cohort-structure classification performance

Model	Avg. Class.	Expert Assessment Summary
Deepseek-r1:8b	2.25	Correctly identifies cohort structures but frequently omits clinically relevant descriptors such as treatment status, molecular alterations, or diagnostic specificity.
Llama3.1-8b	2.75	Achieves high semantic fidelity in some trials but presents inconsistent behaviour, including complete loss of clinical information in specific cases.
Qwen3-8b	2.50	Correctly captures several cohort characteristics but occasionally omits essential molecular or therapeutic information.

Overall, these results indicate that cohort identification is primarily limited by the preservation of the complete clinical definition of each cohort. While structural extraction can be performed reliably, maintaining all clinically relevant eligibility characteristics remains a challenging aspect of LLM-based trial document understanding, as illustrated by the strict accuracy comparison between models (Figure 9.2).

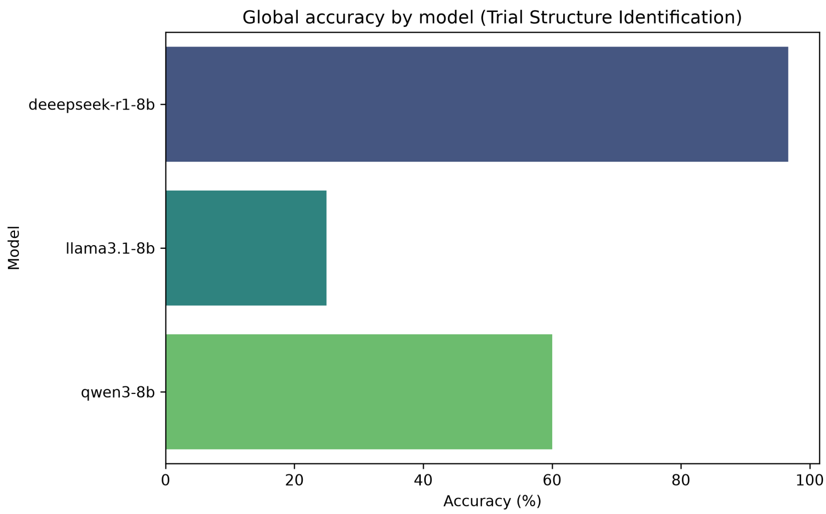


Figure 9.2: Trial Structure Identification - Strict accuracy

Eligibility Criteria Extraction

The third experimental task evaluated the capability of the LLMs to extract individual clinical trial eligibility criteria while preserving their original clinical meaning and identifying whether each criterion was associated with a specific cohort. This step represents a fundamental component of the proposed pipeline, as accurate criteria extraction is required before their conversion into machine-interpretable rules and subsequent patient–trial matching.

The evaluation considered two complementary aspects: (i) the extraction of individual inclusion and exclusion criteria from clinical trial documents, and (ii) the attribution of extracted criteria to the corresponding cohort when cohort-specific eligibility requirements were present. Generated outputs were manually compared against manually annotated gold standards. Two metrics were reported: strict accuracy, considering only completely correct extractions, and partial accuracy, which also accounted for outputs that preserved the majority of the clinical meaning but contained minor structural or semantic deviations.

As presented in Table 9.5, substantial differences were observed between models. Qwen3-8b achieved the strongest overall performance, correctly extracting 823 out of 892 eligibility criteria, resulting in a strict accuracy of 92.3% and a partial accuracy of 95.9%. Furthermore, the model produced only four incorrect extractions, demonstrating a strong capability of preserving the original eligibility information while avoiding hallucinated criteria. Deepseek-r1:8b achieved intermediate performance, with strict and partial accuracies of 83.7% and 88.2%, respectively. In contrast, Llama3.1-8b presented the lowest performance, achieving 71.9% strict accuracy and 75.8% partial accuracy, while also generating the highest number of incorrect criteria.

The second component of this task focused on identifying whether extracted criteria belonged to a specific trial cohort, with the corresponding results presented in Table 9.6. In this scenario, Qwen3-8b again demonstrated superior performance, correctly assigning cohort-specific criteria in 26 out of 27 cases, achieving 96.3% accuracy. Both Deepseek-r1:8b and Llama3.1-8b achieved 66.7% accuracy. The errors observed in these models were mainly associated with incomplete extraction rather than incorrect cohort attribution, suggesting that the main limitation was failure to preserve the complete criterion information before the classification step.

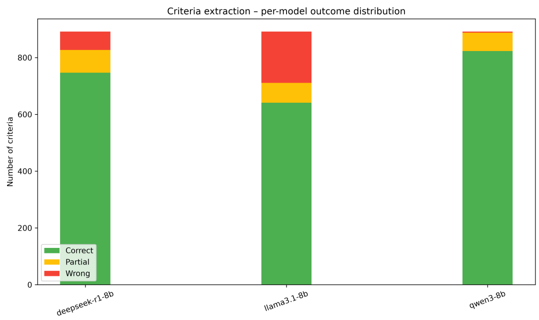
Table 9.5: Global performance of the evaluated models (Eligibility Criteria Extraction)

Model	**Total**	**Correct**	**Partial**	**Wrong**	**Acc./Partial (%)**
Llama3.1-8b	892	641	70	181	71.9/75.8
Deepseek-r1:8b	892	747	80	65	83.7/88.2
Qwen3-8b	892	823	65	4	92.3/95.9

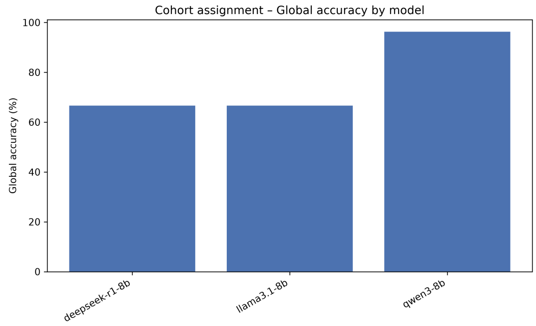
The distribution of extraction outcomes and cohort assignment accuracy across models are further illustrated in Figures 9.3a and 9.3b, respectively.

Table 9.6: Performance on criteria requiring cohort-structure extraction

Model	Total	Correct	Partial	Wrong	Acc./Partial (%)
Llama3.1-8b	27	18	0	9	66.7/66.7
Deepseek-r1:8b	27	18	0	9	66.7/66.7
Qwen3-8b	27	26	0	1	96.3/96.3



(a) Eligibility Criteria Extraction - Outcome distribution



(b) Eligibility Criteria Extraction - Cohort assignment strict accuracy

Figure 9.3: Eligibility Criteria Extraction — Overview of extraction outcomes and cohort assignment accuracy

Although the automatic evaluation provides an objective measurement of extraction performance across the complete dataset, clinical interpretation of eligibility criteria requires assessing whether deviations from the reference representation result in meaningful changes in clinical interpretation. Therefore, a subset of extracted criteria was additionally reviewed by a clinical expert to characterize the severity and nature of possible errors.

The expert assessment confirmed the strong extraction capabilities observed in the automatic evaluation, particularly for Qwen3-8b and Deepseek-r1:8b. Across the evaluated trials, Qwen3-8b consistently preserved the clinical meaning of the eligibility criteria, with observed differences mainly related to acceptable restructuring or expansion of compound criteria. For example, when a broad criterion such as adequate organ function was divided into multiple individual laboratory constraints, the original clinical meaning was maintained and no clinically relevant information was lost.

Deepseek-r1:8b also demonstrated high clinical fidelity in the expert-reviewed subset, achieving complete preservation of inclusion and exclusion criteria in the evaluated trials. The expert analysis confirmed that the model was able to maintain compound requirements, avoid hallucinations, and preserve the relationship between multiple clinical constraints. These findings complement the quantitative results, where Deepseek showed intermediate but consistent extraction performance.

Llama3.1-8b showed a more variable behavior. Although the model successfully extracted most criteria, the expert analysis identified occasional structural errors that could affect downstream processing. In particular, one observed error consisted of splitting a compound tumor-size eligibility criterion into separate entries. Although all relevant information was preserved, this modification altered the original logical structure of the requirement and could potentially affect later conversion into machine-readable rules.

Overall, the expert validation indicated that most model deviations were related to representation differences rather than incorrect clinical interpretation. The main clinically relevant failure modes involved omission of eligibility components, incorrect fragmentation of compound criteria, or loss of relationships between conditions. These observations highlight the importance of maintaining a clinical validation step when transforming natural language eligibility requirements into computational representations.

Eligibility Criteria Conversion

The fourth experimental task evaluated the capability of the LLMs to transform individually extracted clinical trial eligibility criteria into machine-interpretable logical representations while preserving their original clinical meaning. This step represents a critical component of the proposed pipeline, as the generated logical rules enable the deterministic matching engine to perform patient–trial eligibility assessment with reduced dependency on LLM-based reasoning during the final decision process.

The evaluation focused on the conversion of individual inclusion and exclusion criteria previously extracted from clinical trial documents. Generated logical representations were manually evaluated to identify if they kept the clinical meaning of the criteria. Two metrics were reported: strict accuracy, considering only fully correct conversions, and partial accuracy, accounting for cases where the majority of the clinical meaning was preserved but minor structural deviations or incomplete logical representations were present.

It is important to note that the number of evaluated converted criteria differed between models. This difference resulted from occasional generation failures, where models produced explanations, additional text, or invalid outputs instead of the required structured representation, despite explicit instructions to return only the converted criterion. These cases were considered conversion failures and excluded from the correctness assessment, while still reflecting the model's ability to follow the required output format.

As shown in Table 9.7, the conversion task represented a considerably more challenging problem compared with previous extraction tasks. Qwen3-8b achieved the highest strict accuracy, correctly converting 651 criteria, corresponding to 64.1% strict accuracy and 74.4% partial accuracy. However, it also produced the highest number of incorrect conversions (156), indicating that while the model was frequently capable of producing complete logical representations, failures tended to result in clinically incorrect rules rather than partially incomplete ones.

Llama3.1-8b achieved comparable performance, with 62.5% strict accuracy and 73.5% partial accuracy. Despite converting fewer criteria than Qwen3-8b, the model demonstrated a similar balance between correct and partially correct outputs. Deepseek-r1:8b presented the lowest strict accuracy (52.2%), although its partial accuracy increased substantially to 71.2%, indicating that many of its errors were related to incomplete logical formalization rather than complete misunderstanding of the eligibility requirement.

Given the moderate performance observed across all 8b-scale models, an additional experiment was conducted using a significantly larger model, Gpt-oss-120b, to assess whether increased model capacity could improve conversion quality.

Table 9.7: Global performance of the evaluated models (Eligibility Criteria Conversion)

Model	Total	Correct	Partial	Wrong	Acc./Partial (%)
Llama3.1-8b	969	606	212	151	62.5/73.5
Deepseek-r1:8b	1172	612	444	116	52.2/71.2
Qwen3-8b	1016	651	209	156	64.1/74.4
Gpt-oss-120b	1096	713	348	35	65.1/80.9

The distribution of correct, partial, and incorrect conversions across models is further illustrated in Figure 9.4.

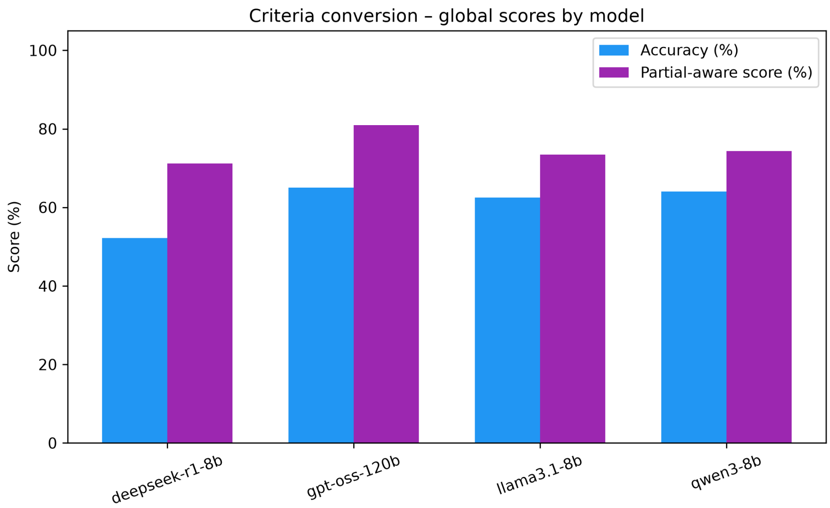


Figure 9.4: Eligibility Criteria Conversion

Gpt-oss-120b achieved the best overall results, with 65.1% strict accuracy and 80.9% partial accuracy, and a substantially lower wrong count (35) compared to the smaller models. Notably, the majority of its partial conversions did not stem from failures in capturing complex logical structures, as observed in the other models, but rather from incorrect normalization of criteria fields to the defined schema, particularly the treatment field, resulting in non-schema fields that would require unnecessary LLM-based evaluation downstream.

A common failure pattern observed across all evaluated models was the difficulty in converting complex compound criteria. Eligibility requirements containing nested logical relationships, particularly combinations of multiple AND/OR conditions, frequently resulted in partially represented rules. Instead of preserving the complete logical structure, models often extracted only the most prominent condition while omitting secondary constraints or incorrectly simplifying relationships between conditions.

Another common problem was related to the conversion of said fields to schema fields, specially to the treatment field, the models consistently failed to identify that certain criteria could easily be represented with the patient schema field of treatments, resulting in new, not schema included fields, that would force an LLM evaluation that wouldn't be necessary.

This limitation was particularly relevant because compound eligibility criteria are common in oncology clinical trials, where patient selection often depends on the simultaneous evaluation of diagnosis, molecular profile, previous treatments, laboratory values, and disease status. Therefore, although the models demonstrated reasonable capability in converting simpler eligibility constraints, additional improvements are required for reliably formalizing complex clinical decision rules.

Patient–Trial Matching

The final experimental task evaluated the capability of the LLMs to assess patient eligibility when confronted with clinical requirements that could not be directly represented within the predefined patient profile schema. This step represents the final decision-support component of the proposed pipeline, where the objective is to extend the deterministic matching engine with LLM-based clinical interpretation while maintaining transparency and expert control.

In this scenario, the LLM receives the patient structured profile, the original clinical diary, relevant laboratory information, and an eligibility criterion represented as a logical rule. The model is then required to determine whether the available patient information satisfies the given criterion and provide a clinically grounded justification for the decision.

The evaluation considered two complementary aspects: (i) correctness of the eligibility assessment, manually evaluated, and (ii) quality of the generated justification. Since this task inherently involves clinical reasoning and interpretation of incomplete medical information, a complementary expert assessment was performed on a representative subset of generated outputs. The expert evaluation focused on clinical reliability, evidence grounding, conservative handling of missing information, and potential impact of incorrect decisions.

As shown in Table 9.8, all models demonstrated strong performance in the eligibility assessment task. Qwen3-8b achieved the highest quantitative performance, correctly evaluating 84 out of 100 criteria, corresponding to a strict and partial accuracy of 84.0%. Deepseek-r1:8b achieved a very similar result with 83.0% accuracy, while Llama3.1-8b obtained 78.0%. No partially correct classifications were observed in this task, as incorrect eligibility decisions generally resulted from an

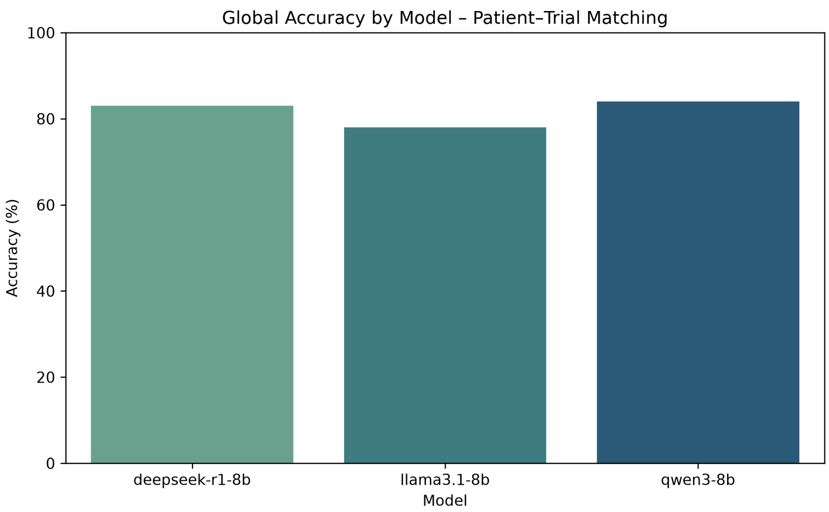


Figure 9.5: Patient–Trial Matching - Strict accuracy by model

incorrect interpretation of the available clinical evidence rather than from partially satisfying the criterion.

Table 9.8: Global performance of the evaluated models (Patient-Trial Matching)

Model	**Total**	**Correct**	**Partial**	**Wrong**	**Acc./Partial (%)**
Llama3.1-8b	100	78	0	22	78.0/78.0
Deepseek-r1:8b	100	83	0	17	83.0/83.0
Qwen3-8b	100	84	0	16	84.0/84.0

The strict accuracy comparison between models is further illustrated in Figure 9.5.

Beyond quantitative accuracy, the expert assessment provided additional insight into the clinical reliability of the generated decisions. Overall, Deepseek-r1:8b and Qwen3-8b demonstrated highly consistent behaviour, with both models grounding their decisions in explicit information available in the patient documentation. These models correctly identified measurable lesions, laboratory thresholds, and molecular alterations when present, while appropriately considering missing information as insufficient evidence rather than assuming eligibility or ineligibility.

Qwen3-8b achieved the highest expert-rated performance, with an average justification score of 4.93/5. The model consistently demonstrated conservative reasoning, avoiding unsupported assumptions and correctly handling incomplete clinical information. In particular, it correctly interpreted measurable disease according to lesion size, laboratory thresholds such as ANC requirements, and molecular eligibility criteria. Missing values, such as unavailable creatinine clearance calculations, were appropriately identified as unknown rather than incorrectly classified.

Deepseek-r1:8b achieved an average expert score of 4.8/5, showing similar strengths. The model consistently produced evidence-based decisions and avoided hallucinating clinical information. Its main advantage was the ability to explicitly distinguish between confirmed absence of a criterion and unavailable information, which is particularly important in a clinical decision-support scenario.

Llama3.1-8b achieved an average expert score of 4.2/5. Although the model showed strong performance when information was explicitly documented, the expert evaluation identified occasional over-interpretation of uncertainty. Examples included incorrectly rejecting criteria despite available laboratory values satisfying the requirement or inferring unsupported conclusions from incomplete documentation. These errors were not caused by hallucinated information, but rather by excessive conservatism or speculative interpretation.

The expert-assigned scores for the generated eligibility justifications are summarized in Table 9.9.

Table 9.9: Expert-assigned average scores for eligibility justification

Model	**Expert Avg. Score**
Deepseek-r1:8b	4.8/5
Llama3.1-8b	4.2/5
Qwen3-8b	4.93/5

Overall, these results suggest that small LLMs are capable of performing clinically aligned eligibility assessments when constrained by structured patient information and explicit evidence grounding. Importantly, the expert evaluation indicates that model reliability is not solely determined by classification accuracy, but also by the ability to justify decisions transparently and avoid unsupported clinical assumptions. This reinforces the proposed clinician-in-the-loop approach, where LLMs act as interpretable decision-support tools rather than autonomous clinical decision makers.

Chapter 10

Discussion

This pipeline demonstrated that small LLMs (8B parameters) are capable of performing relevant tasks in complex clinical contexts, although with different levels of difficulty.

The results reveal a clear tendency: models performed well in direct structured extraction tasks, faced their greatest challenges in logic formalization, where the lowest results were observed, and achieved good performance in conditioned clinical reasoning when explicit context was available.

The performance of the selected trio of models was not homogeneous. Qwen3 excelled in extraction tasks, achieving the highest strict accuracy in both hyper-relevant parameter extraction (78.3

These experiments showed that small models can support parts of a complex clinical pipeline when tasks are properly decomposed and expert oversight is maintained. Rather than indicating that one model is universally superior, the results suggest that different models excel in different components and that a structured pipeline can leverage these complementary strengths.

10.1 Research Hypotheses Assessment

The results obtained allow a direct assessment of the four research hypotheses guiding this study.

H1 is supported by the experimental results. All three models demonstrated the ability to extract and normalize clinically relevant information from heterogeneous oncology diaries while achieving 100% schema compliance. Qwen3-8B achieved the strongest performance with 78.3% strict accuracy, confirming that structured extraction from free-text oncology documentation is feasible with small LLMs, though performance remains dependent on field complexity.

H2 is partially supported. Eligibility criteria extraction yielded strong results, with Qwen3-8B achieving 92.3% strict accuracy, and cohort attribution reaching 96.3% for the same model. However, the subsequent formalization of extracted criteria into machine-interpretable logical rules proved considerably more challenging, with strict accuracies ranging from 52.2% to 64.1% across models. H2 is therefore supported with respect to extraction and preservation of clinical meaning, but only partially supported with respect to the formalization of logical relationships.

H3 is supported by the patient–trial matching results. The hybrid mechanism demonstrated strong performance, with Qwen3-8B and Deepseek-R1-8B achieving 84.0% and 83.0% accuracy respectively, and expert-rated justification quality scores above 4.8 out of 5. The expert evaluation confirmed that decisions were consistently grounded in available clinical evidence, supporting both the effectiveness and the interpretability of the proposed matching strategy.

H4 is broadly supported. Across all evaluated tasks, small open-source models (8B parameters) demonstrated sufficient performance to support the proposed workflow, particularly in extraction and matching tasks. The main exception remains logical formalization, where performance was more limited, suggesting that this specific component may benefit most from larger models or targeted fine-tuning in future work. This was further evidenced by the inclusion of Gpt-oss-120b in the criteria conversion task, where strict accuracy improved to 65.1% and partial accuracy to 80.9%, with a substantially lower wrong count compared to the 8B models.

10.2 Strengths of the Proposed Approach

The results suggest that the proposed approach has significant potential as a clinical decision-support tool for patient recruitment in oncology clinical trials. One of its main strengths lies in its modular design. Rather than treating eligibility assessment as a single end-to-end task, the workflow decomposes the problem into a sequence of specialized stages, namely patient information extraction, trial structure identification, eligibility criteria extraction, criteria formalization, and patient–trial matching.

This decomposition reduces the complexity of individual tasks and allows LLMs to focus on well-defined objectives. Consequently, each component can be evaluated, validated, and improved independently, increasing the overall robustness of the system. As can be seen in the experimental results, it allowed even small models to obtain good results, where an end-to-end model would most likely have had serious difficulties, while also making it harder to identify where failures occurred. These individual evaluations allowed to understand that extraction is not equal to conversion, which also is not equal to matching. A model can be strong on one task and weak on another, Qwen's strength in extraction (92.3% strict accuracy on criteria extraction) and Deepseek's strength in structure identification (96.7%) are a clear example of this. This allows the pipeline to employ the best performing model on the specific task at which it excels, rather than relying on a single model globally.

Another important strength is the high degree of transparency and interpretability provided by the framework. Every intermediate artifact generated by the system, including structured patient profiles, extracted eligibility criteria, logical representations, and matching decisions, remains accessible for inspection. This enables clinicians to understand how eligibility decisions are reached rather than relying on opaque end-to-end model predictions. A system that provides correct outputs but does not explain how is not acceptable in a clinical setting. The expert validation confirmed that Qwen and Deepseek consistently grounded their decisions on available evidence, avoiding hallucinated data, with justification scores of 4.93/5 and 4.8/5, respectively. The clinical justification for a decision is as important as the classification accuracy itself, particularly when the objective is to reduce unnecessary screen time for clinicians.

The clinician-in-the-loop design further contributes to the reliability of the approach. Expert validation is incorporated throughout the workflow, allowing clinicians to review, correct, and approve intermediate outputs before they are used in subsequent stages. This not only increases clinical safety but also mitigates the impact of occasional LLM errors.

Finally, the semi-deterministic matching strategy reduces dependence on probabilistic reasoning by maximizing the use of structured data and rule-based evaluation whenever possible. Unlike many existing approaches that rely on LLMs to directly answer "Is the patient eligible?", the proposed system structures the patient profile, formalizes eligibility criteria into logical rules, and only invokes LLM-based reasoning when the deterministic engine cannot reach a conclusion. This reduces the risk of hallucination, increases explainability, and improves the clinical acceptability of the generated outputs, while preserving the flexibility required to handle complex clinical eligibility requirements.

10.3 Challenges Observed during LLM-based Clinical Processing

10.3.1 Structured Extraction vs Semantic Interpretation

On simple and explicit fields, these models presented excellent results, gender and ECOG-PS achieved 100% accuracy across all three models, whereas on complex fields they faced considerable difficulties, particularly with normalized diagnosis, treatments, dates, and staging. Notably, diagnosis extraction was the most challenging field, with the best result being 46.7% achieved by Deepseek, and a recurrent error pattern in staging where models extracted only the final stage classification (e.g., "IVB") instead of the complete TNM representation (e.g., "cT3N2M1b (IVB)").

These challenges were not solely due to difficulties in extracting the text itself. They were also a consequence of the difficulty in following complex normalization mappings, adhering to clinical ontologies, and performing temporal interpretation, tasks that go well beyond surface-level text extraction.

10.3.2 Preservation of Clinical Meaning is Harder than Extraction

The results also showed that models would consistently identify cohort structures in clinical trials, but would lose clinically relevant details when extracting their descriptions. The expert assessment of trial structure confirmed this: despite Deepseek achieving 96.7% structural accuracy, it received an average clinical classification of only 2.25/5, frequently omitting treatment status, molecular alterations, or diagnostic specificity. Similarly, criteria extraction presented strong results across models, while their conversion into logical rules was considerably worse.

The main difficulty was not identifying information, but preserving the relationships between clinical concepts. Compound criteria such as "KRAS G12C inhibitor-naive", "prior FGFR inhibitor exposure", or "primary platinum refractory disease" involve nested clinical dependencies that models frequently simplified or partially omitted.

10.3.3 Logical Formalization Remains the Bottleneck

The logical formalization of extracted criteria was clearly the main challenge for these small models. While in some cases criteria extraction reached above 90% strict accuracy, Qwen achieving 92.3%, the conversion of these criteria into machine-interpretable logical rules dropped to approximately 64% for the same model. This gap is particularly significant because the conversion step is precisely what enables the deterministic matching engine to operate without LLM-based reasoning.

Small LLMs are effective at interpreting natural language, but face consistent challenges when transforming clinical language into formal logic. They consistently failed to completely represent AND/OR nested rules, dependencies between criteria, and compound eligibility requirements containing multiple simultaneous conditions. Another recurring failure was the inability to map certain criteria to existing schema fields, particularly the treatment field, resulting in the creation of new, schema-excluded fields that would then require additional LLM evaluation, undermining one of the main objectives of the pipeline. This limitation justifies and reinforces the hybrid architecture proposed in this work. The inclusion of Gpt-oss-120b in the conversion task confirmed this bottleneck while also revealing a distinct failure profile: unlike the smaller models, whose partial conversions were predominantly caused by failures in capturing complex logical structures, the larger model's partials were mostly attributable to incorrect normalization of criteria to schema fields, suggesting that logical formalization improves substantially with scale, while schema adherence remains a challenge regardless of model size.

10.4 Limitations

Despite the promising results, the proposed approach presents several limitations.

First, as with most LLM-based systems, output quality is highly dependent on the quality and completeness of the input documentation. Missing, ambiguous, or poorly documented clinical information can negatively affect patient profile extraction and, consequently, all downstream tasks.

Second, the workflow follows a sequential architecture, meaning that errors introduced in earlier stages may propagate to subsequent components. Although the decomposed approach of the pipeline and expert validation substantially reduce this risk, they cannot eliminate it entirely, which still proves to be an advantage over an end-to-end approach. The overall performance of the matching process therefore remains partially dependent on the quality of preceding extraction and conversion steps.

Another limitation arises from the formalization of eligibility criteria. While many criteria can be represented through deterministic logical rules, highly complex clinical requirements often involve contextual reasoning, implicit assumptions, or nuanced clinical judgement. Such criteria may be difficult both for LLMs to formalize and for clinicians to validate within a structured representation.

Furthermore, the effectiveness of the matching process depends on the completeness of the extracted patient profile. When relevant information is absent from the clinical diary, eligibility assessment may become inconclusive, limiting the usefulness of the generated recommendations and potentially increasing the amount of manual review required.

The expert validation, although valuable, was conducted on a reduced subset of the dataset and does not substitute a prospective clinical validation. This limits the strength of conclusions drawn from qualitative assessments and highlights the need for more extensive expert evaluation in future work.

Furthermore, the evaluation was conducted using synthetic clinical diaries and a limited number of clinical trials. Although considerable effort was made to maximize diversity and realism, performance on larger and fully real-world datasets may differ from the results observed in this study.

Another limitation of the current prototype is the absence of direct integration with the hospital information system. Although the proposed architecture includes communication with SClinico as the intended source of clinical documentation, the required Zero Trust infrastructure was not available during the implementation period. Consequently, the evaluation was performed using synthetic clinical diaries generated from real oncology documentation patterns rather than directly retrieved hospital records. While this approach enabled the validation of the complete pipeline while respecting privacy and security constraints, future deployment in a real clinical environment requires the establishment of a secure integration mechanism with existing hospital systems.

10.5 Clinical Implications

The proposed framework addresses a clinically relevant challenge: the identification of potentially eligible patients for oncology clinical trials. This process is traditionally time-consuming and requires clinicians to manually compare extensive free-text clinical documentation against often complex eligibility criteria.

By automatically extracting relevant patient information, structuring eligibility requirements, and providing preliminary matching assessments, the system has the potential to reduce cognitive burden and administrative workload. This may allow clinicians to dedicate more time to direct patient care and clinical decision-making.

Importantly, the framework was designed as a decision-support system rather than a decision-making system. Final responsibility remains entirely with the clinician, who can inspect, validate, and modify every intermediate result and matching decision. In this sense, the LLM serves as an assistive technology that summarizes and organizes information rather than replacing clinical expertise.

The traceability of all generated artifacts further supports clinical adoption by providing a complete audit trail from the original documents to the final eligibility recommendation.

The entire system was continuously evaluated by the clinical stakeholder of the ULS Coimbra oncology department through regular review meetings. These sessions allowed emerging issues and clinical needs to be promptly addressed, ultimately leading to a high level of clinician satisfaction with the system.

10.6 Generalizability

Although the study was conducted in collaboration with the oncology department of ULS Coimbra, the proposed methodology was not designed for a single institution.

The patient profile schema captures clinically meaningful concepts commonly used in oncology practice, while the extraction and matching pipelines operate independently of institution-specific document layouts. Consequently, the framework could potentially be applied to clinical documentation from other hospitals, provided that the required information is present in the source documents.

The approach is also largely language-agnostic. Since the methodology relies on prompt-guided information extraction and logical formalization, adaptation to other languages would primarily require modifications to prompts and validation procedures rather than fundamental architectural changes.

Beyond oncology, the same principles could potentially be extended to other medical specialties where patient recruitment for clinical studies depends on evaluating structured eligibility criteria against heterogeneous clinical documentation.

Chapter 11

Conclusion

This dissertation presented the design, implementation, and experimental evaluation of a modular clinical decision-support system developed in collaboration with the oncology department of ULS Coimbra. The work focused on transforming unstructured clinical data into structured patient profiles and systematically comparing them with eligibility criteria extracted and formalized into logical representations from clinical trial documentation, with the primary goal of reducing the manual workload present in the current workflow and improving the efficiency of the patient-selection process for clinical trials.

11.1 Contributions

The proposed approach decomposes the complex trial–patient matching problem into a sequence of specialized tasks, including the extraction and structuring of patient profiles from free-text clinical diaries, the identification and conversion of clinical trial eligibility criteria into machine-interpretable logical representations, and the final hybrid patient–trial matching process that combines deterministic rule-based evaluation with targeted LLM-assisted reasoning.

The experimental evaluation demonstrated that small open-source LLMs (8B parameters) are capable of performing clinically relevant tasks across all pipeline components, though with varying levels of performance depending on task complexity. Structured extraction tasks yielded the strongest results, with Qwen3-8B achieving 78.3% strict accuracy in patient profile extraction and 92.3% in eligibility criteria extraction. Trial structure identification also produced strong results, particularly for Deepseek-R1-8B, which achieved 96.7% accuracy. In contrast, the logical formalization of eligibility criteria proved to be the most challenging task across all models, with strict accuracies ranging from 52.2% to 64.1%, highlighting that transforming clinical language into formal logic remains an open problem for small LLMs. A supplementary evaluation of Gpt-oss-120b on this task yielded 65.1% strict accuracy and 80.9% partial accuracy, confirming that larger models improve logical formalization performance, though schema normalization challenges persist. Despite this, the final patient–trial matching stage demonstrated that when grounded in structured patient information and explicit clinical context, these models can perform reliable eligibility assessments, with Qwen3-8B and Deepseek-R1-8B achieving accuracy scores of 84.0% and 83.0% respectively, and expert-rated justification quality scores above 4.8 out of 5.

The modular design of the proposed methodology enables increased transparency, traceability, and interpretability of the generated outputs, which are critical requirements in clinical decision-support systems. By translating clinical trial criteria into logical representations and combining deterministic evaluation with targeted LLM assistance only when required, the system reduces dependence on the probabilistic reasoning capabilities of LLMs while preserving their flexibility in handling complex clinical information. Importantly, the evaluation confirmed that decomposing the problem into specialized stages not only improved performance but also allowed the identification of where specific failures occurred, something that would not be possible in an end-to-end approach.

Additionally, to enable a realistic evaluation under data privacy constraints, a clinically validated synthetic dataset was developed based on real oncology clinical documentation patterns, avoiding privacy risks while maintaining clinical realism. This dataset allowed the assessment of lightweight open-source language models under limited computational resources, reflecting realistic deployment constraints in healthcare environments.

11.2 Impact

The proposed system was designed with direct consideration of the needs and workflow constraints of clinical oncologists, positioning the clinician as the final decision-maker rather than replacing expert judgment. Through the incorporation of validation stages, persistent storage of intermediate outputs, and access to the reasoning behind LLM-assisted decisions, the pipeline provides a transparent and auditable framework suitable for clinical decision support. The expert validation conducted throughout the study, including qualitative assessment of clinical justifications, confirmed that the system's outputs are grounded in available evidence and avoid unsupported clinical assumptions.

By reducing the manual effort required to compare extensive clinical documentation against complex trial eligibility criteria, this approach has the potential to decrease clinician cognitive workload and facilitate more efficient identification of suitable clinical trials for patients, directly addressing the challenges identified in the oncology department of ULS Coimbra.

11.3 Personal Reflections on the Development Process

Beyond the technical and experimental outcomes presented above, the development of this dissertation also constituted a significant personal and professional learning experience, which is worth reflecting upon independently of the system's measured performance.

Working directly within a real oncology department, in close collaboration with Dra. Judy, was one of the most valuable aspects of this project. This continuous contact with a clinical stakeholder shaped the system far beyond what would have been possible through a purely theoretical specification of requirements. Recurrent meetings and feedback sessions repeatedly challenged initial assumptions and led to meaningful adjustments throughout the project, such as the reconsideration of a fully deterministic matching strategy in favour of the semi-deterministic approach ultimately adopted, once it became clear that a relevant subset of eligibility criteria could not be faithfully represented within a fixed patient schema. This iterative dialogue between technical design and clinical reality reinforced the importance of grounding decision-support tools in the actual workflow and language of the professionals who will use them, rather than in idealized assumptions about clinical documentation.

The project also represented a steep but rewarding learning curve on multiple fronts. Prior to this work, my experience with Large Language Models was comparatively limited, particularly regarding their application to sensitive, high-stakes domains such as oncology. Understanding the practical trade-offs between deterministic and probabilistic reasoning, the importance of prompt design and schema constraints in obtaining reliable structured outputs, and the specific vocabulary and reasoning patterns used in oncology documentation, required sustained study throughout the project. Engaging with real clinical concepts, such as ECOG performance status, staging systems, and trial cohort structures, demanded an additional effort to bridge the gap between a computer science background and the clinical domain, but this effort proved essential to designing a system that clinicians could trust and meaningfully use.

From a project management perspective, the work was structured as a largely sequential process, organized around weekly status meetings with the supervisor and periodic validation sessions with the stakeholder, complemented by lighter agile practices that allowed previously produced documents and design decisions to be revisited as understanding of the problem matured. This structure proved adequate for a project of this nature, in which an initial phase of problem definition, requirements elicitation, and architectural design was a necessary foundation for the subsequent implementation and experimental phases. At the same time, certain deviations from the initial planning, namely the additional time required to refactor the criteria extraction and matching pipelines once trial cohort structures and unmapped criteria were identified as significant complexity drivers, and the unplanned but ultimately beneficial effort invested in preparing an accompanying scientific article, highlighted the value of maintaining flexibility within an otherwise structured plan. Reserving time for unexpected refinements, and treating the initial schedule as a guideline rather than a rigid constraint, was an important lesson that contributed directly to the quality of the final outcome.

Taken together, these aspects of the project, the sustained collaboration with a clinical expert, the personal effort invested in acquiring both technical and domain-specific knowledge, and the discipline required to manage a long, multi-phase project under real-world constraints, were as central to the successful completion of this dissertation as the technical contributions themselves, and represent a body of experience that extends well beyond the specific system that was built.

11.4 Future Work

Although the proposed methodology demonstrates the feasibility of using lightweight LLMs within a clinical trial matching pipeline, several opportunities for further improvement remain. Due to the time and computational constraints of this study, the evaluation was primarily limited to small-scale language models. A supplementary experiment with Gpt-oss-120b on the criteria conversion task provided initial evidence that larger models can meaningfully improve logical formalization performance, though schema normalization challenges were still observed. Future work should further investigate the impact of larger models across all pipeline tasks, as well as more advanced adaptation techniques such as parameter-efficient fine-tuning and retrieval-augmented generation. The observed gap between extraction and logical formalization performance also suggests that targeted fine-tuning on criterion conversion tasks could yield meaningful improvements.

Furthermore, evaluation with larger and more diverse clinical datasets, including real-world anonymized clinical documentation, would provide additional validation of the system's robustness and generalizability. Finally, integration with hospital information systems, particularly SClinico in the context of ULS Coimbra, represents an important future direction. Such integration requires secure communication infrastructures capable of supporting the processing of real clinical data while maintaining privacy, security, and regulatory compliance.

Overall, this dissertation demonstrates the feasibility of combining structured representations, deterministic logic, and lightweight LLMs to support transparent and clinically aligned patient–trial matching workflows, while also identifying logical formalization as the key remaining bottleneck for future development.

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Appendices

Appendix A

Requirements

In the following chapter, the main requirements of the project at hand will be discussed. These requirements were documented and specified with the clear objective of providing guidance throughout the development of the current project. They were structured using an oyster–sea level analogy: the sea-level requirements represent the highest and broadest level of abstraction, while the oyster-level requirements correspond to the most detailed and system-specific needs. Furthermore, to better organize and prioritize the work, all requirements were tagged using the MoSCoW terminology, as follows:

- Must have (M) – Requirements that are fundamental to the system and without which it would not function properly.
- Should have (S) – Requirements that are important and add significant value, but are not strictly indispensable.
- Could have (C) – Requirements that would benefit the system if implemented, although they are not essential.
- Won't have (WH) – Requirements that are not needed for the system and will not be included.

The requirements were obtained through numerous meetings with the client, and further refined in discussions with the advisor, tutor, and Dra. Margarida. This collaborative process ensured that every requirement is feasible, realistic, and makes sense within the hospital environment, both from a technical and operational perspective.

These requirements can be divided into four main sections, each possibly containing subsections of functional, data, and interface requirements. Below are the four identified sections:

- System requirements, requirements that will allow the proper functioning and overall stability of the system;

- Identification of eligible patients for clinical trials, requirements that will allow the identification of eligible patients (already structured) and match them to a relevant clinical trial;
- Monitoring the quality of care, requirements that will allow the verification and assessment of the level of success that a clinical trial has on the patients involved;
- Automated structured data extraction, requirements that will allow the extraction of structured data from unstructured patient EHRs, storing them for further use.

A.1 Sea Level Requirements

A.1.1 System Requirements

ID	Requirement	MoSCoW	Description
FR0.1	Load historical data	WH	The clinical trials since (a year) should be loaded into the system in order for it to start working.
FR0.2	Pre-process clinical documents	S	The system must clean and normalize the clinical documentation before extraction, removing irrelevant headers, footers, formatting artifacts, and duplicated text to ensure consistent input for the LLM.

Table A.1: System Requirements - Functional Requirements

A.1.2 Eligiblity Criteria Extraction

Functional Requirements

ID	Requirement	MoSCoW	Description
FR1.1	Clinical Trial Text Pre-processing	S	the system must perform the removal of sections that are not relevant to eligibility assessment.
FR1.2	Trial Structure Identification	M	the LLM analyses the clinical trial document to identify its internal structure, namely the existence of trial cohorts.
FR1.3	Criteria Extraction	M	The LLM must extract inclusion and exclusion criteria from a clinical trial individually.
FR1.4	Extracted Criteria Validation	M	The clinician must be able to perform validation on the LLM-extracted criteria, being able to edit, create and remove these artifacts.
FR1.5	Persistence of validated criteria	M	Once approved, validated criteria must be persisted in the database.
FR1.6	Clinical Trial Filtering	C	The clinician must be able to search clinical trials by study name and filter them based on their metada: pathology group, trial status, extraction status, date range and type of file.
FR1.7	Clinical Trial Removal	C	The clinician must be able to remove any clinical trials, and associated artifacts: versions, matches performed, criteria and logical rules

Table A.2: Eligibility Criteria Extraction – Functional Requirements

Interface Requirements

ID	Requirement	MoSCoW	Description
IR1.1	Clinical trial submission	M	The clinician must be able to submit several clinical trials into the system. Including metadata for such documents, as described: Study name, Pathology group, start and end date, and trial's status (Recruiting, Completed, Not Yet Recruiting and Closed)
IR1.2	Clinical trial selection	M	The clinician must be able to select a clinical trial to submit for extraction.
IR1.3	Clinical trial listing	M	The system must display a list of every clinical trial present in the system
IR1.4	Clinical trial details	M	The system must display a page with every detail of a clinical trial, namely: Trial's metadata, versions of the artifact, original content, extracted criteria, logical rules and any matches performed
IR1.5	Extracted criteria validation interface	M	A page must be displayed, allowing the doctor to edit, remove and create the extracted criteria.

Table A.3: Eligibility Criteria Extraction - Interface Requirements

A.1.3 Eligibility Criteria Conversion

Functional Requirements

ID	Requirement	MoSCoW	Description
FR2.1	Retrieval of Extracted Criteria	M	the system must retrieve the respective validated criteria for the trial for further processing.
FR2.2	Eligibility criteria Conversion	M	The LLM must perform the translation of these criteria into structured logical representation.
FR2.3	Logical Representation Validation	M	The clinician must be able to perform validation on the generated logical representations, being able to edit, create and remove these artifacts.
FR2.4	Persistence of Validated Logical Representations	M	The clinician must be able to perform validation on the generated logical representations.

Table A.4: Eligibility Criteria Conversion – Functional Requirements

Data Requirements

ID	Requirement	MoSCoW	Description
DR2.1	Criteria mapping	M	Each criterion must map to the corresponding patient attributes in the database. if the conversion can not be done to an existing schema field, a general field must be created to represent the criterion.

Table A.5: Eligibility Criteria Conversion - Data Requirements

ID	Requirement	MoSCoW	Description
IR2.1	Converted criteria validation interface	M	A page must be displayed, allowing the doctor to edit, remove and create the generated logical representations.

Table A.6: Eligibility Criteria Conversion - Interface Requirements

Interface Requirements

ID	Requirement	Description
IR2.1	Clinical trial submission	The clinician must be able to submit several clinical trials into the system.
IR2.2	Clinical trial selection	The clinician must be able to select a clinical trial and view eligibility results.
IR2.3	Reporting interface	The system must display the justification report in a readable format (PDF/HTML).
IR2.4	Clinical trial validation interface	A page must be displayed, allowing the doctor to edit and validate the extracted criteria.

Table A.7: Eligibility Criteria Conversion - Interface Requirements

A.1.4 Semi-Deterministic Trial-Patient Matching

Functional Requirements

ID	Requirement	MoSCoW	Description
FR3.1	Trial Selection	M	the clinician must be able to select a clinical trial to perform eligibility assessment.
FR3.2	Patient Retrieval	M	the system must retrieve a subset of patient profiles based on the clinical trial's metadata.
FR3.3.1	Eligibility Assessment - Deterministic Assessment	M	each patient retrieved must be evaluated against every validated criterion. if the criterion references a field available in the patient profile schema, the corresponding logical rule is evaluated directly against the structure patient data.
FR3.3.2	Eligibility Assessment - LLM-Assisted Assessment	M	each patient retrieved must be evaluated against every validated criterion. If the criterion references information that is not represented within the structure schema, the LLM receives must receive sufficient context to determine eligibility.
FR3.3.3	Eligibility Assessment - LLM-Assisted Assessment Justification	M	When a eligibility assessment is performed by an LLM, the model must provide a justification to support expert review.
FR3.4	Assessment Validation	M	The clinician must be able to perform a review on all criterion-level assessments. being able to modify any automatically generated decision.
FR3.5	Persistance of Assessment Results	M	The system must persists all criterion assessments, manual corrections, and final eligibility decision.

Table A.8: Semi-Deterministic Trial-Patient Matching – Functional Requirements

Interface Requirements

ID	Requirement	MoSCoW	Description
IR3.1	Eligibility assessment page	M	The system must display every eligibility result, as well as quick metrics for the clinicians
IR3.2	Eligibility assessment inspection	M	The clinician must be able view any criterion-level automatic decision, as well as justification when provided.

Table A.9: Semi-Deterministic Trial-Patient Matching - Interface Requirements

A.1.5 Monitoring the Quality of Care

Functional Requirements

ID	Requirement	MoSCoW	Description
FR4.1	Retrieve treatment data	WH	The system must retrieve patient treatment records, including: • Diagnosis date; • First cycle date; • Progression date / End of treatment; • Death date; • Status (0/1): – 0: censured (patient did not have the event until the end of the study); – 1: event occurred (progression or death).
FR4.2	Compute OS and PFS	WH	The system must calculate Overall Survival (OS) and Progression-Free Survival (PFS).
FR4.3	Apply Kaplan-Meier analysis	WH	The system must use Kaplan-Meier survival estimation to analyze outcomes.
FR4.4	Visualize results	WH	The system must generate survival curves and statistical summaries.

Table A.10: Monitoring the Quality of Care – Functional Requirements

For the requirement FR4.1, the extracted fields from the "Automated Structured Data Extraction" phase are mapped with the necessary fields for calculating the Kaplan-Meier:

Extracted field	Necessary field	Obs
Date of first treatment	First cycle date	
Diagnosis date	Diagnosis date	
Death date	Death date	If it exists: OS event = 1; otherwise OS event = 0.
End date (from treatments)	End of treatment	For PFS, event = 0
-	Progression date	For PFS event = 1

Table A.11: Mapping of Extracted Fields to Necessary Fields

Data Requirements

ID	Requirement	MoSCoW	Description
DR4.1	Required data fields	WH	Diagnosis date, first cycle date, progression/end date, death date, and status (0/1).
DR4.2	Derived variables	WH	OS = (death date - diagnosis date); PFS = (progression date - first cycle date).

Table A.12: Monitoring the Quality of Care - Data Requirements

Interface Requirements

ID	Requirement	MoSCoW	Description
IR4.1	Analysis dashboard	WH	The clinician must be able to select patient cohorts and view survival graphs.
IR4.2	Report export	WH	Results should be exportable to CSV, PDF, or be compatible with SPSS.

Table A.13: Monitoring the Quality of Care - Interface Requirements

A.1.6 Hyper-Relevant Parameter Extraction

Functional Requirements

Table A.14: Hyper-Relevant Parameter Extraction - Functional Requirements

ID	Requirement	MoSCoW	Description
FR5.1	Select Diaries for Extraction	M	The system must allow for the selection of clinical diaries to be submitted for extraction.
FR5.2	Clinical Diary Text Pre-processing	S	The system must preform the pre-processing of the clinical diary text for the removal of headers and footers to reduce context and computational overhead.

ID	Requirement	MoSCoW	Description
FR5.3	Information Extraction	M	The LLM is instructed to indetify and extract clinical relevant parameters according to a predefined schema: • Age: Patient age or date of birth, depending on the information available in the clinical diary; • Gender: Patient gender; • ECOG-PS: ECOG Performance Status score (0–5), describing the patient's functional capacity; • Diagnosis: Primary diagnosis, typically corresponding to the patient's cancer type or major clinical condition; • Diagnosis Date: Date on which the diagnosis was formally established; • Molecular Status: Relevant molecular and biomarker information; • Stage: Disease staging information; • Pathology Group: Clinical category grouping diagnoses according to the organ or system of origin; • Treatments: List of administered treatments: – Name: Treatment name; – Start Date: Date on which the treatment started; – End Date: Date on which the treatment ended; • Control: Information related to disease control, metastases, comorbidities, and other relevant clinical follow-up observations. If a field does not exist in the clinical diary, it must be defined as `None`.

ID	Requirement	MoSCoW	Description
FR5.3.1	Information Extraction - Normalization of Diagnosis	S	The LLM after identifying the diagnosis field of the patient must perform a normalization step according to the diagnosis map provided by the stakeholder.
FR5.4	Retrieving Analysis	S	The system must be able to retrieve analysis associated with the respective patient, allowing for a better eligibility assesment further into the pipeline.
FR5.5	Validation of Extracted Parameters	M	After the LLM extracts the parameters, the clinician must be able to validate said parameters, being able to edit, create and remove parameters.
FR5.6	Persistence of Structure Patient Profiles	M	The system must persists all artifacts: parameters, treatments and analysis associated with the new profile.
FR5.7	Filtering of Clinical Diaries	C	The system must allow the clinician to search clinical diaries by their document name and filter them by: file type, and extraction status.
FR5.8	Removal of Clinical Diaries	S	The system must allow the clinician to remove any clinical diaries and associated artifacts: patient profiles and versions.
FR5.9	Filtering of Structure Patient Profiles	C	The system must allow the clinician to search patient profiles by their molecular status or diagnosis and filter them by: pathology group, molecular status, diagnosis, and stage.
FR5.10	Removal of Structured Patient Profiles	M	The system must allow the clinician to remove patient profiles and associated artifacts: treatments and analysis. Furthermore the diary that generated the removed profile must be reverted to its original status, allowing for a new extraction.

Data Requirements

ID	Requirement	MoSCoW	Description
DR5.1	Extracted fields	M	Age / date of birth, ECOG-PS, Diagnosis, Date of Diagnosis, Stage, Molecular Status, Date of Treatment, Treatments, Treatment start date, and Treatment end date.

Table A.15: Automated Structured Data Extraction - Data Requirements

Interface Requirements

ID	Requirement	MoSCoW	Description
IR5.1	Clinical diary submission	M	The clinician must be able to submit several clinical diaries into the system.
IR5.2	Clinical diary selection	M	The clinician must be able to select a clinical diary to submit for extraction.
IR5.3	Clinical diary listing	M	The system must display a list of every clinical diary present in the system
IR5.4	Clinical diary details	M	The system must display a page with every detail of a clinical diary, namely: diary's metadata, versions of the artifact, original content, generated patient profile, as well as associated artifacts: treatments and analysis.
IR5.5	Extracted parameters validation interface	M	A page must be displayed, allowing the doctor to edit, remove and create the extracted parameters.
IR5.6	Patient profiles listing	M	The system must display a list of every structured patient profile present in the system
IR5.7	Patient profile details	M	The system must display a page with every detail of a patient profile, namely: patient overview, diagnosis information, treatments and laboratory analysis.

Table A.16: Hyper-Relevant Parameter Extraction - Interface Requirements

A.2 Oyster Level Requirements

A.2.1 FR0.1 - Load historical data

All available unstructured clinical diary entries on SClinico for the oncology must be loaded into the PostegreSQL Database.

This will be done through a full load python script where unstructured data is read and stored in the a dedicated table on the DB. Because of the sheer amount of data it is unfeasible for it to be a manual process.

A.2.2 FR0.2 - Pre-process clinical documents

Before any extraction or matching can occur, all clinical documents must undergo a pre-processing phase to ensure that the input provided to the LLM is clean, consistent, and free from structural noise. This step includes the removal of irrelevant headers, footers, duplicated sections, and formatting artifacts commonly present in exported clinical diaries or trial documentation. The pre-processing will be executed automatically as part of the ingestion pipeline. Each document will be parsed, normalized, and transformed into a standardized text format, ensuring that downstream extraction processes operate on uniform and reliable input.

In addition, the pre-processing logic must be tailored to the specific type of document being ingested.

- For clinical trials, the system must isolate and retain only the inclusion and exclusion criteria sections, discarding unrelated protocol content such as study rationale, endpoints, administrative metadata, or regulatory notes.
- For clinical diaries, the system must preserve only the textual content relevant to patient information and remove any extraneous elements, ensuring that the LLM receives clean input for extracting the required clinical parameters: demographic information, ECOG-PS, primary diagnosis, date of diagnosis, molecular status, cancer stage, treatment history , and control information.

A.2.3 FR1.1 - Clinical Trial Text Pre-processing

Before performing eligibility criteria extraction, the system must preprocess the clinical trial document to reduce irrelevant context and improve the reliability of the LLM-based extraction pipeline.

The process must:

- Receive the clinical trial document selected by the clinician;
- Identify the sections relevant for eligibility assessment;
- Remove all sections unrelated to participation criteria, maintaining only the content required for eligibility assessment;
- Prepare the processed document to be used as input for the subsequent LLM-based phases.

This preprocessing step reduces unnecessary context provided to the LLM, improving extraction consistency and reducing computational overhead.

A.2.4 FR1.2 - Trial Structure Identification

After preprocessing the clinical trial document (FR1.1 - Clinical Trial Text Preprocessing), the system must perform a structure identification phase using an LLM.

This phase must:

- Analyze the processed clinical trial text;
- Identify the internal organization of the clinical trial;
- Determine whether the trial contains multiple cohorts or treatment groups;
- Extract structural information required to correctly interpret eligibility criteria;
- Provide the identified structure as additional context for the criteria extraction phase.

Separating structure identification from criteria extraction allows the model to interpret eligibility requirements according to the clinical trial organization, reducing extraction errors in trials with multiple cohorts or different populations.

A.2.5 FR1.3 - Criteria Extraction

After identifying the clinical trial structure (FR1.2 - Trial Structure Identification), the system must use an LLM-based extraction pipeline to convert unstructured clinical trial text into structured eligibility criteria.

The process must:

- Process each clinical trial independently to ensure reproducibility and traceability;
- Provide the LLM with:
  - The preprocessed clinical trial text;
  - The previously identified trial structure;
  - Instructions for extracting eligibility criteria.
- Extract all participation criteria described in the clinical trial;
- Classify each extracted criterion into:
  - Inclusion criteria;
  - Exclusion criteria.
- Determine whether each criterion can be safely decomposed into independent criteria:

  - If separating the criterion preserves the original clinical meaning, it must be divided into multiple individual criteria;
  - Otherwise, the criterion must remain as a compound criterion and be processed as a single logical unit.

- Preserve the original clinical meaning of extracted criteria;
- Generate a structured representation containing:
  - Criterion identifier;
  - Original criterion text;
  - Criterion category (inclusion/exclusion).

The extraction phase produces a validated list of individually represented eligibility criteria that can subsequently be converted into machine-readable logical representations.

A.2.6 FR1.4 - Extracted Criteria Validation

After extracting eligibility criteria (FR1.3 - Criteria Extraction), the clinician must validate the generated criteria before they are used in subsequent processing phases.

The validation process must:

- Display all extracted inclusion and exclusion criteria through a dedicated interface;
- Allow the clinician to:
  - Review extracted criteria;
  - Edit incorrect, incomplete, or ambiguous criteria;
  - Create missing criteria;
  - Remove irrelevant criteria.
- Ensure that the final criteria accurately represent the original clinical trial requirements.

A.2.7 FR1.5 - Persistence of Validated Criteria

After clinician validation, the system must persist the validated eligibility criteria.

The process must:

- Store validated inclusion and exclusion criteria;

- Maintain the association between each criterion and its clinical trial;
- Preserve the original criterion text for traceability;
- Make validated criteria available for the conversion phase.
- Create a new version of the document, for traceability purposes

A.2.8 FR1.6 - Clinical Trial Filtering

To allow efficient data management and quick navigation within the system interface, the system must provide advanced search and filtering capabilities over the ingested clinical trial repository.

The process must:

- Allow the clinician to query clinical trials dynamically by searching directly for the study name;
- Provide filtering mechanics that allow the clinician to narrow down the displayed clinical trials list based on the following specific metadata fields:
  - **Pathology Group:** Clinical category grouping the trials according to the target organ, system of origin, or cancer type;
  - **Trial Status:** The operational status of the trial (e.g., Active, Recruiting, Terminated);
  - **Extraction Status:** The state of the pipeline regarding the document (e.g., Not Processed, In Extraction, Validated);
  - **Date Range:** Temporal constraints reflecting when the trial was ingested, published, or updated;
  - **Type of File:** The file format of the original document (e.g., PDF, DOCX, TXT).

A.2.9 FR1.7 - Clinical Trial Removal

The system must provide secure data deletion operations to maintain database integrity, clean obsolete records, and support workflow corrections.

This process must:

- Allow the clinician to permanently remove any selected clinical trial document from the system repository;
- Trigger an automatic, strict cascade deletion at the database level upon document removal, which must completely delete all associated data and downstream artifacts associated with that specific trial, including:
  - Every generated document version kept for history and auditing;

  - All patient matching processes and compatibility metrics performed using that trial's data;
  - All extracted participation criteria (both raw text and clinician-edited inclusion/exclusion rules);
  - Every generated machine-readable logical representation derived from the criteria.

- Ensure that the system maintains referential integrity throughout the local database post-deletion, ensuring no orphaned data remains from the removed trial.

A.2.10 FR2.1 - Retrieval of Extracted Criteria

Before converting eligibility criteria into logical representations, the system must retrieve the validated criteria associated with the selected clinical trial.

The process must:

- Retrieve validated inclusion and exclusion criteria;
- Preserve the original criterion ordering and classification;
- Provide the extracted criteria as input for the LLM-based conversion phase.

A.2.11 FR2.2 - Eligibility Criteria Conversion

After retrieving validated eligibility criteria (FR2.1 - Retrieval of Extracted Criteria), the system must use an LLM-based conversion pipeline to transform free-text criteria into structured logical representations.

The conversion process must:

- Process each criterion independently;
- Provide the LLM with the complete textual description of each eligibility criterion;
- Instruct the model to convert the criterion into the most appropriate representation while preserving its original clinical meaning;
- Attempt to map each criterion to an existing field available in:
  - The patient profile schema;
  - The laboratory values schema.
- Only when no suitable predefined field exists, generate a general field representation for the criterion;

- Generate logical representations composed of:
  - Fields;
  - Operators;
  - Expected values;
  - Boolean conditions when required.

Simple criteria must be represented as atomic logical rules:

```
{
  "field": "age",
  "operator": ">=",
  "value": "18"
}
```

Compound criteria must be represented using Boolean operators grouping multiple conditions:

```
{
  "operator": "AND",
  "conditions": [
    {
      "field": "ecog_ps",
      "operator": "<=",
      "value": "1"
    },
    {
      "field": "diagnosis",
      "operator": "=",
      "value": "NSCLC"
    }
  ]
}
```

When a criterion cannot be represented using the available schema, the system must generate an unmapped representation:

```
{
  "unmapped": true,
  "source_text": "unmapped part of the criterion"
}
```

All converted criteria must be grouped according to their category:

```
{
  "inclusion_criteria": [
    {
      "id": integer,
      "text": string,
      "logic": object
    }
```

```
  ],
  "exclusion_criteria": [
    {
      "id": integer,
      "text": string,
      "logic": object
    }
  ]
}
```

This representation enables criteria associated with existing patient attributes to be evaluated deterministically, while unmapped criteria remain available for LLM-assisted assessment during the matching phase.

A.2.12 FR2.3 - Logical Representation Validation

After converting eligibility criteria into logical representations (FR2.2 - Eligibility Criteria Conversion), the clinician must validate the generated logical rules before patient eligibility assessment.

The validation process must:

- Display the generated logical representations;
- Allow the clinician to:
  - Review generated rules;
  - Modify fields, operators, and values;
  - Create missing logical representations;
  - Remove incorrect logical rules.
- Ensure that logical representations correctly encode the original eligibility criteria.

A.2.13 FR2.4 - Persistence of Validated Logical Representations

After clinician validation, the system must persist the generated logical representations.

The process must:

- Store validated logical rules associated with their respective clinical trial criteria;
- Maintain traceability between:
  - Original extracted criterion;

  - Converted logical representation;
  - Clinician modifications.

- Create a new version for the document, for traceability;
- Make validated logical representations available for the semi-deterministic eligibility assessment phase.

A.2.14 FR3.1 - Trial Selection

The clinician must be able to select a clinical trial to perform the semi-deterministic eligibility assessment.

The process must:

- Allow the clinician to select a clinical trial available in the system;
- Retrieve the validated eligibility criteria associated with the selected trial;
- Retrieve the validated logical representations generated during the criteria conversion phase;
- Prepare the required information for patient retrieval and eligibility assessment.

The selected clinical trial acts as the reference context for the complete patient matching pipeline.

A.2.15 FR3.2 - Patient Retrieval

After selecting a clinical trial (FR3.1 - Trial Selection), the system must retrieve the subset of patients that will be evaluated during the eligibility assessment process.

The process must:

- Retrieve structured patient profiles from the database;
- Use the clinical trial metadata to restrict the initial search space;
- Filter patients according to available metadata, namely the pathology group associated with the clinical trial;
- Provide the retrieved patient profiles as input for criterion-level assessment.

This filtering step reduces the number of patient-criterion comparisons and minimizes unnecessary LLM-assisted assessments.

A.2.16 FR3.3.1 - Deterministic Assessment

For every retrieved patient, the system must evaluate validated eligibility criteria through deterministic logical evaluation whenever the required information is available in the structured patient schema.

The process must:

- Process each patient against every applicable validated eligibility criterion;
- Identify criteria whose logical representation references available structured patient fields;
- Evaluate logical rules directly against the stored patient data;
- Support:
  - Comparison operations;
  - Boolean conditions;
  - Inclusion and exclusion criterion evaluation.
- Produce a criterion-level assessment containing:
  - The evaluated criterion;
  - The assessment result;
  - The data used during evaluation.
- Avoid LLM inference whenever the required information can be deterministically evaluated.

This approach ensures transparent and reproducible evaluation for criteria represented in the structured patient schema.

A.2.17 FR3.3.2 - LLM-Assisted Assessment

When an eligibility criterion cannot be represented or evaluated using the available structured patient schema, the system must perform an LLM-assisted assessment.

The process must:

- Identify criteria requiring information unavailable in the deterministic evaluation pipeline;
- Provide the LLM with sufficient clinical context to assess the criterion;
- Include, whenever available:
  - Structured patient profile information;

  - Clinical diary information;
  - Laboratory results;
  - The corresponding eligibility criterion.

- Instruct the LLM to determine whether the patient satisfies the criterion;
- Produce the same criterion-level assessment output format used by deterministic evaluation.

Although different assessment mechanisms are employed, both deterministic and LLM-assisted approaches must generate equivalent criterion-level eligibility decisions.

A.2.18 FR3.3.3 - LLM-Assisted Assessment Justification

Whenever a criterion is evaluated using LLM-assisted assessment (FR3.3.2 - LLM-Assisted Assessment), the system must generate a justification supporting the produced decision.

The justification generation process must:

- Provide an explanation of the reasoning behind the criterion-level assessment;
- Reference the clinical information considered during evaluation;
- Support clinician review and validation of automatically generated decisions.

The generated justification must be stored together with the corresponding criterion-level assessment to ensure traceability.

A.2.19 FR3.4 - Assessment Validation

After completing the criterion-level eligibility assessment, the clinician must validate the generated results before final eligibility decisions are persisted.

The validation process must:

- Display all criterion-level assessments for each evaluated patient;
- Present the assessment mechanism used:
  - Deterministic evaluation;
  - LLM-assisted assessment.
- Display LLM-generated justifications whenever available;

- Allow the clinician to:
  - Review automatic decisions;
  - Modify incorrect assessments;
  - Validate uncertain or inconclusive cases.
- Preserve all clinician modifications for audit purposes.

A.2.20 FR3.5 - Persistence of Assessment Results

After clinician validation, the system must persist the final patient-trial eligibility assessment results.

The persistence process must:

- Store every criterion-level assessment;
- Store generated justifications from LLM-assisted assessments;
- Store clinician modifications and final validated decisions;
- Maintain the association between:
  - Patient;
  - Clinical trial;
  - Evaluated eligibility criteria.

The final eligibility decision must be computed by aggregating all criterion-level assessments.

The aggregation process must:

- Require all inclusion criteria to be satisfied;
- Require no exclusion criteria to be triggered;
- Avoid weighting, ranking, or probabilistic scoring mechanisms.

Formally:

```
Eligible = (All Inclusion Criteria Satisfied)
           AND
           (No Exclusion Criteria Triggered)
```

For clinical trials containing multiple cohorts, the system must evaluate eligibility at both trial and cohort levels. A patient is considered eligible for the clinical trial when:

- General trial eligibility criteria are satisfied;
- At least one cohort-specific eligibility assessment returns an eligible result.

When a criterion cannot be reliably assessed due to missing information, unavailable values, ambiguous definitions, or evaluation failures, the criterion must be classified as inconclusive.

For conservative clinical decision support:

- Inconclusive inclusion criteria must be considered as not satisfied;
- Inconclusive exclusion criteria must be considered as not triggered.

This ensures that final eligibility decisions remain conservative and fully traceable.

Upon completion of this phase, the system must generate:

- Criterion Assessments: Individual criterion-level eligibility decisions, including supporting justifications whenever LLM-assisted assessment is used;
- Eligibility Results: The final patient eligibility classification for the selected clinical trial and, when applicable, individual cohorts.

A.2.21 FR4.1 - Retrieve treatment data

The system must retrieve all patient treatment-related records required for survival analysis. This process must:

- Load, for each patient, the following structured fields from the PostgreSQL database (previously extracted and validated in the "Automated Structured Data Extraction" phase):
  - Diagnosis date;
  - Date of first treatment (mapped to First cycle date);
  - Progression date (if present);
  - End date of treatment (if no progression occurred);
  - Death date (if applicable).
- Automatically derive the Status (0/1) required for Kaplan-Meier calculations based on the available data:
  - OS status rules:
    * 1 - if a death date exists;
    * 0 - otherwise (censored).

- PFS status rules
  - 1 - if a progression date exists;
  - 0 - if progression did not occur and only end-of-treatment is available (censored)

- Apply the following deterministic mapping rules between extracted fields on the "Automated Structured Data Extraction" phase and Kaplan-Meier required fields:

Extracted field	**Required KM field**	**Notes**
Date of first treatment	First cycle date	Direct mapping
Diagnosis date	Diagnosis date	Direct mapping
Death date	Death date	OS event = 1 if present, else event = 0.
End date (from treatments)	End of treatment	Used when no progression; PFS event = 0
-	Progression date	if present then PFS event = 1

Table A.17: Mapping of Extracted Fields to Necessary Fields

A.2.22 FR4.2 - Compute OS and PFS

After retrieving and mapping all necessary fields in FR2.1, the system must compute the survival metrics Overall Survival (OS) and Progression-Free Survival (PFS) for each patient. This process must:

- Use standardized date formats;
- Apply the following logic for Overall Survival (OS):
  - If death date exists:
    - $OS = (\text{death date} - \text{diagnosis date})$
- Apply the following logic for Progression-Free Survival (PFS):
  - If progression date exists:
    - $PFS = (\text{progression date} - \text{first cycle date})$

A.2.23 FR4.3 - Apply Kaplan-Meier analysis

After computing OS and PFS for each patient (FR2.2), the system must generate survival estimations using the Kaplan–Meier method. This process must:

- Compute the Kaplan–Meier survival function deterministically using the standard formula:

$$S(t) = \prod_{t_i < t} (1 - \frac{d_i}{n_i})$$

Where:

  - $S(t)$, probability of surviving beyond time t;
  - t_i, specific time points at which an event occurs;
  - d_i,number of events that occur at time t_i;
  - n_i, number of individuals who are still at risk of the event just before time t_i.

- Ensure implementation includes:
  - Grouping of patients by survival time to compute di and ni;
  - Proper handling of censored observations (included in the risk set but not counted as events);
  - Stepwise survival curves (piecewise constant function).

- The Kaplan-Meier estimates must be outputted into a structured format suitable for visualization (FR2.4 - Persistence of Validated Logical Representations).
  - Time points;
  - Survival probabilities;
  - Number at risk;
  - Number of events;
  - Censoring information.

A.2.24 FR4.4 - Visualize results

After computing the Kaplan–Meier survival estimates (FR2.3 - Logical Representation Validation), the system must generate interactive and exportable visualizations for clinician analysis. This process must:

- Produce visualization-ready survival curves with the following characteristics:
  - Stepwise Kaplan-Meier survival function;
  - Distinct curves for OS and PFS;
  - Markers for censored patients (e.g., "+" symbols);
  - Time displayed on the x-axis and survival probability on the y-axis.

- Allow clinicians to:

  - Select the patient cohort or subgroup to be visualized
  - Toggle between OS and PFS views
  - Overlay multiple curves (e.g., by treatment group or stage)
  - Display underlying survival tables (number at risk per time interval)

- Ensure the graphical output can be:
  - Rendered interactively in the system's dashboard;
  - Exported in standard formats (PNG/SVG for images, PDF for reports);
  - Accompanied by summary statistics (median survival, events count, censored count).

A.2.25 FR5.1 - Select Diaries for extraction

The system must allow the clinician to select one or multiple diaries to undergo structured data extraction. This process must:

- Retrieve and display all available unstructured diary entries stored in the database.
- Provide an interface allowing the clinician to:
  - Select individual diaries manually;
- Validate the selection to ensure:
  - Only diaries that have not yet undergone structured extraction, or those explicitly reverted to their original status after profile removal (FR5.10 - Removal of Structured Patient Profiles), can be selected;
  - Duplicate active extractions are prevented;
  - Any missing, unreadable, or corrupted diaries trigger a system warning.
- Forward selected diaries to the structured extraction pipeline (FR5.2 - Clinical Diary Text Pre-processing-FR5.5 - Validation of Extracted Parameters) ensuring the system:
  - Records the diary status to "`In Extraction`" in the database.

A.2.26 FR5.2 - Clinical Diary Text Pre-processing

The system must perform automated pre-processing on any clinical diary selected and submitted for extraction. This process must:

- Parse the raw unstructured text, identify and strip out irrelevant non-clinical sections, specifically page headers, footers, page numbers, and system-generated metadata;

- Reduce unnecessary context length and noise before the data is transmitted to the language model, thereby minimizing computational overhead and preventing potential extraction artifacts.

A.2.27 FR5.3 - Information Extraction

After text pre-processing (FR5.2 - Clinical Diary Text Pre-processing), the system must execute an automated LLM-based extraction pipeline to convert the free-text diary content into structured patient data. The process must:

- Process each selected diary independently to ensure reproducibility and traceability;
- Inject the pre-processed text into the LLM through a deterministic zero-shot prompting pipeline implemented via a Python script;
- Construct a standardized prompt template, replacing only the free-text diary content to ensure consistent outputs;
- The prompt must include clear extraction instructions, a fixed target schema, and the raw text inserted verbatim in a designated section using structural delimiters to avoid metadata leaks;
- The full prompt template should be structured like this:

```
You are a clinical information extraction assistant.
Your task is to extract structured clinical data from
   an unstructured free-text clinical diary.

Follow these rules:
- Extract only the information explicitly present in
   the diary.
- If a field or an entire array is missing, set its
   value to null.
- Do not infer, guess, or extrapolate clinical data.
- Output only valid JSON, respecting the specified
   schema exactly.

Schema (all fields must appear in the output, null if
   missing):
{
    "age_or_birthdate": ...,
    "gender": ...,
    "ecog_ps": ...,
    "diagnosis": ...,
    "diagnosis_date": ...,
    "molecular_status": ...,
    "stage": ...,
    "pathology_group": ...,
    "treatments": [
        {
```

```
                "name": ...,
                "start_date": ...,
                "end_date": ...
            }
        ],
        "control": ...
    }

    Clinical diary to process:
    <DIARY>
    {{DIARY_TEXT}}
    </DIARY>

    Return only the JSON object.
```

- Ensure extraction robustness and error handling by implementing automated recovery mechanisms:
  - **Invalid JSON Structure:** If the LLM generates conversational text alongside JSON, the system attempts to programmatically isolate the JSON block. If recovery fails, the extraction request is automatically repeated.
  - Missing or Truncated Outputs: If no valid JSON is returned or if the response is cut short due to token context limits, the request is retried.
  - Missing Information: If data for a specific field does not exist within the text, the model must assign a `null` value (interpreted as `None`).

A.2.28 FR5.3.1 - Information Extraction - Normalization of Diagnosis

Still within the extraction of clinical parameters from the clinical diary, the system must perform a dedicated normalization step focusing exclusively on the patient's primary diagnosis. The process must:

- Enforce that the LLM maps the extracted free-text primary diagnosis entry directly to a standardized clinical term, utilizing the official diagnosis reference map provided by the stakeholder;
- Ensure that this mapping step standardizes the terminology across patient profiles to maintain cross-record consistency, preventing spelling variations or alternative medical nomenclatures from disrupting downstream data aggregation;
- Prohibit any modifications, structural overrides, or clinical inferences to the remaining extracted parameters, ensuring the integrity of the surrounding JSON payload is fully preserved.

If it is impossible to map the diagnosis to an existing entry in the document provided by the stakeholder the LLM must maintain the original diagnosis.

A.2.29 FR5.4 - Retrieving Analysis

To support comprehensive clinical assessments downstream in the matching pipeline, the system must handle laboratory and clinical analysis artifacts independently. The process must:

- Parse, extract, and associate detailed clinical and laboratory analysis entries from the normalized patient data structure;
- Make these analysis records retrievable alongside the patient's profile overview to allow clinicians a highly granular review of the patient's clinical state during final eligibility evaluations.

A.2.30 FR5.5 - Validation of Extracted Parameters

After automated extraction, the system must provide an interactive clinician validation interface to review, edit, and approve the data before permanent database persistence. This process must:

- Display all parameters in a comprehensive interface organized into dedicated sections: Demographics (Age/Date of birth, Gender); Performance Status (ECOG-PS); Diagnosis Information (diagnosis, date of diagnosis, stage, pathology group, molecular status); Treatments (name, start date, end date); Laboratory Analysis (analysis name, result, date); and Control;
- Allow the authenticated clinician to manually edit incorrectly extracted values, add missing attributes missed by the LLM, or completely remove faulty blocks (such as erroneous treatments or analyses);
- Track and secure validation metadata by assigning a boolean `validated = TRUE` flag to the verified diary, logging the explicit timestamp, and archiving changes in a versioned log for complete auditability.

A.2.31 FR5.6 - Persistence of Structured Patient Profiles

Following explicit clinician approval (FR5.5 - Validation of Extracted Parameters), the system must permanently commit the unified patient record into the local database. This process must:

- Persist the data across relational tables, including `patient_profiles`, `treatments`, and `analysis`, linking all artifacts seamlessly under unique patient identifiers;
- Enforce that fields missing from the source text are explicitly stored as `NULL` values;
- Ensure all temporal fields are written into standard ISO 8601 format;

- Store the complete lifecycle history—including raw unstructured text, initial extraction JSON, and manual edits—within a versioned logging table named `version` alongside their respective timestamps;
- Update the root clinical diary status entry within the database to "`Extracted`".

A.2.32 FR5.7 - Filtering of Clinical Diaries

To allow efficient data management within the document ingestion repository, the system interface must provide specific filtering capabilities for unstructured artifacts. The process must:

- Allow the clinician to search through uploaded clinical diaries directly by their document name;
- Provide filtering options allowing the clinician to narrow down the clinical diaries list by: file type (e.g., PDF, TXT) and current extraction status (e.g., "Not Extracted", "In Extraction", "Validated").

A.2.33 FR5.8 - Removal of Clinical Diaries

The system must provide secure data deletion operations to clear raw entries and maintain database integrity. This process must:

- Allow the clinician to permanently remove any selected clinical diary from the system;
- Trigger an automatic cascade deletion upon removal, purging all derived child artifacts associated with that diary, including any generated structured patient profiles, historical version logs, and related database constraints.

A.2.34 FR5.9 - Filtering of Structured Patient Profiles

The system must provide specialized query interfaces to search through normalized clinical records swiftly during matching workflows. This process must:

- Allow the clinician to perform text searches across the repository of patient records by querying their molecular status or primary diagnosis text fields;
- Provide multi-parameter filtering for the profiles directory, allowing constraints based on: pathology group, molecular status, diagnosis, and disease stage.

A.2.35 FR5.10 - Removal of Structured Patient Profiles

The system must allow the deletion of extracted profiles independently from the source documents to permit workflow corrections. This process must enforce the following rules:

- Allow the clinician to remove a specific structured patient profile from the system repository;
- Trigger a database cascade to completely delete all associated artifacts associated with the profile, namely its relational records within the treatments and laboratory analysis tables;
- Automatically revert the extraction state of the original unstructured clinical diary that generated the deleted profile back to its initial status, making it fully available for a structured extraction cycle.

Appendix B

Abstract Templates

B.0.1 Clinical Trial Template

Listing B.1: Canonical structure of a clinical trial protocol

```
Study Objectives
- Primary objective:
  Description of the main clinical question addressed by the trial
- Secondary objectives:
  Description of additional clinical outcomes of interest
  (e.g., quality of life, functional status, safety profiles).
- Exploratory objectives:
  Optional objectives such as biomarker discovery or subgroup analysis.

Study Design
- Study phase (e.g., Phase II, Phase III)
- Study type (e.g., randomized, open-label, double-blind)
- Allocation ratio (e.g., 1:1, 2:1)
- Control arm and experimental arm description
- Stratification factors (e.g., ECOG status, histology)

Number of Patients
- Planned total sample size
- Allocation per study arm
- Expected number of screened patients (if applicable)

Number of Sites
- Number of participating clinical centers
- Geographic distribution (e.g., countries or regions)

Eligibility Criteria

Inclusion Criteria:
- Demographic requirements (e.g., age, sex)
- Disease characteristics (e.g., diagnosis, stage)
```

```
- Molecular or biomarker status (if applicable)
- Functional status (e.g., ECOG performance status)
- Prior treatment history and timing constraints
- Laboratory and organ function thresholds
- Informed consent and regulatory requirements
Exclusion Criteria:
- Alternative diagnoses or disease subtypes
- Contraindicated comorbidities
- Active infections or autoimmune conditions
- Prior or concomitant therapies not permitted
- Safety-related laboratory abnormalities
- Other conditions posing excessive risk

Treatment Description
- Investigational product(s)
  - Dose, route of administration, schedule
- Control treatment(s)
  - Dose, route of administration, schedule
- Treatment duration and discontinuation rules

Study Endpoints
- Primary endpoint definition
- Secondary endpoint definitions
- Safety endpoints
- Timing of assessments
```

B.0.2 Clinical Diary Template

Listing B.2: Abstract template for clinical diaries in research

```
Patient Identification
- Age, sex
- Date of consultation

Diagnosis and Disease Characteristics
- Primary diagnosis
- Disease stage or classification
- Molecular or biomarker status (if applicable)
- Relevant comorbidities or prior malignancies

Clinical History
- History of present illness (HPI)
  - Symptom onset, duration, severity
  - Diagnostic imaging or lab results
  - Previous interventions (surgical, pharmacological)
- Past medical history
- Surgical history
- Medication history
```

```
– Family medical history

Assessment at Consultation
– ECOG Performance Status or equivalent functional assessment
– Vital signs (weight, blood pressure, etc.)
– Relevant laboratory values or biomarkers
– Physical exam findings
– Imaging studies and results

Planned Interventions or Treatment
– Proposed pharmacological therapy
  – Drug names, doses, schedules
– Proposed surgical or procedural interventions
– Radiotherapy plans (if applicable)
– Monitoring or follow-up procedures

Treatment Rationale
– Explanation for treatment choice based on:
  – Disease stage and biology
  – Patient functional status
  – Safety considerations
  – Evidence from guidelines or prior studies

Follow-up and Outcomes Tracking
– Planned follow-up dates and assessments
– Metrics to track disease progression (e.g., tumor size, metastasis, lab
– Adverse events and safety monitoring
– Patient-reported outcomes (optional)

Comments / Notes
– Additional considerations or observations by clinician
– Any relevant patient preferences or social considerations
```

Appendix C

Background Knowledge

C.1 Symbolic and Connectionist AI

This section is the study and interpretation of the document [Douglas, 2023].

Symbolic AI systems are explicitly designed and programmed, that is, human experts encode their domain knowledge directly into the system. The model receives a set of rules and an input, applying those rules to generate an appropriate output.

For example, in a grammar-based system:

Rule set:
Sentence $\rightarrow$ Subject + Predicate
Subject $\rightarrow$ Article + Noun
Article $\rightarrow$ "the" | "a"
Predicate $\rightarrow$ Verb + Object

Input: “The cat chases the mouse”

Application of rules:
“The” $\rightarrow$ Article
“cat” $\rightarrow$ Noun $\rightarrow$ together form the Subject
“chases” $\rightarrow$ Verb
“the” $\rightarrow$ Article
“mouse” $\rightarrow$ Noun $\rightarrow$ together form the Predicate

Thus, the system determines that the sentence is grammatically correct.

However, this rule-based approach relies on fixed and absolute rules, which makes it difficult to apply to real-world situations where such absolutes rarely exist. Moreover, as problem complexity increases, the number of required rules grows exponentially, making maintenance and scalability impractical.

To overcome these limitations, probabilistic reasoning was introduced, along with the development of extensive rule sets designed by experts, giving rise to Expert Systems.

In contrast, the Connectionist approach is inspired by the structure and functioning of the human brain, particularly the interconnected network of neurons, hence the term connectionist. This paradigm relies on statistics and information theory rather than explicitly defined rules. Connectionism represents the foundation of Machine Learning (ML) and, consequently, of Large Language Models (LLMs). In modern ML, a task is framed as a statistical inference problem based on a large dataset.

There are three main learning paradigms:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning, in which the task is formulated as an objective function to be optimized.

According to Occam's Razor, simpler models tend to generalize better, [Soklakov, 2002]. Therefore, a model should include only the minimum number of parameters necessary to fit the dataset effectively.

Terms such as Deep Learning and Connectionism refer to the use of Neural Networks as the underlying computational structure.

Nonetheless, the Interpretability Problem remains one of the main challenges in deep learning. The large number of parameters in neural networks makes it extremely difficult to understand how the model reaches its conclusions or why a given input leads to a particular output.

C.2 Language Models

Large Language Models (LLMs) are statistical models designed to encode a probability distribution over sequences of words. Formally, the probability of a sentence consisting of c words can be represented as:

$$P(W_1, W_2, \ldots, W_c)$$

where W_1 denotes the first word in the sequence.

C.2.1 From Simple Models to Probabilistic Language Understanding

The simplest case, a 1-gram (unigram) model, assumes that each word is independent of all others. In such a case, the probability of the entire sentence is simply the product of the individual word probabilities:

$$P(W_1, W_2, \ldots, W_L) = \prod_{i=1}^{L} P(W_i)$$

where

$$P(W) = \frac{\text{Number of occurrences of } W \text{ in the corpus}}{\text{Total number of words in the corpus}}$$

Here, the corpus refers to a large collection of text used for estimating these probabilities.

However, modern LLMs are generative models, which means they can sample from this learned probability distribution to generate new sequences. Given a context, i.e., a sequence of words, the model estimates the probability of the most likely next word:

$$P(\text{next word}|\text{context})$$

C.2.2 Examples of Prediction Tasks

- Masked Word Prediction Example:
  $P(\text{went}|\text{The cat [BLANK] outside}) = 0.5$
  $P(\text{sat}|\text{The cat [BLANK] outside}) = 0.2$
  $P(\text{is}|\text{The cat [BLANK] outside}) = 0.1$
  Each probability contributes to a total of 1 (100%). The input is "The cat ... outside" and the output is the missing token.
- Next Word Prediction
  $P(\text{outside}|\text{The cat went}) = 0.7$
  $P(\text{home}|\text{The cat went}) = 0.2$
  $P(\text{to}|\text{The cat went}) = 0.1$

By sampling from this distribution and appending the predicted word to the existing sequence, the model can generate text one token at a time:

$$P(W_{n+1}|W_1, W_2, \ldots, W_n)$$

Example:

$$P(W|\text{the cat}) \to W = \text{went}$$

$$P(W|\text{the cat went}) \to W = \text{outside}$$

$$P(W|\text{the cat went outside}) \to W = \text{and}$$

This iterative use of the model's own output as the next input defines an autoregressive model, represented by:

$$\begin{aligned} P(\text{the cat went outside}) =& P(\text{the}) \\ & \times P(\text{cat}|\text{the}) \\ & \times P(\text{went}|\text{the cat}) \\ & \times P(\text{outside}|\text{the cat went}) \end{aligned}$$

C.2.3 Evaluating Language Models

To assess how well a language model approximates the true probability distribution of a corpus, the standard metric used is cross-entropy loss:

$$L = -\frac{1}{N}\sum_{i=1}^{N-n} \log P(W_{i+n}|W_1, W_{i+1}, \ldots, W_{i+n-1})$$

The summation captures the surprise of each word in the training corpus, except for the first n words that serve as context. The term $\frac{1}{N}$ averages the total surprise across the dataset. The logarithm penalizes confident but incorrect predictions: for example, if $P = 0.01$, then $\log(P) = -4.6$; if $P = 0.99$, then $\log(P) = -0.01$. The negative sign transforms this measure into a loss function to be minimized during training.

$$P(W_{i+n}|W_1, W_{i+1}, \ldots, W_{i+n-1})$$

represents the probability the model assigns to the next word. For example, if $i = 1$ and $n = 3$, for the sentence "The cat hunts the mouse", we would be computing $P(\text{the}|\text{The cat hunts})$.

C.2.4 Training Process

The training process of a language model can be summarized as follows:

1. A sequence of words is provided to the model, which predicts the next word and assigns probabilities to all vocabulary tokens.
2. The loss L is calculated to measure how far the prediction is from the true target.
3. The error is propagated backward through the network (backpropagation).
4. Model parameters are updated via gradient descent.
5. The process repeats iteratively until convergence.

C.3 Simpler Generative Language Models (N-gram)

Through the statistical analysis of neighbouring words, it is possible to capture quantitative properties, and even aspects of meaning, of words. The simplest expression of this idea is the co-occurrence matrix.

Language models operate on tokens rather than whole words. For example:

- "Supersymmetrization" $\rightarrow$ "Super", "Symmetry", "ization"

This process, called tokenization, helps models to better understand the structure and meaning of language.

C.3.1 Co-occurrence Matrix

Let w be the set of words in a corpus, and let $|w|$ denote its cardinality. We define the N-gram co-occurrence matrix M_N, which is of size $|w| \times |w|$. Each row and column represents a word, and each entry (w, w') counts the number of N-grams in the corpus that contain both words.

Example corpus: "O gato caça o rato. O cão guarda a casa"

To create a 3-gram co-occurrence matrix, we extract trigrams from the corpus. For instance, checking for "o" and "gato":

- "O gato caça" $\rightarrow$ contains both
- "gato caça o" $\rightarrow$ contains both
- "caça o rato" $\rightarrow$ does not contain both

The resulting matrix is represented below:

	o	gato	caça	rato	cão	guarda	a	casa
o	-	2	2	3	2	1	0	0
gato	2	-	2	0	0	0	0	0
caça	2	2	-	1	0	0	0	0
rato	3	0	1	-	1	0	0	0
cão	2	0	0	1	-	2	1	0
guarda	1	0	0	0	2	-	2	1
a	0	0	0	0	1	2	-	1
casa	0	0	0	0	0	1	1	-

Table C.1: 3-gram Co-occurrence Matrix

Once this matrix is built, word embeddings can be derived simply by taking the corresponding column (or row) vector of the word:

$$\text{gato} \Rightarrow [2, 0, 2, 0, 0, 0, 0, 0]$$

This vector provides a distributional representation of the word.

However, such vectors tend to have very high dimensionality. To address this, dimensionality reduction techniques such as Principal Component Analysis (PCA) are applied. The resulting lower-dimensional vectors are then fed into the model.

C.3.2 Predicting the Next Word

To predict the most probable next word, the model outputs a vector v, and the system selects the word w that maximizes the inner product between v and its embedding:

$$v \cdot \iota(w)$$

where $\iota(w)$ is the embedding of word w derived from the co-occurrence matrix.

The vector v itself is generated by a function F, which may be implemented as a Neural Network, Transformer, or other architectures.

To convert the similarity scores into a probability distribution, the Boltzmann distribution is applied:

$$P(w) = \frac{e^{\beta v \cdot \iota(w)}}{\sum_{w'} e^{\beta v \cdot \iota(w')}}$$

- $v \cdot \iota(w)$: measures similarity between the word and the predicted vector.
- $e^{\beta v \cdot \iota(w)}$: transforms scores into positive values and amplifies differences.
- $\sum_{w'} e^{\beta v \cdot \iota(w')}$: ensures probabilities sum to 1.

C.3.3 Semantic Properties of Word Embeddings

It can be observed that word embeddings capture semantic relationships. For example:

$$\iota(\text{king}) - \iota(\text{man}) + \iota(\text{woman}) \approx \iota(\text{queen})$$

This phenomenon arises from the statistical regularities of word co-occurrences, and requires a large corpus to emerge clearly.

C.3.4 Generalized Autoregressive L-gram Model

From these ideas, we can define a general autoregressive L-gram model:

1. Map the L input words w_i to their embedding vectors $\iota(w_i)$.
2. Apply the function F to this list of vectors to produce a prediction vector v.
3. Use the Boltzmann distribution to obtain a probability distribution over all possible words.

Regarding the function F:

- A simple Feedforward Network can be used, but such networks expect a single input, whereas we have L vectors, one per word.
- One possible solution is to concatenate these vectors into a single input.

However, this approach proved limited, as transformer architectures later emerged and surpassed it. A major challenge is that predicting the next word requires remembering previous context.

A simple improvement involves adding memory, which leads to Recurrent Neural Networks (RNNs). RNNs maintain a state vector s_i that updates with each word:

$$(v_{i+1}, s_{i+1}) = F(s_i, v_i, v_{i-1}, v_{i-2}, \ldots, v_{i-k+1})$$

where:

- (v_{i+1}, s_{i+1}) is the output of F; v_{i+1} predicts the next word, and s_{i+1} is the updated memory state.
- $F(s_i, v_i, v_{i-1}, \ldots, v_{i-k+1})$ is the input to F; s_i encodes memory so far, while $v_i, v_{i-1}, \ldots$ provide immediate context within a window of k words.

C.4 Recipe for a Large Language Model

A Transformer can be defined as a function F that takes a list of vectors as input and produces an output vector. This represents a generalization of Feedforward Networks (FFNs), with an important distinction: Transformers account for permutation symmetry, meaning they can handle sequences where the order of inputs matters.

A Transformer is composed of two key types of layers:

C.4.1 Feedforward Layers

A Feedforward Neural Network (FFN) is applied independently to each embedding vector. For each input vector u_i, the model applies a learned transformation to produce an output vector:

$$v_i = F_{\text{FFN}}(u_i)$$

C.4.2 Attention Mechanism

The Attention mechanism allows each output vector to attend to, or focus on, other input vectors in the sequence. This enables the model to determine which parts of the input are most relevant for each word at a given moment.

The output vector v_i is computed as a weighted sum of all input vectors u_j, each multiplied by a learnable linear transformation matrix w:

$$v_i = w \sum_{j=1}^{i} c_{i,j} u_j$$

Where:

- v_i is the output vector.

- w is the transformation matrix.
- $c_{i,j}$ are the attention weights.
- u_j are the input vectors.

$$c_{i,j} \equiv \frac{e^{u_i \cdot B \cdot u_j}}{\sum_{j=1}^{i} e^{u_i \cdot B \cdot u_j}}$$

The attention weights $c_{i,j}$ determine how much attention the vector at position i gives to the vector at position j. These weights are computed using a softmax function, which converts numerical scores into normalized probabilities, ensuring that the sum of all $c_{i,j}$ for a given i equals 1.

The matrix B is a learned parameter during training. It allows the model to learn which relational aspects between vectors are most relevant.

- When $B = I$, the dot product measures only similarity.
- When B is learnable, it can capture more complex relationships, such as those between verbs and subjects, even when they are not semantically similar.

C.4.3 Attention Heads

Transformers use multiple attention heads operating in parallel rather than relying on a single mechanism. Each head has its own learnable matrices B and W, and produces output vectors of smaller dimension q (compared to the original embedding dimension p).

If there are H heads, their outputs are concatenated, resulting in an overall embedding dimension:

$$p = H \times q$$

Analogy: Each attention head may focus on a different aspect of the input:

- Head 1 $\rightarrow$ object colour
- Head 2 $\rightarrow$ shape
- Head 3 $\rightarrow$ spatial relationships
- Head 4 $\rightarrow$ texture

This multi-head attention enables the model to integrate diverse types of information simultaneously.

C.4.4 Positional Encodings

A pure attention mechanism is invariant to input order, meaning it cannot inherently distinguish between permutations of the same words. To address this, Transformers introduce Positional Encodings, numerical representations that encode the position of each word in the sequence.

These encodings are generated using trigonometric functions:

$$(e_{2i-1}, e_{2i}) = \left(\cos \frac{\text{position}}{10000^{\frac{2i}{d_{\text{pos}}}}}, \sin \frac{\text{position}}{10000^{\frac{2i}{d_{\text{pos}}}}}\right), \quad i \in \left\{1, \ldots, \frac{d_{\text{pos}}}{2}\right\}$$

Where:

- $2i$ is the index within the vector.
- d_{pos} is the dimensionality of the positional embedding (equal to the word embedding dimension, e.g., $d = 512$).
- $\frac{d_{\text{pos}}}{2}$ ensures the sine and cosine functions alternate along half the embedding dimension.

This mechanism provides positional context, ensuring that word order affects meaning, for instance: "The cat hunts" is not equal to "Hunts the cat". Although both sentences contain the same words, their meanings differ due to positional encoding. Each word's embedding is summed with its positional vector:

- Word embedding: $[0.3, 0.9, 0.1, 0.4]$
- Positional encoding: $[0.84, 0.54, 0.01, 1.00]$
- Combined input: $[1.14, 1.44, 0.11, 1.40]$

Thus, even repeated words at different positions in the text will receive unique positional embeddings, enabling the model to preserve sentence structure and meaning.

C.5 Attention Is All You Need

C.5.1 Model Architecture

The Transformer architecture is composed of two main components: an encoder and a decoder. The encoder maps a sequence of symbolic representations $(x_1, \ldots, x_n)$ (e.g., words) into a sequence of continuous representations:

$$Z = (Z_1, \ldots, Z_n)$$

The decoder then generates a sequence of output symbols $(y_1, \ldots, y_m)$ one element at a time, consuming previously generated symbols as additional input, making the model autoregressive.

C.5.2 Encoder

The encoder consists of a stack of $N = 6$ layers, each containing two sub-layers:

1. A multi-head self-attention mechanism
2. A feedforward neural network (FFN)

A layer normalization step is applied between sub-layers to stabilize and accelerate training.

C.5.3 Decoder

The decoder also consists of $N = 6$ layers. In addition to the two sub-layers found in the encoder, the decoder introduces a third sub-layer that performs multi-head attention over the encoder's output. Layer normalization is again applied between all sub-layers.

A key component in the decoder is the Masked Multi-Head Attention mechanism, which prevents each position from attending to subsequent (future) words. This ensures that predictions for position i depend only on positions $< i$.

Example:

- Input (to the decoder): $[< s >, \text{The}, \text{cat}, \text{chases}, \text{the}]$
- Target sequence: $[\text{The}, \text{cat}, \text{chases}, \text{the}, \text{mouse}]$

ID	Decoder Input	Target Output
1	$[< s >]$	The
2	$[< s >, \text{The}]$	cat

Table C.2: Decoder Time Steps and Target Outputs

This design ensures that each word at position i can only see previous words, maintaining the causal structure required for autoregressive generation.

C.5.4 The Attention Mechanism

The Attention mechanism maps a query and a set of key-value pairs to an output, computed as a weighted sum of the values. Each weight represents how compatible the query is with the corresponding key.

Example context:

- Sentence: "The cat chases"

- Query: "chases"
- Keys: ["The", "cat", "chases"]

Mathematically:

$$Q = W_q \times \text{embedding(chases)}$$

$$K = W_k \times [\text{embedding(The)}, \text{embedding(cat)}, \text{embedding(chases)}]$$

$$V = W_v \times [\text{embedding(The)}, \text{embedding(cat)}, \text{embedding(chases)}]$$

This allows the model to build contextualized and dynamic representations, capturing long-range dependencies and context-dependent meanings. Because the matrices W_q, W_k, and W_v are learnable parameters initialized randomly, the model can learn diverse linguistic relations in parallel during training.

C.5.5 Scaled Dot-Product Attention

The Scaled Dot-Product Attention function is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

where Q and K are matrices with dimension d_k.

- QK^T: computes similarity scores between each query and key (compatibility matrix).
- $\text{softmax}(\cdot)$: converts these scores into normalized probabilities.

This function processes multiple queries simultaneously, enabling efficient parallel computation.

C.5.6 Applications of Attention in the Transformer

There are three types of attention mechanisms in the Transformer model:

1. Encoder–Decoder Attention
   - Queries (Q): come from the decoder (previous decoder output).
   - Keys (K) and Values (V): come from the encoder's final output.
   - Example: To generate "mouse", Q = chases (Decoder) and $K, V = [\text{The}, \text{cat}, \text{chases}, \text{the}, \text{mouse}]$ (Encoder).
2. Self-Attention in the Encoder
   - Q, K, V all come from the same source, the encoder's previous layer output.

- Each word can attend to all words in the input.

3. Masked Self-Attention in the Decoder

    - Q, K, V all come from the decoder.
    - Future words are masked to prevent information leakage.
    - Example: For "chases", the model can only attend to [The, cat, chases].

C.6 Prompting Strategies

Prompting is the practice of guiding a pretrained language model at inference time using instructions or examples, allowing it to perform new tasks without updating its internal parameters, [Hooper]. It has become the standard method for adapting LLMs to specific applications.

Prompt engineering techniques fall into three main categories:

- Manual prompting, zero-shot, pure instruction, one-shot, few-shot templates, where examples are embedded into the prompt to demonstrate the desired behavior, [Chen et al., 2025].
- Automatic prompt optimization, these methods search or optimize prompts to improve performance, techniques include discrete search over prompt variants, continuous optimization where prompts are learned as vectors, or LLMs generate/refine prompts, [Zhou et al., 2023].
- hybrid approaches, combine manual intuition with automatic tools. Human-in-the-loop prompt refinement allows iterative improvement based on model outputs, [Zhou et al., 2023].

These methods exist because LLM behavior is highly sensitive to prompt design.

A key phenomenon underlying prompting is In-Context-Learning (ICL), this happens when a LLM is given examples of input–output pairs in a prompt, it can perform a new task without any parameter updates. This capacity distinguishes modern LLMs from traditional classifiers and enables task adaptation purely via prompt context, [Zhou et al., 2024].

Some works try to explain these behavior through: Implicit Gradient-like Behavior, where the Transformer architecture may emulate gradientbased learning on the provided examples, Bayesian Inference Framework, where the model implicitly infers latent task variables conditioned on the examples within the prompt, and Structure Induction and Pattern Matching, where the model may leverage learned structural regularities in language data, recombining compositional patterns to generalize from prompt examples to new inputs, [Hahn and Goyal, 2023].

More advance prompting strategies exist, such as:

- Chain-of-Thought prompting, this type of prompts guide models to generate intermediate reasoning steps before producing a final answer [Wei et al., 2023].
- Self-correction and Task decomposition, Techniques like iterative self-critique or problem decomposition encourage the model to review and refine its outputs or to break down complex tasks into simpler sub-tasks, [Yang et al., 2024] and [Liu et al., 2025].
- Retrieval-augmented prompting, [Son et al., 2025].

Despite its effectiveness, prompting faces limitations: sensitivity to wording, context-length constraints, hallucinations and bias, vulnerability to adversarial prompts, and weaker robustness compared to fine-tuned models, [Hooper].

Prompt engineering matters because it enables rapid task adaptation, reveals emergent capabilities like ICL, and provides insight into how LLMs reason, generalize, and use their pretrained knowledge

Prompting allows models to perform a wide range of tasks without task-specific training, enabling access to AI systems for users without deep technical knowledge.

C.7 Information Extraction

Information Extraction (IE), aims to convert unstructured text into structured representations such as entities, relations, and attributes. Traditionally, IE relied on rule-based systems and later statistical models like Hidden Markov Models, [Su], Maximum Entropy classifiers, [Osborne, 2002], and Conditional Random Fields, [Ding et al.], which required extensive feature engineering and linguistic preprocessing, .

Large Language Models fundamentally change this landscape. LLMs enable end-to-end information extraction directly from raw text, reducing or eliminating the need for task-specific architectures and extensive preprocessing pipelines, [Xu et al., 2024]. Through prompting and in-context learning, LLMs can perform entity and relation extraction in zero-shot or few-shot settings, adapting to new schemas without retraining, [Wei et al., 2022]. This capability has been demonstrated across multiple IE tasks, including named entity recognition, relation extraction, and event extraction, often achieving competitive or superior performance compared to traditional supervised models, .

Moreover, LLMs naturally integrate semantic knowledge acquired during pretraining, allowing them to leverage world knowledge and long-range context, which is particularly beneficial for relation extraction and complex document understanding.

However, LLM based IE still faces issues such as prompt sensitivity, hallucinations, and limited control over output structure. As a result, hybrid approaches combining LLMs with symbolic constraints or post-processing are increasingly explored to improve reliability, [Xu et al., 2024].

Appendix D

Mockups

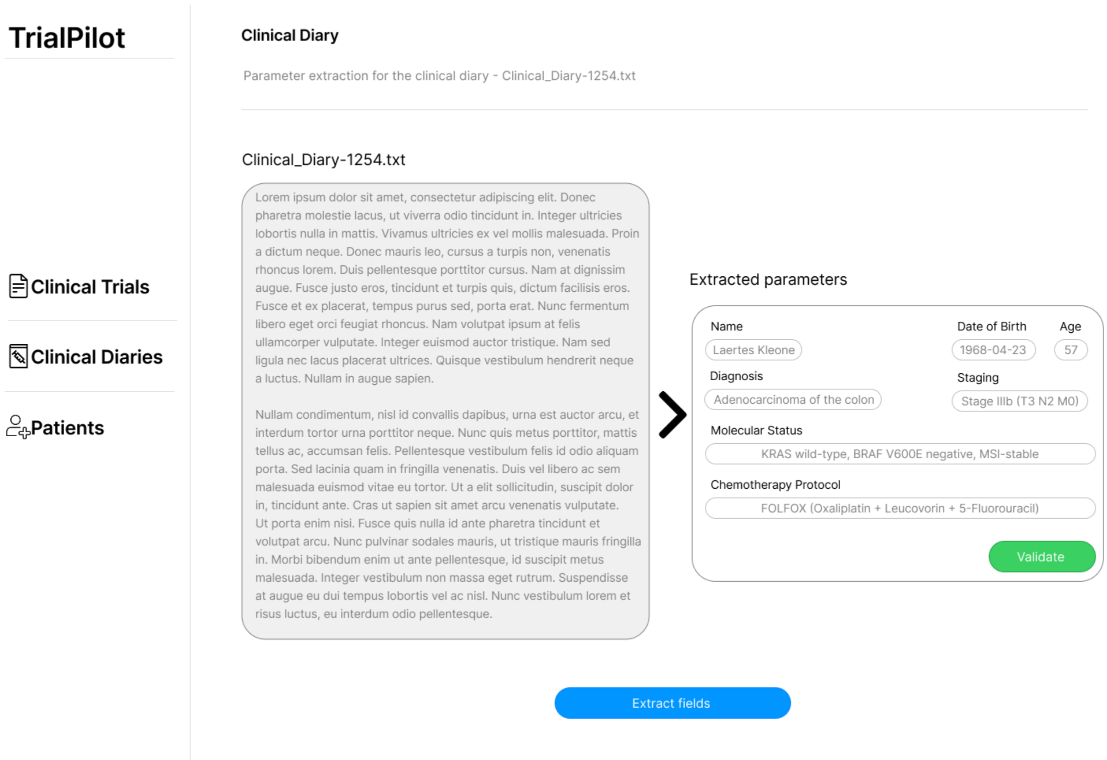


Figure D.1: Clinical diary field extraction page

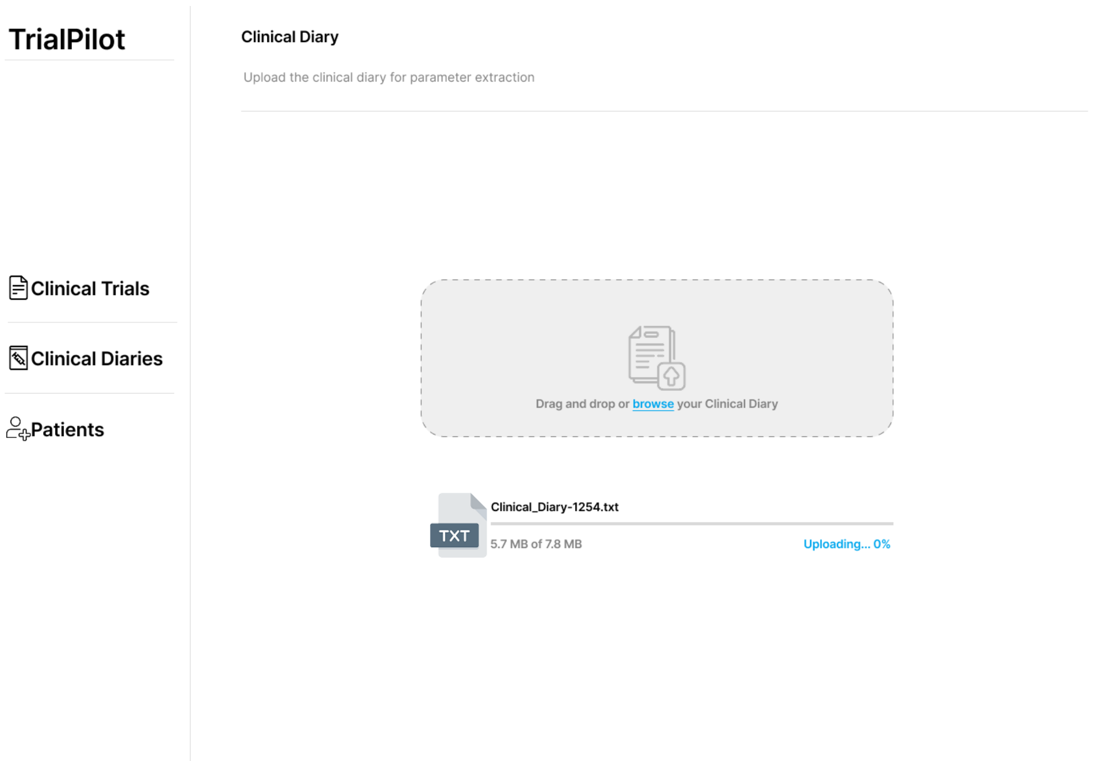


Figure D.2: Clinical diary upload page

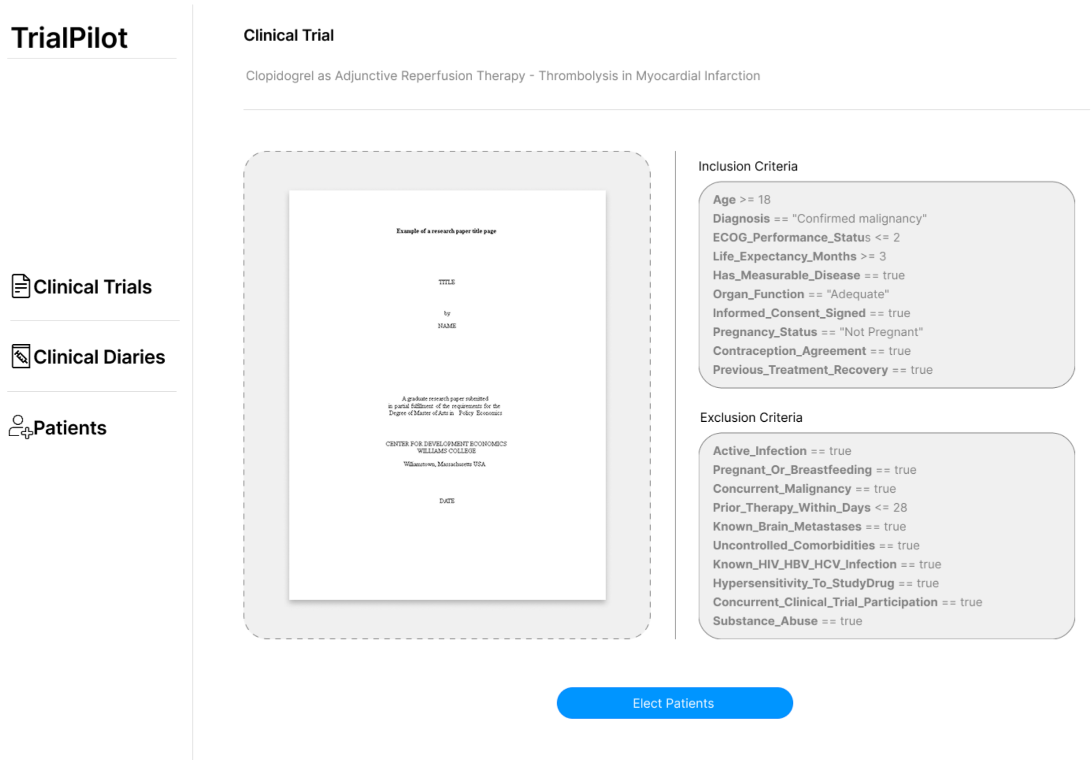


Figure D.3: Clinical trial detail page

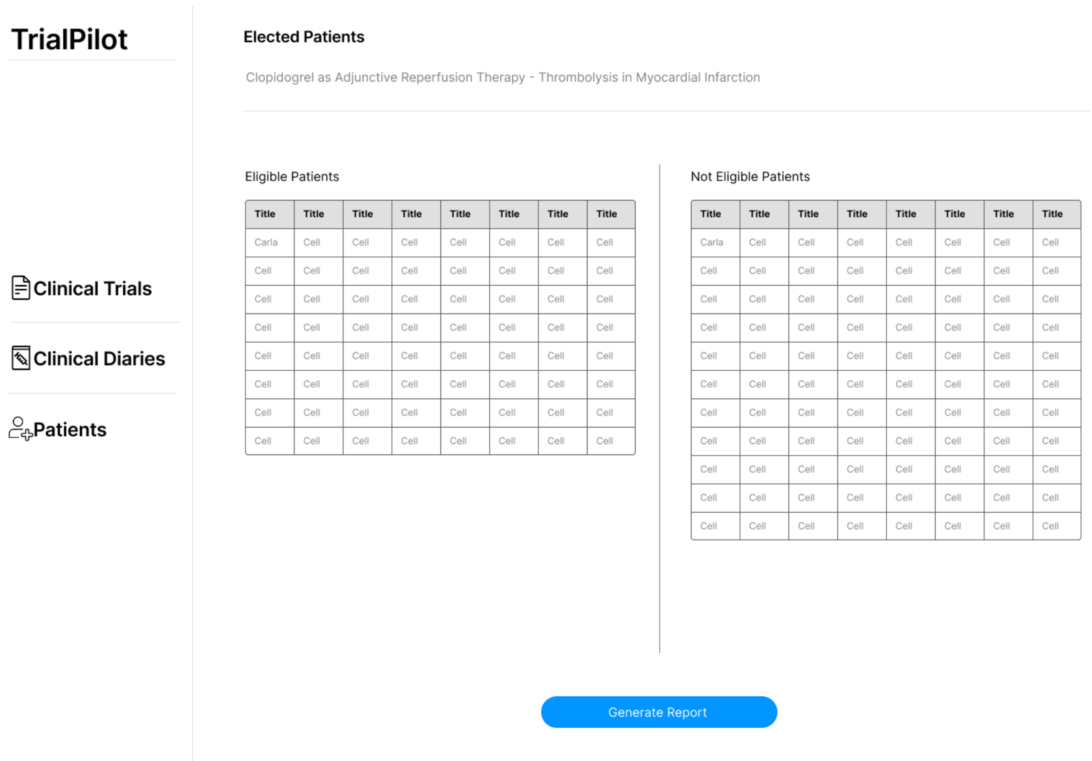


Figure D.4: Elected patients page

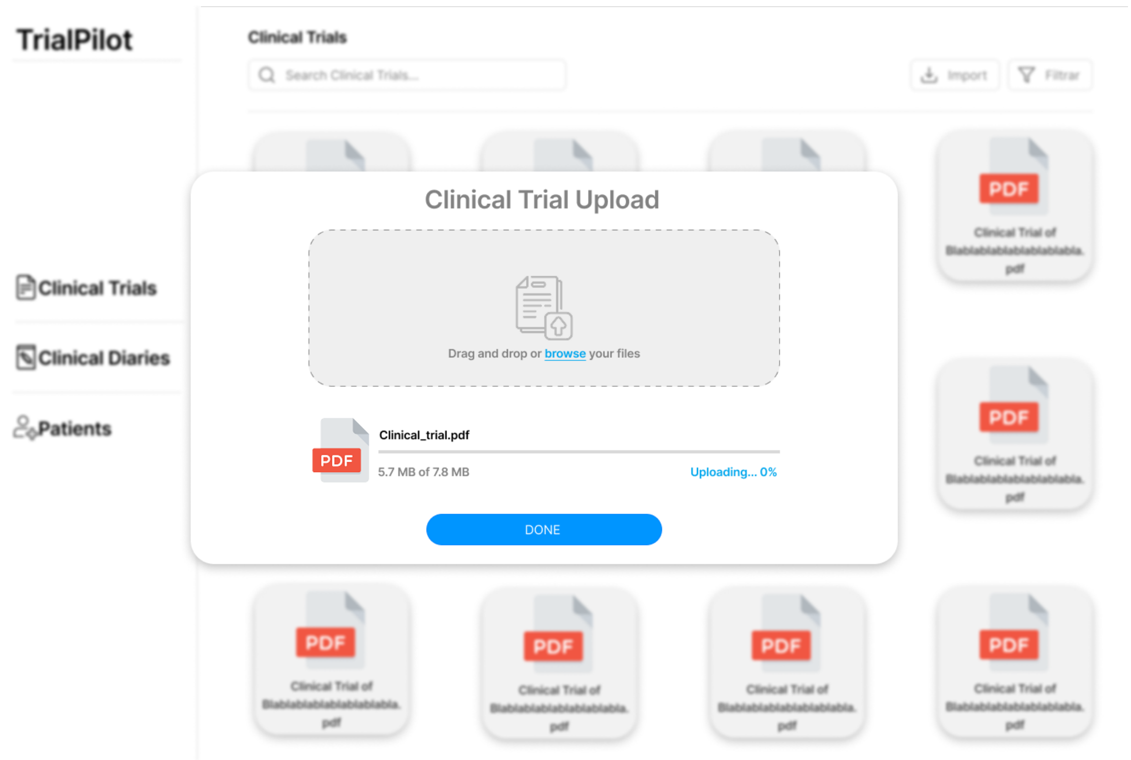


Figure D.5: Import clinical trial page

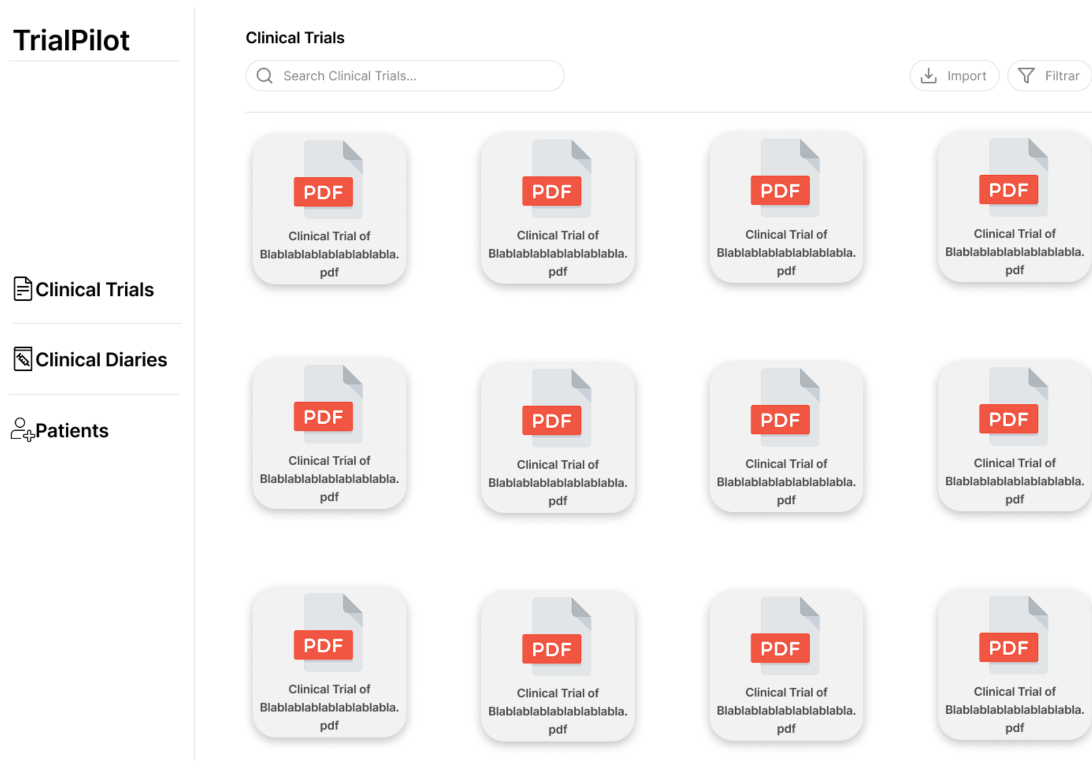


Figure D.6: Main-page

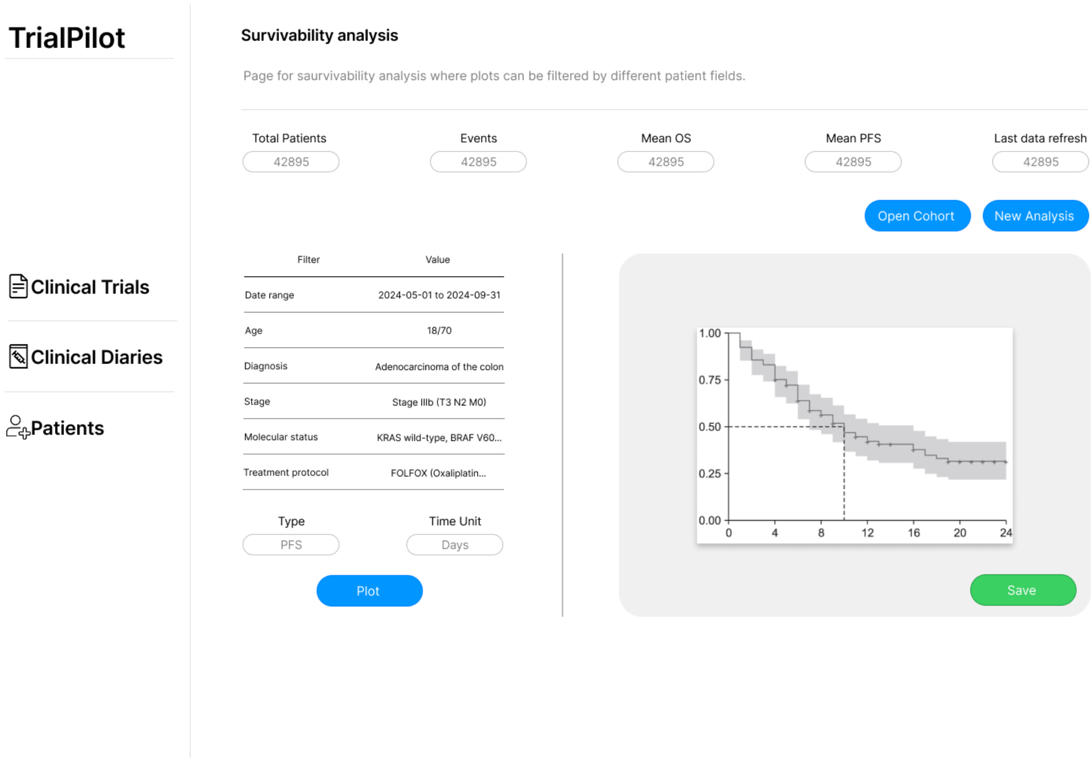


Figure D.7: Survivability analysis page

Appendix E

User Interface Design

The elaboration of this study would not be complete without integrating the entire pipeline into a system capable of connecting every core component of the proposed solution. For this reason, a web application was developed to support the study and to provide a tangible interface for stakeholders, ensuring their full engagement throughout the process.

The application itself was guided by the requirements previously described, which proved to be an excellent roadmap for the development of the system. These requirements helped structure the work, ensuring that the application would be robust and of high quality. In addition, the ER diagram presented in earlier sections played an important role by offering a clear overview of the system even before it existed. It guided not only the design of the models but also the overall reasoning behind the views, ensuring alignment with the database structure defined from the start.

The system was designed as a web application because this format best suits the clinical context, offering portability and integrating naturally with the workflow already in place. The application was developed using Django, a Python-based framework used to build both the frontend and backend. This technology was chosen due to its simplicity, its smooth learning curve, and existing experience with the framework.

All templates were created following the previously designed low-fidelity prototypes, which again proved to be a significant time-saver, since the layout and structure of each page had already been planned. Some adjustments were made during development, as the prototypes were created early in the project. A UI library, Bootstrap, was also used to improve the visual appearance of the system, reducing the effort required to refine the interface.

Every page in the platform follows the same structure to ensure a simple and coherent visual experience, improving usability and making the learning process of the system more intuitive.

E.1 Home

The home page serves as the central entry point to the entire system, providing access to every feature and offering an immediate overview of the platform's current state. This page presents several key metrics that summarise the progress and activity of the system:

- For clinical diaries, the total number of artefacts is displayed, along with how many have already been extracted and how many remain pending. A progress bar is included to give a clearer visual representation of the overall extraction progress.
- For patients, the total number of structured patient profiles is shown, together with the average age and the number of ongoing treatments. The most frequent diagnoses are also presented, which is particularly important given the mapping required during the parameter extraction process.
- For clinical trials, the total number of artefacts is displayed, as well as how many have already been processed, how many remain pending, and how many patient matches have been performed across all trials. A progress bar is again included to help visualise the completion of trial extraction and conversion.

In addition to these metrics, the home page provides quick access to the most recent unprocessed clinical trial, the latest unprocessed clinical diaries, and the most recently structured patient profiles. This ensures that clinicians can begin working immediately upon entering the application, without needing to navigate through multiple pages to find the most relevant items.

Everything described can be seen below in the Figure E.1

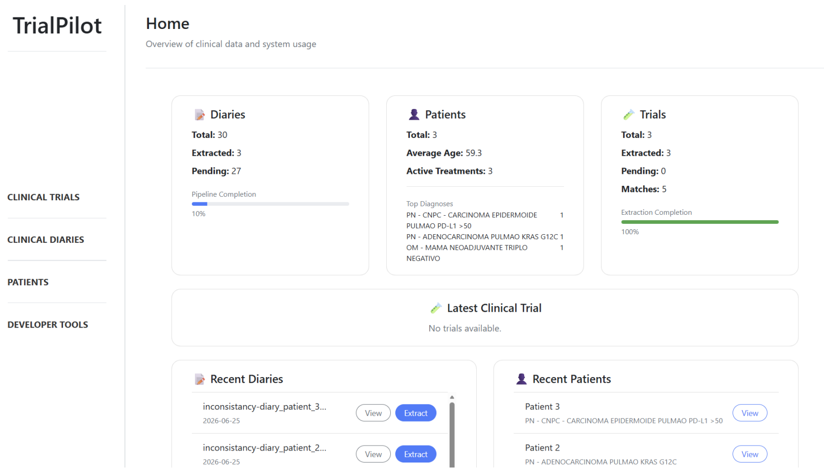


Figure E.1: Home page

E.2 Developer Tools

This page is not intended for clinicians, but it provides valuable metrics regarding the overall health of the system. These metrics include the number of documents stored in the platform and the progress of their extraction, the total number of versions created and the average number of versions per document, the number of structured patient profiles, and the total number of patient–trial matches already performed.

There are also specific metrics dedicated to assessing both the versioning health of the system and the performance of the matching engine. These include:

- The total number of versions and the average number of versions per file, as well as how many files exist without any version. This is important because every file must have at least one version, the RAW version.
- The total number of matches performed and the distribution of their outcomes, namely how many patients were classified as eligible, ineligible, or inconclusive.

Finally, this page includes an overview of the entire system, describing each page and its purpose. Although primarily designed for internal use, this overview can also be helpful for clinicians who may feel lost or unfamiliar with the structure of the application.

Everything described can be seen below in the Figure E.2

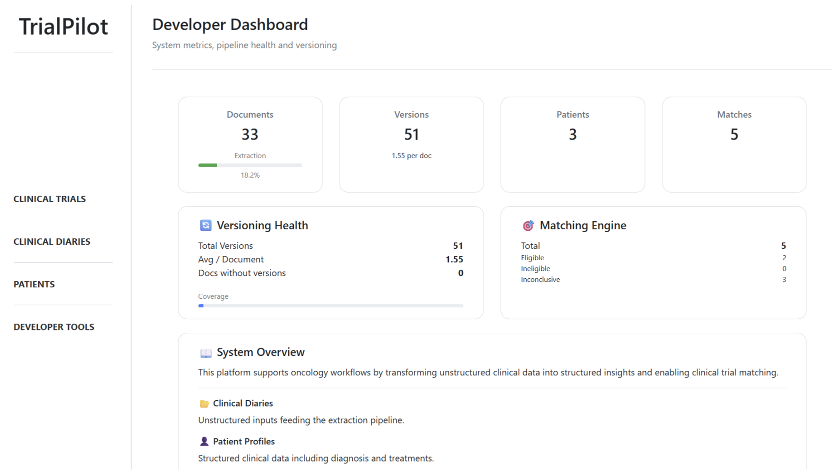


Figure E.2: Developer tools page

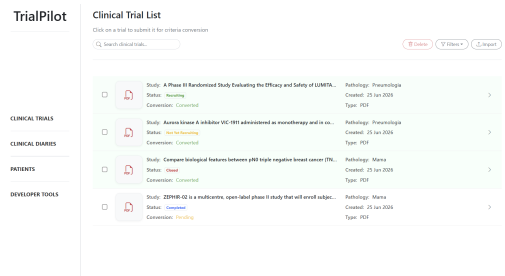
E.3 Clinical Trials

This page presents the list of all uploaded clinical trials in the system. Here it is possible to verify whether a clinical trial has already been processed or is still pending, as well as to quickly access essential information that helps identify each trial. This is also the page where new clinical trials can be manually uploaded into the system.

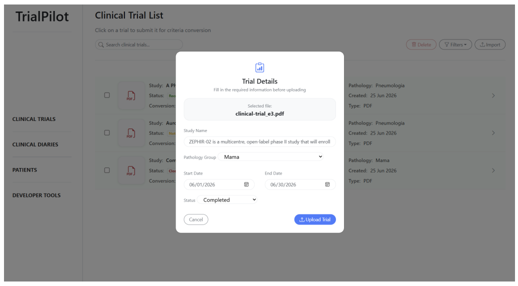
To support efficient navigation and make it easier to locate a specific document, several filtering options are available. Trials can be filtered by extraction status, trial status, pathology group, file type, and date range. In addition, a search bar allows users to search directly by study name.

From this page, users can also access the detailed view of any clinical trial, delete a trial if necessary, or submit it to the eligibility criteria extraction and conversion pipeline. This ensures that all actions related to clinical trials are centralised in a single, coherent interface.

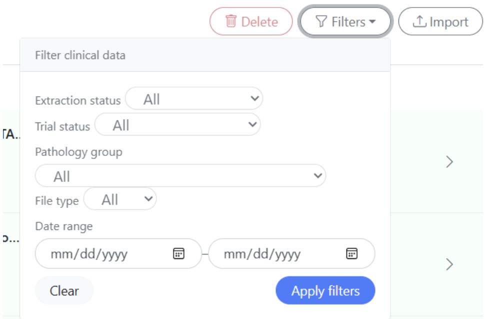
Everything described can be seen below in the Figure E.3



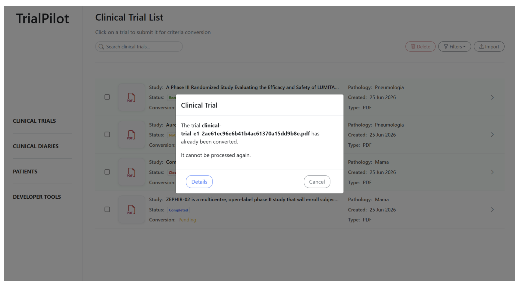
(a) Listing



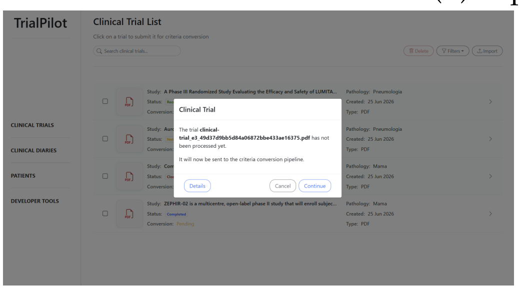
(b) Upload metadata



(c) Filters



(d) Popup (extracted)



(e) Popup (not extracted)

Figure E.3: Clinical Trials Interface Overview

E.4 Clinical Trial Details

The clinical trial details page presents all metadata associated with a given clinical trial, as well as every clinical artefact linked to it. On this page it is possible to access the trial's metadata, including pathology group, start and end dates, extraction status, and trial status. It also shows whether the trial contains cohorts and, if so, how many inclusion and exclusion criteria belong to each cohort. The original content of the uploaded document can be inspected, together with all versions that have been created throughout the pipeline.

This page also displays whether any patient-trial matches have already been performed. In this section, users can review the eligibility decision for each patient and the corresponding justification, criterion by criterion, providing full transparency over the matching process.

If the clinical trial has already been processed, the action button in the upper-right corner becomes available, allowing the user to initiate the matching process directly from this page.

Everything described can be seen below in Figure E.4.

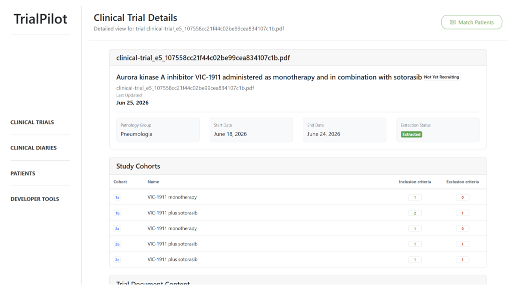
E.5 Match Patients

As mentioned earlier in the clinical trial details page, the matching of patients can only be performed once the trial has been fully extracted and converted. When the matching process is initiated, the user is first shown a loading screen while both the deterministic and LLM-based evaluations are executed. Once this process finishes, the user is redirected to the match patients page.

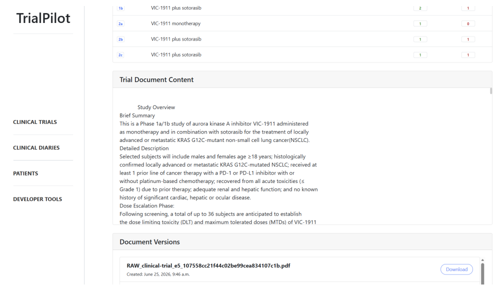
On this page, the user has access to a concise summary of the matching results, including how many patients were processed, how many were considered eligible, and how many were not. The expert can then inspect the detailed evaluation for each patient, reviewing the automatic result for every criterion and, when necessary, manually overriding these decisions. If the clinical trial contains cohorts, the criteria are clearly separated into general and cohort-specific sections, ensuring that the expert can easily understand how each decision was reached.

Finally, once all overrides and validations are complete, the results can be saved. The user is then redirected back to the trial's detail page, where the final matching outcomes and their justifications remain accessible for future consultation.

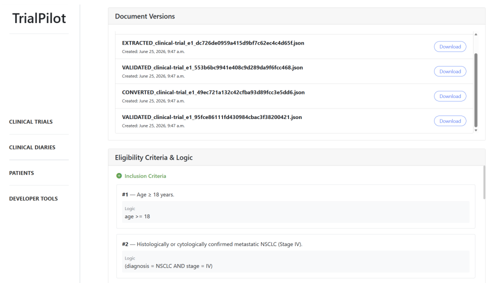
Everything described can be seen below in Figure E.5.



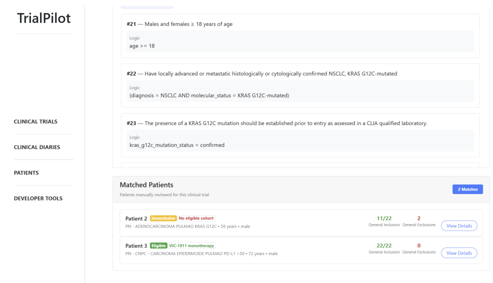
(a) Clinical trial details (1)



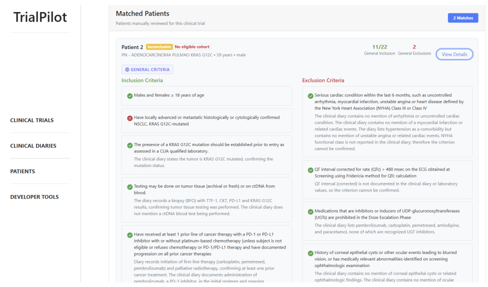
(b) Clinical trial details (2)



(c) Clinical trial details (3)



(d) Clinical trial details (4)



(e) Clinical trial details (5)

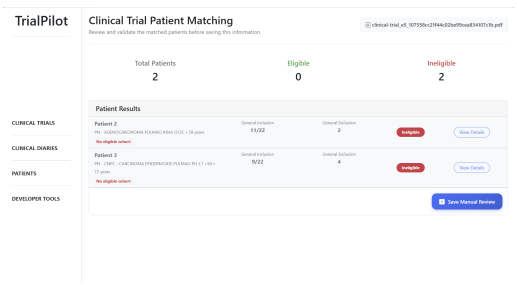
Figure E.4: Clinical Trial Details Interface Overview

E.6 Clinical trial criteria extraction

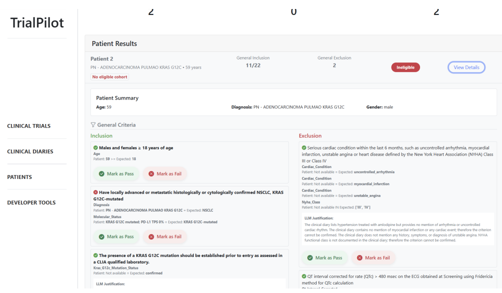
Through the Clinical Trials page, a user can select a trial to submit to the extraction and conversion pipeline. Once a trial is submitted, a loading page is shown while the LLM performs the extraction. When this process finishes, the user is presented with the original contents of the trial alongside the LLM-extracted criteria. At this stage, the user can freely edit, remove, or create new criteria as needed. If the trial contains cohorts, they are also identified during this phase, and each cohort can be populated with new criteria, edited, or cleared. For clarity, all criteria are separated into the main inclusion and exclusion sections, ensuring a more intuitive visual organisation.

When the user is satisfied with the extracted and edited criteria, the process can continue. The results are saved, and the LLM is prompted again to convert these criteria into their logical form, advancing the trial to the next stage of the pipeline.

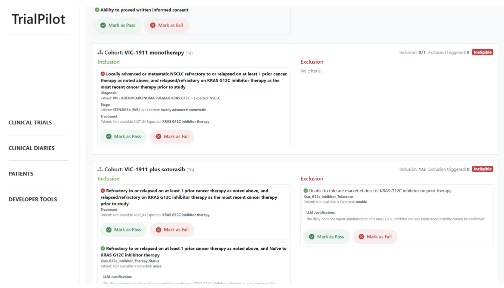
Everything described can be seen below in Figure E.6.



(a) Matching patients overview

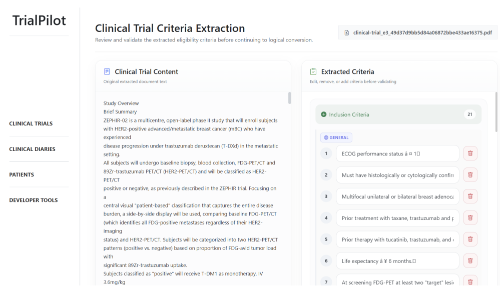


(b) Matching patients details

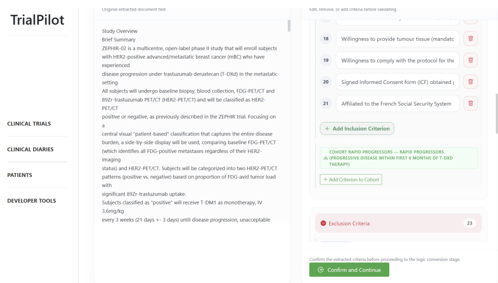


(c) Matching patients cohort assessment

Figure E.5: Matching Patients Interface Overview



(a) Clinical trial extraction (1)



(b) Clinical trial extraction (2)

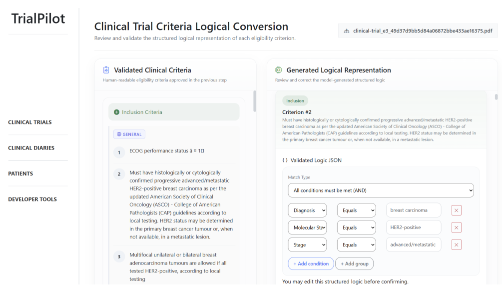
Figure E.6: Clinical Trial Extraction Interface Overview

E.7 Clinical trial criteria conversion

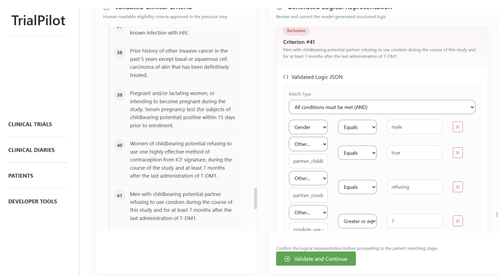
Following the Clinical Trial Criteria Extraction page, the system prompts the LLM with the extracted criteria and performs their conversion into logical rules. While this process is running, the user is shown a loading page. Once the conversion is complete, the user is presented with two columns: on one side, the previously extracted criteria, and on the other, their corresponding converted versions. At this stage, the expert can again edit, remove, or create new rules as needed. Each criterion is uniquely identified and correctly ordered, ensuring a clear and intuitive comparison between the textual criterion and its logical representation.

When the expert is satisfied with the converted rules, the pipeline can be finalised. The results are saved, storing every converted criterion and enabling the matching of patients for the processed clinical trial. After saving, the user is redirected to the Clinical Trials page, where a notification confirms that the artefact has been successfully processed.

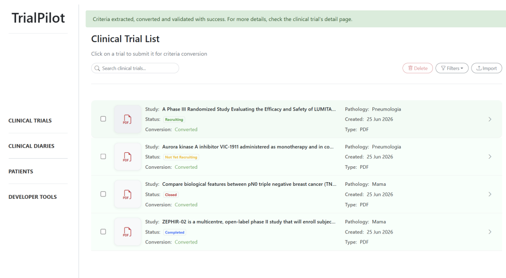
Everything described can be seen below in Figure E.7.



(a) Clinical trial criteria conversion (1)



(b) Clinical trial criteria conversion (2)



(c) Successful conversion notification

Figure E.7: Clinical Trial Criteria Conversion Interface Overview

E.8 Clinical diaries

This page presents the list of all uploaded clinical diaries in the system. Here it is possible to verify whether a diary has already been processed or is still pending, as well as to quickly access essential information that helps identify each document. This is also the page where new clinical diaries can be manually uploaded.

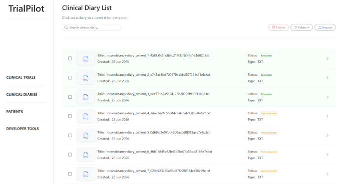
To support efficient navigation and make it easier to locate a specific diary, several filtering options are available. Diaries can be filtered by extraction status and file type, and a search bar allows users to search directly by document name.

From this page, users can also access the detailed view of any clinical diary, remove a diary if necessary, or submit it to the parameter extraction pipeline. This centralises all actions related to clinical diaries in a single, coherent interface.

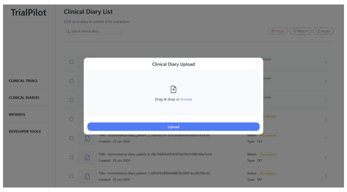
Everything described can be seen below in Figure E.8.

E.9 Clinical diary details

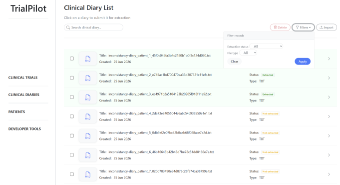
The clinical diary details page presents all metadata associated with a given clinical diary, as well as every clinical artefact linked to it. On this page it is possible to access the diary's metadata, including document name, start and end dates, and extraction status. The original content of the uploaded document can also



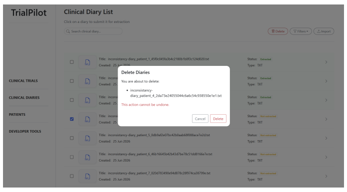
(a) Clinical diary listing



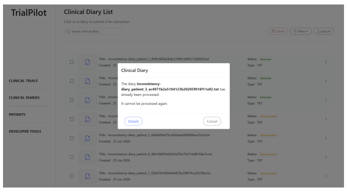
(b) Clinical diary upload



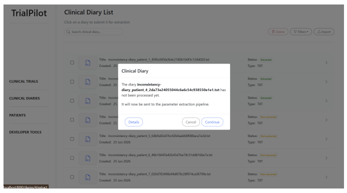
(c) Clinical diary filters



(d) Clinical diary removal



(e) Popup (already extracted)



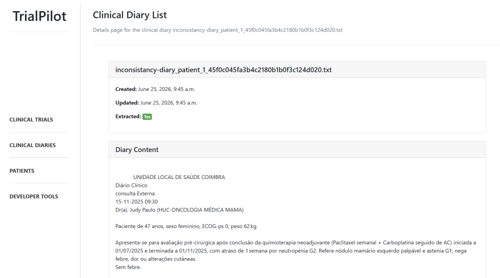
(f) Popup (not extracted)

Figure E.8: Clinical Diary Interface Overview

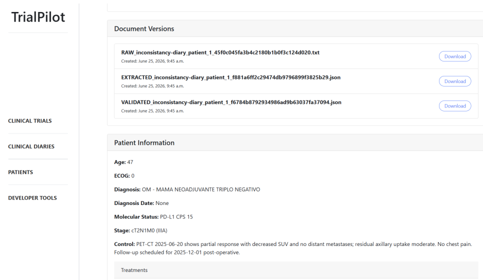
be inspected, together with all versions that have been created throughout the pipeline.

Finally, this page shows whether parameter extraction has already been performed. Through the patient overview section, the user can inspect the complete patient object produced from the diary's extraction, including the clinical overview, clinical details, treatments, and laboratory analysis. This provides a clear and structured representation of the information derived from the diary.

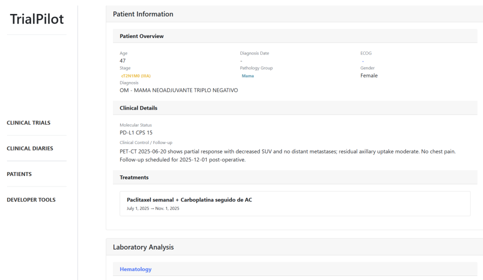
Everything described can be seen below in Figure E.9.



(a) Clinical diary details (1)



(b) Clinical diary details (2)



(c) Clinical diary details (3)

Figure E.9: Clinical Diary Details Interface Overview

E.10 Clinical diary parameter extraction

Through the Clinical Diaries page, a user can select a diary to submit to the extraction pipeline. Once a diary is submitted, a loading page is shown while the LLM performs the extraction. When this process finishes, the user is presented with the original contents of the diary alongside the LLM-extracted parameters, which can be manually edited as needed.

The extraction section follows the structure of the patient schema previously described. It is organised into four main areas:

- Patient Overview, where age, gender, ECOG-PS and clinical follow-up information are displayed
- Diagnosis Information, containing the patient's clinical and diagnostic details
- Treatments, listing both past and ongoing treatments, which can be manually removed or created
- Laboratory Analysis, presenting all laboratory results associated with the patient, including value and unit

Once the expert finishes validating and adjusting the extracted information, the results can be saved. All changes are stored, and the user is redirected back to the Clinical Diaries page, where a notification confirms that the artefact has been successfully processed. The extracted information then becomes available in the diary's detail page and in the Patient page described in the following section.

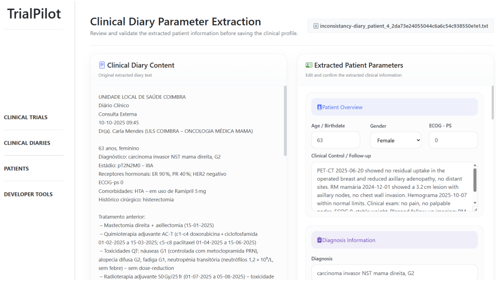
Everything described can be seen below in Figure E.10.

E.11 Patients

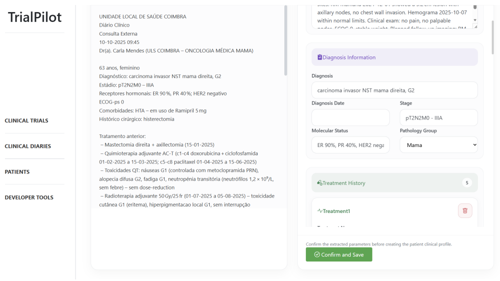
This page presents the list of processed clinical diaries, which result in structured patient profiles within the system. Here it is possible to view every patient created and quickly access the essential information needed to identify each one. To make navigation easier, several filtering options are available. Patients can be filtered by stage, pathology group, and, when selected, by normalised diagnosis, as well as by molecular status. It is also possible to search directly for a patient by entering a diagnosis or molecular status.

From this page, users can also inspect the details of any patient. When selecting a patient, a pop-up window appears displaying all relevant sections of the patient profile: clinical overview, clinical details, treatments, and laboratory analyses associated with the artefact. Furthermore, this page allows a patient to be reset, returning the diary that generated the profile to its original state, updating its extraction status, and removing any previously created versions.

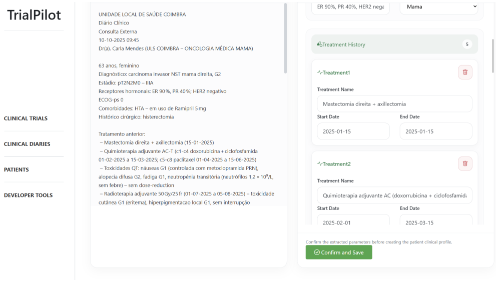
Everything described can be seen below in Figure E.11.



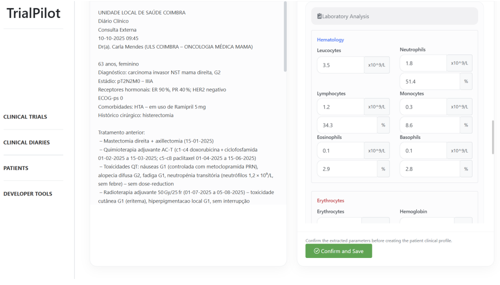
(a) Parameter extraction (1)



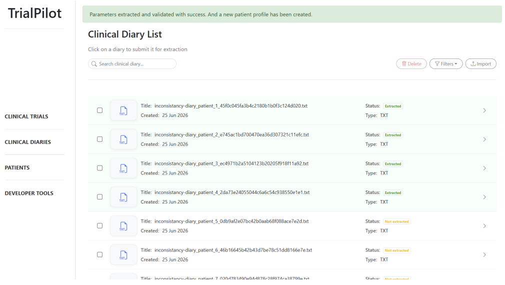
(b) Parameter extraction (2)



(c) Parameter extraction (3)

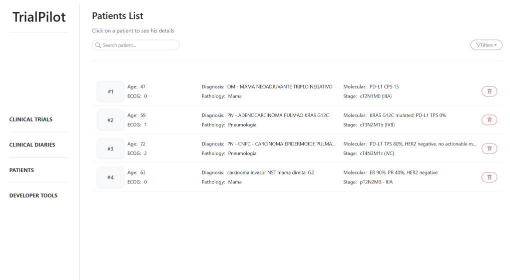


(d) Parameter extraction (4)

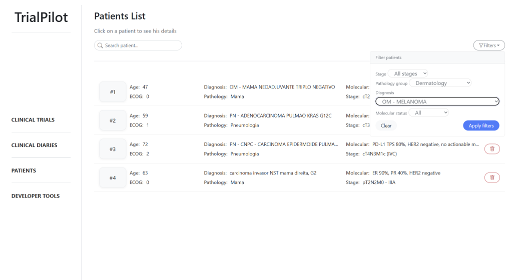


(e) Successful extraction notification

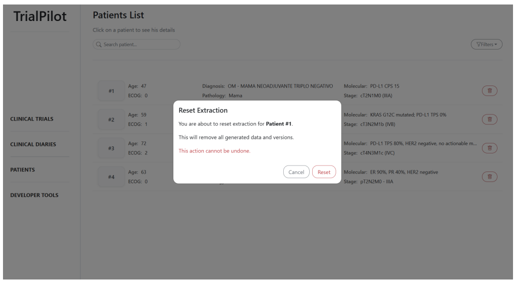
Figure E.10: Clinical Diary Parameter Extraction Interface Overview



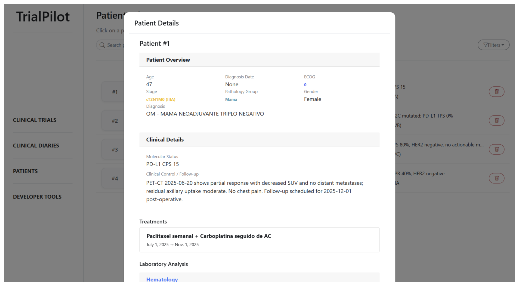
(a) Patient listing



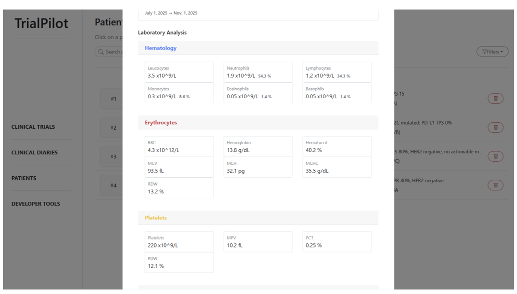
(b) Patient filters



(c) Patient removal



(d) Patient details (1)



(e) Patient details (2)

Figure E.11: Patients Interface Overview

Appendix F

Unit Testing

In order to guarantee that the system's web application works as intended extensive unitary tests were performed. These tests had the objective of ensuring correct functionality and error handling of every single view present in the system, ranging all the way from: helpers and auxiliary views, to the main pipeline views of parameter extraction, trial structuring identification, criteria extraction, criteria conversion and patient-trial matching.

A total of 388 tests were performed, which served to ensure the correct behavior of the app and to correct any bugs and errors that weren't expected.

F.1 Models

The first stage of the validation process involved unit testing the models that populate the system's database. These tests were designed to ensure the correct storage of information, the integrity of relationships between entities (such as foreign keys), and the enforcement of uniqueness constraints. This is particularly important because several stages of the pipeline depend on artifacts generated and stored in previous steps, making data consistency a critical requirement.

All models were individually tested:

- Document Model - Tested to ensure that document creation behaves as expected and that the initial state of newly created documents (specifically `extracted=False`) is correctly enforced. This is essential for the parameter-extraction pipeline, which relies on the distinction between processed and unprocessed documents.
- Clinical Trial Model - Tested to confirm that the model correctly accepts enumerated values and that its foreign-key relationship with the `Document` model is properly maintained.
- Version Model - Tested to validate the document versioning mechanism, which is fundamental for tracking the evolution of clinical diaries and clinical trial protocols.

- Patient Profile Model - Tested to ensure that patient profiles are instantiated correctly, forming the foundation of the patient-trial matching module.
- Treatment Model - Tested to validate the representation of therapeutic history, which is required for criteria that depend on prior treatments, and to ensure accurate storage of patient treatment records.
- Trial Cohort Model - Tested to confirm that the cohort structure is correctly instantiated and linked to clinical trials, supporting the trial-structure identification module.
- Trial Criteria Model - Tested to ensure that eligibility criteria are correctly created before being processed by the criteria-conversion module.
- Logic Criteria Model - Tested to validate that each criterion can be associated with a single logical rule, forming the basis of the criteria-conversion process.
- Patient Trial Match Model - Tested to ensure that a patient cannot have multiple matching records for the same trial, a constraint that is fundamental for maintaining consistency in the patient-trial matching module.
- Criterion Evaluation Model - Tested to verify that each (`match`, `criterion`) pair can only have one evaluation, which is essential for auditability, reproducibility, and clinical validation of the system's outputs.

Overall, the unit tests for the database models ensure that the structural backbone of the system is robust, and aligned with the requirements of the clinical trial matching pipeline. By validating relationships, constraints, and expected behaviors, these tests provide a strong foundation upon which the more complex extraction, conversion, and matching modules can operate.

F.2 Helper and Auxiliaries

The second stage of the validation process focused on unit testing the helper and auxiliary functions. Each major pipeline view relies on a set of auxiliary utilities designed to ensure modularity, clarity, and separation of concerns. Without these helper functions, the core logic of the system would become entangled within large views, reducing reproducibility, maintainability, and overall system robustness. The purpose of these tests is therefore to verify that each helper behaves correctly and consistently, and that none of them introduce unintended side effects that could compromise the main pipelines. This is particularly important given the sequential and interdependent nature of the system, where incorrect behavior in a helper function could propagate downstream and affect multiple stages of the pipeline.

- Normalize Text Test – Ensures that textual inputs are cleaned, normalized, and made consistent before being sent to LLMs or used for parsing. This prevents textual noise that could negatively impact parsing, matching, or criteria extraction.

  - Examples include: removal of duplicated spaces, removal of multiple line breaks, trimming leading and trailing whitespace, Unicode normalization, and conversion of en dashes/em dashes into hyphens.

- Parse Gender Test – Converts gender strings into consistent boolean values, which is essential for parameter extraction and for criteria that depend on patient gender.
  - Examples include: handling of male/female and masculino/feminino, treatment of invalid inputs (`None`), and normalization of strings containing extra whitespace.
- Clean Value Test – Ensures that empty or null like values are converted into `None`, preventing downstream errors and maintaining consistency in patient profiles.
  - Examples include: converting `None`, `""`, and `null` into `None`, preserving `None` as `None`, and returning valid values unchanged.
- Format Logic Test – Converts logical rule structures (dictionaries) into readable strings, which is useful for presenting converted rules to clinicians during validation.
  - Examples include: formatting simple conditions, nested AND/OR structures, and handling `None` logic.
- Extract JSON From Response Test – Ensures robust extraction of JSON from LLM outputs, even when wrapped in markdown or surrounded by additional text. This is critical across all pipeline stages, including parameter extraction, trial structure identification, criteria extraction, criteria conversion, and patient trial matching.
  - Examples include: extracting raw JSON, JSON inside ‘‘‘`json` fences, JSON embedded in text, and handling empty or invalid JSON strings.
- Deduplicate Test – Ensures correct removal of duplicated or near duplicated criteria, preventing redundancy in criteria extracted by LLMs.
  - Examples include: removing exact duplicates, removing near identical criteria, preserving distinct entries, and supporting lists of dictionaries.
- Split Text Into Chunks Test – Ensures that long clinical trial free text is correctly split into manageable chunks for LLM processing, which is essential for criteria conversion where criteria may expand into complex logical structures.
  - Examples include: verifying that short text produces a single chunk and that long text is split into multiple chunks.

- Safe Float Test – Ensures safe conversion of numeric strings into floats without raising exceptions, which is important when normalizing laboratory values and clinical parameters.
  - Examples include: converting integers and floats, returning `None` for invalid inputs, and handling `None` gracefully.
- Parse Date Test – Ensures correct interpretation of dates in common formats, used when extracting temporal information from clinical diaries.
  - Examples include: parsing ISO dates, handling `None`, and returning `None` for invalid formats.
- Parse Relative Date Test – Ensures correct interpretation of expressions such as `"3 years ago"`, which is crucial for understanding temporal references in clinical diaries.
  - Examples include: interpreting years, months, and weeks ago, and handling invalid or null inputs.
- Normalize Unit Test – Ensures normalization of laboratory units (e.g., `g/dL`, `x10`^{9}`/L`), which is fundamental for comparing laboratory values in the patient trial matching module.
  - Examples include: removing spaces, removing leading `"x"`, handling `None`, and ensuring case insensitive normalization.
- Parse Possible List Test – Ensures correct interpretation of strings that may contain JSON lists, which is important for parsing multi value fields.
  - Examples include: passing through boolean values, converting JSON strings into lists, and returning plain strings unchanged.
- Evaluate Condition Test – One of the most critical auxiliary functions, responsible for evaluating logical rules against patient attributes during deterministic matching. This function forms the core of the deterministic patient trial matching module.
  - Examples include: returning `True` when no logic is provided, enforcing operators such as `>=`, `=`, `!=`, `IN`, `NOT_IN`, `CONTAINS`, evaluating AND/OR logic, and returning `False` when required patient values are missing.
- Extract Lab Parameters Test – Ensures correct extraction of laboratory values from JSON structures, which is essential for constructing patient profiles during parameter extraction.
  - Examples include: extracting specific laboratory parameters and returning an empty list when the input JSON is empty or `None`.

The unit tests for helper and auxiliary functions play a crucial role in validating the reliability and stability of the entire system. By ensuring that each helper behaves in predictable way, these tests safeguard the integrity of the pipeline, preventing subtle errors from propagating across modules. Collectively, they contribute to a robust, maintainable, and clinically trustworthy system, reinforcing the correctness of both deterministic and LLM-assisted components of the platform.

F.3 Views

The third stage of the validation process focused on unit testing the system's views. In addition to the major pipeline views, the application includes a set of auxiliary views that support navigation, validation, inspection, and management of the various artifacts produced throughout the system. These views are essential for providing a coherent, unified, and user-friendly interface. Without them, the application would neither feel complete nor offer the necessary transparency and control expected from a clinical decision-support system.

The purpose of these tests is therefore to verify that each view behaves correctly and consistently, and that none of them introduce unintended issues that could compromise the usability, reliability, or overall coherence of the system. This is particularly important because the system is not defined solely by its core pipelines, but by the entire ecosystem of views that enable users to interact with the underlying data and processes.

- Index View Test – As the entry point of the system, these tests ensure that the homepage loads correctly, uses the appropriate template, and provides essential system wide statistics. This guarantees that the initial dashboard accurately presents the global indicators of the platform.
  - Examples include: verifying that the view returns HTTP 200, confirming the correct template, and ensuring that key metrics are present in the context.
- Diary List View Test – This page displays the list of available clinical diaries. The tests validate that the listing behaves correctly, retrieves the expected diaries, and supports the provided filters, ensuring that users can navigate and refine diary selection before initiating processing.
  - Examples include: verifying HTTP 200 responses, confirming the correct template, and testing search and status filters.
- Trial List View Test – This page presents the list of clinical trials stored in the system. The tests ensure that the listing works correctly, retrieves the expected trials, and supports the available filters, enabling users to navigate and filter trials before extracting criteria.
  - Examples include: verifying HTTP 200 responses, confirming the correct template, and testing search, pathology group, and trial status filters.
- Patient List View Test – This page displays the structured patient profiles. The tests ensure that the listing functions correctly and that the provided

filters behave as expected, which is essential for navigating and managing patient profiles prior to matching.

  – Examples include: verifying HTTP 200 responses, confirming the correct template, and testing search and stage filters.

- Diary Details View Test – This page presents the details of a clinical diary. The tests verify that the view behaves correctly when opening a diary, preventing access to invalid or incorrectly typed documents.
  – Examples include: handling non existent diaries, handling documents of the wrong type, and returning HTTP 200 for valid diaries.
- Trial Details View Test – This page presents the details of a clinical trial. The tests ensure that the view behaves correctly when opening a trial, guaranteeing that only valid clinical trials are displayed.
  – Examples include: handling non existent trials and returning HTTP 200 for valid trials.
- Diary Remove View Test – This view enables the removal of clinical diaries from the system. The tests validate that deletion behaves correctly, preventing inconsistencies when removing documents. Additional tests will ensure that filesystem cleanup and related artifacts are also handled correctly.
  – Examples include: verifying successful deletion, ensuring removal from the database, and confirming removal of associated files and extracted data.
- Trial Remove View Test – This view provides the functionality to remove clinical trials. The tests ensure that deletion behaves correctly, preserving system integrity when removing trial documents. Future tests will also validate the removal of associated cohorts, criteria, and extracted structures.
  – Examples include: verifying successful deletion, ensuring removal from the database, and confirming cleanup of related artifacts.
- Patient Reset View Test – This view allows resetting or removing a patient profile. Its purpose is to validate the correct deletion of patient records, associated document versions, and extracted values, which is essential for reprocessing diaries and correcting errors. Additional tests will ensure that all dependent objects are removed and that the system state is fully reset.
  – Examples include: verifying successful reset, ensuring that the patient profile is removed from the database, and confirming removal of associated versions and extracted parameters.
- Document Upload View Test – This view handles the upload of clinical diaries. The tests validate that the upload process behaves correctly, ensuring that only valid files enter the pipeline.

  - Examples include: verifying that GET displays the form, that valid POST requests redirect, and that invalid file extensions are correctly rejected.

- Clinical Trial Upload View Test – This view handles the upload of clinical trial documents. The tests validate that the upload process behaves correctly and that only complete and valid submissions are accepted.
  - Examples include: verifying redirection on valid POST requests and returning errors when mandatory fields are missing.

- Dev Tools View Test – This view provides internal metrics and debugging information. The tests ensure that the page loads correctly and presents the expected system statistics, supporting internal validation and development workflows.
  - Examples include: verifying HTTP 200 responses, confirming the correct template, and ensuring that the context contains the expected metrics.

Overall, the unit tests for the auxiliary views ensure that the system's user interface behaves reliably, consistently, and transparently. By validating navigation, filtering, detail inspection, uploads, deletions, and internal tools, these tests guarantee that the application functions as a cohesive whole rather than a collection of isolated pipelines. This contributes to a robust and user-friendly platform, reinforcing both the technical integrity and the practical usability of the system.

F.4 Hyper-Relevant Parameter Extraction

The fourth stage of the validation process focuses on the first main module of the system: the hyper-relevant parameter extraction. This represents the initial step of the entire pipeline and is of major importance, as it produces the clinical artifacts that form the foundation of the patient–trial matching process. Ensuring the correct behavior of this module is therefore essential to prevent downstream inconsistencies or corrupted outputs.

The tests designed for this module follow a clear and structured organization, as described below:

- Guard Clauses - ensuring that the view correctly rejects invalid requests before any pipeline execution begins.
- Happy Path - ensuring that a successful GET request correctly prepares and triggers the pipeline.
- Creation of Patient Profiles - validating that POST requests correctly create the required clinical artefacts.

- Auxiliary Pipelines - ensuring that internal functions behave robustly and consistently.
- Laboratory Analysis Processing - ensuring that laboratory values are processed and stored correctly.

F.4.1 Guard Clauses

The primary objective of these tests is to ensure that any invalid request sent to this core module is handled safely and predictably. If the diary does not exist, the view must return an error instead of crashing. If the diary has already been extracted, the view must prevent further reprocessing. If the document is not a clinical diary, it must be rejected. Finally, if the diary exists but is empty or unreadable, the pipeline must not be executed. These checks protect the system from invalid states and unnecessary computation.

F.4.2 Happy Path

These tests ensure that, when all conditions are met, the pipeline behaves as expected. They verify that the LLM is indeed invoked and that the rendered context includes all required sections. In essence, they confirm that the module performs correctly under ideal conditions.

F.4.3 Creation of Patient Profiles

The objective here is to validate the creation of clinical artefacts, particularly patient profiles. The tests ensure that a minimal POST request successfully creates a profile and redirects the user to the appropriate page. When valid laboratory analyses or treatments are included, they are correctly created and stored in the database, whether individually or in multiples. Invalid treatments are ignored. The tests also confirm that, after a successful POST, the document is marked as extracted and that the appropriate validated version of the file is created.

F.4.4 Auxiliary Pipeline

These tests ensure that all auxiliary functions supporting this core module behave robustly. They verify successful LLM calls, test retry mechanisms when invalid JSON is returned, and ensure that an error is raised when the maximum number of retries is reached. They also validate the normalization of clinical diaries and the loading of laboratory analyses, both for valid and invalid JSON inputs.

F.4.5 Laboratory Analysis Processing

These tests ensure that laboratory analyses are processed correctly. They verify that grouped analyses containing both value and unit are saved properly, and that laboratory fields without a value do not result in the creation of invalid database records.

Overall, the unit tests for this first main module ensure that the initial step of the pipeline is structurally sound and capable of handling both expected and adverse scenarios. By validating guard clauses, artefact creation, auxiliary logic, and laboratory processing, these tests prevent errors from propagating downstream and compromising subsequent modules. In doing so, they contribute to a robust, and well-designed pipeline that supports the clinical validity and operational stability of the entire system.

F.5 Eligibility Criteria Extraction & Trial Structure Identification

The fifth stage of the validation process focuses on the second main module of the system, which is composed of two tightly connected components: trial structure identification and eligibility criteria extraction. This represents the initial step in the processing workflow for clinical trials and, as a core module, is of major importance. It outputs the structural organization of participation criteria for a given trial and extracts individual criteria, assigning them to their respective cohorts when applicable. Ensuring the correct behavior of this module is therefore essential to prevent corruption of the subsequent criteria-conversion step and to guarantee that a strong and reliable rule structure is produced.

Because this stage combines two modules, it naturally contains multiple happy paths and branching behaviors. The tests designed for this module follow a clear and structured organization, as described below:

- Guard Clauses - ensuring that the view correctly rejects invalid requests before any pipeline execution begins.
- Happy Path, without cohorts - ensuring that the pipeline behaves correctly when no cohorts are present; with cohorts - validating the flow of the pipeline when cohorts are identified in the trial.
- Extraction Pipeline, without cohorts - ensuring that chunking works correctly and that extraction remains robust; with cohorts - validating extraction when cohorts are present.
- Split & Chunk Helpers - ensuring that section-splitting and chunking helpers work as intended.
- Deduplication - ensuring that duplicated entries are correctly removed, particularly due to chunking.

- Criteria Validation - validating that POST requests correctly update and create criteria.

F.5.1 Guard Clauses

The primary objective of these tests is to ensure that any invalid request sent to this core module is handled safely and predictably. If the trial does not exist, the view must return an error instead of crashing. If the trial has already been extracted, the view must prevent further reprocessing. If the document is not a clinical trial, it must be rejected. Finally, if the trial exists but its extracted text is empty or unreadable, the pipeline must not be executed. These checks protect the system from invalid states and unnecessary computation.

F.5.2 Happy Path

These tests ensure that, when all conditions are met, the pipeline behaves as expected for both branches of the view. If no cohorts are identified, the tests verify that the correct template is rendered, that the context includes all required sections, that the LLM receives the appropriate `has_cohorts=False` flag, that both inclusion and exclusion criteria are created in the database, and that a new `EXTRACTED` version is created for the document. If cohorts are identified, the tests verify that the cohorts are created in the database, that the LLM is prompted with the cohort information and the flag set to true, and that the extracted criteria are correctly associated with the corresponding cohorts.

F.5.3 Extraction Pipeline

These tests validate the robustness of the chunking and extraction process for both branches of the view. When no cohorts are present, the tests verify that a single chunk results in two LLM calls (one for inclusion and one for exclusion), while multiple chunks result in multiple calls. They also ensure that missing participation-criteria sections raise an error, that duplicated criteria across chunks are correctly deduplicated, and that the merging of results combines lists from multiple chunks correctly. When cohorts are present, the tests verify that the pipeline again performs the expected number of LLM calls (one per section), that criteria are associated with the correct cohort identifier, that identical criteria belonging to different cohorts are not deduplicated, and that empty sections are ignored without failure.

F.5.4 Split & Chunk Helpers

These tests ensure that the helpers used during this step are structurally sound and behave as expected. They validate the correct splitting of participation-criteria sections (inclusion and exclusion), confirm that missing sections return empty strings, and test chunking behavior for both short and long texts, ensuring that the correct number of chunks is produced.

F.5.5 Deduplication

These tests ensure that the deduplication functions behave correctly, removing duplicated entries from LLM outputs. They verify that identical or highly similar criteria (similarity threshold above 0.92) are deduplicated into a single entry, that distinct criteria are all preserved, and that identical criteria belonging to different cohorts are kept separate, while identical criteria within the same cohort are deduplicated.

F.5.6 Criteria Validation

These final tests validate that the POST request correctly updates and creates criteria in the database. They ensure that existing criteria are updated in the `validated_criterion` field and marked as validated, that invalid criterion identifiers are ignored without errors, that newly expert-created criteria are processed using the correct dictionary structure, that empty entries are ignored, and that new expert criteria associated with cohorts are correctly created. They also verify that the view redirects to the next pipeline step (criteria conversion) with the correct trial identifier and that a new `VALIDATED` version is created.

Overall, the unit tests for this second main module ensure that the system can reliably identify trial structure, extract eligibility criteria, and organize them into a coherent format. By validating guard clauses, extraction logic, chunking behavior, cohort handling, deduplication, and criteria validation, these tests guarantee that the module behaves robustly under both ideal and adverse conditions. This prevents structural inconsistencies from propagating into the criteria-conversion stage and contributes to a stable clinical-trial processing pipeline.

F.6 Eligibility Criteria Conversion

The sixth stage of the validation process focuses on the third main module of the system: the eligibility criteria conversion. This represents the second step in the processing of clinical trial eligibility criteria and is of major importance, as it produces the logical rules that will be used in the subsequent patient–trial matching stage. Ensuring the correct behavior of this module is therefore essential to guarantee accurate and clinically meaningful matching outcomes.

The tests designed for this module follow a clear and structured organization, as described below:

- Guard Clauses - ensuring that the view correctly rejects invalid requests before any pipeline execution begins.

- Happy Path - ensuring that a successful GET request correctly prepares and triggers the conversion pipeline.
- Batching & Chunking - ensuring that batching and criteria chunking behave as expected.
- Processing & Ordering Helpers - validating that the processing of individual and grouped conditions is correct and that the ordering of converted rules is coherent for expert validation.
- Logic Criteria Validation - validating that the POST request correctly updates and creates logical rules.

F.6.1 Guard Clauses

The primary objective of these tests is to ensure that any invalid request sent to this core module is handled safely and predictably. If the document does not exist, the view must return an error instead of crashing. If the trial has already been converted, the view must prevent further reprocessing. If the document is not a clinical trial, it must be rejected. These checks protect the system from invalid states and unnecessary computation.

F.6.2 Happy Path

These tests ensure that, when all conditions are met, the pipeline behaves as expected. They verify that the correct template is rendered and that the context contains all required keys, that the LLM-generated inclusion and exclusion logic is stored in the database, and that a new version of the file is created with status `CONVERTED`. They also ensure that criteria returned by the LLM with invalid identifiers are ignored without error, that both simple and compound logic structures (field/operator/value or grouped conditions) are processed correctly, that missing fields or conditions result in an empty placeholder, and that the ordering of criteria is coherent and predictable for expert review.

F.6.3 Batching & Chunking

These tests ensure that batching and chunking behave as intended and do not introduce unnecessary errors into the main pipeline. They validate that the correct number of batches and LLM calls are made for a given set of criteria, that empty batches do not trigger unnecessary LLM requests, that multi-batch outputs are concatenated correctly, that duplicates across batches are removed, and that the payload sent to the LLM is structurally correct for each batch.

F.6.4 Processing & Ordering Helpers

These tests ensure that the processing of conditions is structurally sound and that the ordering of criteria is correct. They validate that known fields (those present in the patient profile schema) are correctly tagged, while unknown fields are handled as custom fields. They also ensure that grouped conditions are processed recursively and that the correct flags are applied depending on whether a condition is simple or compound. Finally, they verify that the ordering of criteria, general inclusion, cohort-specific inclusion, general exclusion, cohort-specific exclusion, is preserved for coherent expert validation.

F.6.5 Logic Criteria Validation

These final tests validate that the POST request correctly updates and creates logical rules in the database. They ensure that simple conditions generate logic entries with the correct field, operator, and value, while multiple conditions are grouped under a logical operator (typically AND). They verify that custom fields are correctly handled when the field is not part of the predefined schema, and that units are included when provided. Invalid logic identifiers are ignored without raising errors, ensuring robustness.

The tests also confirm that expert-validated logic is correctly stored and updated, that validated rules are marked as such, and that the trial document is flagged as converted. A new `VALIDATED` version is created with the appropriate payload, and the view redirects the user to the trial listing page. Overall, these tests ensure that the validation step produces a clean and interpretable set of logical rules ready for the matching module.

In summary, the unit tests for the eligibility criteria conversion module ensure that the transformation from textual criteria to structured logical rules is reliable and consistent. By validating guard clauses, happy-path behavior, batching mechanisms, condition processing, ordering logic, and the final validation of logical rules, this test suite guarantees that the module behaves robustly under both ideal and adverse conditions. This prevents malformed or inconsistent logic from propagating into the patient–trial matching stage and contributes to a stable clinical decision-support pipeline.

F.7 Patient–Trial Matching

The final stage of the validation process focuses on the last and largest module of the entire pipeline: patient–trial matching. As the final step, it is also the most sensitive, since it produces the actual eligibility decisions that determine whether a patient can or cannot participate in a given clinical trial. Ensuring the correct behavior of this module is therefore essential to guarantee that the matching outcomes are reliable, clinically trustworthy, and aligned with the logic produced in the previous stages.

The tests designed for this module follow a clear and structured organization, as described below:

- Guard Clauses - ensuring that the view correctly rejects invalid requests before any matching logic is executed.
- Happy Path - ensuring that a successful GET request correctly prepares and triggers the matching pipeline.
- Matching Patients - validating the flow of the matching process both with and without cohorts.
- Condition Evaluation - ensuring that all types of conditions are evaluated correctly, including rule operators, temporal fields, and laboratory values.
- Laboratory Helpers & Evidence Extractors - ensuring that laboratory values are correctly retrieved and serialized, and that evidence extraction behaves as expected.
- LLM Matching - ensuring that out-of-schema fields are correctly handled by the LLM.
- Manual Overrides - ensuring that clinicians can update and replace automatic results, whether deterministic or LLM-generated.

F.7.1 Guard Clauses

The primary objective of these tests is to ensure that any invalid request sent to this core module is handled safely and predictably. If the document does not exist, the view must return an error instead of crashing. If the trial has not yet been converted, the view must prevent further matching. If the document is not a clinical trial, it must be rejected. These checks protect the system from invalid states and unnecessary computation.

F.7.2 Happy Path

These tests ensure that, when all conditions are met, the pipeline behaves as expected. They verify that the correct template is rendered and that the context contains all required keys, that only patients belonging to the same pathology group as the clinical trial are evaluated, that a matching record is created for each eligible patient, and that a criterion-evaluation record is created for each evaluable criterion. They also ensure that criteria without logic are ignored, preventing unnecessary or invalid evaluations.

F.7.3 Matching Patients

These tests validate that the matching pipeline behaves correctly both with and without cohorts. When no cohorts exist, the tests ensure that the matching logic is applied globally, that inclusion and exclusion criteria behave as expected, and that the final eligibility decision reflects the combination of these results. When cohorts are present, the tests ensure that criteria are correctly grouped by cohort, that patients must satisfy both general and cohort-specific criteria to be considered eligible, and that eligibility is correctly computed even when multiple cohorts are available. This ensures that the system can handle both simple and more complex trial structures.

F.7.4 Condition Evaluation

These tests ensure that the evaluation of conditions is robust and consistent across all types of rules. They validate the behaviour of all supported operators (numeric, equality, inequality, membership, containment), the correct handling of grouped conditions (AND/OR), and the behaviour when patient values are missing or incompatible. They also ensure that temporal conditions (e.g., "diagnosis within 3 years") and laboratory values (including unit handling) are evaluated correctly. This guarantees that the system can interpret a wide range of clinical criteria in a deterministic and clinically meaningful way.

F.7.5 Laboratory Helpers & Evidence Extractors

These tests ensure that laboratory values are correctly retrieved, serialised, and made available to the matching logic. They validate that laboratory analyses are returned in a consistent structure, that missing or empty analyses are handled gracefully, and that evidence extraction produces clear and interpretable explanations for each evaluated condition. This contributes to the transparency of the matching process, allowing clinicians to understand why a patient passed or failed a given criterion.

F.7.6 LLM Matching

These tests ensure that when a criterion refers to a field outside the patient profile schema, the system correctly falls back to the LLM. They validate that the LLM receives the appropriate prompt, that its output is correctly interpreted, and that failures or exceptions are handled safely. This ensures that the system remains flexible and capable of handling criteria that cannot be evaluated deterministically.

F.7.7 Manual Overrides

These final tests validate that clinicians can manually override automatic results when necessary. They ensure that overrides correctly update the evaluation of individual criteria, that the final eligibility decision is recalculated accordingly, and that invalid overrides are ignored without error. They also verify that the system redirects to the appropriate page and that the updated results are stored consistently. This ensures that the system supports clinical judgement and remains aligned with real-world decision-making workflows.

Overall, the unit tests for the patient–trial matching module ensure that the final and most critical stage of the pipeline behaves reliably and transparently. By validating guard clauses, matching flows, condition evaluation, laboratory helpers, LLM fallbacks, and manual overrides, these tests guarantee that eligibility decisions are both technically sound and clinically interpretable. This contributes to a robust and trustworthy matching system, capable of supporting clinicians in identifying suitable patients for clinical trials with confidence and clarity.

Appendix G

Database Specification

To ensure a robust and well-designed database, an Entity–Relationship (ER) diagram was created to guide the construction of the system's data layer. The design process always kept in mind the versioning of processed documents, the distinction between clinical trials and clinical diaries, and the need for full traceability of results. Every relevant artefact, LLM outputs, clinician-validated fields, and intermediate processing steps, is stored to guarantee accountability and to ensure that all information remains queriable whenever needed, which is fundamental in a clinical setting.

The final database consists of 11 tables, some of which act as linking structures to maintain relationships between artefacts, while others represent core clinical entities essential to the pipeline. Together, they support the entire workflow: from document ingestion, to criteria extraction and conversion, to patient–trial matching and evaluation.

The tables are the following:

- Document - stores the main documents of the system, namely clinical diaries and clinical trials.
- Version - stores every version of every document, ensuring full version control and reproducibility.
- Clinical Trial - stores metadata for each clinical trial and acts as the central node linking cohorts, eligibility criteria, and patient matches.
- Patient Profile - stores the structured patient profile extracted from clinical diaries, including all previously discussed fields, as well as associated analyses and treatments.
- Trial Cohort - stores the cohorts defined within a clinical trial.
- Trial Criteria - stores all eligibility criteria (inclusion and exclusion) associated with a clinical trial, and links to any corresponding logical representation.

- Logic Criteria - stores the logical rules generated from each criterion after the conversion step.
- Patient Trial Match - stores the matching result between a patient and a clinical trial, including the final decision and supporting metadata.
- Criterion Evaluation - stores the evaluation of each patient against each individual criterion, capturing both automatic and manually overridden results.
- Analysis - stores every laboratory analysis of each patient
- Treatment - stores every treatment received and ongoing for each patient

The whole diagram can be seen on the Figure G.1

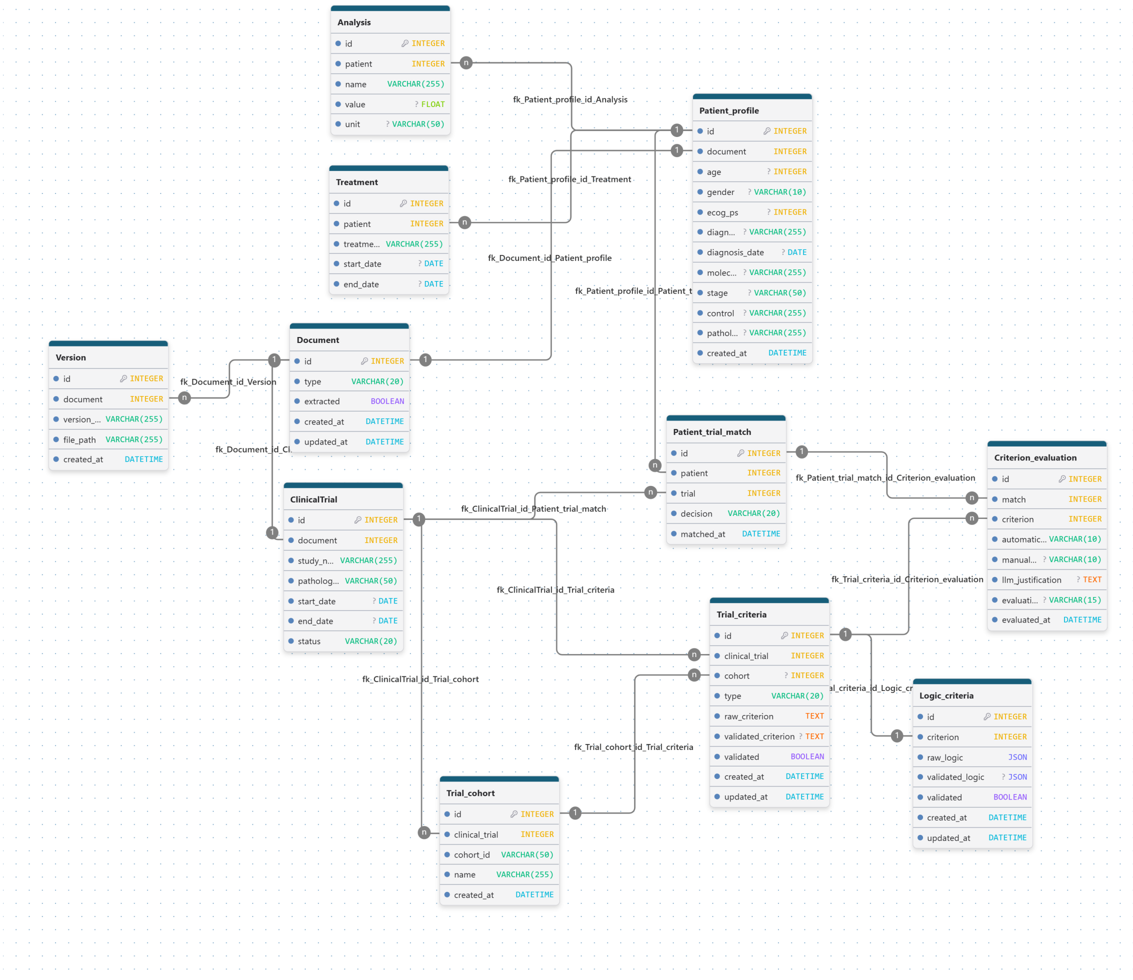


Figure G.1: Overall ER Diagram

G.1 Document

As mentioned before, this table has the main objective of storing and keeping track of the main clinical documents in the system, namely clinical diaries and clinical trials. It is the core table of the entire architecture, and many other entities branch out from it.

It contains the metadata fields that are common to both types of documents. All fields are mandatory and are the following:

- id - integer that uniquely identifies every document in the system
- type - varchar field that specifies the type of document, which can only be `diary` or `trial`
- extracted - boolean flag that indicates whether the document has already been processed
- created_at - datetime field that specifies when the record was created
- updated_at - datetime field that specifies when the record was last modified

Everything described above can be seen on the Figure G.2.

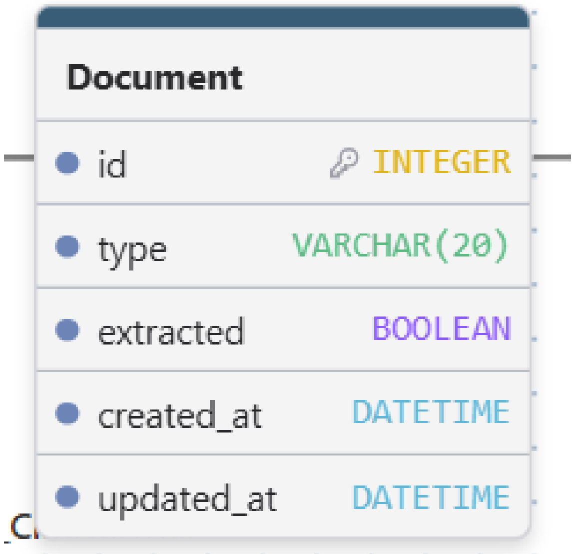


Figure G.2: Document Table

This entity has the following relationships:

- Document-Version - this is a one to many relationship indicating that a document can have multiple versions, allowing full traceability of changes
- Document-Patient Profile - this is a one to one relationship indicating that a clinical diary can only produce one patient profile
- Document-Clinical Trial - this is a one to one relationship indicating that a clinical trial document can only have one corresponding clinical trial entry

G.2 Version

As mentioned before, this table has the main objective of storing every version of every document, ensuring full versioning and traceability, which are fundamental requirements for the system.

It contains the fields needed to identify the document and store the version file. All fields are mandatory and are the following:

- id - integer that uniquely identifies every version in the system
- document - foreign key for the document to which the version belongs
- version_name - varchar field that identifies the version
- file_path - varchar field that specifies where the version file is stored in the filesystem
- created_at - datetime field that specifies when the record was created

Everything described above can be seen on the Figure G.3.

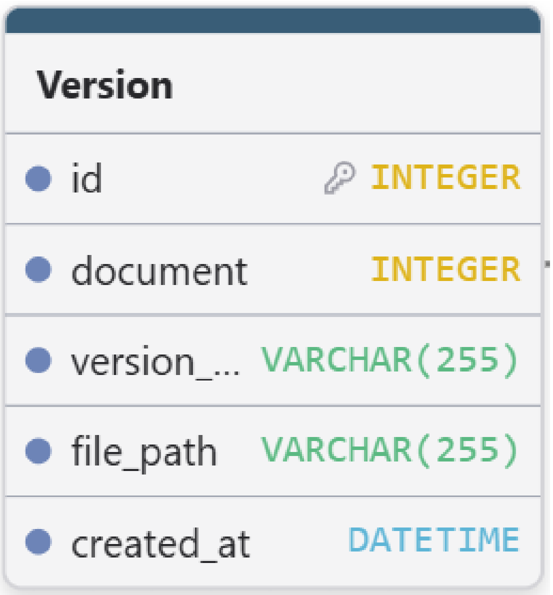


Figure G.3: Version - table

This entity has one relationship:

- Document-Version - this is a one to many relationship indicating that a document can have multiple versions, ensuring traceability and storage of every modification

G.3 Clinical Trial

As mentioned before, this table has the main objective of storing clinical trial metadata. It also acts as a central node that links several other clinical artefacts, allowing clear identification of the trial and supporting a safe and robust patient matching process.

It contains the fields that compose a clinical trial. All fields are mandatory and are the following:

- id - integer that uniquely identifies every clinical trial in the system
- document - foreign key for the document to which this clinical trial metadata belongs
- study_name - varchar field that identifies the clinical trial
- pathology_group - varchar field that specifies the pathology group of the trial, restricted to predefined clinical categories
- start_date - datetime field that specifies when the clinical trial starts
- end_date - datetime field that specifies when the clinical trial ends
- status - varchar field that identifies the current status of the clinical trial, following the terminology used in ClinicalTrials.gov, such as `recruiting`, `closed`, `not yet recruiting`, and `completed`

Everything described above can be seen on the Figure G.4.

This entity has the following relationships:

- Document-Clinical Trial - this is a one to one relationship indicating that a clinical trial document can only have one corresponding clinical trial entry
- Clinical Trial-Trial Cohort - this is a one to many relationship listing all cohorts belonging to a given clinical trial
- Clinical Trial-Trial Criteria - this is a one to many relationship listing all inclusion and exclusion criteria associated with a clinical trial
- Clinical Trial-Patient Trial Match - this is a one to many relationship listing all patient matches performed for a given clinical trial

G.4 Patient Profile

As mentioned before, this table has the main objective of storing every LLM-generated and subsequently clinician-validated patient profile used for matching. Being a

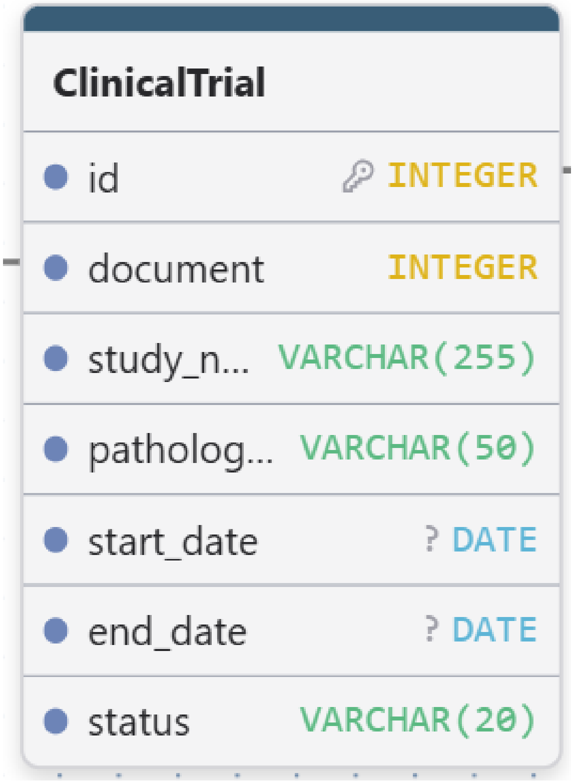


Figure G.4: Clinical Trial - Table

core table, it also acts as a central node that connects several other clinical artefacts, supporting a robust and trustworthy matching process with clinical trials.

It contains the fields that compose a patient profile. Since this table is populated with information extracted from free-text clinical diaries, none of the clinical fields can be guaranteed, meaning they may be null or missing. The only mandatory fields are the creation timestamp and the document to which the profile belongs. The fields are the following:

- id - integer that uniquely identifies every patient profile in the system
- document - foreign key for the document to which this patient profile belongs, this field is mandatory
- age - integer field that specifies the patient's age
- gender - varchar field that specifies the patient's gender, which can only be `male` or `female`
- ecog_ps - integer field that specifies the patient's ECOG-PS score
- diagnosis - varchar field that specifies the patient's diagnosis
- diagnosis_date - datetime field that specifies when the patient was diagnosed
- molecular_status - varchar field that specifies the patient's molecular alterations

- stage - varchar field that specifies the patient's disease stage
- control - varchar field that contains control information such as disease status, metastases, comorbidities, or follow-up details
- pathology_group - varchar field that specifies the patient's pathology group, restricted to predefined clinical categories
- created_at - datetime field that specifies when the record was created

Everything described above can be seen on the Figure G.5.

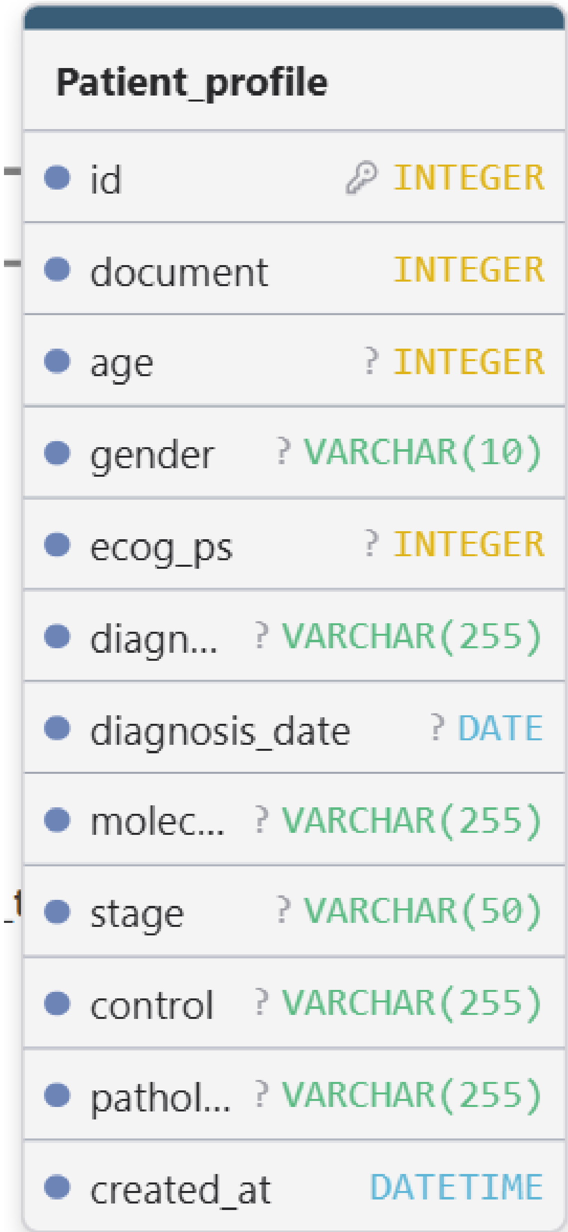


Figure G.5: Patient Profile - Table

This entity has several relationships:

- Document-Patient Profile - this is a one to one relationship indicating that a clinical diary produces exactly one patient profile
- Patient Profile-Treatment - this is a one to many relationship listing all treatments the patient has received or is currently receiving
- Patient Profile-Analysis - this is a one to many relationship listing all laboratory analyses associated with the patient
- Patient Profile-Patient Trial Match - this is a one to many relationship listing all clinical trials the patient has been matched with, including the decision for each match

G.5 Trial Cohort

As mentioned before, this table has the main objective of storing LLM-generated cohort metadata belonging to a given clinical trial. This allows for a more accurate matching process by separating general eligibility criteria from cohort-specific ones.

It contains the fields that describe a clinical trial cohort:

- id - integer that uniquely identifies every clinical trial cohort in the system
- clinical_trial - foreign key for the clinical trial to which this cohort belongs
- cohort_id - varchar field containing the original cohort identifier extracted from the trial's free text
- name - varchar field containing a clinically meaningful name for the cohort
- created_at - datetime field that specifies when the record was created

Everything described above can be seen on the Figure G.6.

This entity has the following relationships:

- Clinical Trial-Trial Cohort - this is a one to many relationship listing all cohorts belonging to a given clinical trial
- Trial Cohort - Trial Criteria - this is a one to many relationship listing all criteria that belong to a specific cohort

G.6 Trial Criteria

As mentioned before, this table has the main objective of storing every individually extracted eligibility criterion, whether inclusion or exclusion, associated with

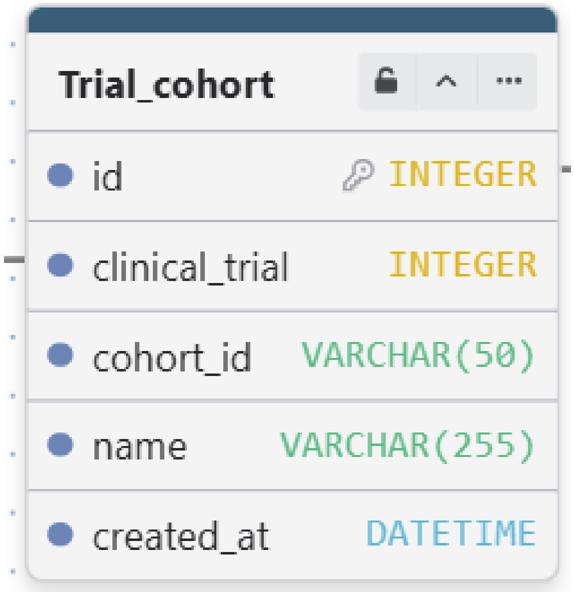


Figure G.6: Trial Cohort - Table

a clinical trial. It also links each criterion to its corresponding logical representation, enabling a smooth conversion process.

It contains the fields required to ensure traceability and clear identification of each criterion:

- id - integer that uniquely identifies every criterion in the system
- clinical_trial - foreign key for the clinical trial to which this criterion belongs
- cohort - foreign key for the cohort to which the criterion may belong; this field is optional because some criteria are general
- type - varchar field that specifies the type of criterion, either `inclusion` or `exclusion`
- raw_criterion - text field containing the original LLM-extracted criterion
- validated_criterion - text field containing the clinician-validated version of the criterion; optional to avoid redundancy when no changes are made
- validated - boolean field indicating whether the criterion was validated by a clinician
- created_at - datetime field that specifies when the record was created
- updated_at - datetime field that specifies when the record was last modified

Everything described above can be seen on the Figure G.7.

This entity has several relationships:

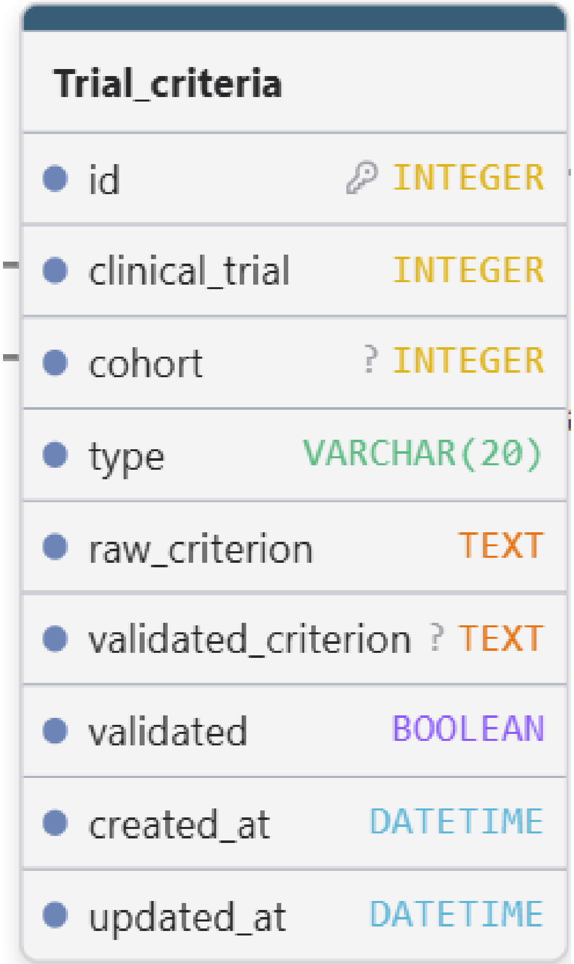


Figure G.7: Trial Criteria - Table

- Trial Cohort-Trial Criteria - this is a one to many relationship listing all criteria belonging to a given cohort
- Clinical Trial-Trial Criteria - this is a one to many relationship listing all inclusion and exclusion criteria associated with a clinical trial
- Trial Criteria-Logic Criteria - this is a one to one relationship linking each criterion to its converted logical representation
- Trial Criteria-Criterion Evaluation - this is a one to many relationship listing all evaluations performed for a given criterion

G.7 Logic Criteria

As mentioned before, this table has the main objective of storing the converted logical representation of a clinical trial criterion. It provides a structured and machine-interpretable version of the criterion.

It contains the fields required to ensure traceability and clear identification of each logical rule:

- id - integer that uniquely identifies every logical rule in the system
- criterion - foreign key for the criterion to which this logical rule belongs
- raw_logic - json field representing the LLM-generated logical structure
- validated_logic - json field representing the clinician-validated logical structure; optional to avoid redundancy when no changes are made
- validated - boolean field indicating whether the logic was validated by a clinician
- created_at - datetime field that specifies when the record was created
- updated_at - datetime field that specifies when the record was last modified

Everything described above can be seen on the Figure G.8.

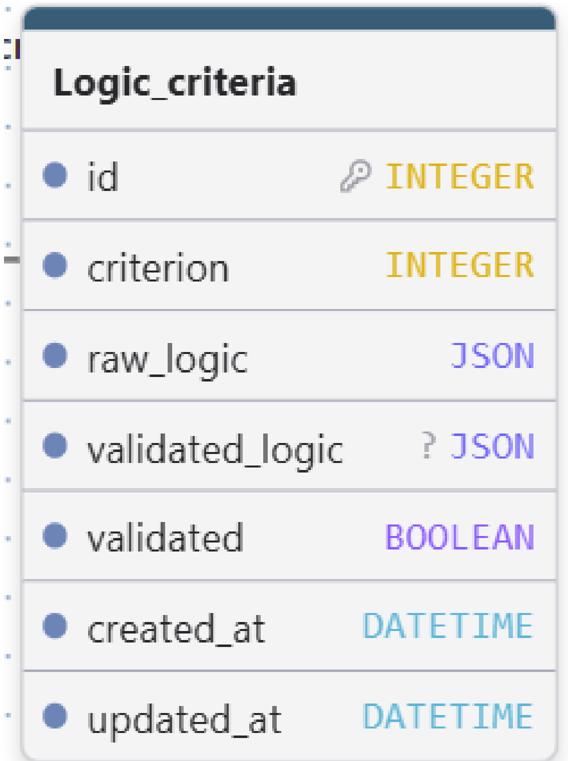


Figure G.8: Logic Criteria Table

This entity has one relationship:

- Trial Criteria - Logic Criteria - this is a one to one relationship linking each criterion to its logical representation

G.8 Patient Trial Match

As mentioned before, this table has the main objective of storing the matching result between a patient and a clinical trial, including the final decision and all supporting metadata. It ensures explainability of the results obtained and full traceability throughout the matching process.

It keeps the required fields to clearly identify each match:

- id - integer that uniquely identifies every patient trial match in the system
- patient - foreign key that specifies which patient was matched
- trial - foreign key that specifies which clinical trial the patient was matched with
- decision - varchar field that specifies the final matching result for that patient and trial
- matched_at - datetime field that specifies when the matching was performed

Everything described above can be seen on the Figure G.9.

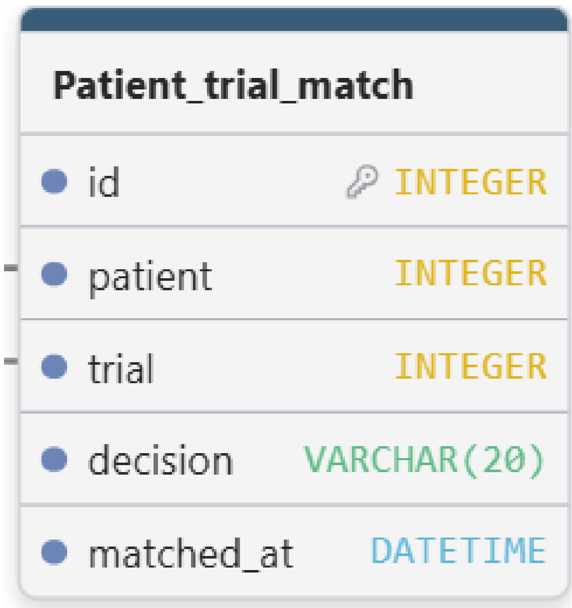


Figure G.9: Patient Trial Match - Table

This entity has several relationships, all of which are straightforward:

- Patient Profile - Patient Trial Match - this is a one to many relationship that indicates all the clinical trials a given patient has been matched with, while keeping record of the decision for each match
- Clinical Trial - Patient Trial Match - this is a one to many relationship that lists every patient who has been matched to a given clinical trial
- Patient Trial Match - Criterion Evaluation - this is a one to many relationship that identifies all the criteria that were evaluated when the match was performed

G.9 Criterion Evaluation

As mentioned before, this table has the main objective of storing the evaluation result for each criterion when matching a patient to a clinical trial. It enables full explainability and traceability of the eligibility decision, including the justification for each evaluated criterion.

It keeps the required fields to clearly identify each evaluation:

- id - integer that uniquely identifies every criterion evaluation in the system
- match - foreign key that identifies the specific patient-trial match record
- criterion - foreign key that identifies which criterion was evaluated
- automatic_result - varchar field that specifies whether the patient passed or failed the criterion, regardless of whether the evaluation was deterministic or LLM-based; this value can only be `Pass` or `Fail`
- manual_result - varchar field that specifies a manual override performed by the expert; it can only be `Pass` or `Fail` and is optional, since the expert may choose not to override the automatic result
- llm_justification - text field that contains the justification for the evaluation when the method was LLM-based; this field is optional because deterministic evaluations do not require a justification
- evaluation_method - varchar field that specifies the evaluation method; it can only be `Rule` for deterministic evaluation or `LLM` for LLM-based evaluation
- evaluated_at - datetime field that specifies when the evaluation was performed

Everything described above can be seen on the Figure G.10.

This entity has two relationships, both already discussed:

- Patient Trial Match-Criterion Evaluation - this is a one to many relationship that lists all the criteria evaluated for a given match
- Trial Criteria-Criterion Evaluation - this is a one to many relationship that identifies every evaluation performed for a given criterion

G.10 Analysis

As mentioned earlier, this table serves the primary purpose of storing every laboratory analysis associated with a given patient. These values play a crucial role in

Figure G.10: Criterion Evaluation - Table

the matching pipeline, since several eligibility criteria depend directly on specific laboratory measurements.

The table includes the essential fields required to clearly identify each laboratory entry:

- id - integer that uniquely identifies each analysis in the system
- patient - foreign key linking the analysis to the corresponding patient
- name - varchar field specifying the type of laboratory test
- value - float field representing the measured value
- unit - varchar field indicating the measurement unit

Everything described above can be seen on the Figure G.11.

This entity has a single relationship, already discussed:

- Patient Profile-Analysis - a one-to-many relationship indicating that each patient may have multiple laboratory analyses recorded.

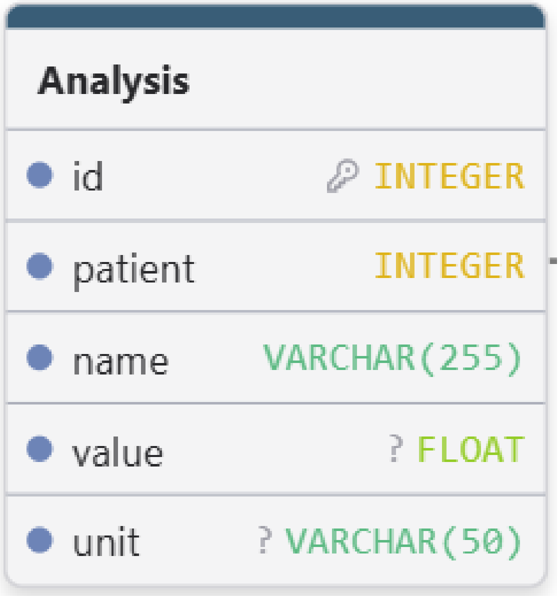


Figure G.11: Analysis - Table

G.11 Treatment

As previously described, this table is responsible for storing all treatments a patient has received or is currently receiving. This information is essential for the matching process, as certain eligibility criteria depend on treatment history or ongoing therapies.

The table includes the necessary fields to ensure clear identification of each treatment:

- id - integer that uniquely identifies each treatment
- patient - foreign key linking the treatment to the corresponding patient
- treatment_name - varchar field specifying the name of the treatment
- start_date - date field indicating when the treatment began; this value is extracted from free-text clinical diaries and may be missing
- end_date - date field indicating when the treatment ended; also extracted from free-text diaries and therefore optional

Everything described above can be seen on the Figure G.12.

This entity also has a single relationship:

- Patient Profile-Treatment - a one-to-many relationship specifying all treatments a patient has undergone or is currently undergoing.

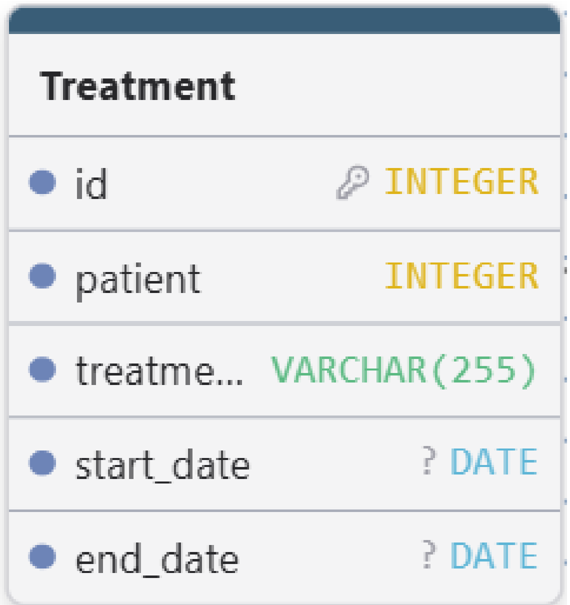


Figure G.12: Treatment - Table

Appendix H

Scientific Article

The following appendix contains the scientific article developed during this thesis. This work presents the experimental evaluation and main findings obtained throughout the project.

A CLINICIAN-VALIDATED LLM PIPELINE FOR STRUCTURING ONCOLOGY DATA AND CLINICAL TRIAL ELIGIBILITY MATCHING

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***Abstract*—Oncology departments generate large volumes of unstructured clinical data, complicating patient–trial matching and contributing to the under-enrolment of eligible patients in clinical trials. While Large Language Models (LLMs) offer promising solutions, existing approaches often rely exclusively on probabilistic reasoning, limiting transparency and reproducibility. This work proposes an integrated, hybrid system that leverages LLMs to transform unstructured oncology notes into structured, clinician-validated profiles and support transparent eligibility assessment.**

The system operates in two core phases: (i) an LLM-based pipeline that extracts and normalizes clinical parameters from free-text notes for expert validation, and (ii) an automated module that converts trial criteria into machine-readable logical rules. Eligibility is then evaluated through a hybrid mechanism, combining deterministic rule-based execution for structured fields with LLM-assisted reasoning for complex, unstructured clinical context. This architecture balances the auditability of rule-based logic with the flexibility of generative models.

The system was evaluated using three open-source 8B-parameter LLMs across five pipeline tasks. Structured extraction yielded strong results, with the best model achieving 92.3% accuracy in criteria extraction and 78.3% in patient profile extraction. Transitioning clinical language into formal logic proved the most challenging task, with strict accuracies between 52.2% and 64.1%. Nevertheless, the final patient–trial matching stage demonstrated clinically reliable performance; the top models achieved up to 84.0% accuracy and expert-rated justification quality scores of 4.93 out of 5, ensuring decisions were grounded in clinical evidence. These findings demonstrate that modular pipelines combining structured representations, deterministic logic, and lightweight LLMs can support transparent, efficient, and clinically aligned patient recruitment in real-world oncology settings.

***Index Terms*—Large Language Models; Information Extraction; Clinical Trial Eligibility; Un structured Clinical Data; Clinician-in-the-Loop; Deterministic Eligibility Evaluation; Survival Analysis; Structured Patient Data**

INTRODUCTION

Oncology departments generate large volumes of unstructured clinical data through free-text notes, making it difficult to efficiently retrieve clinically relevant information for decision support and research workflows. This challenge is particularly evident in the context of clinical trial enrollment, where clinicians must manually extract, interpret, and compare patient information against extensive eligibility criteria. The process is time-consuming, cognitively demanding, and frequently interrupted by routine clinical activities, contributing to the persistent under-enrollment of eligible patients in clinical trials and limiting access to innovative therapeutic options.

Recent advances in Large Language Models (LLMs) have shown promise for clinical information extraction and patient–trial matching. However, existing approaches typically address these tasks independently, often relying on partially structured data or probabilistic LLM reasoning to assess eligibility. Such methods may perform well in controlled settings but lack the transparency, reproducibility, and auditability required in real-world healthcare environments. Moreover, limited attention has been given to integrated workflows that combine extraction from highly unstructured oncology diaries, clinician validation, logical formalization of eligibility criteria, and transparent eligibility assessment mechanisms that combine structured evaluation with LLM-assisted reasoning within operational hospital systems.

To address these limitations, this work proposes an integrated clinical decision-support system that leverages LLMs to transform unstructured oncology notes into structured, clinician-validated patient profiles and to enable transparent and reproducible eligibility assessment. The system includes an LLM-based extraction pipeline for identifying and normalizing clinically relevant parameters, a semi-automated process for extracting and formalizing trial inclusion and exclusion criteria into machine-readable logical rules, and a hybrid matching mechanism that evaluates validated patient profiles against validated eligibility criteria through a combination of deterministic rule-based assessment and LLM-assisted reasoning for criteria that cannot be directly represented within the structured patient schema. Clinician-in-the-loop validation is incorporated throughout the workflow to ensure correctness, interpretability, and trustworthiness.

This design allows structured and objectively measurable criteria to be evaluated transparently through logical rules while preserving the ability to assess complex or schema-independent eligibility requirements using clinical context extracted from patient records.

The experimental evaluation, conducted using three lightweight open-source LLMs (8B parameters), demonstrated the feasibility of the proposed approach across all pipeline components. Results indicate a consistent performance pattern: models performed strongest in structured extraction tasks, faced greater difficulty in the logical formalization of eligibility criteria, and recovered competitive performance in the final eligibility assessment stage when grounded in explicit clinical context. Notably, expert clinical evaluation of the generated outputs confirmed that model decisions were consistently grounded in available evidence, with justification quality scores above 4.8 out of 5 for the best-performing models, supporting the clinical reliability of the proposed framework.

This article presents the system architecture, describes the extraction and matching pipelines, and reports an experimental evaluation of multiple LLMs in a real hospital context. The proposed approach aims to reduce administrative burden, improve traceability, and enhance the efficiency and reproducibility of clinical trial enrollment in oncology settings.

Related Work

Research on clinical information extraction using Large Language Models (LLMs) has largely focused on transforming unstructured clinical narratives into structured representations through tasks such as Named Entity Recognition, [11], and attribute extraction, [9] and [1]. Existing approaches commonly rely on prompt-engineering strategies, including zero-shot, and few-shot prompting, as well as parameter-efficient fine-tuning techniques such as PEFT and Retrieval-Augmented Generation, [12], [10], [2], [11] and [8]. Hybrid frameworks that combine multiple techniques have also emerged, demonstrating improved robustness and extraction accuracy when evaluated against expert-annotated gold standards, [3] and [9]. Overall, the literature shows that LLMs can outperform traditional NLP pipelines and rule-based systems, offering greater flexibility and reduced development effort, [6] and [12]. However, performance often degrades in noisy or highly contextual real-world clinical environments, and hallucinations, inconsistent outputs, and limited robustness remain significant barriers to clinical deployment, [11], [12] and [9].

Parallel work has explored the use of LLMs for patient–trial matching, ranging from zero-shot prompting approaches to multi-stage retrieval and re-ranking pipelines, [12], [4] and [8]. These systems typically represent patients or trials as queries, simplify eligibility criteria into binary questions, [8], [3] and [7], or classify eligibility using categorical or percentage-based scoring, [5], [4] and [3]. Results consistently show that LLMs can match or surpass expert performance, particularly when enhanced with ontologies, retrieval-augmented architectures, or task-aligned fine-tuning, [7], [8], [4] and [5]. Open-source models have also demonstrated competitive performance relative to proprietary systems, while offering substantial reductions in screening time and computational cost, [5], [3] and [4]. Nonetheless, these approaches face limitations, including high false-negative rates caused by non-standard abbreviations and ontology gaps, reliance on synthetic datasets that may introduce bias, sensitivity to prompt design, [5], [2] and [12], and challenges related to privacy and regulatory compliance, [4], [12] and [6].

More recent studies have begun to explore hybrid pipelines that integrate information extraction and patient–trial matching within unified frameworks, [4] and [3]. These systems typically follow an extraction-first paradigm, [7], or employ multi-agent LLM setups that combine extraction and reasoning, [5] and [4]. However, most hybrid approaches rely on end-to-end probabilistic LLM reasoning or assume access to structured electronic health records, [12] and [4], blurring the distinction between deterministic rule-based evaluation and LLM-assisted interpretation. As a result, few systems explicitly separate the interpretation of unstructured clinical text from the transparent, reproducible evaluation of eligibility criteria.

Taken together, the literature highlights several gaps that remain insufficiently addressed. Clinical information extraction and patient–trial matching are often treated as independent tasks, [11] and [9], and evaluations frequently rely on curated or synthetic datasets that do not reflect the complexity of real-world oncology workflows, [6], [4], [12] and [3]. Eligibility assessment is commonly performed through probabilistic LLM reasoning, limiting transparency, auditability, and reproducibility, [4] and [12]. Only a small number of studies propose integrated workflows that combine clinician-validated extraction, structured patient representation, formalization of eligibility criteria into machine-readable rules, [3], and hybrid eligibility assessment mechanisms capable of combining deterministic evaluation with LLM-assisted reasoning when necessary. This gap motivates the need for approaches that bridge the flexibility of LLMs with the transparency of rule-based systems while operating directly on real-world unstructured oncology data.

System Design and Architecture

As described above the present approach aims to reduce work-overload and screen time of oncologist experts by leveraging the use of LLMs to extract hyper-relevant parameters of free-text clinical diaries forming this way validated structured patient profiles, that in conjunction with the criteria extraction and conversion, will allow for a faster, automatic and semi-deterministic matching of potentially eligible patients for given clinical trials. Given the non-deterministic nature of LLMs and the critical clinical context, all automated steps require expert validation by the clinician.

At a methodological level, the system pipeline is divided into interconnected modules phases:

- Phase 1: Hyper-relevant Parameter Extraction;
- Phase 2: Extraction and Conversion of Clinical Criteria;
- Phase 3: Semi-Deterministic Trial-Patients Matching.

Architecture

Flow

The proposed architecture follows a sequential workflow in which each phase produces artifacts required by subsequent stages. The Hyper-Relevant Parameter Extraction phase generates clinician-validated structured patient profiles from free-text clinical diaries. In parallel, the Extraction and Conversion of Clinical Criteria phase processes clinical trial documentation to produce validated eligibility criteria and their corresponding machine-readable logical representations.

The communication between phases is performed through the system database, where all intermediate artifacts are persisted after clinician validation. This design ensures traceability, reproducibility, and allows experts to review and modify the outputs of each phase before they are consumed by downstream components.

Once both structured patient profiles and validated eligibility rules are available, the Semi-Deterministic Trial–Patient Matching phase can be executed. This final phase retrieves the artifacts produced by the previous stages and evaluates patient eligibility for a selected clinical trial, producing a ranked list of potentially eligible patients together with criterion-level assessment results.

Hyper-Relevant Parameter Extraction

The objective of the first phase is to transform unstructured free-text clinical information into normalized and structured patient data. The input consists of clinical diaries written in free-text format, which are selected by the clinician through the system interface.

The extraction process comprises the following steps:

1) Text Pre-processing: A clinical diary uploaded to the system is selected and pre-processed. During this stage, irrelevant sections such as headers and footers are removed to reduce unnecessary context and computational overhead.
2) Information Extraction and Normalization: The pre-processed text is provided to the LLM together with a task-specific prompt. The model is instructed to identify and extract clinically relevant parameters according to a predefined schema. During the same step, diagnosis normalization is performed using a stakeholder-provided mapping to ensure consistency across patient records.
3) Validation and Structured Output Generation: The extracted information is presented to the clinician for expert validation and potential correction. Once approved, the validated patient profile is persisted in the database and becomes available for subsequent phases of the workflow.

A formalization of this workflow can be seen on the Figure 1 below.

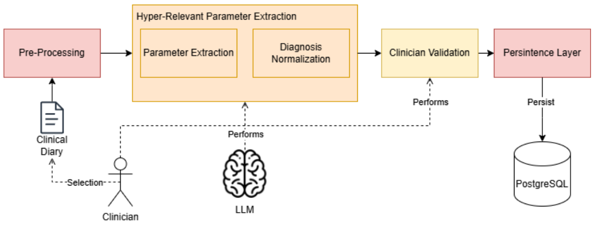

Fig. 1. Hyper-Relevant Parameter Extraction

Upon completion of this phase, the system produces a structured repository of validated patient profiles containing the key clinical parameters required for eligibility assessment.

Patient Profile Schema: The extraction schema follows a JSON structure composed of the following fields:

- Age: Patient age or date of birth, depending on the information available in the clinical diary;
- Gender: Patient gender;
- ECOG-PS: ECOG Performance Status score (0–5), describing the patient's functional capacity;
- Diagnosis: Primary diagnosis, typically corresponding to the patient's cancer type or major clinical condition;
- Diagnosis Date: Date on which the diagnosis was formally established;
- Molecular Status: Relevant molecular and biomarker information;
- Stage: Disease staging information;
- Pathology Group: Clinical category grouping diagnoses according to the organ or system of origin;
- Treatments: List of administered treatments:
  - Name: Treatment name;
  - Start Date: Date on which the treatment started;
  - End Date: Date on which the treatment ended;
- Control: Information related to disease control, metastases, comorbidities, and other relevant clinical follow-up

observations.

All schema fields are provided to the LLM as target extraction parameters. However, as clinical diaries may not contain all required information, the model is allowed to return null values for unavailable fields. Since all extracted information undergoes clinician validation before persistence, missing or uncertain values can be reviewed and corrected when necessary.

Error Handling and Robustness: To improve robustness, several error scenarios are explicitly handled during extraction:

- Invalid JSON Structure: If the LLM generates explanatory text together with the JSON output, the system attempts to isolate and recover the valid JSON structure. If recovery is unsuccessful, the extraction request is automatically repeated.
- Missing JSON Output: If no valid JSON structure is generated, the extraction request is repeated.
- Truncated Responses: If the generated output is incomplete due to context or generation limits, the extraction request is repeated.

The objective of these recovery mechanisms is to ensure that a valid and structurally consistent JSON representation is obtained before the extracted information is presented for clinician validation.

Produced Artifacts: Upon completion of this phase, a single primary artifact is generated and made available through the system interface:

- Structured Patient Profile: A clinician-validated representation of the patient containing the extracted clinical parameters required for subsequent eligibility assessment.

The resulting structured profiles provide a normalized representation of patient information that can be consistently queried and evaluated during the matching process, while maintaining traceability through clinician validation.

Extraction and Conversion of Clinical Criteria

The objective of the second phase is to transform clinical trial eligibility requirements into machine-interpretable representations that can later be used during patient eligibility assessment. The input to this phase consists of clinical trial documentation uploaded, managed, and selected through the system interface.

The process comprises the following steps:

1) Text Pre-processing: The clinician selects a clinical trial document, which undergoes pre-processing to remove sections that are not relevant to eligibility assessment. Similar to the clinical diary pipeline, this step reduces unnecessary context and computational overhead.
2) Trial Structure Identification: Before extracting eligibility criteria, the LLM analyses the clinical trial document to identify its internal structure, namely the existence of trial cohorts. This information is subsequently used during criteria extraction and patient eligibility assessment.
3) Criteria Extraction: The LLM receives the processed trial documentation together with the identified trial structure and extracts inclusion and exclusion criteria. Whenever possible, criteria are extracted individually and associated with their corresponding cohort. However, some eligibility requirements must remain grouped to preserve their original clinical meaning. The system therefore supports both simple and compound criteria.
4) Extracted Criteria Validation and Persistence: The extracted criteria are presented to the clinician for expert validation. Once approved, they are persisted in the database, ensuring traceability and providing the input required for the next step of the pipeline.
5) Criteria Conversion: Following validation, each criterion is translated into a structured logical representation. The objective of this step is to maximize the use of deterministic evaluation and reduce reliance on probabilistic LLM reasoning during eligibility assessment.
6) Logical Representation Validation and Persistence: The generated logical representations are presented to the clinician for validation to ensure that they accurately reflect the meaning of the original eligibility criteria. Once approved, they are stored in the database and become available for patient eligibility assessment.

A formalization of this workflow can be seen on the Figure 2 below.

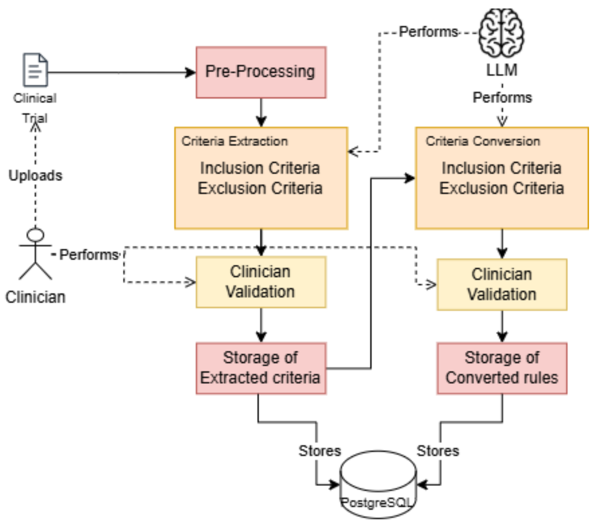

Fig. 2. Extraction and Conversion of Clinical Criteria Workflow

Initially, criteria extraction and logical conversion were designed as a single task. However, previous studies have shown that decomposing complex tasks into simpler sequential steps often leads to improved performance and reliability.

For this reason, the structure identification, extraction and conversion stages were implemented as separate phases.

Conversion of Criteria: The LLM is provided with a prompt that guides it to perform the best possible conversion for the system. It tries at all costs, while maintaining its meaning to convert the criterion into a valid field available in the patient profile schema, or in the laboratory values schema. Only if it is incapable of such, the LLM is instructed to generate a general field for it. These incompatible fields will be discussed forward in the sections and

Complex Criteria Handling: The identification of simple and compound criteria is performed by the LLM. The model receives the complete textual description of each eligibility criterion and determines whether it can be safely decomposed into independent criteria without altering its clinical meaning. If separation preserves the original semantics, the criterion is divided into multiple individual criteria. Otherwise, it is maintained as a compound criterion and processed as a single logical unit.

Logical Representation of Eligibility Criteria: To enable machine-interpretable and partially deterministic eligibility assessment, each validated criterion is converted into a JSON-based logical representation.

Simple criteria are represented as atomic rules composed of a field, an operator, and a value. For example:

```
{
  "field": "age",
  "operator": ">=",
  "value": "18"
}
```

More complex criteria involving multiple conditions are represented through Boolean operators such as AND and OR, where each operator groups a set of sub-conditions. For example:

```
{
  "operator": "AND",
  "conditions": [
    {
      "field": "ecog_ps",
      "operator": "<=",
      "value": "1"
    },
    {
      "field": "diagnosis",
      "operator": "=",
      "value": "NSCLC"
    }
  ]
}
```

In some cases, a criterion cannot be automatically mapped to the predefined logical schema. In these situations, the following representation is generated, allowing the clinician to manually define the corresponding logical rule:

```
{
  "unmapped": true,
  "source_text": "unmapped part of the criterion"
}
```

Finally, all converted criteria are grouped according to their category, resulting in the following structure:

```
{
  "inclusion_criteria": [
    {
      "id": integer,
      "text": string,
      "logic": object
    }
  ],
  "exclusion_criteria": [
    {
      "id": integer,
      "text": string,
      "logic": object
    }
  ]
}
```

Produced Artifacts: Upon completion of this phase, two primary artifacts are generated and made available through the system interface:

- **Clinical Trial Criteria:** A validated list of individually extracted eligibility criteria, organized into inclusion and exclusion categories;
- **Logical Rules:** The corresponding machine-readable logical representations of those criteria, also organized into inclusion and exclusion categories.

This representation allows the system to uniformly encode both simple and compound eligibility requirements. Criteria that reference fields available in the patient profile schema can subsequently be evaluated deterministically, while criteria that cannot be represented through the schema remain available for LLM-assisted assessment during the matching phase. This hybrid approach increases transparency and auditability while preserving the flexibility required to handle complex clinical requirements.

Semi-Deterministic Trial-Patient Matching

The objective of the final phase is to identify potentially eligible patients for a given clinical trial through a semi-deterministic eligibility assessment process. This phase consumes the structured patient profiles generated during the first phase and the validated logical representations of eligibility criteria produced during the second phase.

The matching process comprises the following steps:

1) Trial Selection and Patient Retrieval: A clinician selects the clinical trial to be assessed. Based on the metadata defined during trial registration, a subset of patient profiles is retrieved from the database. Patient selection is initially restricted to the pathology group associated with the clinical trial, reducing the number of comparisons and LLM calls required during eligibility assessment.
2) Eligibility Assessment: Each patient is evaluated against every validated eligibility criterion. Two assessment paths are available:
   - **Deterministic Assessment:** If the criterion references a field available in the patient profile schema, the corresponding logical rule is evaluated directly against the structured patient data.
   - **LLM-Assisted Assessment:** If the criterion references information that is not represented within the structured schema, the LLM receives the patient profile, clinical diary, laboratory results, and the corresponding criterion to determine eligibility. In addition to the assessment result, the model provides a justification to support expert review.

   Although different evaluation mechanisms are employed, both paths produce the same output: a criterion-level eligibility assessment that is subsequently used during final eligibility determination.
3) Assessment Validation and Persistence: All criterion-level assessments are presented to the clinician for review. Experts may modify any automatically generated decision, regardless of whether it originated from deterministic evaluation or LLM-assisted reasoning. Following validation, all criterion assessments, manual corrections, and final eligibility decisions are persisted in the database to ensure traceability and auditability.

A formalization of this workflow can be seen on the Figure 3 below.

Matching Strategy: The proposed matching approach is semi-deterministic. Whenever a criterion references information available in the structured patient profile schema, eligibility is assessed through deterministic logical evaluation. Conversely, criteria that cannot be directly represented within the schema are evaluated through LLM-assisted reasoning.

This design maximizes the use of transparent and reproducible rule-based assessment while preserving the flexibility required to evaluate complex clinical requirements that cannot be formally encoded.

Eligibility Aggregation: The final patient eligibility decision is computed through a strict rule-based aggregation of all criterion-level assessments. Criteria are divided into inclusion and exclusion criteria and are evaluated independently before aggregation.

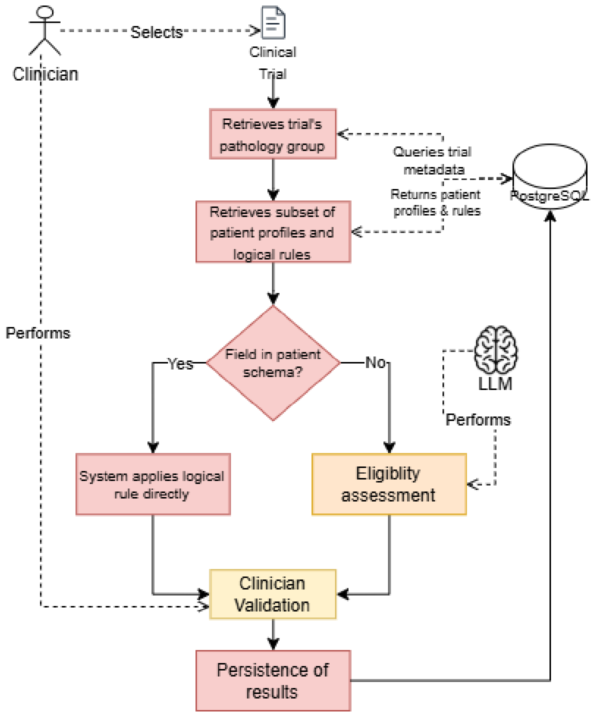

Fig. 3. Semi-Deterministic Trial–Patient Matching

A patient is considered eligible if:

- All inclusion criteria are satisfied; and
- No exclusion criterion is triggered.

Formally:

```
Eligible = (All Inclusion Criteria Satisfied)
           AND
           (No Exclusion Criteria Triggered)
```

No weighting, ranking, or probabilistic scoring mechanism is employed. The final decision relies exclusively on logical consistency across all eligibility criteria.

For clinical trials containing multiple cohorts, eligibility is evaluated at both the trial and cohort levels. A patient is considered eligible for the trial if the general eligibility requirements are satisfied and at least one cohort-specific eligibility assessment returns an eligible result.

Formally, a patient is considered eligible if:

- The general trial criteria are satisfied; and
- At least one cohort satisfies its respective inclusion and exclusion requirements.

Inconclusive Assessments: Certain criteria may be impossible to evaluate reliably. This can occur due to missing clinical information, unavailable laboratory values, unit mismatches, parsing failures, ambiguous criterion definitions, or criteria that cannot be resolved through either deterministic evaluation or LLM-assisted assessment.

In these situations, the corresponding criterion is classified as inconclusive.

To ensure a conservative clinical decision-support strategy:

- Inconclusive inclusion criteria are treated as not satisfied;
- Inconclusive exclusion criteria are treated as not triggered.

This approach prioritizes patient safety by avoiding positive eligibility decisions when essential information is unavailable or uncertain.

Produced Artifacts: Upon completion of this phase, two primary artifacts are generated and made available through the system interface:

- Criterion Assessments: Individual eligibility decisions for each criterion, together with supporting justifications whenever LLM-assisted assessment is employed;
- Eligibility Results: A list of patients classified according to their eligibility status for the selected clinical trial and, when applicable, for individual cohorts.

The resulting assessments provide full traceability from the final eligibility decision down to the individual criterion evaluations, allowing clinicians to review, validate, and audit every step of the matching process.

Materials and Methods

Documents as Inputs

This section describes the two document types processed by the proposed system and the abstractions adopted to model them within the pipeline.

The proposed framework operates on two primary document sources: clinical diaries and clinical trial protocols. Although both are written in free-text format, they differ substantially in their structure, purpose, and degree of standardization. Understanding these characteristics is essential for the design of the extraction and matching pipelines.

Clinical Trials: Clinical trial protocols are predominantly narrative documents describing study objectives, methodology, treatment arms, safety procedures, and participant selection requirements. Despite variations across studies, trial protocols generally follow a well-defined organizational structure.

Among all protocol sections, eligibility criteria constitute the only component directly involved in patient selection. Inclusion and exclusion criteria are expressed in natural language but fundamentally represent constraints over patient attributes, such as demographic characteristics, diagnoses, biomarkers, treatment history, laboratory values, or performance status.

However, eligibility structures vary considerably between trials. Some studies define a single set of eligibility requirements, while others contain multiple cohorts, each with its own inclusion and exclusion criteria. Furthermore, eligibility requirements may range from simple atomic conditions to highly complex compound criteria involving multiple logical dependencies.

To enable automated processing, the proposed system abstracts clinical trial documents into two components: trial structure and eligibility criteria. This separation allows descriptive protocol information to remain independent from the patient selection logic. Consequently, LLM-based extraction can focus exclusively on eligibility requirements, which are subsequently converted into machine-readable logical representations suitable for automated evaluation.

All clinical trials used throughout this study were obtained in English.

Clinical Diaries: Clinical diaries represent the primary source of patient information used by the system. Unlike clinical trial protocols, these documents exhibit substantial variability in writing style, organization, terminology, and level of detail. Such variability arises from differences between institutions, clinical specialties, individual physicians, and documentation practices.

The design of the extraction pipeline was guided by a structured template provided by the clinical stakeholder and derived from four representative oncology clinical diaries routinely used within the Oncology Department of Coimbra University Hospital Centre (HUC). Rather than defining a fixed document structure, these diaries were analysed to identify recurring clinical concepts and information categories required for patient eligibility assessment.

From this analysis, a patient profile schema was defined, containing the hyper-relevant clinical parameters considered essential for trial matching. These parameters serve as the target representation of the extraction pipeline and provide a consistent structured format regardless of how information is expressed in the original diary.

To avoid overfitting the system to a specific documentation template, the abstraction process focused on preserving semantic information rather than document layout or wording. Structural elements, section ordering, formatting conventions, and stylistic patterns were intentionally disregarded whenever they were not clinically relevant. Only information categories consistently required for eligibility assessment, such as diagnosis, staging, treatments, biomarkers, ECOG performance status, and disease evolution, were retained in the final schema.

This abstraction enables the system to process heterogeneous clinical documentation while producing a standardized representation suitable for automated eligibility assessment. Although the proposed schema was derived from documents used within a specific oncology department, it was designed to capture clinically meaningful concepts rather than institution-specific formatting conventions.

All clinical diaries used throughout this study were written in European Portuguese.

Datasets

This section describes the datasets used to evaluate the components of the proposed system, including synthetic oncology clinical diaries, clinical trial documentation, and manually curated gold standards. A quantitative summary of these normalized datasets is provided in Table I. All data used in this study were fully anonymized; no identifiable patient information was used, and every synthetic diary contains no real clinical content.

Synthetic Oncology Clinical Diaries: The system evaluation was conducted using a dataset of 30 synthetic free-text oncology clinical diaries. These diaries were generated using the open-source `gpt-oss-120b` model, selected for its high generation capacity, and were grounded on four representative clinical diaries provided by the clinical stakeholder to serve as references for writing style, structure, and clinical content. The dataset size balances clinical diversity with the feasibility of manual annotation.

To ensure representativeness, 30 unique patient profiles were first created, each describing a distinct oncological case. The generation pipeline consisted of two sequential stages designed to reproduce the variability observed in real-world clinical documentation:

1) **Idealized Generation:** The LLM generated a clinically coherent and internally consistent Portuguese oncology note from the patient profile, maintaining alignment between diagnosis, staging, treatments, imaging findings, ECOG performance status, comorbidities, and disease evolution.
2) **Noise and Artifact Insertion:** To resemble routine hospital records, the generated diary was transformed by introducing realistic imperfections while preserving complete clinical correctness. For each diary, two generation factors were randomly assigned: a *length mode* (short, medium, or long) and a *style mode* (narrative-dominant, telegraphic hospital style, exam-and-imaging focused, toxicity focused, or psychosocial emphasis). The model introduced variations in sentence length, irregular formatting, heterogeneous abbreviation usage, minor spelling imperfections, structural asymmetries (e.g., missing headers), and administrative workflow notes.

To guide this process, the prompts incorporated examples of authentic clinical documentation exclusively for stylistic learning, with explicit instructions not to reproduce original clinical content. Following generation, an experienced oncologist reviewed and validated all diaries for clinical plausibility and consistency.

After normalization (removing headers, footers, and irrelevant sections), the clinical diary dataset contained a total of 64,953 tokens across the 30 documents, averaging 2,165 tokens per diary.

Clinical Trials Dataset: A dataset of 30 clinical trials was assembled from *ClinicalTrials.gov* to evaluate the criteria extraction, conversion, and patient–trial matching pipelines, seeking to maximize clinical diversity and minimize selection bias.

The trial collection covers a broad range of tumor types (including breast, lung, pancreatic, colorectal, melanoma, among others), disease stages, molecular profiles, and therapeutic settings. The selected trials encompass different designs (with and without cohort structures) and exhibit substantial variability in eligibility complexity, ranging from simple demographic or diagnostic constraints to highly specialized molecular biomarkers and prior treatment histories.

After normalization, the clinical trial dataset contained a total of 120,786 tokens across the 30 protocols, averaging 4,026 tokens per trial, reflecting the complexity of oncology requirements. The eligibility criteria corpus comprises a total of 892 individual criteria (480 inclusion and 412 exclusion criteria), averaging approximately 30 eligibility criteria per trial.

Manually Curated Gold Standards: All gold standards were manually annotated to provide reliable reference data for system evaluation across four core tasks:

- **Hyper-Relevant Parameter Extraction:** All fields defined in the patient profile schema were manually identified and annotated within the 30 clinical diaries to enable automated, deterministic comparisons with the LLM outputs.
- **Trial Structure Identification:** The organization of eligibility requirements and the presence of cohorts were manually mapped following the system's formal representation.
- **Criteria Extraction:** All 892 inclusion and exclusion criteria were manually extracted as individual requirements to evaluate the completeness of the automated extraction.
- **Criteria Conversion:** Expected logical representations were defined according to the system's conversion schema. Criteria mapping to predefined patient profile fields were annotated with logical rules for deterministic evaluation. Criteria depending on unstructured clinical context were assessed qualitatively to ensure semantic consistency.

Proposed Method

The proposed method aims to support clinical trial recruitment by combining clinical information extraction and patient–trial matching within a single LLM-assisted framework. Rather than relying exclusively on probabilistic reasoning, the

TABLE I
QUANTITATIVE SUMMARY OF THE NORMALIZED DATASETS USED IN THIS STUDY

Dataset Component	Total	Average per Document
Clinical Diaries (tokens)	64,953	2,165
Clinical Trials (tokens)	120,786	4,026
Inclusion Criteria	480	16
Exclusion Criteria	412	14
Total Criteria	892	30

approach seeks to maximize deterministic processing whenever possible by transforming both patient information and trial eligibility criteria into structured representations that can be automatically compared.

The method is composed of three sequential tasks. First, relevant clinical information is extracted from free-text oncology clinical diaries and transformed into structured patient profiles. Second, eligibility criteria are extracted from clinical trial documentation and converted into machine-interpretable logical representations. Finally, patient profiles and trial criteria are combined through a semi-deterministic matching process to identify potentially eligible patients.

The central premise of the proposed approach is that eligibility assessment can be significantly improved by separating complex reasoning into multiple specialized tasks. Instead of asking a model to directly determine patient eligibility from raw documents, the system decomposes the problem into information extraction, criteria formalization, and eligibility assessment stages. This decomposition aims to improve interpretability, traceability, and clinician oversight while reducing reliance on end-to-end LLM reasoning.

The overall system was designed with real clinical environments in mind and was iteratively discussed with domain experts to ensure practical applicability. As a consequence, every LLM-assisted step was built to guarantee interpretability and auditability by allowing direct comparison between outputs and their source documents, while storing and clearly labeling all intermediate results. Throughout the pipeline, the clinician remains the final approver and editor of all outputs, ensuring clinical safety while leveraging LLMs to accelerate the workflow.

A secondary objective of this study is to evaluate whether relatively small, non-proprietary open-source language models can achieve satisfactory performance across the different stages of the pipeline. Demonstrating competitive performance with such models would increase the feasibility of deploying the proposed framework in environments with limited computational resources and stricter privacy requirements.

The proposed method is evaluated through the following tasks, each task is evaluated independently to isolate error sources and quantify the contribution of each module:

- Extraction and normalization of hyper-relevant patient information from free-text clinical diaries;
- Identification of clinical trial eligibility structures, (cohort detection);
- Extraction of individual inclusion and exclusion criteria;
- Conversion of eligibility criteria into machine-interpretable logical representations;
- Semi-deterministic patient–trial matching through the combination of deterministic and LLM-assisted assessments.

The study is guided by the following research hypotheses:

- H1: Large Language Models can accurately extract and normalize hyper-relevant clinical information from heterogeneous oncology clinical diaries while adhering to a predefined structured schema.
- H2: Large Language Models can accurately identify, extract, and formalize clinical trial eligibility criteria while preserving their original clinical meaning and logical relationships.
- H3: A hybrid eligibility assessment strategy combining deterministic rule evaluation and LLM-assisted reasoning can effectively support patient–trial matching while maintaining interpretability and clinician control.
- H4: Small open-source language models can achieve performance levels sufficient to support the proposed workflow across the evaluated tasks.

Experimental Setup

As previously described, the proposed framework consists of three LLM-assisted pipelines:

- Hyper-Relevant Parameter Extraction;
- Extraction and Conversion of Clinical Criteria;
- Semi-Deterministic Patient–Trial Matching.

The experimental evaluation was designed to assess the performance of the LLM-dependent components of each pipeline independently. Evaluating each stage separately allows the identification of potential error sources and provides a clearer understanding of the contribution of each component to the overall system performance.

The experiments aim to evaluate both the effectiveness and robustness of the proposed approach, as well as its suitability for integration into a semi-automated clinical decision support workflow. Particular emphasis is placed on accuracy, reliability, interpretability, and the ability of the system to operate under clinician supervision.

Evaluation Scenarios: To evaluate the different stages of the proposed framework, two primary experimental scenarios were defined.

Scenario 1 – Clinical Diary Processing

This scenario evaluates the Hyper-Relevant Parameter Extraction pipeline. Each model receives an individual clinical diary and is tasked with extracting and normalizing the information required to populate the predefined patient profile schema.

The objective is to assess the ability of the models to identify clinically relevant information from heterogeneous free-text oncology documentation while generating outputs that comply with the required structured representation.

Scenario 2 – Clinical Trial Processing

This scenario evaluates the Extraction and Conversion of Clinical Criteria pipeline. Each model receives an individual clinical trial protocol and performs the following tasks:

- Identification of the trial eligibility structure, including cohort detection when applicable;
- Extraction of inclusion and exclusion criteria;
- Conversion of extracted criteria into machine-interpretable logical representations.

The objective is to assess whether the models can preserve the clinical meaning of eligibility requirements while transforming them into structured representations suitable for automated evaluation.

Scenario 3 - Patient-Trial Matching This scenario evaluates the LLM-assisted component of the Semi-Deterministic Patient–Trial Matching pipeline. In cases where eligibility criteria cannot be directly evaluated through deterministic logical rules, either due to incomplete formalization or the presence of schema-independent requirements, the model receives the structured patient profile, the original clinical diary, relevant laboratory information, and the eligibility criterion represented as a logical rule. The model is then tasked with determining whether the available patient information satisfies the given criterion and providing a clinically grounded justification for the decision.

The objective is to assess the ability of the models to perform reliable eligibility assessments when operating under constrained and explicit clinical context, as well as the quality and transparency of the generated clinical justifications.

All models are evaluated under identical experimental conditions to ensure fair and reproducible comparisons.

To further improve reproducibility, prompts are version-controlled and preserved throughout the experiments, and all model configuration parameters remain fixed across evaluation runs.

Evaluation Metrics: Different evaluation metrics are employed depending on the task under analysis.

Hyper-Relevant Parameter Extraction

The extraction of structured patient information is evaluated using:

- Field-level Accuracy: proportion of correctly extracted fields relative to the manually annotated gold standard;
- Missing Field Rate: proportion of schema fields returned as null or omitted;
- Schema Compliance Rate: percentage of outputs that fully comply with the predefined JSON schema.

Trial Structure Identification

The identification of clinical trial structure is evaluated through:

- Cohort Detection Accuracy: correctness of identifying whether cohort structures are present;
- Structure Description Correctness: expert assessment of the generated trial structure representation.

Eligibility Criteria Extraction

The extraction of inclusion and exclusion criteria is evaluated through:

- Criteria Extraction Correctness: manual and expert assessment of whether extracted criteria faithfully represent the original trial requirements.

Eligibility Criteria Conversion

The conversion of eligibility criteria into logical rules is evaluated through:

- Logical Representation Correctness: Manual assessment indicating whether the generated logical representation accurately reflects the clinical meaning of the original criterion.

Patient–Trial Matching

The semi-deterministic matching process is evaluated through:

- Eligibility Assessment Accuracy: correctness of the generated eligibility decision relative to the reference documentation of the patient (clinical diary, laboratory values, and patient profile);
- Justification Quality: expert evaluation of the clinical coherence and usefulness of generated explanations using a five-point Likert scale.

Evaluated Models: The experiments compare multiple open-source Large Language Models representing different model families and architectural approaches:

- Qwen3 8B Instruct;
- Llama 3.1 8B Instruct;
- DeepSeek-R1 8B.

These models were selected because they provide a representative sample of state-of-the-art open-source LLMs that can be deployed with relatively modest computational resources.

All models are evaluated using deterministic inference settings to minimize output variability and maximize reproducibility. No model fine-tuning is performed, and all experiments are conducted under a zero-shot setting. This configuration reflects the intended deployment scenario of the proposed framework, where task-specific training data may not be available and rapid adaptation to new clinical contexts is desirable.

Each experiment was executed once under deterministic inference settings, on the datasets of each 30 clinical diaries and 30 clinical trials.

RESULTS

As mentioned before the selected language models were evaluated in 5 tasks: Hyper-Relevant Parameter Extraction; Trial Structure Identification; Eligibility Criteria Extraction; Eligibility Criteria Conversion; and Patient–Trial Matching.

Hyper-Relevant Parameter Extraction

The first experimental task evaluated the capability of the LLMs to extract clinically relevant information from free-text oncology clinical diaries while adhering to a predefined JSON schema. The evaluation focused on three main aspects: field-level extraction accuracy, schema compliance, and missing field rate. Field-level accuracy was computed by manually comparing the generated patient profiles against the manually annotated gold standards. Two values are reported: strict accuracy, considering only fully correct extractions, and a partial accuracy score, which also accounts for partially correct extractions.

As shown in Table II, all evaluated models were able to consistently follow the predefined output structure, achieving 100% schema compliance. This demonstrates that the evaluated models were capable of generating structurally valid outputs according to the required JSON representation, regardless of their differences in extraction performance.

Regarding extraction accuracy, Qwen3-8B achieved the highest overall performance, obtaining a strict accuracy of 78.3% and a partial accuracy score of 83.3%. DeepSeek-R1-8B followed with 74.3% strict accuracy and 80.5% partial accuracy. Llama3.1-8B presented the lowest performance, achieving 62.3% strict accuracy and 70.3% partial accuracy, with a higher number of incorrect and partially extracted fields. These results indicate that although all models were capable of producing valid structured outputs, their ability to correctly identify and populate clinical information varied considerably. Figure 6 illustrates the comparison between strict and partial accuracy across models.

The missing field rate followed a similar trend. Qwen3-8B and DeepSeek-R1-8B presented lower rates of missing information (11.6% and 13.3%, respectively), whereas Llama3.1-8B showed a considerably higher missing field rate of 21.7%. This suggests that weaker models had more difficulty identifying whether relevant clinical information was present within the diary and subsequently extracting it into the required schema fields.

Table III provides a more detailed field-level analysis across the patient profile schema. Some fields were consistently extracted with high accuracy across all models. Gender and ECOG-PS achieved 100% accuracy for every evaluated model, indicating that these parameters are generally explicit and unambiguous within oncology clinical documentation. Figure 4 further illustrates the performance distribution across schema fields.

Conversely, more complex fields requiring interpretation, normalization, or structured reasoning presented lower performance. Diagnosis extraction represented one of the most challenging fields, with DeepSeek-R1-8B achieving the highest accuracy at 46.7%. However, manual inspection revealed that most errors were not related to failure in identifying the diagnosis itself, but rather to difficulties in applying the required normalization mapping. In several cases, models extracted the correct clinical diagnosis but failed to convert it into the expected normalized representation.

Treatment extraction was another challenging task due to its hierarchical structure, requiring the identification of multiple sub-fields, including treatment name, start date, and end date. Qwen3-8B achieved the best performance in this field, with 70.0% strict accuracy and an 85.0% partial accuracy score, while Llama3.1-8B achieved substantially lower results.

Disease staging also proved challenging, particularly due to the need to preserve the complete staging representation. A recurrent error pattern was the extraction of only the final stage classification (e.g., ”IVB”) instead of the complete TNM representation (e.g., ”cT3N2M1b (IVB)”). Although the clinically relevant information was partially captured, these outputs were considered incomplete according to the defined schema.

Date-related fields, particularly diagnosis dates and treatment periods, also represented a source of errors across models. These errors consisted mainly of incorrect extraction, incomplete temporal information, or returning null values when dates were present in the clinical diary.

Overall, the results demonstrate that lightweight LLMs are capable of performing structured clinical information extraction from heterogeneous oncology documentation. However, performance is strongly influenced by the complexity of the target field, with explicit clinical attributes being extracted reliably, while fields requiring normalization, temporal reasoning, or hierarchical interpretation remain more challenging. Figure 5 summarises the schema compliance rate across all evaluated models.

TABLE II
GLOBAL PERFORMANCE OF THE EVALUATED MODELS (HYPER-RELEVANT PARAMETER EXTRACTION)

Models	Total	Correct	Partial	Wrong	Acc./Partial (%)	Schema (%)	Missing (%)
Llama3.1-8b	300	187	48	65	62.3/70.3	100.0	21.7
Deepseek-r1:8b	300	223	37	40	74.3/80.5	100.0	13.3
Qwen3-8b	300	235	30	35	78.3/83.3	100.0	11.6

TABLE III
FIELD-LEVEL PERFORMANCE ACROSS SCHEMA COMPONENTS

Schema Fields	Llama3.1-8b	Deepseek-r1:8b	Qwen3-8b
age_or_birthdate	46.6/46.6	100.0/100.0	100.0/100.0
gender	100.0/100.0	100.0/100.0	100.0/100.0
ecog_ps	100.0/100.0	100.0/100.0	100.0/100.0
diagnosis	36.7/36.7	46.7/46.7	43.3/43.3
diagnosis_date	56.7/56.7	63.3/63.3	76.7/76.7
molecular_status	60/71.7	66.7/73.3	63.3/71.7
stage	23.3/61.7	23.3/61.7	46.7/73.3
pathology_group	63.3/63.3	90.0/90.0	83.3/83.3
treatments	36.7/66.7	56.7/73.3	70.0/85.0

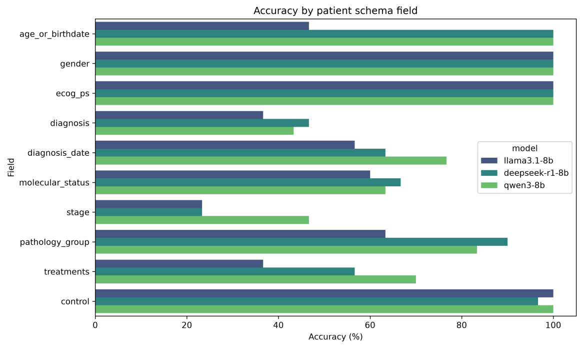

Fig. 4. Hyper-Relevant Parameter Extraction - Accuracy by schema field

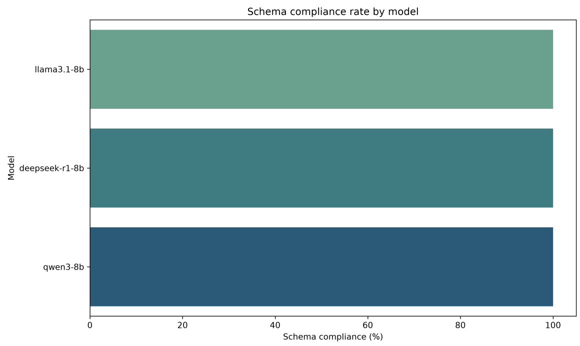

Fig. 5. Hyper-Relevant Parameter Extraction - Schema compliance rate by model

Trial Structure Identification

The second experimental task evaluated the capability of the LLMs to identify and represent the internal structure of clinical trials according to a predefined JSON schema. This step is required for subsequent eligibility assessment, as cohort-specific requirements may define different patient populations within the same clinical trial.

The evaluation considered two aspects: (i) identification of whether formal cohort structures were present, and (ii) correctness of the generated cohort representation. The generated outputs were compared against manually annotated gold standards. Strict accuracy, considering only fully correct outputs, and partial accuracy, accounting for partially correct representations, were reported.

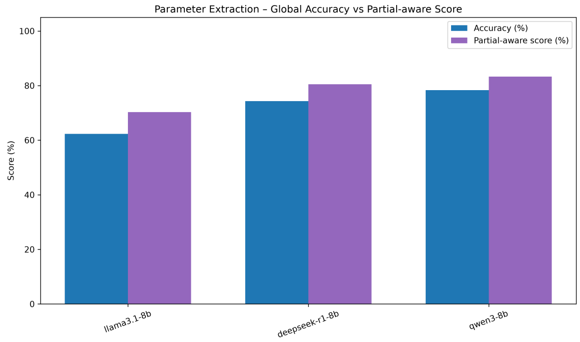

Fig. 6. Hyper-Relevant Parameter Extraction - strict vs partial accuracy

As shown in Table IV, substantial differences were observed between models. Deepseek-r1:8b achieved the best performance, correctly identifying 58 out of 60 evaluated fields, corresponding to 96.7% strict accuracy. In contrast, Llama3.1-8b achieved considerably lower performance (25.0%), mainly due to difficulties distinguishing formal trial cohorts from descriptive patient groups. Qwen3-8b obtained intermediate results, achieving 60.0% strict accuracy and 75.0% partial accuracy.

TABLE IV
GLOBAL PERFORMANCE OF THE EVALUATED MODELS (TRIAL STRUCTURE IDENTIFICATION)

Model	Total	Correct	Partial	Wrong	Acc./Partial (%)
Llama3.1-8b	60	15	0	45	25.0/25.0
Deepseek-r1:8b	60	58	0	2	96.7/96.7
Qwen3-8b	60	36	18	6	60.0/75.0

Further analysis showed that the main limitation of Llama3.1-8b was the incorrect interpretation of descriptive clinical groups as formal cohorts. Phrases describing patient characteristics, such as disease stage or molecular status, were frequently interpreted as independent cohort structures, leading to false cohort identification.

For Qwen3-8b, most errors were related to schema compliance rather than semantic understanding. In several cases, the model correctly identified the absence of cohorts but returned incomplete JSON structures, omitting required fields such as the empty cohort array. Conversely, when explicit cohort structures were present, all models demonstrated substantially higher performance, indicating that the main challenge was distinguishing cohort absence from descriptive patient categories.

Beyond structural correctness, a qualitative expert assessment was conducted on a subset of four clinical trials (Trials 5, 16, 26, and 30) to evaluate whether generated cohort descriptions preserved clinically relevant information. The assessment focused on the maintenance of diagnosis, molecular alterations, previous treatment exposure, and therapeutic requirements. The summarized expert evaluation of the evaluated models is presented in Table V.

The expert evaluation demonstrated that correct cohort identification does not necessarily guarantee preservation of clinical meaning. Across the analyzed trials, the main errors were related to omission of eligibility-defining information rather than incorrect cohort structures.

TABLE V
EXPERT ASSESSMENT OF COHORT-STRUCTURE CLASSIFICATION PERFORMANCE

Model	Avg. Class.	Expert Assessment Summary
Deepseek-r1:8b	2.25	Correctly identifies cohort structures but frequently omits clinically relevant descriptors such as treatment status, molecular alterations, or diagnostic specificity.
Llama3.1-8b	2.75	Achieves high semantic fidelity in some trials but presents inconsistent behaviour, including complete loss of clinical information in specific cases.
Qwen3-8b	2.50	Correctly captures several cohort characteristics but occasionally omits essential molecular or therapeutic information.

Overall, these results indicate that cohort identification is primarily limited by the preservation of the complete clinical definition of each cohort. While structural extraction can be performed reliably, maintaining all clinically relevant eligibility characteristics remains a challenging aspect of LLM-based trial document understanding, as illustrated by the strict accuracy comparison between models (Figure 7).

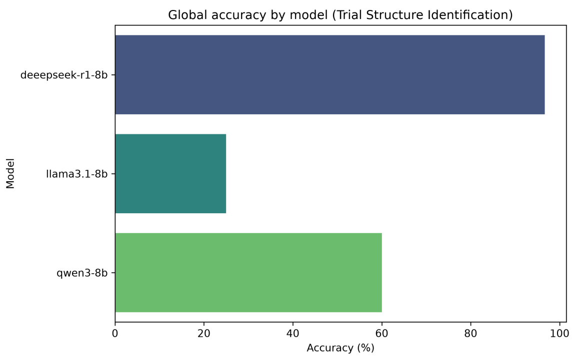

Fig. 7. Trial Structure Identification - Strict accuracy

Eligibility Criteria Extraction

The third experimental task evaluated the capability of the LLMs to extract individual clinical trial eligibility criteria while preserving their original clinical meaning and identifying whether each criterion was associated with a specific cohort. This step represents a fundamental component of the proposed pipeline, as accurate criteria extraction is required before their conversion into machine-interpretable rules and subsequent patient–trial matching.

The evaluation considered two complementary aspects: (i) the extraction of individual inclusion and exclusion criteria from clinical trial documents, and (ii) the attribution of extracted criteria to the corresponding cohort when cohort-specific eligibility requirements were present. Generated outputs were manually compared against manually annotated gold standards. Two metrics were reported: strict accuracy, considering only completely correct extractions, and partial accuracy, which also accounted for outputs that preserved the majority of the clinical meaning but contained minor structural or semantic deviations.

As presented in Table VI, substantial differences were observed between models. Qwen3-8b achieved the strongest overall performance, correctly extracting 823 out of 892 eligibility criteria, resulting in a strict accuracy of 92.3% and a partial accuracy of 95.9%. Furthermore, the model produced only four incorrect extractions, demonstrating a strong capability of preserving the original eligibility information while avoiding hallucinated criteria. Deepseek-r1:8b achieved intermediate performance, with strict and partial accuracies of 83.7% and 88.2%, respectively. In contrast, Llama3.1-8b presented the lowest performance, achieving 71.9% strict accuracy and 75.8% partial accuracy, while also generating the highest number of incorrect criteria.

The second component of this task focused on identifying whether extracted criteria belonged to a specific trial cohort, with the corresponding results presented in Table VII. In this scenario, Qwen3-8b again demonstrated superior performance, correctly assigning cohort-specific criteria in 26 out of 27 cases, achieving 96.3% accuracy. Both Deepseek-r1:8b and Llama3.1-8b achieved 66.7% accuracy. The errors observed in these models were mainly associated with incomplete extraction rather than incorrect cohort attribution, suggesting that the main limitation was failure to preserve the complete criterion information before the classification step.

TABLE VI
GLOBAL PERFORMANCE OF THE EVALUATED MODELS (ELIGIBILITY CRITERIA EXTRACTION)

Model	Total	Correct	Partial	Wrong	Acc./Partial (%)
Llama3.1-8b	892	641	70	181	71.9/75.8
Deepseek-r1:8b	892	747	80	65	83.7/88.2
Qwen3-8b	892	823	65	4	92.3/95.9

TABLE VII
PERFORMANCE ON CRITERIA REQUIRING COHORT-STRUCTURE EXTRACTION

Model	Total	Correct	Partial	Wrong	Acc./Partial (%)
Llama3.1-8b	27	18	0	9	66.7/66.7
Deepseek-r1:8b	27	18	0	9	66.7/66.7
Qwen3-8b	27	26	0	1	96.3/96.3

The distribution of extraction outcomes and cohort assignment accuracy across models are further illustrated in Figures 8 and 9, respectively.

Although the automatic evaluation provides an objective measurement of extraction performance across the complete

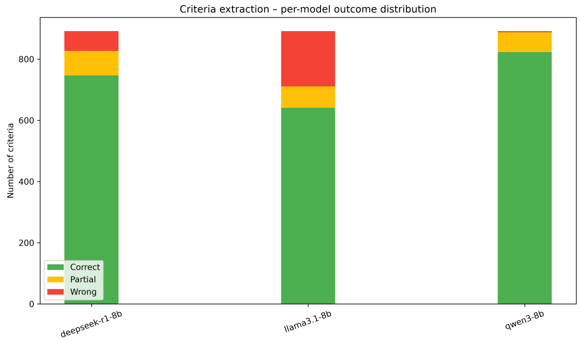

Fig. 8. Eligibility Criteria Extraction - Outcome distribution

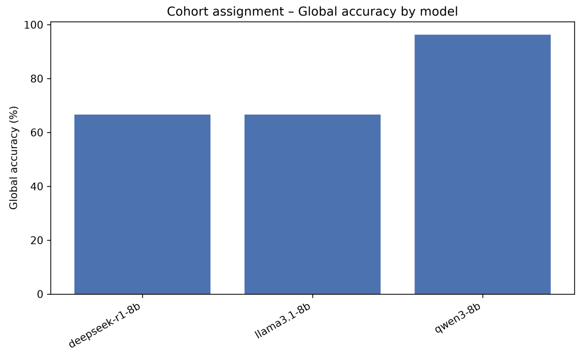

Fig. 9. Eligibility Criteria Extraction - Cohort assignment strict accuracy

dataset, clinical interpretation of eligibility criteria requires assessing whether deviations from the reference representation result in meaningful changes in clinical interpretation. Therefore, a subset of extracted criteria was additionally reviewed by a clinical expert to characterize the severity and nature of possible errors.

The expert assessment confirmed the strong extraction capabilities observed in the automatic evaluation, particularly for Qwen3-8b and Deepseek-r1:8b. Across the evaluated trials, Qwen3-8b consistently preserved the clinical meaning of the eligibility criteria, with observed differences mainly related to acceptable restructuring or expansion of compound criteria. For example, when a broad criterion such as adequate organ function was divided into multiple individual laboratory constraints, the original clinical meaning was maintained and no clinically relevant information was lost.

Deepseek-r1:8b also demonstrated high clinical fidelity in the expert-reviewed subset, achieving complete preservation of inclusion and exclusion criteria in the evaluated trials. The expert analysis confirmed that the model was able to maintain compound requirements, avoid hallucinations, and preserve the relationship between multiple clinical constraints. These findings complement the quantitative results, where Deepseek showed intermediate but consistent extraction performance.

Llama3.1-8b showed a more variable behavior. Although the model successfully extracted most criteria, the expert analysis identified occasional structural errors that could affect downstream processing. In particular, one observed error consisted of splitting a compound tumor-size eligibility criterion into separate entries. Although all relevant information was preserved, this modification altered the original logical structure of the requirement and could potentially affect later conversion into machine-readable rules.

Overall, the expert validation indicated that most model deviations were related to representation differences rather than incorrect clinical interpretation. The main clinically relevant failure modes involved omission of eligibility components, incorrect fragmentation of compound criteria, or loss of relationships between conditions. These observations highlight the importance of maintaining a clinical validation step when transforming natural language eligibility requirements into computational representations.

Eligibility Criteria Conversion

The fourth experimental task evaluated the capability of the LLMs to transform individually extracted clinical trial eligibility criteria into machine-interpretable logical representations while preserving their original clinical meaning. This step represents a critical component of the proposed pipeline, as the generated logical rules enable the deterministic matching engine to perform patient–trial eligibility assessment with reduced dependency on LLM-based reasoning during the final decision process.

The evaluation focused on the conversion of individual inclusion and exclusion criteria previously extracted from clinical trial documents. Generated logical representations were manually evaluated to identify if they kept the clinical meaning of the criteria. Two metrics were reported: strict accuracy, considering only fully correct conversions, and partial accuracy, accounting for cases where the majority of the clinical meaning was preserved but minor structural deviations or incomplete logical representations were present.

It is important to note that the number of evaluated converted criteria differed between models. This difference resulted from occasional generation failures, where models produced explanations, additional text, or invalid outputs instead of the required structured representation, despite explicit instructions to return only the converted criterion. These cases were considered conversion failures and excluded from the correctness assessment, while still reflecting the model's ability to follow the required output format.

As shown in Table VIII, the conversion task represented a considerably more challenging problem compared with previous extraction tasks. Qwen3-8b achieved the highest strict accuracy, correctly converting 651 criteria, corresponding to 64.1% strict accuracy and 74.4% partial accuracy. However, it also produced the highest number of incorrect conversions (156), indicating that while the model was frequently capable of producing complete logical representations, failures tended to result in clinically incorrect rules rather than partially incomplete ones.

Llama3.1-8b achieved comparable performance, with 62.5% strict accuracy and 73.5% partial accuracy. Despite converting fewer criteria than Qwen3-8b, the model demonstrated a similar balance between correct and partially correct outputs. Deepseek-r1:8b presented the lowest strict accuracy (52.2%), although its partial accuracy increased substantially to 71.2%, indicating that many of its errors were related to incomplete logical formalization rather than complete misunderstanding of the eligibility requirement.

Given the moderate performance observed across all 8b-scale models, an additional experiment was conducted using a significantly larger model, Gpt-oss-120b, to assess whether increased model capacity could improve conversion quality.

TABLE VIII
GLOBAL PERFORMANCE OF THE EVALUATED MODELS (ELIGIBILITY CRITERIA CONVERSION)

Model	Total	Correct	Partial	Wrong	Acc./Partial (%)
Llama3.1-8b	969	606	212	151	62.5/73.5
Deepseek-r1:8b	1172	612	444	116	52.2/71.2
Qwen3-8b	1016	651	209	156	64.1/74.4
Gpt-oss-120b	1096	713	348	35	65.1/80.9

The distribution of correct, partial, and incorrect conversions across models is further illustrated in Figure 10.

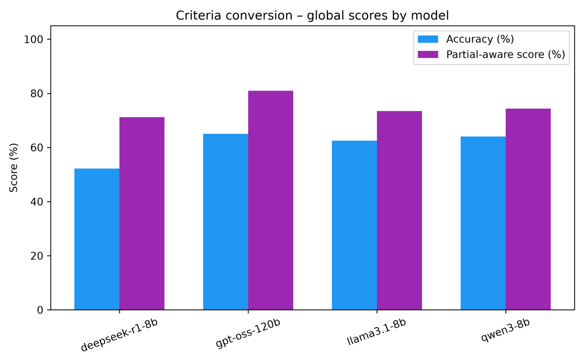

Fig. 10. Eligibility Criteria Conversion

Gpt-oss-120b achieved the best overall results, with 65.1% strict accuracy and 80.9% partial accuracy, and a substantially lower wrong count (35) compared to the smaller models. Notably, the majority of its partial conversions did not stem from failures in capturing complex logical structures, as observed in the other models, but rather from incorrect normalization of criteria fields to the defined schema, particularly the treatment field, resulting in non-schema fields that would require unnecessary LLM-based evaluation downstream.

A common failure pattern observed across all evaluated models was the difficulty in converting complex compound criteria. Eligibility requirements containing nested logical relationships, particularly combinations of multiple AND/OR conditions, frequently resulted in partially represented rules. Instead of preserving the complete logical structure, models often extracted only the most prominent condition while omitting secondary constraints or incorrectly simplifying relationships between conditions.

Another common problem was related to the conversion of said fields to schema fields, specially to the treatment field, the models consistently failed to identify that certain criteria could easily be represented with the patient schema field of treatments, resulting in new, not schema included fields, that would force an LLM evaluation that wouldn't be necessary.

This limitation was particularly relevant because compound eligibility criteria are common in oncology clinical trials, where patient selection often depends on the simultaneous evaluation of diagnosis, molecular profile, previous treatments, laboratory values, and disease status. Therefore, although the models demonstrated reasonable capability in converting simpler eligibility constraints, additional improvements are required for reliably formalizing complex clinical decision rules.

Patient–Trial Matching

The final experimental task evaluated the capability of the LLMs to assess patient eligibility when confronted with clinical requirements that could not be directly represented within the predefined patient profile schema. This step represents the final decision-support component of the proposed pipeline, where the objective is to extend the deterministic matching engine with LLM-based clinical interpretation while maintaining transparency and expert control.

In this scenario, the LLM receives the patient structured profile, the original clinical diary, relevant laboratory information, and an eligibility criterion represented as a logical rule. The model is then required to determine whether the available patient information satisfies the given criterion and provide a clinically grounded justification for the decision.

The evaluation considered two complementary aspects: (i) correctness of the eligibility assessment, manually evaluated, and (ii) quality of the generated justification. Since this task inherently involves clinical reasoning and interpretation of incomplete medical information, a complementary expert assessment was performed on a representative subset of generated outputs. The expert evaluation focused on clinical reliability, evidence grounding, conservative handling of missing information, and potential impact of incorrect decisions.

As shown in Table IX, all models demonstrated strong performance in the eligibility assessment task. Qwen3-8b achieved the highest quantitative performance, correctly evaluating 84 out of 100 criteria, corresponding to a strict and partial

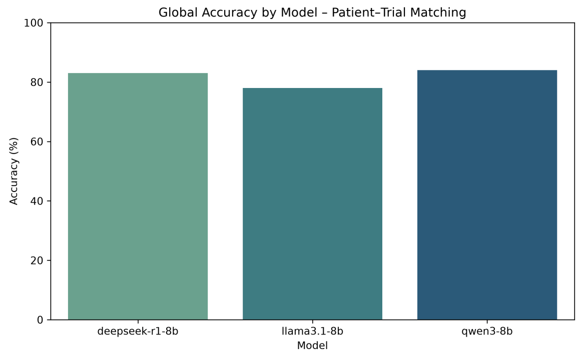

Fig. 11. Patient–Trial Matching - Strict accuracy by model

accuracy of 84.0%. Deepseek-r1:8b achieved a very similar result with 83.0% accuracy, while Llama3.1-8b obtained 78.0%. No partially correct classifications were observed in this task, as incorrect eligibility decisions generally resulted from an incorrect interpretation of the available clinical evidence rather than from partially satisfying the criterion.

TABLE IX
GLOBAL PERFORMANCE OF THE EVALUATED MODELS (PATIENT-TRIAL MATCHING)

Model	Total	Correct	Partial	Wrong	Acc./Partial (%)
Llama3.1-8b	100	78	0	22	78.0/78.0
Deepseek-r1:8b	100	83	0	17	83.0/83.0
Qwen3-8b	100	84	0	16	84.0/84.0

The strict accuracy comparison between models is further illustrated in Figure 11.

Beyond quantitative accuracy, the expert assessment provided additional insight into the clinical reliability of the generated decisions. Overall, Deepseek-r1:8b and Qwen3-8b demonstrated highly consistent behaviour, with both models grounding their decisions in explicit information available in the patient documentation. These models correctly identified measurable lesions, laboratory thresholds, and molecular alterations when present, while appropriately considering missing information as insufficient evidence rather than assuming eligibility or ineligibility.

Qwen3-8b achieved the highest expert-rated performance, with an average justification score of 4.93/5. The model consistently demonstrated conservative reasoning, avoiding unsupported assumptions and correctly handling incomplete clinical information. In particular, it correctly interpreted measurable disease according to lesion size, laboratory thresholds such as ANC requirements, and molecular eligibility criteria. Missing values, such as unavailable creatinine clearance calculations, were appropriately identified as unknown rather than incorrectly classified.

Deepseek-r1:8b achieved an average expert score of 4.8/5, showing similar strengths. The model consistently produced evidence-based decisions and avoided hallucinating clinical information. Its main advantage was the ability to explicitly distinguish between confirmed absence of a criterion and unavailable information, which is particularly important in a clinical decision-support scenario.

Llama3.1-8b achieved an average expert score of 4.2/5. Although the model showed strong performance when information was explicitly documented, the expert evaluation identified occasional over-interpretation of uncertainty. Examples included incorrectly rejecting criteria despite available laboratory values satisfying the requirement or inferring unsupported conclusions from incomplete documentation. These errors were not caused by hallucinated information, but rather by excessive conservatism or speculative interpretation.

The expert-assigned scores for the generated eligibility justifications are summarized in Table X.

TABLE X
EXPERT-ASSIGNED AVERAGE SCORES FOR ELIGIBILITY JUSTIFICATION

Model	Expert Avg. Score
Deepseek-r1:8b	4.8/5
Llama3.1-8b	4.2/5
Qwen3-8b	4.93/5

Overall, these results suggest that small LLMs are capable of performing clinically aligned eligibility assessments when constrained by structured patient information and explicit evidence grounding. Importantly, the expert evaluation indicates that model reliability is not solely determined by classification accuracy, but also by the ability to justify decisions transparently and avoid unsupported clinical assumptions. This reinforces the proposed clinician-in-the-loop approach, where LLMs act as interpretable decision-support tools rather than autonomous clinical decision makers.

DISCUSSION

This pipeline demonstrated that small LLMs (8B parameters) are capable of performing relevant tasks in a complex clinical context, although with different levels of difficulty.

The results obtained present a clear tendency: models performed well in tasks of direct structured extraction, clearly faced their greatest challenges in the formalization of logic, where the worst results were observed across all models, and presented good results in tasks involving conditioned clinical reasoning when explicit context was available.

The performance of the selected trio of models was not homogeneous. Qwen3 clearly shined on extraction tasks, achieving the highest strict accuracy across both hyper-relevant parameter extraction (78.3%) and eligibility criteria extraction (92.3%). Deepseek distinguished itself in trial structure identification (96.7%) and presented strong conservative reasoning in patient–trial matching, while Llama presented the highest variability of results across all tasks.

These experiments showed that small models can support parts of a complex clinical pipeline when the task is properly decomposed and when there is expert oversight. The conclusion is not that one model is universally better, but rather that different models excel in different components, and a properly structured pipeline can leverage these complementary strengths.

Research Hypotheses Assessment

The results obtained allow a direct assessment of the four research hypotheses guiding this study.

H1 is supported by the experimental results. All three models demonstrated the ability to extract and normalize clinically relevant information from heterogeneous oncology diaries while achieving 100% schema compliance. Qwen3-8B achieved the strongest performance with 78.3% strict accuracy, confirming that structured extraction from free-text oncology documentation is feasible with small LLMs, though performance remains dependent on field complexity.

H2 is partially supported. Eligibility criteria extraction yielded strong results, with Qwen3-8B achieving 92.3% strict accuracy, and cohort attribution reaching 96.3% for the same model. However, the subsequent formalization of extracted criteria into machine-interpretable logical rules proved considerably more challenging, with strict accuracies ranging from 52.2% to 64.1% across models. H2 is therefore supported with respect to extraction and preservation of clinical meaning, but only partially supported with respect to the formalization of logical relationships.

H3 is supported by the patient–trial matching results. The hybrid mechanism demonstrated strong performance, with Qwen3-8B and Deepseek-R1-8B achieving 84.0% and 83.0% accuracy respectively, and expert-rated justification quality scores above 4.8 out of 5. The expert evaluation confirmed that decisions were consistently grounded in available clinical evidence, supporting both the effectiveness and the interpretability of the proposed matching strategy.

H4 is broadly supported. Across all evaluated tasks, small open-source models (8B parameters) demonstrated sufficient performance to support the proposed workflow, particularly in extraction and matching tasks. The main exception remains logical formalization, where performance was more limited, suggesting that this specific component may benefit most from larger models or targeted fine-tuning in future work. This was further evidenced by the inclusion of Gpt-oss-120b in the criteria conversion task, where strict accuracy improved to 65.1% and partial accuracy to 80.9%, with a substantially lower wrong count compared to the 8b models.

Strengths of the Proposed Approach

The results suggest that the proposed approach has significant potential as a clinical decision-support tool for patient recruitment in oncology clinical trials. One of its main strengths lies in its modular design. Rather than treating eligibility assessment as a single end-to-end task, the workflow decomposes the problem into a sequence of specialized stages, namely patient information extraction, trial structure identification, eligibility criteria extraction, criteria formalization, and patient–trial matching.

This decomposition reduces the complexity of individual tasks and allows LLMs to focus on well-defined objectives. Consequently, each component can be evaluated, validated, and improved independently, increasing the overall robustness of the system. As can be seen in the experimental results, it allowed even small models to obtain good results, where an end-to-end model would most likely have had serious difficulties, while also making it harder to identify where failures occurred in the first place. These individual evaluations allowed to understand that extraction is not equal to conversion, which also is not equal to matching. A model can be strong on one task and weak on another, Qwen's strength in extraction (92.3% strict accuracy on criteria extraction) and Deepseek's strength in structure identification (96.7%) are a clear example of this. This allows the pipeline to employ the best performing model on the specific task at which it excels, rather than relying on a single model globally.

Another important strength is the high degree of transparency and interpretability provided by the framework. Every intermediate artifact generated by the system, including structured patient profiles, extracted eligibility criteria, logical representations, and matching decisions, remains accessible for inspection. This enables clinicians to understand how eligibility decisions are reached rather than relying on opaque end-to-end model predictions. A system that provides correct outputs but does not explain how is not acceptable in a clinical setting. The expert validation confirmed that Qwen and Deepseek consistently grounded their decisions on available evidence, avoiding hallucinated data, with justification scores of 4.93/5 and 4.8/5, respectively. The clinical justification for a decision is as important as the classification accuracy itself, particularly when the objective is to reduce unnecessary screen time for clinicians.

The clinician-in-the-loop design further contributes to the reliability of the approach. Expert validation is incorporated throughout the workflow, allowing clinicians to review, correct, and approve intermediate outputs before they are used in subsequent stages. This not only increases clinical safety but also mitigates the impact of occasional LLM errors.

Finally, the semi-deterministic matching strategy reduces dependence on probabilistic reasoning by maximizing the use of structured data and rule-based evaluation whenever possible. Unlike many existing approaches that rely on LLMs to directly answer "Is the patient eligible?", the proposed system structures the patient profile, formalizes eligibility criteria into logical rules, and only invokes LLM-based reasoning when the deterministic engine cannot reach a conclusion. This reduces the risk of hallucination, increases explainability, and improves the clinical acceptability of the generated outputs. This approach improves consistency, reproducibility, and auditability while preserving the flexibility required to handle complex clinical eligibility requirements.

Challenges observed during LLM-based clinical processing

Structured extraction vs semantic interpretation

On simple and explicit fields, these models presented excellent results, gender and ECOG-PS achieved 100% accuracy across all three models, whereas on complex fields they faced considerable difficulties, particularly with normalized diagnosis, treatments, dates, and staging. Notably, diagnosis extraction was the most challenging field, with the best result being 46.7% achieved by Deepseek, and a recurrent error pattern in staging where models extracted only the final stage classification (e.g., "IVB") instead of the complete TNM representation (e.g., "cT3N2M1b (IVB)").

These challenges were not solely due to difficulties in extracting the text itself. They were also a consequence of the difficulty in following complex normalization mappings, adhering to clinical ontologies, and performing temporal interpretation, tasks that go well beyond surface-level text extraction.

Preservation of clinical meaning is harder than extraction

The results also showed that models would consistently identify cohort structures in clinical trials, but would lose clinically relevant details when extracting their descriptions. The expert assessment of trial structure confirmed this: despite Deepseek achieving 96.7% structural accuracy, it received an average clinical classification of only 2.25/5, frequently omitting treatment status, molecular alterations, or diagnostic specificity. Similarly, criteria extraction presented strong results across models, while their conversion into logical rules was considerably worse.

The main difficulty was not identifying information, but preserving the relationships between clinical concepts. Compound criteria such as "KRAS G12C inhibitor-naive", "prior FGFR inhibitor exposure", or "primary platinum refractory disease" involve nested clinical dependencies that models frequently simplified or partially omitted.

Logical formalization remains the bottleneck

The logical formalisation of extracted criteria was clearly the main challenge for these small models. While in some cases criteria extraction reached above 90% strict accuracy, Qwen achieving 92.3%, the conversion of these criteria into machine-interpretable logical rules dropped to approximately 64% for the same model. This gap is particularly significant because the conversion step is precisely what enables the deterministic matching engine to operate without LLM-based reasoning.

Small LLMs are effective at interpreting natural language, but face consistent challenges when transforming clinical language into formal logic. They consistently failed to completely represent AND/OR nested rules, dependencies between criteria, and compound eligibility requirements containing multiple simultaneous conditions. Another recurring failure was the inability to map certain criteria to existing schema fields, particularly the treatment field, resulting in the creation of new, schema-excluded fields that would then require additional LLM evaluation, undermining one of the main objectives of the pipeline. This limitation justifies and reinforces the hybrid architecture proposed in this work. The inclusion of Gpt-oss-120b in the conversion task confirmed this bottleneck while also revealing a distinct failure profile: unlike the smaller models, whose partial conversions were predominantly caused by failures in capturing complex logical structures, the larger model's partials were mostly attributable to incorrect normalization of criteria to schema fields, suggesting that logical formalization improves substantially with scale, while schema adherence remains a challenge regardless of model size.

Limitations

Despite the promising results, the proposed approach presents several limitations.

First, as with most LLM-based systems, output quality is highly dependent on the quality and completeness of the input documentation. Missing, ambiguous, or poorly documented clinical information can negatively affect patient profile extraction and, consequently, all downstream tasks.

Second, the workflow follows a sequential architecture, meaning that errors introduced in earlier stages may propagate to subsequent components. Although the decomposed approach of the pipeline and expert validation substantially reduce this risk, they cannot eliminate it entirely, which still proves to be an advantage over an end-to-end approach. The overall performance of the matching process therefore remains partially dependent on the quality of preceding extraction and conversion steps.

Another limitation arises from the formalization of eligibility criteria. While many criteria can be represented through deterministic logical rules, highly complex clinical requirements often involve contextual reasoning, implicit assumptions, or nuanced clinical judgement. Such criteria may be difficult both for LLMs to formalize and for clinicians to validate within a structured representation.

Furthermore, the effectiveness of the matching process depends on the completeness of the extracted patient profile. When relevant information is absent from the clinical diary, eligibility assessment may become inconclusive, limiting the usefulness of the generated recommendations and potentially increasing the amount of manual review required.

The expert validation, although valuable, was conducted on a reduced subset of the dataset and does not substitute a prospective clinical validation. This limits the strength of conclusions drawn from qualitative assessments and highlights the need for more extensive expert evaluation in future work.

Finally, the evaluation was conducted using synthetic clinical diaries and a limited number of clinical trials. Although considerable effort was made to maximize diversity and realism, performance on larger and fully real-world datasets may differ from the results observed in this study.

Clinical Implications

The proposed framework addresses a clinically relevant challenge: the identification of potentially eligible patients for oncology clinical trials. This process is traditionally time-consuming and requires clinicians to manually compare extensive free-text clinical documentation against often complex eligibility criteria.

By automatically extracting relevant patient information, structuring eligibility requirements, and providing preliminary matching assessments, the system has the potential to reduce cognitive burden and administrative workload. This may allow clinicians to dedicate more time to direct patient care and clinical decision-making.

Importantly, the framework was designed as a decision-support system rather than a decision-making system. Final responsibility remains entirely with the clinician, who can inspect, validate, and modify every intermediate result and matching decision. In this sense, the LLM serves as an assistive technology that summarizes and organizes information rather than replacing clinical expertise.

The traceability of all generated artifacts further supports clinical adoption by providing a complete audit trail from the original documents to the final eligibility recommendation.

The entire system was continuously evaluated by oncologists from the Oncology Department of Coimbra University Hospital Centre (HUC) through review meetings. These sessions allowed emerging issues and clinical needs to be promptly addressed, ultimately leading to a high level of clinician satisfaction with the system.

Generalizability

Although the study was conducted in collaboration with the Oncology Department of Coimbra University Hospital Centre (HUC), the proposed methodology was not designed specifically for a single institution.

The patient profile schema captures clinically meaningful concepts commonly used in oncology practice, while the extraction and matching pipelines operate independently of institution-specific document layouts. Consequently, the framework could potentially be applied to clinical documentation originating from other hospitals, provided that the required information is present within the source documents.

The approach is also largely language-agnostic. Since the underlying methodology relies on prompt-guided information extraction and logical formalization, adaptation to other languages would primarily require modifications to the prompts and validation procedures rather than fundamental architectural changes.

Beyond oncology, the same methodological principles could potentially be extended to other medical specialties where patient recruitment for clinical studies depends on the evaluation of structured eligibility criteria against heterogeneous clinical documentation.

Future Work

Several opportunities exist for extending and improving the proposed framework.

First, this study focused exclusively on prompt engineering techniques. Future work could investigate parameter-efficient fine-tuning approaches, such as PEFT methods, as well as Retrieval-Augmented Generation (RAG) architectures to improve both extraction accuracy and eligibility assessment performance.

Second, although the prompts employed throughout the study underwent extensive manual refinement, they were not systematically optimized. More rigorous prompt optimization strategies could potentially yield further improvements across all evaluated tasks.

Third, the patient profile schema was defined collaboratively with the clinical stakeholder to capture the information considered most relevant for eligibility assessment. Future studies may explore richer and more comprehensive representations incorporating additional clinical variables, laboratory measurements, imaging findings, and longitudinal patient information.

Another important direction involves evaluating larger and more capable language models. Due to hardware constraints, the main experiments were limited to open-source models with approximately eight billion parameters. A preliminary evaluation of Gpt-oss-120b on the criteria conversion task suggested that larger models can meaningfully improve logical formalization, though schema normalization challenges persist. Broader evaluation of larger models across all pipeline tasks remains an important direction for future work.

Finally, future validation should be conducted on larger collections of real-world clinical documentation and clinical trials to further assess the robustness, scalability, and clinical applicability of the proposed approach in routine healthcare environments.

CONCLUSION

A. Contributions

This study presents a modular LLM-based pipeline for the identification of potentially eligible patients for clinical trials. The proposed approach decomposes the complex trial–patient matching problem into a sequence of specialized tasks, including the extraction and structuring of patient profiles from free-text clinical diaries, the identification and conversion of clinical trial eligibility criteria into machine-interpretable logical representations, and the final semi-deterministic patient–trial matching process.

The experimental evaluation demonstrated that small open-source LLMs (8B parameters) are capable of performing clinically relevant tasks across all pipeline components, though with varying levels of performance depending on task complexity. Structured extraction tasks yielded the strongest results, with Qwen3-8B achieving 78.3% strict accuracy in patient profile extraction and 92.3% in eligibility criteria extraction. Trial structure identification also produced strong results, particularly for Deepseek-R1-8B, which achieved 96.7% accuracy. In contrast, the logical formalization of eligibility criteria proved to be the most challenging task across all models, with strict accuracies ranging from 52.2% to 64.1%, highlighting that transforming clinical language into formal logic remains an open problem for small LLMs. A supplementary evaluation of Gpt-oss-120b on this task yielded 65.1% strict accuracy and 80.9% partial accuracy, confirming that larger models improve logical formalization, though schema normalization challenges persist. Despite this, the final patient–trial matching stage demonstrated that when grounded in structured patient information and explicit clinical context, these models can perform reliable eligibility assessments, with Qwen3-8B and Deepseek-R1-8B achieving accuracy scores of 84.0% and 83.0% respectively, and expert-rated justification quality scores above 4.8 out of 5.

The modular design of the proposed methodology enables increased transparency, traceability, and interpretability of the generated outputs, which are critical requirements in clinical decision-support systems. By translating clinical trial criteria into logical representations and combining deterministic evaluation with targeted LLM assistance only when required, the system aims to reduce the dependence on the probabilistic reasoning capabilities of LLMs while preserving their flexibility in handling complex clinical information. Importantly, the evaluation confirmed that decomposing the problem into specialized stages not only improved performance but also allowed the identification of where specific failures occurred, something that would not be possible in an end-to-end approach.

Additionally, to enable a realistic evaluation under data privacy constraints, a clinically validated synthetic dataset was developed based on real oncology clinical documentation patterns, avoiding privacy risks while maintaining clinical realism. This dataset allowed the assessment of lightweight open-source language models under limited computational resources, reflecting realistic deployment constraints in healthcare environments.

B. Impact

The proposed system was designed with direct consideration of the needs and workflow constraints of clinical oncologists, positioning the clinician as the final decision-maker rather than replacing expert judgment. Through the incorporation of validation stages, persistent storage of intermediate outputs, and access to the reasoning behind LLM-assisted decisions, the pipeline provides a transparent and auditable framework suitable for clinical decision support. The expert validation conducted throughout the study, including qualitative assessment of clinical justifications, confirmed that the system's outputs are grounded in available evidence and avoid unsupported clinical assumptions.

By reducing the manual effort required to compare extensive clinical documentation against complex trial eligibility criteria, this approach has the potential to decrease clinician cognitive workload and facilitate more efficient identification of suitable clinical trials for patients.

C. Future Work

Although the proposed methodology demonstrates the feasibility of using lightweight LLMs within a clinical trial matching pipeline, several opportunities for further improvement remain. Due to the time and computational constraints of this study, the evaluation was primarily limited to small-scale language models. A supplementary experiment with Gpt-oss-120b on the criteria conversion task provided initial evidence that larger models can meaningfully improve logical formalization performance, though schema normalization challenges were still observed. Future work should further investigate the impact of larger models across all pipeline tasks, as well as more advanced adaptation techniques such as parameter-efficient fine-tuning and retrieval-augmented generation. The observed gap between extraction and logical formalization performance also suggests that targeted fine-tuning on criterion conversion tasks could yield meaningful improvements.

Furthermore, evaluation with larger and more diverse clinical datasets, including real-world anonymized clinical documentation, would provide additional validation of the system's robustness and generalizability. Finally, integration with hospital information systems represents an important future direction, requiring secure infrastructures capable of supporting the processing of real clinical data while maintaining privacy and regulatory compliance.

Overall, this study demonstrates the feasibility of combining structured representations, deterministic logic, and lightweight LLMs to support transparent and clinically aligned patient–trial matching workflows, while also identifying logical formalization as the key remaining bottleneck for future development.

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